
Functional Analysis

— *EYNTKA* Series —

NATHANAEL CHWOJKO-SRAWLEY

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Contents

Contents	3
I Functional Analysis	5
1 Banach and Fréchet Spaces	9
1.1 Banach Spaces	9
1.2 Baire Category Theorem	21
1.3 Dual Spaces	29
1.4 Weak Topologies and TVS's	33
1.5 Fréchet Spaces	38
2 Lebesgue Spaces	43
2.1 Basics	43
2.1.1 Relating Lebesgue spaces: Interpolation	58
2.2 The dual of Lebesgue Space	64
2.3 Some Useful Inequalities	65
2.4 Interpolation: The Riesz-Thorin Theorem	71
2.5 Convolution and Approximate Identities	82
3 Radon Measures and Dual Theory	93
3.1 Positive Linear Functionals on Compact Support Continuous Functions	93
3.2 Dual of Continuous Decaying Functions	98

4	Distribution Theory	101
4.1	Intuition	102
4.2	Definitions	102
4.3	Operations on Distributions	107
4.4	Density Results	110
4.5	Distributional Derivatives of Classical Functions	112
5	Sobolev Spaces	115
5.1	Weak Derivatives and the Spaces $W^{k,p}$	115
5.2	Approximation by Smooth Functions	118
5.3	Extensions and Traces	125
5.4	Sobolev Embeddings	135
5.5	Compactness and the Poincaré Inequality	144
6	Hilbert Spaces	151
6.1	Basic definition and properties	158
6.2	Operators on Hilbert Spaces	172
6.3	Compact Operators	183
6.4	Trace	189
6.5	*Intuition to add	189
7	Spectral Theory	191
7.1	Banach Algebras	191
7.2	Continuous Functional Calculus	195

Part I

Functional Analysis

The transition from Calculus and Linear Algebra to Functional Analysis marks a fundamental shift in mathematical perspective. In classical analysis, we study individual functions, their derivatives, and their integrals. In linear algebra, we study vectors in finite dimensions. *Functional Analysis* we treat the function f itself as a point and look at transformations between spaces of these functions.

Note the term functional historically meant a function F that takes a function and returns a number. We shall show that most linear functionals are integrating against a function as we'll show in section 2.2 and chapter 3), and many important exceptions will be shown to still integrating against distributions.

The Historical Imperative: Taming the Infinite

Historically, the field grew out of a crisis in physics and differential equations. By the early 20th century, physicists dealing with Quantum Mechanics and classical wave equations realized that their problems could not be solved by looking at individual coordinates.

- In Quantum Mechanics, the state of a particle is not a position in $\mathbb{R}^3$, but a wave function ψ . The superposition principle—the fact that if ψ_1 and ψ_2 are states, then $\alpha\psi_1 + \beta\psi_2$ is a state—suggests that these wave functions live in a linear vector space.
- In Partial Differential Equations, viewing the heat or wave equation as an operation on an infinite-dimensional space allows us to decouple the “geometry” of the operator from the specific details of the function.

However, unlike the finite-dimensional world of $\mathbb{R}^n$, where linear algebra is rigid and predictable, infinite-dimensional spaces are wild. The intuitions of Euclidean geometry breaks down: bounded sets are no longer compact, linear maps are not necessarily continuous, and the notion of “convergence” splinters into multiple, distinct topologies.

Functional Analysis is the rigorous attempt to tame this infinite wilderness. It builds a hierarchy of structures to recover the tools of calculus and algebra in the infinite setting.

The Three Pillars of the Theory

The narrative of this course is driven by the tension between three mathematical forces:

1. The Algebraic Structure (Linearity): At its core, we are doing linear algebra. We want to solve equations like $Lx = y$. If L is a matrix, we invert it. If L is a differential operator, we want to do the same. The dream of Functional Analysis is to treat differential operators as “infinite matrices” and apply the Spectral Theorem to diagonalize them. When this works (as in Hilbert Spaces), we can solve complex PDEs with the same ease as a 2×2 matrix equation.
2. The Topological Structure (Continuity): Algebra alone is insufficient in infinite dimensions. In $\mathbb{R}^n$, all norms are equivalent, and continuity is automatic for linear maps. In infinite dimensions, this rigidity evaporates. We must introduce *Topology* via norms and metrics to distinguish between operators that are “well-behaved” (bounded) and those that are explosive (unbounded, like differentiation). The interplay between the algebraic dual space (all linear functionals) and the topological dual space (continuous functionals) becomes a central theme of the theory.

3. The Geometric Structure (Distance and Angle): This is where the “Zoo” of spaces emerges.

- (a) *Hilbert Spaces* (L^2): These are “correct” generalization of finite-dimensional euclidean space. They possess an inner product, allowing us to measure angles and define orthogonality.
- (b) *Banach Spaces* ($L^p, C(K)$): They have a notion of size (norm) but lack a notion of angle (i.e. a linear notion of relatedness). The geometry here can be strange – unit balls can be “pointy” or “flat,” and the shortest path to a subspace might not be unique.
- (c) *Fréchet and Topological Vector Spaces*: necessary for distributions and generalized functions, where the concept of a “norm” is abandoned for families of semi-norms. They are equivalent to invariant metric spaces

There are more general spaces, but they are beyond an introductory text.

Ultimately, Functional Analysis is the study of *Dualities*. It is the duality between a space and the (linear) functions acting on it. We will see that by lifting concrete problems into these abstract spaces, we gain a new vantage point. The goal of this Part is to map this universe, understanding how the shape of the space dictates the physics of the operators within it.

Banach and Fréchet Spaces

(these are all spaces key spaces to consider with normed
(alternative title: Fréchet, Banach, and Hilbert Spaces)

1.1 Banach Spaces

Let k be either $\mathbb{R}$ or $\mathbb{C}$, let V or X denote a vector-space over k , with 0 representing the zero vector. Let kx denote the subspace of X spanned by x . If $M, N \subseteq X$, then $M + N = \{m + n \mid m \in M, n \in N\}$.

Definition 1.1.1: Semi-Norm And Norm

Let V be a vector-space. A *semi-norm* on V is a function $x \mapsto \|x\|$ from V to $[0, \infty)$ satisfying the properties:

1. $\|x + y\| \leq \|x\| + \|y\|$ (the triangle inequality)
2. $\|\lambda x\| = |\lambda| \|x\|$ (homogeneity of scalars)

If furthermore:

- (3) $\|x\| = 0$ if and only if $x = 0$

then the function is called a *norm*

Definition 1.1.2: Normed Vector-Space

Let V be a vector space. Then if there exists a function $\|\cdot\|$ that is defined on V , then $(V, \|\cdot\|)$ is called a *normed vector-space*.

If V is a normed vector-space, we can induce a metric $\rho(x, y) = \|x - y\|$, namely since:

$$\|x - z\| \leq \|x - y\| + \|y - z\| \quad \|x - y\| = \|(-1)(y - x)\| = \|y - x\|$$

Since any metric induces a topology, the induced topology by the metric induced by a norm $\|\cdot\|$ is called the *norm topology* on V . A nice consequence of the induced norm topology is that addition and scalar multiplication is continuous. From the axioms of a semi-norm, $\|0\| = 0$. There are norms for which $x \neq 0$ but $\|x\| = 0$, as we'll soon show. Note that the norm $\|\cdot\| : V \rightarrow \mathbb{F}$ is evidently continuous in the topology it induces¹. If $x_n \rightarrow x$ in X , then for all $\epsilon > 0$, there is an $N \in \mathbb{N}$ such that for all $n \geq N$, $\|x_n - x\| < \epsilon$. By the reverse triangle inequality:

$$|\|x_n\| - \|x\|| \leq \|x_n - x\| < \epsilon$$

showing that $\|x_n\| - \|x\|$ get's arbitrarily close, and so $\|x_n\| \rightarrow \|x\|$, i.e. $\|\cdot\|$ is continuous. This means that the pre-image of $\{0\}$ of $\|\cdot\|$ is closed. Furthermore, it is a vector-space. Denoting

$$K = \{x \in V \mid \|x\| = 0\}$$

Then if $x, y \in K$, $0 \leq \|x + y\| \leq \|x\| + \|y\| = 0$, and $\|cx\| = |c|\|x\| = |c|0 = 0$. Hence, we may consider V/K . What norm to put on V/K that is natural with respect to the projection map $\pi : V \rightarrow V/K$ will be addressed soon.

Two norms $\|\cdot\|_1$ and $\|\cdot\|_2$ are said to be *equivalent* if there exists some $C_1, C_2 > 0$ st

$$C_1\|\cdot\|_1 \leq \|\cdot\|_2 \leq C_2\|\cdot\|_1$$

Equivalent norms define equivalent metrics and the same topology, and so the same Cauchy sequences. Recall that in finite dimensions, all norms are equivalent, meaning there is only one notion of distance. As we'll see, this is not the case in infinite dimensions.

Since we have a metric, we have a notion of completeness. As is the case when working with analysis, we will require our spaces to be complete:

Definition 1.1.3: Banach Space

Let V be a normed vector space. Then V is a *Banach space* if it is complete with respect to it's induced metric.

Example 1.1.4: Banach Spaces

1. Every finite dimensional vector space over $\mathbb{R}$ or $\mathbb{C}$ is a Banach space. In this space, all linear map are continuous.
2. Let K be a compact space, and take the set of all continuous function, $C_0(K)$. Then $C(K)$ is a Banach space with the supremum norm $\|f\|_\infty = \sup_{x \in K} |f(x)|$ (also called uniform

¹Note that in the metric space topology, continuous if and only if sequentially continuous

norm). We will later show (Banach–Mazur theorem) that all real separable Banach spaces are isometric (the isomorphisms in the category of Banach spaces) to a subspace of $C([0, 1])$, giving us a theoretical universal way of thinking about Banach spaces. We shall later see that $C([0, 1])$ has the Approximation property and has a Schauder basis, but there exists Banach spaces without these properties. This shows that for Banach spaces, it is possible to have nice total spaces with “bad” subspaces. In particular, if X is an isometric embedding into $C([0, 1])$ there are not always a nice projection $P : C([0, 1]) \rightarrow X$.

If the space is not separable, if the smallest dense set D of a Banach space X has density of cardinality α , then X is isometric to $C_0([0, 1]^\alpha, \mathbb{R})$ ^a

3. If X is a topological space, then $B(X, k)$ and $BC(X, k)$ (all bounded functionals and bounded continuous functionals) form a Banach space with the uniform norm $\|f\|_u = \sup_{x \in X} |f(x)|$.
4. For p where $1 \leq p \leq \infty$, the space $L^p(X)$ for some measurable space X is a Banach space with the L^p -norm (recall that $L^p(\mu)$ is the set of equivalence classes functions that agree a.e.; if $L^p(\mu)$ represented only functions, then $\|\cdot\|_p$ would be a semi-norm)
5. Two other common examples of Banach spaces we will encounter later will be special subspaces of differentiable functions: Banach Algebras, and Hilbert Spaces.

There are also examples of spaces that are not Banach Spaces:

1. Let X be a locally compact Hausdorff space and consider $C_c(X)$ with the ∞ -norm. Then it is not a Banach space, not being complete. It is easy to see that $C_c(X)$, the set of continuous bounded functions, is a Banach space and contains $C_c(X)$, and so it suffices to find the closure of $C_c(X)$ in $C_b(X)$. The closure turns out to be $C_0(X)$.
2. Let $U \subseteq \mathbb{R}^n$ be open and take $C_0(U, \mathbb{C})$. Then This space is *not* a Banach space
3. The space $H(U, \mathbb{C})$ of all holomorphic functions from U to $\mathbb{C}$ is *not* a Banach Space
4. The space $C_c^\infty(\mathbb{R}^n)$ is *not* a Banach space
5. (If you know some PDE’s) The space of distributions $\mathcal{D}'$ is *not* a Banach Space

These are all examples of a weaker structure known as a *topological vector space* which we shall cover in section 1.4.

^aAn interesting result using Banach-Mazur’s theorem by Luis Rodríguez-Piazza shows that the space of smooth functions with the uniform norm is isometric to the space of nowhere differentiable functions

There is a nice way to check whether a vector space V is a Banach space which requires the following definition (this essentially follows from the fact that continuity if and only if sequential continuity in a metric space and that $\|\cdot\|$ is continuous)

Definition 1.1.5: Absolutely Convergent

Let V be a normed vector space. and $\{x_k\}_k^\infty \subseteq V$ a sequence. Then the sequence is said to *converge absolutely* if

$$\sum_k \|x_k\| < \infty$$

Lemma 1.1.6: Cauchy and Absolute Convergence

Let V be a normed vector space. Then V is complete if and only if every absolutely convergent series in V converges.

The idea is that $\mathbb{R}$ has an order, making it easier to produce many arguments about sequences.

Proof:

Let's first say that X is complete and that $\sum_{n=1}^\infty \|x\| < \infty$. To show it converges, we take advantage of completeness and show there is a Cauchy sequence. In particular, let $S_n = \sum_{k=1}^n x_k$ be the partial sum. Then since the series is absolutely convergent, there for all $n > m$,

$$\|S_n - S_m\| \leq \sum_{m+1}^n \|x_k\| \rightarrow 0 \quad \text{as } m, n \rightarrow \infty$$

so the sequence $\{S_n\}$ is Cauchy, and so converges, so $\{x_k\}$ converges.

Conversely, let's say an absolutely convergent series converges, and choose any Cauchy sequence $\{x_k\}$. Since it's a Cauchy sequence, choose a subsequence such that

$$\|x_{k_{i+1}} - x_{k_i}\| < 2^{-i}$$

Take $y_1 = x_{k_1}$ and $y_i = x_{k_{i+1}} - x_{k_i}$ for $i > 1$ so that $\sum_{i=1}^n y_i = x_{k_n}$. Then notice that $\{y_i\}$ is in fact absolutely convergent:

$$\sum_{i=1}^\infty \|y_i\| \leq \|y_1\| + \sum_{i=1}^\infty 2^{-i} = \|y_1\| + 1 < \infty$$

So $\sum_{i=1}^\infty y_i = \lim x_{k_n}$ exists. Since the subsequence of every Cauchy sequence approaches the same "point" (just like convergent sequences), we see that $\{x_n\}$ also converges to the same point, showing that Cauchy sequences are in fact convergent sequences, as we sought to show.

Next, we would want to study which linear functions between two vector-spaces are *continuous*. It will *not* be the case that all linear functions will be continuous. The following builds up to the classification of continuous linear functions:

Definition 1.1.7: Bounded Linear Map

Let $T : X \rightarrow Y$ be a linear map between two normed vector space. Then if there exists some C such that for all $x \in X$

$$\|T(x)\| \leq C\|x\|$$

Then T is said to be *bounded*, or is a *bounded linear operator*.

Note how this is different then boundedness of a set (ex. the range is bounded if $\|T(x)\| \leq C$). This definition is would be useless to us since no nonzero linear map would satisfy it. All linear map over finite dimensional vector spaces clearly bounded (can you find the C ?), so this concept gets interesting in infinite dimesion

Proposition 1.1.8: Bounded $\Leftrightarrow$ Continuous

Let $T : X \rightarrow Y$ be a linear map between two normed vector spaces. Then the following are equivalent:

1. T is continuous
2. T is continuous at 0
3. T is bounded

Proof:

If T is the zero map, then we're done, so assume $T \neq 0$

(1) implies (2) by definition. For (2) implies (3), Since T is continuous at 0, for for all $\epsilon > 0$, there exists a $\delta > 0$ such that if $\|x - 0\| < \|x\| < \delta$, then $\|T(x) - T(0)\| = \|T(x) - 0\| = \|T(x)\| < \epsilon$. Choosing $\epsilon = 1$, we get that there exists a δ_1 such that

$$\|x\| < \delta_1 \rightarrow \|T(x)\| < 1$$

we know want to show there exists a C such that $\|T(y)\| \leq C\|y\|$ for all y in the domain. We will do this by taking advantage of linearity and shrinking element so that we can take advantage of continuity.

Define $x(y) = \frac{\delta_1}{2\|y\|}y$ so that

$$\|x(y)\| = \left\| \frac{\delta_1}{2\|y\|}y \right\| = \frac{\delta_1}{2} \frac{1}{\|y\|} \|y\| = \frac{\delta_1}{2} < \delta_1$$

Therefore, by continuity and linearity:

$$\|T(x(y))\| = \left\| T \left(\frac{\delta_1}{2\|y\|}y \right) \right\| = \frac{\delta_1}{2\|y\|} \|T(y)\| < 1$$

since this is true for all y , we if we re-arrange we have:

$$\|T(y)\| < \frac{2}{\delta_1} \|y\|$$

showing that T is indeed bounded.

Finally, if $\|T(x)\| \leq C\|x\|$, then for all $\epsilon > 0$, choose $\delta = C^{-1}\epsilon$ so that

$$\|T(y) - T(x)\| = \|T(y - x)\| \leq \epsilon$$

completing the proof.

Note that for $T \in B(X, Y)$ to be an isomorphism, we require that T^{-1} also be bounded². If $\|T(x)\| = \|x\|$ for all x , then T is called an *isometry*.

Since the topology is that of norms, then continuous linear functions between normed vector-spaces are the homomorphisms, meaning they preserve the norm (to the degree of boundedness mentioned in the above proposition) and the vector-space structure (to the degree the map is injective). We will denote the set of bounded linear functions (which we will call linear operators) as $B(X, Y)$, $\mathcal{L}(X, Y)$, or $\text{Hom}_k(X, Y)$, where we understand that since X and Y are normed spaces $\text{Hom}_k(X, Y)$ refers to continuous linear maps.

Like with most fields of mathematics, we are more often more interested in the functions between spaces rather than the spaces itself, for it is function that ultimately give us information about the structure we are working with. Thus, if we can use the techniques we will be developing on $B(X, Y)$, we may gain many interesting results. For this to be the case, we require to introduce a norm on $B(X, Y)$. Naturally, we want that the norm on $B(X, Y)$ to be compatible with X and Y in some way, especially the codomain Y (as we usually investigate the structure of the image of a map). Since we are working with bounded operators, we have that $\|T(x)\| \leq C\|x\|$. Furthermore, since we are working with linear operators, by some scaling we see that we only need to consider when $\|x\| = 1$. This motivates the following norm on $B(X, Y)$:

Definition 1.1.9: Operator Norm

$B(X, Y)$ is a normed vector space with norm^a

$$\begin{aligned} \|T\| &= \sup_{\|x\|=1} \|T(x)\| \\ &= \inf \{C \mid \|T(x)\| \leq C\|x\| \text{ for all } x\} \\ &= \sup_{\|x\| \neq 0} \frac{\|T(x)\|}{\|x\|} \end{aligned}$$

If $\|T(x)\| = \|x\|$, then T is called an *isometry*

^athe third equality holds if $V \neq \{0\}$

The operator norm can be thought of as the smallest constant that will satisfy $\|T(x)\| \leq C\|x\|$ when T is bounded, that is $\|T(x)\| \leq \|T\|\|x\|$. In the special case where we have $B(X, k)$, we write V^* and call it the dual space with the *norm topology*. It is useful to establish when is $B(X, Y)$ a Banach space

²We will show using the open mapping theorem that it suffices to show T is continuous and bijective

Proposition 1.1.10: Completeness Of $B(X, Y)$

Let X, Y be normed vector-space and $B(X, Y)$ the normed vector-space of all bounded linear functions from X to Y . If Y is complete, so is $B(X, Y)$.

Proof:

Let (T_n) be a cauchy sequence in $B(X, Y)$. For each $x \in X$, we have a cauchy sequence $(T_n(x)) \subseteq Y$, and thus a limit vector which we will denote $T(x)$. This defines a function $T : X \rightarrow Y$ which is clearly linear. The inequality

$$\begin{aligned} \|T(x) - T_n(x)\| &= \lim_m \|T_m(x) - T_n(x)\| \\ &\leq (\limsup_m \|T_m - T_n\|) \|x\| \end{aligned}$$

shows that T is bounded and that $T_n \rightarrow T$.

Hence, V^* is always a Banach space, since $\mathbb{R}$ and $\mathbb{C}$ are complete. Next, we see what happens with common vector-space constructions like products and quotients.

Subspaces, Product spaces, Quotient Spaces

There are 3 equivalent norms we can put on the product of two normed vector-spaces:

1. $\|(x, y)\| = \max(\|x\|, \|y\|)$
2. $\|(x, y)\| = \|x\| + \|y\|$
3. $\|(x, y)\| = (\|x\|^2 + \|y\|^2)^{1/2}$

The key to these being equivalent is that we are only multiplying by a finite number of spaces, a different strategy would have to be implemented for norms on infinite dimensional vector spaces (recall the different ℓ^p -norms put on the set of all sequences). For subspace, it is clear that if $W \subseteq V$ is a subspace of a normed vector space V , then W has an induced norm with respect to V . What is more interesting is if V is a Banach space and we want W to be a Banach space. We then require all sequences to converge. Since V is Hausdorff (even metrizable since there is a norm on V), we see that W is complete if W is a *closed subspace* with respect to the topology of V . For quotients V/W , we would like to define a new norm on V/W so that the natural projection map $\pi : V \rightarrow V/W$ is continuous. This leads us to the following definition:

Proposition 1.1.11: Banach Quotient Space

Let X be a normed vector space and $Y \subseteq X$ a subspace of X . Then $\pi : X \rightarrow X/Y$ is the quotient map. This induces the topology on X/Y by defining:

$$\|\pi(x)\| = \|x + Y\| = \inf_{y \in Y} \|x - y\|$$

which is a *seminorm* on X/Y . If Y is closed, then this is in fact a norm, and if furthermore X is Banach, then X/Y is a Banach space.

The intuition in my mind for this definition of quotient norm is to “eliminate” any effect of the quotienting space Y on any element of $x + Y$. Take for example $\mathbb{R}^2/\mathbb{R}$. Then the the norm of $\|(x, y) + \mathbb{R}\|$ should really be $|x|$. Furthermore, π is naturally a continuous map since it is norm-decreasing, meaning $\|\pi(x)\| \leq \|x\|$ (simply choose $C = 1$).

Proof:

For subadditivity, we take advantage of the definition of infimum: for every $x_1, x_2 \in X$, there exists an $\epsilon > 0$ and elements $y_1, y_2 \in Y$ such that

$$\begin{aligned} \|\pi(x_1)\| + \|\pi(x_2)\| + \epsilon &\geq \|x_1 - y_1\| + \|x_2 - y_2\| \\ &\geq \|x_1 + x_2 - (y_1 + y_2)\| \\ &\geq \|\pi(x_1 + x_2)\| \end{aligned}$$

Since ϵ was arbitrary, we have subadditivity. Homogeneity is immediate since π is linear, and hence we have a seminorm.

If $\|\pi(x)\| = \|x + Y\| = 0$, there is a sequence $(y_n) \subseteq Y$ such that $\|x - y_n\| \rightarrow 0$. If Y is closed, $\lim_n(y_n) = x \in Y$, meaning $\pi(x) = 0$. It follows that X/Y is a normed space if and only if Y is closed in X .

Now, say $(z_n) \subseteq X/Y$ is a cauchy sequence. Then we can find a subsequence (z'_n) were

$$\|z'_{n+1} - z'_n\| < 2^{-n}$$

By induction we can choose $x_n \in X$ such that $\pi(x_n) = z'_n$ and $\|x_{n+1} - x_n\| < 2^{-n}$. If X is banach, (x_n) converges to an element $x \in X$, and since π is “norm decreasing” (the norm the same or smaller), then (z'_n) converges to $\pi(x)$. Hence, (z_n) converges to $\pi(x)$, and so X/Y is complete, as we sought to show.

Proposition 1.1.12: Universal Property of Quotient Maps

If $T \in B(X, Y)$ where X and Y are normed spaces, and if $W \subseteq X$ is a closed subspace of X containing $\ker(T)$, then then there exists an operator $\tilde{T} \in B(X/W, Y)$ defined pointwise as $\tilde{T}(\pi(x)) := T(x)$ such that

$$\|\tilde{T}\| = \|T\|$$

in particular, the following diagram commutes in **Ban**:

$$\begin{array}{ccc} X & \xrightarrow{T} & Y \\ \pi \downarrow & \nearrow \tilde{T} & \\ X/W & & \end{array}$$

Proof:

First, by elementary linear algebra we have that $\tilde{T}$ is an operator on X/W , which is a normed space by the earlier proposition. Since

$$\|\tilde{T}(\pi(x))\| = \|T(x)\| = \|T(x - z)\| \leq \|T\| \|x - z\|$$

for every $z \in W$, it follows from the definition of the quotient norm that

$$\|\tilde{T}(\pi(x))\| \leq \|T\| \|\pi(x)\|$$

hence, $\|\tilde{T}\| \leq \|T\|$. The reverse inequality $\|\tilde{T}\| \geq \|T\|$ is clear since Q is norm decreasing.

Proposition 1.1.13: Extending Banach Spaces

Let X be a normed space and Y a closed subspace. Then if Y and X/Y are Banach spaces, then X is a Banach space.

Proof:

Let $(x_n) \subseteq X$ be a Cauchy sequence. Since $\pi : X \rightarrow X/Y$ is norm-decreasing, $(\pi(x_n)) = (y_n) \subseteq X/Y$ is a Cauchy sequence. Since X/Y is complete, it converges to a point $\bar{x} \in X/Y$. Next, by the definition of the quotient norm, for every x_n , there exists an n st

$$\|(x - x_n) - y_n\| \leq \frac{1}{n} + \|\pi(x - x_n)\|$$

Then we get:

$$\begin{aligned} \|y_n - y_m\| &= \|(y_n + x - x_n) - (y_m + x - x_m) + x_n - x_m\| \\ &\geq \frac{1}{n} \|\pi(x - x_n)\| + \frac{1}{m} \|\pi(x - x_m)\| + \|x_n - x_m\| \end{aligned}$$

which is clearly Cauchy, and hence since Y is Banach it is complete and so this sequence converges to y . Finally, it is easy to see that

$$\|x_n - (x + y)\| \leq \|x_n - (x - y_n)\| + \|y - y_n\|$$

showing that $x - n \rightarrow x + y$, as we sought to show.

Finite spaces naturally have none of the problems introduced in infinite dimensions and preserve all of their nice properties even when subspace of infinite dimensional vector space:

Proposition 1.1.14: Finite Dimensional Normed Space

Every finite-dimensional subspace Y of a normed space X is a Banach space, and consequently is closed in X . Moreover, if $\dim(Y) = n$, every linear isomorphism of $\mathbb{F}^n$ onto Y is a homeomorphism

Proof:

p. 46 Analysis now.

Working with infinite dimensional spaces dense subsets become of key interest. As a simple example, the smooth bump functions form a dense subset of L^p for all $1 \leq p < \infty$. The following simplifies for us how we may define functions from spaces given a known dense set:

Proposition 1.1.15: Defining Operator using Dense Sets

Let X and Y be Banach spaces and X_0 is a dense subspace of X . Then every operator $T_0 \in B(X_0, Y)$ has a unique extension to an operator $T \in B(X, Y)$ and $\|T\| = \|T_0\|$

Proof:

For each $x \in X$, there is a sequence $(x_n) \subseteq X_0$ that converges to x : $x_n \rightarrow x$. Since T_0 is a bounded operator ($\|T_0(x_n)\| \leq C\|x_n\|$), it is clear that $(T_0(x_n))$ is a convergent sequence in Y ; let's say it converges to $T(x)$. It should be clear that if we choose another sequence converging to x and do this procedure, then we will get the same value of $T(x)$. Thus, the map $x \rightarrow T(x)$ is a operator that is linear, extends T_0 , and by construction

$$\|T\| = \|T_0\|$$

an excellent example would be polynomial over a compact set are dense over continuous functions over compact sets, which themselves are dense over all L^p spaces. Hence in the compact case, for real or complex functions, it suffices to define our function over the polynomials. For the case of locally compact spaces, Note that smooth bump functions are dense in L^p , and since they have compact support a polynomial sequence uniformly converges to them. Hence, we may define a lot of operators on different real/complex functions spaces over locally compact spaces using polynomials.

For future reference, it will be important to remember that all normed spaces can be completed and that the norm between a normed space X and its completion $\overline{X}$ is isometric:

Proposition 1.1.16: Unique Extension To Banach Space

For each normed space X , there is a Banach space $\overline{X}$, uniquely determined up to isometric isomorphism, such that $\overline{X}$ contains X as a dense subspace

Proof:

p. 47 Analysis now

An important concept in any infinite dimensional space is to figure out which sets are compact. For example, you may recall in real analysis that we proved the Stone-Weierstrass theorem, or even simpler that in $\mathbb{R}^n$ a set is compact if it is closed and bounded. The following shows that subsets that are “almost finite-dimensional” are compact.

Proposition 1.1.17: Compact Sets In Banach Spaces

Let $K \subseteq V$ be a subset of a Banach Space V . Then the following are equivalent:

1. K is compact
2. K is sequentially compact
3. K is closed and bounded, and for every $\epsilon > 0$, K lies in the ϵ -neighborhood

$$\{x \in V \mid \|x - y\| < \epsilon \text{ for some } y \in W\}$$

of a finite dimensional subspace W of V .

Suppose further that there is a nested sequence $V_1 \subseteq V_2 \subseteq \dots$ of finite dimensional subspaces of V such that $\bigcup_{n=1}^{\infty} V_n$ is dense. Then the following statement is equivalent to the first three:

4. K is closed and bounded, and for every $\epsilon > 0$, there exists an n such that K lies in the ϵ -neighborhood of V_n

Proof:
exercise

Example 1.1.18: Compact Sets In Banach Spaces

Let $1 \leq p < \infty$ and take a set $K \subseteq \ell^p(\mathbb{N})$. In order for it to be compact in the norm topology, it needs to be closed, bounded, and also uniformly p^{th} -power integrable at spatial infinity, that is for every $\epsilon > 0$, there exists an $n > 0$ such that

$$\left(\sum_{m>n} |f(m)|^p \right)^{1/p} \leq \epsilon$$

for all $f \in K$. This shows that the moving bump set is not uniformly p^{th} power integrable and thus not a compact subset, even though it's closed and bounded.

This condition doesn't translate to the continuous equivalent $L^p(\mathbb{R})$. We also need some uniform integrability at a "finite scale", which would be described using harmonic analysis tools like Fourier transforms, and hence we simply mention it here without developing it further.

Given the Banach space $\text{End}_k(V)$, it is natural to have a second binary operation of composition and ask if it is also continuous (hence bounded). More generally, if $T \in B(X, Y)$, and $S \in L(Y, Z)$, then

$$\|ST(x)\| \leq \|S\|\|T(x)\| \leq \|S\|\|T\|\|x\|$$

and so $ST \in B(X, Z)$. In particular, that makes $B(X, X) = \text{End}_k(X)$ into an algebra, and when X is complete, $B(X, X)$ is a *Banach Algebra*. More generally:

Definition 1.1.19: Banach Algebra

Let X be a normed k -algebra where addition, multiplication, and scalar multiplication are continuous with respect to the induced topology. Then if X is complete, X is a *Banach Algebra*.

For the multiplication to be continuous, it suffices to check that $\|xy\| \leq \|x\|\|y\|$ (can you see why?). If X has a multiplicative unit, it is called *unital*. Not all Banach algebra's have a multiplicative unit, $C_0(X)$ where X is a locally compact (hausdorff) space and $C_0(X)$ is the set of functions from X to $\mathbb{R}$ (or $\mathbb{C}$) that vanish at infinity is *not* unital. On the otherhand, $C_b(X)$, the set of bounded functions with pointwise operations with the supremum norm *is* unital. We will work more with Banach Algebras later.

The Local Geometry of Banach Spaces

While infinite-dimensional Banach spaces can be geometrically wild (as we will see with L^1 and L^∞), their finite-dimensional subspaces exhibit a surprising regularity. We previously noted that all norms on a finite-dimensional space are equivalent. The following theorem, a cornerstone of the “Local Theory of Banach Spaces,” makes a much stronger claim: Euclidean geometry is “locally universal”.

Theorem 1.1.20: Dvoretzky's Theorem

Let $(X, \|\cdot\|)$ be an infinite-dimensional Banach space. For every integer $n \geq 1$ and every $\epsilon > 0$, there exists a subspace $E_n \subset X$ with $\dim(E_n) = n$ such that E_n is $(1 + \epsilon)$ -isomorphic to the Euclidean space ℓ_2^n .

In terms of the Banach-Mazur distance, this means:

$$d(E_n, \ell_2^n) = \inf \|T\| \|T^{-1}\| \leq 1 + \epsilon$$

where the infimum is taken over all isomorphism $T : E_n \rightarrow \ell_2^n$.

Geometrically, this is often described as the “slicing problem.” If you visualize the unit ball of a Banach space X as a high-dimensional convex body K , Dvoretzky's theorem states that there is always a slice through the center of K (a subspace) that is nearly spherical.

This theorem highlights a tension between local and global structure:

1. *Globally*, X might be very far from a Hilbert space (like ℓ_∞ or the Enflo space).
2. *Locally*, X contains “copies” of Hilbert spaces of all finite dimensions.

Thus, while the “global” geometry of Banach spaces is a vast zoo of structures, the “atomic” local geometry is always Euclidean.

Exercise 1.1.1

1. Let $A, B \subseteq V$ are two closed subspaces. Show that $A + B$ need not be a closed subspace! However, if one of A or B is finite dimensional, then $A + B$ is closed.

2. Show that the open unit ball in X maps onto (by π) the open unit ball of X/Y , but a similar statement about closed unit balls holds only in special cases.
3. For the categorically inclined, it should be verified that the topology on X/Y is the quotient topology corresponding to the equivalent relation $\sim$ where $x_1 \sim x_2$ if and only if $x_1 - x_2 \in Y$.

1.2 Baire Category Theorem

In this section, we introduce some of the basic tools that allow us to work with functions between Banach spaces. Of keen interest to us is the *open mapping theorem*, the *closed graph theorem*, and the *uniform boundedness theorem*. The open mapping theorem makes it sufficient that a surjective continuous linear operator is open, which in particular means that if it is also bijective it is an isomorphism. The closed graph theorem allows us to show that a linear operator $T : X \rightarrow Y$ is continuous if the graph $\Gamma(T)$ is closed, i.e. if all sequences in the graph converge. This will lead to great simplifications in showing a function is continuous. Finally, the uniform boundedness theorem will give a powerful criterion for upgrading pointwise boundedness of a family of operators to uniform boundedness.

The underpinning results we need are about dense sets. Let's say we have a countable family of dense sets, which can be thought of as our "generalized basis". Then it is natural that their intersection may be empty, for example taking linearly independent transcendental elements $r_1, r_2, \dots \in \mathbb{R}$, then $\bigcap_r \mathbb{Q} + r = \emptyset$. What we need is to ensure that the intersection is non-empty. For that, we require that the limit of an infinite intersection exists (ex. our space is LCH or a complete metric space), and that our sets have a nice property that ensures non-emptiness. The following deals with the case of the space being a complete metric space:

Proposition 1.2.1: Baire Category Theorem I

Let (A_n) be a sequence of open dense subset of a complete metric space (X, d) . Then $\bigcap_n A_n$ is dense in X .

Proof:

We want to show that $\bigcap_n A_n$ is dense, so for every ball $B_0 \subseteq X$, $B_0 \cap \bigcap_n A_n \neq \emptyset$.

Let B_0 be any closed ball in X with radius $r > 0$. Then since A_1 is dense in X and is open, the set $A_1 \cap \text{int}(B_0)$ is a nonempty open set, and hence contains a closed ball B_1 with radius at most $r/2$. Since A_2 is dense and open, we can repeat this with $A_2 \cap \text{int}(B_1)$ and find a closed ball B_2 with radius at most $r/2^2$. By induction, we get a sequence of closed balls (B_k) in X such that $B_k \subseteq A_k \cap \text{int}(B_{k-1})$, each with radius at most $r/2^k$. The sequence of centers is Cauchy (since the radii tend to zero and the balls are nested), and since X is complete, there exists a unique point $\{x\}$ such that

$$\{x\} = \bigcap_k B_k \subseteq B_0 \cap \left(\bigcap_k A_k \right)$$

This shows that $\bigcap_k A_k$ intersects every nontrivial ball in X , which implies density, completing the proof.

Note that this is a fully topological proof, and hence is invariant under homeomorphism. In particular,

completeness is *not* a topological property (it is a uniform property), however $(0, 1) \cong_{\text{Top}} \mathbb{R}$. Furthermore, since each U_n is open, its complement must be nowhere dense. For if some non-empty open set O satisfies $O \subseteq \overline{U_n^c} = U_n^c$ (since U_n^c is closed), then $U_n \subseteq O^c$, and so

$$X = \overline{U_n} \subseteq O^c \subsetneq X$$

a contradiction. This leads to the following very useful corollary:

Corollary 1.2.2: Countable union of Empty Interior Sets

The countable union of closed sets with empty interior has empty interior. If X is a complete metric space, $X \neq \bigcup_i \overline{X_i}$ for nowhere dense sets X_i .

As a consequence, if $X = \bigcup_n X_n$, then there exists some n_0 such that X_{n_0} has nonempty interior.

Proof:

Let (F_n) be a countable collection of closed sets with empty interior. Then each F_n^c is open and dense (open since F_n is closed; dense since F_n has empty interior, meaning F_n^c is dense). By the Baire Category Theorem, $\bigcap_n F_n^c$ is dense in X , and in particular nonempty. Hence $\bigcup_n F_n \neq X$, and since $\bigcup_n F_n$ is contained in a countable union of closed sets with empty interior, it itself has empty interior.

If each X_i is nowhere dense, then each $\overline{X_i}$ is closed with empty interior, so $X \neq \bigcup_i \overline{X_i} \supseteq \bigcup_i X_i$. Contrapositively, if $X = \bigcup_n X_n$, then some X_{n_0} must have nonempty interior, for otherwise each $\overline{X_n}$ would be closed with empty interior, contradicting the above.

This consequence is very interesting, since it shows that through a countable process, we require a set with nonempty interior to form our entire space. This motivates the following definition:

Definition 1.2.3: Meager and Generic Set

Let X be a topological space. Then $A \subseteq X$ is said to be *meager* (or of first-category) if it is a countable union of nowhere dense sets. The complement of a meager set is a *residual* (or co-meager, or generic, or of second-category).

Example 1.2.4: Interesting Meager Sets

1. The Cantor set $\mathcal{C}$ is meager in $\mathbb{R}$. Indeed, $\mathcal{C}$ is closed (as the intersection of closed sets in its construction) and has empty interior (it contains no intervals, since at each stage of the construction every remaining interval is split and its middle third removed). Hence $\mathcal{C}$ is nowhere dense, and in particular meager (as a single-term union of a nowhere dense set).
2. Take $C([0, 1])$ with the uniform convergence topology. Let E be the set of all continuous functions that are differentiable somewhere. Then E is in fact a *meager* set in $C([0, 1])$. To see this, take:

$$E_n = \{f \mid \exists x_0 \in [0, 1], |f(x) - f(x_0)| < n|x - x_0| \forall x \in [0, 1]\}$$

Then all differentiable functions are contained in $\bigcup_n E_n$, and it should be shown that this is a

closed set and that there exists functions g such that $\|f - g\|_u < \epsilon$ for any $f \in E_n$ but $g \notin E_n$. Hence, we satisfied the conditions of the Baire Category theorem. As a consequence, the set of continuous nowhere differentiable functions is residual! Thus more generally, smooth and analytic functions are meager.

3. $\mathbb{Q}$ is not a G_δ subset of $\mathbb{R}$. Suppose for contradiction that $\mathbb{Q} = \bigcap_n U_n$ for open sets U_n . Since $\mathbb{Q}$ is dense, each U_n is open and dense. The irrationals $\mathbb{R} \setminus \mathbb{Q}$ can also be written as a countable intersection of open dense sets: $\mathbb{R} \setminus \mathbb{Q} = \bigcap_q (\mathbb{R} \setminus \{q\})$ where q ranges over $\mathbb{Q}$. Then

$$\emptyset = \mathbb{Q} \cap (\mathbb{R} \setminus \mathbb{Q}) = \left(\bigcap_n U_n \right) \cap \left(\bigcap_q (\mathbb{R} \setminus \{q\}) \right)$$

would be a countable intersection of open dense subsets of the complete metric space $\mathbb{R}$. By the Baire Category Theorem, this intersection must be dense, contradicting the fact that it is empty. Hence $\mathbb{Q}$ is *not* a G_δ . Note however that $\mathbb{R} \setminus \mathbb{Q}$ is a G_δ , i.e. the irrationals form a residual set.

The 2nd example above is worth proving in full: the set of continuous functions on $[0, 1]$ that are differentiable at *even one point* is meager in $C([0, 1])$. Equivalently, the “generic” continuous function is nowhere differentiable. Compare this with ??, which gives an explicit construction of a single such function; the Baire category argument produces no explicit example, but shows that *almost every* continuous function (in the sense of category) has this pathological behaviour.

Theorem 1.2.5: Genericity of Nowhere Differentiable Functions

Let $X = C([0, 1])$ equipped with the uniform norm $\|\cdot\|_u$. Then the set of functions in X that are differentiable at some point of $[0, 1]$ is meager. Consequently, the set of continuous nowhere differentiable functions is residual in X .

Proof:

For each $n \in \mathbb{N}$, define:

$$E_n = \{f \in X \mid \exists x_0 \in [0, 1], |f(x) - f(x_0)| \leq n|x - x_0| \forall x \in [0, 1]\}$$

We first check that every function differentiable at some point belongs to $\bigcup_n E_n$. If $f'(x_0)$ exists with $|f'(x_0)| = L$, then there exists $\delta > 0$ such that $|f(x) - f(x_0)| \leq (L + 1)|x - x_0|$ whenever $|x - x_0| < \delta$. For $|x - x_0| \geq \delta$, the bound $|f(x) - f(x_0)| \leq 2\|f\|_u \leq (2\|f\|_u/\delta)|x - x_0|$ holds trivially. Taking $n > \max(L + 1, 2\|f\|_u/\delta)$ gives $f \in E_n$. It now suffices to show each E_n is closed with empty interior.

E_n is closed. Let $(f_k) \subseteq E_n$ converge uniformly to $f \in X$. For each k , choose $x_k \in [0, 1]$ witnessing $f_k \in E_n$, i.e. $|f_k(x) - f_k(x_k)| \leq n|x - x_k|$ for all x . By compactness of $[0, 1]$, pass to a subsequence (still called f_k) with $x_k \rightarrow x_0$. Fix any $x \in [0, 1]$. Then:

$$\begin{aligned} |f(x) - f(x_0)| &\leq |f(x) - f_k(x)| + |f_k(x) - f_k(x_k)| + |f_k(x_k) - f(x_0)| \\ &\leq \|f - f_k\|_u + n|x - x_k| + |f_k(x_k) - f(x_0)| \end{aligned}$$

As $k \rightarrow \infty$, $\|f - f_k\|_u \rightarrow 0$ and $|f_k(x_k) - f(x_0)| \leq \|f_k - f\|_u + |f(x_k) - f(x_0)| \rightarrow 0$ by uniform convergence and continuity of f . Hence $|f(x) - f(x_0)| \leq n|x - x_0|$ for all x , showing $f \in E_n$.

E_n has empty interior. Let $f \in X$ and $\epsilon > 0$. We construct g with $\|f - g\|_u < \epsilon$ and $g \notin E_n$. Since f is uniformly continuous on $[0, 1]$, there exists $\delta > 0$ such that $|f(x) - f(y)| < \epsilon/8$ whenever $|x - y| < \delta$. Choose an integer $M > 4n$ large enough that $w := \epsilon/(2M) < \delta$. Define the zigzag function $\varphi : [0, 1] \rightarrow \mathbb{R}$ to be the continuous piecewise linear function with slope alternating between $+M$ and $-M$ on consecutive intervals of width w , starting from $\varphi(0) = 0$. Since φ oscillates between 0 and $Mw = \epsilon/2$, we have $\|\varphi\|_u \leq \epsilon/2$.

Set $g = f + \varphi$, so $\|f - g\|_u = \|\varphi\|_u \leq \epsilon/2 < \epsilon$. Suppose for contradiction that $g \in E_n$, witnessed by some $x_0 \in [0, 1]$. Then x_0 lies in some monotone piece of φ of width w . Pick x to be the endpoint of this piece farther from x_0 , so that $|x - x_0| \geq w/2$ and φ has constant slope $\pm M$ on the segment from x_0 to x . Since $|x - x_0| \leq w < \delta$:

$$\begin{aligned} |g(x) - g(x_0)| &\geq |\varphi(x) - \varphi(x_0)| - |f(x) - f(x_0)| \\ &= M|x - x_0| - |f(x) - f(x_0)| \\ &> M \cdot \frac{w}{2} - \frac{\epsilon}{8} = \frac{\epsilon}{4} - \frac{\epsilon}{8} = \frac{\epsilon}{8} \end{aligned}$$

On the other hand, $n|x - x_0| \leq nw = n\epsilon/(2M) < \epsilon/8$ since $M > 4n$. Hence $|g(x) - g(x_0)| > n|x - x_0|$, contradicting $g \in E_n$.

Since each E_n is closed with empty interior, $\bigcup_n E_n$ is meager. The set of nowhere differentiable functions contains the complement of $\bigcup_n E_n$ and is therefore residual, completing the proof.

Hence the vast majority (in the sense of Baire category) of continuous functions have this “wild” behaviour: they oscillate so violently at every scale that no tangent line can be drawn at any point. The zigzag perturbation in the proof is a shadow of this oscillation, forced into existence by the Baire Category Theorem without any explicit fractal construction.

The following lemma is the key technical ingredient for the open mapping theorem. It shows that if the image of the unit ball is *dense* in some ball, then it actually *contains* a slightly smaller ball:

Proposition 1.2.6: Open Mapping Lemma

Let $T : V \rightarrow W$ be a bounded linear operator between two Banach spaces, $T \in B(V, W)$. Then if the image of the unit ball $B_1(0)$ is dense in some ball $B_r(0)$ with $r > 0$, i.e. $B_r(0) \subseteq \overline{T(B_1(0))}$, then

$$B_{(1-\epsilon)r}(0) \subseteq T(B_1(0))$$

for every $\epsilon > 0$.

Proof:

Let $U = T(B_1(0))$. Since $T(B_1(0))$ is dense in $B_r(0)$, for any $y \in B_r(0)$ and $\epsilon > 0$, there is a $y_1 \in U$ such that $\|y - y_1\| < \epsilon r$. Consider now ϵU . This set is dense in $B_{\epsilon r}(0)$. Hence, there exists a $y_2 \in \epsilon U$ such that $\|(y - y_1) - y_2\| < \epsilon^2 r$. Continuing by induction, we get a sequence (y_n) where $y_n \in \epsilon^{n-1}U$ and

$$\left\| y - \sum_{k=1}^n y_k \right\| < \epsilon^n r$$

Choose now $x_n \in B_1(0)$ such that $T(x_n) = y_n$ and $\|x_n\| \leq \epsilon^{n-1}$. Then $\sum x_n = x \in X$ converges,

and by definition $T(x) = y$. Since $\|x\| \leq \sum \epsilon^{n-1} = (1 - \epsilon)^{-1}$ ^a we have

$$(1 - \epsilon)^{-1}U \supseteq B_r(0)$$

completing the proof.

^arecall that $1 - b^n = (1 - b)(1 + b + b^2 + \dots + b^{n-1})$

Theorem 1.2.7: Open Mapping Theorem

Let X and Y be Banach spaces and $T \in B(X, Y)$ be a bounded linear operator with $T(X) = Y$. Then T is an open map (i.e. T maps open sets to open sets)

Proof:

Since T is surjective:

$$Y = T(X) = \bigcup_n T(B_n(0)) \subseteq \bigcup_n \overline{T(B_n(0))} = \bigcup_n \overline{nT(B_1(0))}$$

Then by corollary 1.2.2, not all closed sets $\overline{nT(B_1(0))}$ have an empty interior. Since scaling by n is a homeomorphism, it follows that $\overline{T(B_1(0))}$ itself has nonempty interior. Hence, there exist $y_0 \in Y$ and $\delta > 0$ such that

$$B_\delta(y_0) \subseteq \overline{T(B_1(0))}$$

We now show $B_{\delta/2}(0) \subseteq \overline{T(B_1(0))}$. For any z with $\|z\| < \delta$, both y_0 and $y_0 + z$ lie in $B_\delta(y_0) \subseteq \overline{T(B_1(0))}$. Hence:

$$z = (y_0 + z) - y_0 \in \overline{T(B_1(0))} - \overline{T(B_1(0))} \subseteq \overline{T(B_1(0)) - T(B_1(0))} \subseteq \overline{T(B_2(0))} = 2\overline{T(B_1(0))}$$

where we used linearity of T in the last two steps. It follows that $B_\delta(0) \subseteq 2\overline{T(B_1(0))}$, i.e. $B_{\delta/2}(0) \subseteq \overline{T(B_1(0))}$, so $T(B_1(0))$ is dense in $B_{\delta/2}(0)$.

By the Open Mapping Lemma (proposition 1.2.6), $B_s(0) \subseteq T(B_1(0))$ for every $s < \delta/2$. Since every open set $U \subseteq X$ is a union of open balls, we conclude from linearity of T that $T(U)$ contains a neighborhood around each of its points, thus $T(U)$ is open, completing the proof.

Corollary 1.2.8: Bijective And Continuous

Let $T : X \rightarrow Y$ be a bijective bounded operator. Then it has a bounded inverse, that is T is in fact a Banach-space isomorphism

The following also lets us recover a result similar to the one showing us that there is only one norm on a finite dimensional vector space up to equivalence, namely no (non-equivalent) norm on a Banach space can dominate another:

Corollary 1.2.9: Equivalent Banach Spaces

Let X be a vector-space, and a Banach space under two norms $\|\cdot\|_a$ and $\|\cdot\|_b$. Then if $\|\cdot\|_a \leq \alpha\|\cdot\|_b$ for some $\alpha > 0$, then there is a $\beta > 0$ such that $\|\cdot\|_b \leq \beta\|\cdot\|_a$

Proof:

Take $\text{id} : (X, \|\cdot\|_b) \rightarrow (X, \|\cdot\|_a)$. This is certainly bijective, continuous by the boundedness assumption, and open by the open mapping theorem. Hence it is a homeomorphism, meaning it has a bounded inverse, giving the desired result.

The next powerful result we get is a quick way of proving a function is bounded:

Theorem 1.2.10: Closed Graph Theorem

Let $T : X \rightarrow Y$ be an operator between Banach spaces. Then if the graph

$$\Gamma(T) = \{(x, y) \in X \times Y \mid T(x) = y\}$$

is closed in $X \times Y$, then T is bounded

Proof:

Using the ∞ -norm $\|(x, y)\| = \max(\|x\|_X, \|y\|_Y)$ on $X \times Y$, assume that $\Gamma(T)$ is closed in $X \times Y$. Since $X \times Y$ is a Banach space (as both X and Y are), and $\Gamma(T)$ is a closed linear subspace, $\Gamma(T)$ is itself a Banach space. Now consider the projection map $\pi_1 : \Gamma(T) \rightarrow X$ given by $\pi_1(x, T(x)) = x$. This is bijective (since T is a function) and norm-decreasing: $\|\pi_1(x, T(x))\| = \|x\| \leq \max(\|x\|, \|T(x)\|)$. Since π_1 is a bounded bijection between Banach spaces, the Banach isomorphism theorem (corollary 1.2.8) gives that π_1^{-1} is bounded. The other projection $\pi_2 : \Gamma(T) \rightarrow Y$ given by $\pi_2(x, T(x)) = T(x)$ is also norm-decreasing by the same reasoning. Hence:

$$T = \pi_2 \circ \pi_1^{-1}$$

is a composition of bounded operators, and therefore bounded, completing the proof.

The key take-away is that if we are verifying the continuity of a linear map $T : X \rightarrow Y$ (so $x_n \rightarrow x$ implies $T(x_n) \rightarrow T(x)$), we just need to verify the closedness of the graph. The closedness of the graph implies that if $x_n \rightarrow x$ and $T(x_n) \rightarrow y$, then $T(x) = y$. Notice how this simplifies proving a function is continuous, since we did not need to show that $\lim_n T(x_n) = T(x)$, simply that it converges to *something*, namely if $T(x_n) \rightarrow y$, then $T(x) = y$. Note how completeness is crucial in both these theorems (exercise 29-31 folland for counter-examples).

The converse of the closed graph theorem, that if $T : X \rightarrow Y$ is bounded between Banach spaces, then $\Gamma(T)$ is closed, is elementary: let's say T is bounded and that $T(x_n) \rightarrow y$. Since $x_n \rightarrow x$, $\lim_n T(x_n) = T(x)$. Then:

$$\begin{aligned} \|T(x) - y\|_Y &\leq \|T(x) - y + T(x_n) - T(x_n)\|_Y \\ &\leq \|T(x - x_n)\|_Y + \|T(x_n) - y\|_Y \\ &\leq C\|x - x_n\|_X + \|T(x_n) - y\| \xrightarrow{n \rightarrow \infty} 0 \end{aligned}$$

hence, $\|T(x) - y\| = 0$, implying $T(x) = y$ and showing $\Gamma(T)$ is closed.

Finally, the following result is a way to pass from pointwise boundedness to uniform boundedness, only requiring the completeness of our domain! It can be thought of as one of the “pillars” of functional analysis:

Theorem 1.2.11: Principle Of Uniform Boundedness

Let X, Y be normed vector spaces and $A \subseteq L(X, Y)$. Then if X is a Banach space and $\sup_{T \in A} \|T(x)\| < \infty$ for all $x \in X$, then $\sup_{T \in A} \|T\| < \infty$

This theorem is also called by some the Banach–Steinhaus Theorem.

Proof:

We want to upgrade from a pointwise bound to a uniform bound, showing that $\sup_{T \in A} \|T\| < \infty$. Let

$$E_n = \left\{ x \in X \mid \sup_{T \in A} \|T(x)\| \leq n \right\} = \bigcap_{T \in A} \{x \in X \mid \|T(x)\| \leq n\}$$

Each E_n is closed (as an intersection of closed sets, since each T is continuous) and $X = \bigcup_n E_n$ (since $\sup_{T \in A} \|T(x)\| < \infty$ for each x). Then by the Baire Category Theorem, there must exist some $n \in \mathbb{N}$ such that E_n has nonempty interior. Hence, for some $x_0 \in E_n$ and $r > 0$, $\overline{B_r(x_0)} \subseteq E_n$.

Now, consider $u \in X$ such that $\|u\| \leq 1$ and take any $T \in A$. Since $x_0 + ru \in \overline{B_r(x_0)} \subseteq E_n$, we have $\|T(x_0 + ru)\| \leq n$. Also $x_0 \in E_n$, so $\|T(x_0)\| \leq n$. Then:

$$\begin{aligned} \|T(u)\|_Y &= r^{-1} \|T(ru)\|_Y = r^{-1} \|T(x_0 + ru) - T(x_0)\|_Y \\ &\leq r^{-1} (\|T(x_0 + ru)\|_Y + \|T(x_0)\|_Y) \\ &\leq r^{-1} (n + n) \end{aligned}$$

Hence, taking the supremum over u in the unit ball of X and over $T \in A$, we get:

$$\sup_{T \in A} \|T\|_{B(X, Y)} \leq 2r^{-1}n < \infty$$

completing the proof.

If X is simply a normed vector-space, we may keep this theorem by simply taking the set supremum of the set of nonmeager points. The key realization is that $\sup_{T \in A} \|T(x)\|$ is a different supremum for each x , but is finite for all x ! That is, we are not working with $\sup_{T \in A} \sup_{x \in X} \|T(x)\|$ since that would make the result trivial:

$$\sup_{T \in A} \|T\| = \sup_{\substack{T \in A \\ \|x\|=1}} \|T(x)\| \leq \sup_{\substack{T \in A \\ x \in X}} \|T(x)\|$$

This is what makes the Uniform boundedness principle so strong, we got a *uniform* bound by having a *local* bound for all x !

Example 1.2.12: Divergence of Fourier Series

One of the most celebrated applications of the uniform boundedness principle is the existence of continuous functions whose Fourier series diverges at a given point. Consider $X = C(\mathbb{T})$, the Banach space of continuous 2π -periodic functions with the supremum norm. For each n , define the functional $\Lambda_n : C(\mathbb{T}) \rightarrow \mathbb{C}$ by

$$\Lambda_n(f) = S_n f(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) D_n(t) dt$$

where $S_n f(0)$ is the n -th partial sum of the Fourier series of f evaluated at 0, and $D_n(t) = \sum_{k=-n}^n e^{ikt}$ is the Dirichlet kernel. Each Λ_n is a bounded linear functional with $\|\Lambda_n\| = \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_n(t)| dt = \|D_n\|_{L^1}$. It is a classical computation that $\|D_n\|_{L^1} \sim \frac{4}{\pi^2} \log n \rightarrow \infty$. Hence $\sup_n \|\Lambda_n\| = \infty$. By the *contrapositive* of the uniform boundedness principle, there must exist some $f \in C(\mathbb{T})$ for which $\sup_n |\Lambda_n(f)| = \sup_n |S_n f(0)| = \infty$, i.e. the Fourier series of f diverges at 0. By translation, the same holds at any prescribed point.

Note how the principle is used here: we showed $\sup_n \|\Lambda_n\| = \infty$, and concluded that the pointwise boundedness hypothesis must fail for some f , even though we never constructed such an f explicitly. This is a quintessential non-constructive existence proof via Baire category methods.

Corollary 1.2.13: Pointwise Convergence of Operators

Let X be a Banach space, Y a normed vector space, and $(T_n)_{n \in \mathbb{N}} \subseteq B(X, Y)$ a sequence of bounded operators such that $(T_n(x))_{n \in \mathbb{N}}$ converges in Y for every $x \in X$. Define $T(x) = \lim_n T_n(x)$. Then $T \in B(X, Y)$, i.e. the pointwise limit is itself a bounded linear operator.

Proof:

First, T is linear: for $x, y \in X$ and α, β scalars, $T(\alpha x + \beta y) = \lim_n T_n(\alpha x + \beta y) = \alpha \lim_n T_n(x) + \beta \lim_n T_n(y) = \alpha T(x) + \beta T(y)$.

Since $(T_n(x))$ converges for each x , it is bounded: $\sup_n \|T_n(x)\| < \infty$ for all $x \in X$. Hence, by the principle of uniform boundedness, $\sup_n \|T_n\| = C < \infty$. That is, $\|T_n(x)\| \leq C\|x\|$ for all n and all $x \in X$, where C is independent of n . Taking $n \rightarrow \infty$ we get:

$$\|T(x)\| \leq C\|x\|$$

which shows $T \in B(X, Y)$, completing the proof.

More generally, if $(T_\lambda)_{\lambda \in \Lambda} \subseteq B(X, Y)$ is a net such that each $(T_\lambda(x))$ is bounded and converges for every $x \in X$, the same conclusion holds. For sequences, convergence automatically implies boundedness. It also suffices in the principle of uniform boundedness to know that each sequence $(T_n(x))$ is bounded and that the sequence converges for a dense set of elements in X .

We conclude this section with an application of the Baire Category Theorem that does not involve linear operators. It shows that the set of discontinuities of a pointwise limit of continuous functions is, in a precise sense, small:

Theorem 1.2.14: Continuity of Pointwise Limits

Let (f_n) be a sequence of continuous (complex-valued) functions on a complete metric space X such that $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ exists for every $x \in X$. Then the set of points at which f is continuous is a dense G_δ (and hence residual) in X .

Proof:

Recall that f is continuous at x if and only if the *oscillation* of f at x is zero, where

$$\omega_f(x) = \inf_{\delta > 0} \sup_{y, z \in B_\delta(x)} |f(y) - f(z)|$$

Thus the set of continuity points of f is $\{x : \omega_f(x) = 0\} = \bigcap_{m=1}^{\infty} \{x : \omega_f(x) < 1/m\}$. Each set $O_m = \{x : \omega_f(x) < 1/m\}$ is open (this follows from the definition of oscillation), so the set of continuity points is a G_δ . It remains to show that each O_m is dense.

Fix $m \geq 1$ and let B be any nonempty open ball in X . We must find a point in $B \cap O_m$. For each $n, k \in \mathbb{N}$, define

$$F_{n,k} = \left\{ x \in \overline{B} \mid |f_n(x) - f_k(x)| \leq \frac{1}{3m} \right\}$$

Each $F_{n,k}$ is closed (since $f_n - f_k$ is continuous). Since $f_n(x) \rightarrow f(x)$ for every x , for each $x \in \overline{B}$ there exists N such that $|f_n(x) - f_k(x)| \leq 1/(3m)$ for all $n, k \geq N$. Hence:

$$\overline{B} = \bigcup_{N=1}^{\infty} \bigcap_{n,k \geq N} F_{n,k}$$

The sets $G_N = \bigcap_{n,k \geq N} F_{n,k}$ are closed in $\overline{B}$. Since $\overline{B}$ is a complete metric space (closed subset of X), the Baire Category Theorem implies that some G_{N_0} has nonempty interior in $\overline{B}$. Hence there exists an open ball $B' \subseteq B$ such that $|f_n(x) - f_k(x)| \leq 1/(3m)$ for all $n, k \geq N_0$ and all $x \in B'$.

Taking $k \rightarrow \infty$, we get $|f_n(x) - f(x)| \leq 1/(3m)$ for all $n \geq N_0$ and $x \in B'$. In particular, f_{N_0} is continuous, so we may shrink B' so that f_{N_0} varies by at most $1/(3m)$ on B' . Then for any $y, z \in B'$:

$$|f(y) - f(z)| \leq |f(y) - f_{N_0}(y)| + |f_{N_0}(y) - f_{N_0}(z)| + |f_{N_0}(z) - f(z)| \leq \frac{1}{3m} + \frac{1}{3m} + \frac{1}{3m} = \frac{1}{m}$$

Hence any point of B' lies in O_m , showing that O_m is dense. Since each O_m is open and dense, $\bigcap_m O_m$ is a dense G_δ by the Baire Category Theorem.

As a consequence, if $f = \lim f_n$ is a pointwise limit of continuous functions (such an f is called a *Baire class 1* function), then f cannot be discontinuous everywhere: its set of continuity points is always residual. This provides another perspective on why “most” functions are badly behaved: a generic continuous function is nowhere differentiable (example 1.2.4), but any pointwise limit of continuous functions must still be continuous on a large set.

1.3 Dual Spaces

Let V be a Banach space over k (k being either $\mathbb{R}$ or $\mathbb{C}$). Since we are in the category of Banach spaces, we'll consider $V^* := \text{Hom}(V, k)$ to be the collection of all continuous linear functionals (i.e. bounded linear functionals). If V is finite dimensional, then it is clear what the structure of V^* , namely since $V \cong V^*$. In infinite dimensions, it is in fact hard to identify when a functional is continuous and bounded. Famously, d/dx is an unbounded operator, and $\int$ is bounded given an appropriate restriction of the domain. It would be good if we can find some easier type of function that we know can be extended to a functional. We start by doing just that, in particular we start by proving the Hahn-Banach extension theorem, which shows that there exists “enough” linear functions in V^* to make their study interesting (this theorem is the equivalent of the Uryohn's Theorem from topology).

The key “simpler” function is the following:

Definition 1.3.1: Linear Subfunctional

Let $m : V \rightarrow \mathbb{R}$ be a subadditive function $m(x+y) \leq m(x) + m(y)$ that is positive homogeneous $m(tx) = tm(x)$ for all $t \geq 0$ in $\mathbb{R}$. Then m is called a *linear subfunctional*.

Note that it is not a semi-norm since homogeneity only needs to hold for positive scalars. On the otherhand, every semi-norm (and hence norms) are example of minkowski functionals. Note that the result works just as well over real or complex fields:

Proposition 1.3.2: Converting Complex And Real Linear Functionals

Let X be a vector space over $\mathbb{C}$. If f is a complex linear functional on X , and $u = \operatorname{Re} f$, the nu is a real linear functional, and $f(x) = u(x) - iu(ix)$ for all $x \in X$. Conversely, if u is a real linear functional on X and $f : X \rightarrow \mathbb{C}$ is defined by $f(x) = u(x) - iu(ix)$, then f is a complex linear functional. In this case, if X is normed, $\|u\| = \|f\|$

Proof:

If f is complex linear then $u = \operatorname{Re} f$ is clearly real linear and $\operatorname{Im} f(x) = -\operatorname{Re}[if(x)] = -u(ix)$, so $f(x) = u(x) - iu(ix)$. On the other hand, if u is real linear, then $f(x) = u(x) - iu(ix)$ is clearly $\mathbb{C}$ linear and

$$f(ix) = u(ix) - iu(-x) = u(ix) + iu(x) = if(x)$$

thus f is $\mathbb{C}$ -linear. Finally, if X is normed since $|u(x)| = |\operatorname{Re} f(x)| \leq |f(x)|$, $\|u\| \leq \|f\|$. On the other hand, if $f(x) \neq 0$, let $\alpha = \operatorname{sgn}(f(x))$. Then

$$|f(x)| = \alpha f(x) = f(\alpha x) = u(\alpha x)$$

since $f(\alpha x)$ is real. Hence $|f(x)| \leq \|u\| \|\alpha x\| = \|u\| \|x\|$, thus $\|f\| \leq \|u\|$, hence

$$\|f\| = \|u\|$$

completing the proof.

Theorem 1.3.3: Hahn-Banach Extension theorem

Let X be a real vector space, p a sublinear functional on X , $M \subseteq X$ a subspace of X , and f a linear functional on M such that $f(x) \leq p(x)$ for all $x \in M$. Then there exists a linear functional F on X such that $F(x) \leq p(x)$ for all $x \in X$ and $F|_M = f$

Proof:

We shall start with a “base case” and then use Zorn’s lemma (in this context, think of it as transfinite induction). Let $x \in X \setminus M$. We shall extend f to a functional g on $M + \mathbb{R}x$ that satisfies $g(y) \leq p(y)$ on $M + \mathbb{R}x$. First, for any $y_1, y_2 \in M$, we have

$$f(y_1) + f(y_2) = f(y_1 + y_2) \leq p(y_1 + y_2) \leq p(y_1 - x) + p(x + y_2)$$

where x is our above chosen $x \in X$. Re-arranging, we get:

$$f(y_1) - p(y_1 - x) \leq p(x + y_2) - f(y_2)$$

Since this is true for all $y_1, y_2 \in M$, we have:

$$\sup_{y \in M} \{f(y) - p(y - x)\} \leq \inf_{y \in M} \{p(x + y) - f(y)\}$$

Thus, we can find an α satisfying:

$$\sup_{y \in M} \{f(y) - p(y - x)\} \leq \alpha \leq \inf_{y \in M} \{p(x + y) - f(y)\}$$

Define

$$g : M + \mathbb{R}x \rightarrow \mathbb{R} \quad g(y + \lambda x) = f(y) + \lambda \alpha$$

Clearly g is linear, $g|_M = f$, and on M $g(y) \leq p(y)$ (equivalently when $\lambda = 0$). When $\lambda > 0$, then for any $y \in M$:

$$g(y + \lambda x) = \lambda[f(y/\lambda) + \alpha] \leq \lambda[f(y/\lambda) + p(x + (y/\lambda)) - f(y/\lambda)] = p(y + \lambda x)$$

and when $\lambda = -\mu < 0$:

$$g(y + \lambda x) = \mu[f(y/\mu) - \alpha] \leq \mu[f(y/\mu) - f(y/\mu) + p((y/\mu) - x)] = p(y + \lambda x)$$

Thus, $g(z) \leq p(z)$ for all $z \in M + \mathbb{R}x$.

But now, notice that g satisfies the same condition as in the hypothesis: g is a linear functional on a subspace $M + \mathbb{R}x \subseteq X$ such that $g(z) \leq p(z)$ for all $z \in M + \mathbb{R}x$. We can thus finish off by noticing that we can create a partial order given by inclusion of the domains (since the domains restrict appropriately), and we clearly see that this partial is closed under unions of chains. Hence, it has a maximal, which through an easy proof by contradiction can be shown to be X , completing the proof.

Before moving on to some of the fascinating consequences, we first insure that we understand how to move this result to complex functionals. For order to be well-defined, we must insure that we are using seminorms instead of general linear subfunctionals, that is if p is a seminorm and $f : X \rightarrow \mathbb{R}$ is linear, then the inequality $f \leq p$ is equivalent to $|f| \leq p$ since $|f(x)| = \pm f(x) = f(\pm x)$ and $p(-x) = p(x)$.

Proposition 1.3.4: Complex Hahn-Banach Theorem

Let X be a vector space over $\mathbb{C}$, p a seminorm on X , $M \subseteq X$ a subspace of X and f a complex linear functional on M such that $|f(x)| \leq p(x)$ for all $x \in M$. Then there exists a complex linear functional F on X such that $|F(x)| \leq p(x)$ for all $x \in X$ and $F|_M = f$.

Proof:

Let $u = \operatorname{Re} f$. By The Hahn-Banach Theorem there exists a linear extension of u to X such that $|U(x)| \leq p(x)$ for all $x \in X$. Let $F(x) = \overline{U(x)} - iU(ix)$. Then F is a complex linear extension of f , and by proposition 1.3.2, if $\alpha = \operatorname{sgn}(f(x))$ we have $|f(x)| = \alpha F(x) = F(\alpha x) = U(\alpha x) \leq p(\alpha x) = p(x)$, completing the proof.

Proposition 1.3.5: Hahn-Banach Consequences

Let X be a normed vector-space. Then:

1. If $M \subseteq X$ is a closed subspace and $x \in X \setminus M$, there exists a $f \in X^*$ such that $f(x) \neq 0$ and $f|_M = 0$. In particular, if $\delta = \inf_{y \in M} \|x - y\|$, there exists an f such that

$$\|f\| = 1 \quad f(x) = \delta$$

2. If $0 \neq x \in X$, there exists $f \in X^*$ such that $\|f\| = 1$ and $f(x) = \|x\|$
3. The bounded linear functionals on X separate points
4. if $x \in X$, the double dual $\hat{x} : X^* \rightarrow \mathbb{C}$ given by $\hat{x}(f) = f(x)$ is a linear isometry from X to X^{**}

Proof:

1. $M \subseteq X$ be a closed subspace and $\delta = \inf_{y \in M} \|x - y\|$. Define

$$f : M + \mathbb{C}x \rightarrow \mathbb{C} \quad f(y + \lambda x) = \lambda \delta$$

Then $f(x) = \delta$, $f|_M = 0$, and for $\lambda \neq 0$ by the infimum property:

$$|f(y + \lambda x)| = |\lambda| \delta \leq |\lambda| \|\lambda^{-1}y + x\| = \|y + \lambda x\|$$

Hence, taking $p(x) = \|x\|$ to be the norm (which is a linear subfunctional defined on X), by the Hahn-Banach theorem F on X where $F(x) = \delta$ and $F|_M = 0$. Finally, if $\|y + \lambda x\| = 1$, then certainly the supremum of all such values when plugged into F must be 1, hence $\|F\| = 1$

Alternatively, apply the result from part (a) to X/Y , then note that functionals in $(X/Y)^*$ may be regarded as elements in X^* that annihilate Y

2. In the above, let $M = \{0\}$ be the trivial vector-space and define $f(x) = \|x\|$.
3. If $x \neq y$, by (b) there exists a $f \in X^*$ such that $f(x - y) \neq 0$, hence $f(x) \neq f(y)$
4. Certainly $\hat{x}$ is a linear functional on X^* and $x \mapsto \hat{x}$ is linear. Furthermore

$$|\hat{x}(f)| = |f(x)| \leq \|f\| \|x\|$$

hence $\|\hat{x}\| \leq \|x\|$. On the other hand, by (b), we get $\|\hat{x}\| \geq \|x\|$, hence we have an isometry

The first (and second) point can be thought of as a generalization of Uryshon's lemma, and also inspires the following definition:

Definition 1.3.6: Annihilator

For a subspace $Y \subseteq X$ of a normed space X , the *annihilator* of Y is defined as:

$$Y^\perp = \{\varphi \in X^* \mid \varphi(y) = 0, \forall y \in Y\}$$

Similarly, we define the annihilator of a subspace $Y^* \subseteq X^*$ to be

$$(Y^*)^\perp = \{x \in X \mid \varphi(x) = 0, \forall \varphi \in Y^*\}$$

It is immediate that $Y \subseteq (Y^\perp)^\perp$, and it follows that in fact $Y = (Y^\perp)^\perp$ for every norm closed subspace $Y^* \subseteq X^*$.

The third point is useful if we have setup a scenario where for all $f \in X^*$:

$$f(T(x)) = f(y) \quad f \in X^*$$

Since the f separate points, it must be that

$$T(x) = y$$

in other words, separation of points can be thought of as a sort of “injectivity”. As an exercise, show that if $T : X \rightarrow Y$ is a linear operator and for each $f \in Y^*$, $f \circ T \in X^*$, then T is bounded (hint: use closed graph theorem and the above remark).

The 4th point inspires a quick look at double duals. If we let $\hat{X} = \{\hat{x} \mid x \in X\}$ and let X^{**} be the double dual, then since X^{**} is always complete, the closure of $\hat{X}$ in X^{**} , $\overline{\hat{X}}$ is a Banach space, and $x \mapsto \hat{x}$ embeds X into $\overline{\hat{X}}$ as a dense subspace. The space $\overline{\hat{X}}$ is called the *completion* of X , and if X is itself Banach then $\overline{\hat{X}} = \hat{X}$. If X is finite dimensional, then $\hat{X} = X^{**}$, since they have the same dimension. This is not guaranteed in infinite dimension, hence if $\hat{X} = X^{**}$, we shall call X a *reflexive space*. For example, $C_c^\infty([0, 1])$ will be shown to not be reflexive. We shall see later that L^p spaces are reflexive.

1.4 Weak Topologies and TVS's

(This post seems like an excellent post to follow! link [here](#))

There are many topologies on which the operations are continuous (i.e. + and scalar multiplication is continuous) that do not arise from norms. The topology on a vector-space induced by a norm is in fact quite strong. In general, such a space (in infinite dimensions) doesn't have many compact sets, and so we lose many of the nice results of working on compact or pre-compact sets. This is naturally unfavourable in analysis, since many arguments rely on properties of sequences converging. Looking at functional analysis as the study of sequences, norms induce a particular definition of convergence: $\|x - x_n\|_V \rightarrow 0$. However, there are many other notions of convergence that are weaker that are quite useful, for example:

1. pointwise convergence
2. convergence in measure

3. smooth convergence
4. convergence on compact sets

None of these types of convergence are induced by a norm. We will instead be induced new types of topologies on our vector-spaces. The two types of topologies we will usually impose on V will be known as the *weak topology* and the *weak* topology*. We will define analogous topologies on $\text{Hom}_k(V, W)$ which somewhat confusingly will be named *strong operator topology* and *weak operator topology*.

Definition 1.4.1: Topological Vector Space (TVS)

Let V be a vector-space over $\mathbb{R}$ or $\mathbb{C}$. Then V is a *topological vector space* (TVS) if the operation of addition and scalar multiplication are continuous. A topological vector-space is said to be *locally convex* if there is a base for the topology consisting of convex sets.

It should be verified that the translation map $x \mapsto x_0 + x$ and the dilation map $x \mapsto \lambda x$ are both homeomorphisms for any scalar λ and vector x_0 .

Example 1.4.2: Topological Vector Spaces

1. Let V be a normed vector-space with the topology induced by the norm. Then $\|c \cdot x\| \leq |c| \|x\|$ (in fact $=$) and $\|(x, y)\| \leq 2(\|x\| + \|y\|)$ where $\leq$ comes from the triangle inequality and the fact that $\|(x, y)\| = \max(\|x\|, \|y\|)$. Hence, V is a topological vector space. If $\| - \|$ was a quasi-norm so that we replace the triangle inequality with $\|x + y\| \leq K(\|x\| + \|y\|)$ then a quasi-normed space V is still a topological vector space.
2. Let V be a semi-normed space so that $\|x\| = 0$ does not necessarily imply $x = 0$. Then V is a topological vector space with the topology induced by the semi-norm. V is Hausdorff if and only if the semi-norm is a norm.
3. Any subspace $W \subseteq V$ of a topological vector space V is a topological vector space with the subspace topology. Similarly for products and quotients.
4. Let V be a vector-space and $(\mathcal{T}_\alpha)_{\alpha \in A}$ be a collection of topologies on V that makes V a topological vector space. Let $\mathcal{T} = \langle \mathcal{T}_\alpha \rangle_{\alpha \in A}$ be the topology generated by $\bigcup_{\alpha \in A} \mathcal{T}_\alpha$ (this is the weakest topology that contains each $\mathcal{T}_\alpha$). Then $(V, \mathcal{T})$ is also a topological vector space. Importantly, a sequence (resp. net) $x_n \rightarrow x$ in $\mathcal{T}$ if and only if $x_n \rightarrow x$ in $\mathcal{T}_\alpha$ for all $\alpha \in A$ (recall that $x_n \rightarrow x$ in a general topology space if for any neighborhood U of x , there exists an N such that for all $n \geq N$, $x_n \in U$). Commonly, we will have $(\mathcal{T}_\alpha)_{i=1}^\infty$ be generated by countable collection of semi-norms. A complete space generated by a countable collection of semi-norms is called a *Fréchet space*.
5. (Pointwise Convergence) Let X be a set and $\mathbb{C}^X$ the collection of all complex valued functions $f : X \rightarrow \mathbb{C}$. Then $\mathbb{C}^X$ is a complex-vector-space. For each $x \in X$, define the seminorm:

$$\|f\|_x := |f(x)|$$

Then the topology generated by all of these seminorms is the *topology of pointwise convergence* on $\mathbb{C}^X$ (it is also the product topology on this space). A sequence $(f_n) \subseteq \mathbb{C}^X$ converges to

f in this topology if and only if it converges pointwise. Note that if X has more than one point, then none of the semi-norms individually generate a Hausdorff topology, but taken all together the induced topology is Hausdorff.

6. (Uniform Convergence) Let X be a topological space and $C(X)$ the set of complex-valued continuous function $f : X \rightarrow \mathbb{C}$. If X is not compact, then in general we would not expect functions in $C(X)$ to be bounded, so the sup-norm will not necessarily make $C(X)$ into a normed vector-space. Nonetheless, we can still define “balls” $B_r(f) \subseteq C(X)$ by:

$$B_r(f) := \left\{ g \in C(X) \mid \sup_{x \in X} |f(x) - g(x)| \leq r \right\}$$

these do form a topological structure on $C(X)$, but not a topological vector space structure since scalar multiplication is no longer continuous. Nonetheless, we have that $(f_n) \subseteq C(X)$ converges in this topology to a limit $f \in C(X)$ if and only if f_n converge uniformly to f . Thus $\sup_{x \in X} |f_n(x) - f(x)|$ is finite for sufficiently large n and converges to 0 as $n \rightarrow \infty$.

7. (Uniform Convergence on Compact sets) Let X and $C(X)$ be as in the above example. For every compact subset $K \subseteq X$, define a seminorm:

$$\|f\|_{C(K)} = \sup_{x \in K} |f(x)|$$

The topology generated by all these seminorms as K ranges over all compact subsets of X is called the *topology of uniform convergence on compact sets*. It is stronger than the topology of pointwise convergence but weaker than the topology of uniform convergence, and is frequently used in complex analysis. A sequence $(f_n) \subseteq C(X)$ converges to $f \in C(X)$ in this topology if and only if f_n converges uniformly to f on each compact set. Is the resulting space a topological vector space?

8. (Smooth Convergence) Let $C^\infty([0, 1])$ be the space of smooth functions $f : [0, 1] \rightarrow \mathbb{C}$. Then we can define the C^k -norm on this space for any non-negative integer k by the formula:

$$\|f\|_{C^k} := \sum_{i=1}^k \sup_{x \in [0, 1]} |f^{(i)}(x)|$$

where $f^{(i)}$ is the i th derivative of f . The topology generated by all C^k norms $k = 0, 1, 2, \dots$, is the *smooth topology*. A sequence f_n converges in this topology to a limit f if $f_n^{(i)}$ converges uniformly to $f^{(i)}$ for each $i \geq 0$. Is this a topological vector space?

9. (Convergence in measure) Let $(X, \mathcal{M}, \mu)$ be a measure space and $L(X)$ the set of measurable functions $f : X \rightarrow \mathbb{C}$. Then the sets

$$B_{r, \epsilon}(f) := \{g \in L(X) \mid \mu(\{x \mid |f(x) - g(x)| \geq r\}) < \epsilon\}$$

for $f \in L(X)$, $\epsilon, r > 0$ forms the basis for a topology that makes $L(X)$ into a topological vector space. A sequence $(f_n) \subseteq L(X)$ converges to f in this topology if and only if it converges in measure.

10. Let $[0, 1]$ be given the usual Lebesgue measure. Then show that $L^\infty([0, 1])$ cannot be given a topological vector space structure in which a sequence $(f_n) \subseteq L^\infty([0, 1])$ converges to f in this topology if and only if it converges a.e.

11. Let V be an $\mathbb{R}$ -vector space. a subset $U \subseteq V$ is called *algebraically open* if

$$\{t \in \mathbb{R} \mid x + tv \in U\}$$

is open for all $x, v \in V$. Then any open set in a topological vector space is algebraically open. Note that there are algebraically open sets that are not open (take any dense subset of S^1 , say D , and $\mathbb{R} \setminus \{D\}$.) This larger set of algebraically open sets forms a topology, but this topology does not give the $\mathbb{R}^2$ structure of a topological vector space.

12. Let X be a LCH space. On $\mathbb{C}^X$, the topology of uniform convergence on compact sets is defined by the seminorms $p_K(f) = \sup_{x \in K} |f(x)|$ as K ranges over compact subsets of X . If X is σ -compact and $\{U_n\}$ are as in proposition ref:HERE (Folland p. 168), this topology is defined by the seminorms $p_n(f) = \sup_{x \in \overline{U_n}} |f(x)|$. In this case, $\mathbb{C}^X$ is easily seen to be complete, so it is a Fréchet space. By proposition ref:HERE, so is $C(X)$.
13. The space $L^1_{\text{loc}}(\mathbb{R}^n)$ is a Fréchet space with the topology defined by the seminorms $p_k(f) = \int_{|x| \leq k} |f(x)| dx$ (completeness follows from completeness of L^1).
14. Here is another useful place where TVS are defined. Let's say we want to study a space in which d/dx is continuous (and hence bounded) on $C^\infty([0, 1])$. However, this is essentially impossible to define the topology through a norm. Indeed, if $f_\lambda(x) = e^{\lambda x}$, then $(d/dx)f_\lambda = \lambda f_\lambda$, and so $\|d/dx\| \geq \lambda$ for all λ , no matter what norm is used on $C^\infty([0, 1])$. There are a few ways to rectify this:
- (a) One can consider differentiation as an unbounded operator from V to W where W is a suitable Banach space and X is a dense subspace of W (exercise 30 in Folland)
 - (b) One can consider differentiation as a bounded linear map from one Banach space V to a different one W such that $V = C^k([0, 1])$ and $W = C^{k-1}([0, 1])$ (exercise 9)
 - (c) One can consider differentiation as a continuous operator on a locally convex space V whose topology is not given by a norm. It is easy to construct families of seminorms on the space of smooth functions so that differentiation becomes continuous almost by definition. For example, the seminorms:

$$p_k(f) = \sup_{0 \leq x \leq 1} |f^{(k)}(x)|$$

where $k \in \{0, 1, 2, \dots\}$ makes $C^\infty([0, 1])$ into a Fréchet space (the completeness is proved in exercise 9), and d/dx is continuous by proposition 5.15, since $p_k(f') = p_{k+1}(f)$. Other examples are in exercise 45 in chapter 9 of Folland.

15. The space $\mathbb{R}^\infty = \varprojlim \mathbb{R}^n$ is a Fréchet space; can you see the norms?

Proposition 1.4.3: Topological Vector Space And Hausdorff

Let V be a topological vector space. Then V is Hausdorff if and only if $\{0\}$ is closed

Proof:
exercise

We may ask what properties of Banach spaces extend to topological vector spaces. Some nice properties of Banach spaces include the uniform boundedness principle and the open mapping theorem. These can be extended to the slightly larger class of *Fréchet spaces* that we mentioned in the examples above. We will single them out and talk about them in the next section.

If V is a topological vector space, we can take V^* to be the set of all continuous functionals from V to the base field $k \in \{\mathbb{R}, \mathbb{C}\}$. Since V has no norm, V^* has no norm topology. We can however still define interesting topologies on it. This topology is dual to a weaker topology we can put on V :

Definition 1.4.4: Weak And Weak* Topologies

Let V be a topological vector space and V^* its dual. Then

1. the *weak* topology* on V^* is the topology generated by the seminorms $\|\lambda\|_x := |\lambda(x)|$ for all $x \in V$
2. the *weak topology* on V is the topology generated by the seminorms $\|x\|_\lambda = |\lambda(x)|$ for all $\lambda \in V^*$

If V is also a normed vector-space, then the weak topology on V is weaker than the norm topology. If V is a normed space, then so is V^* and $(V^*)^*$. the weak* topology on V^* makes V^* into a topological vector space, and is a topology weaker than the weak topology on V^* (which is defined using the double dual $(V^*)^*$). When V is reflexive ($V \cong V^{**}$), then the weak and weak* topologies on V^* are in fact equivalent.

Definition 1.4.5: Weak Convergence

Let V a vector-space endowed with the weak topology. Then a sequence $(x_n) \subseteq V$ that converges in V with the weak topology is called *weakly convergent* and is said to *weakly converge* to a point x . A sequence weakly converges $x_n \rightarrow x$ if and only if

$$\lambda(x_n) \rightarrow \lambda(x) \quad \forall \lambda \in V^*$$

and is usually denoted as $x_n \rightharpoonup x$. Similarly, $\lambda_n \rightharpoonup \lambda$ if $\lambda_n(x) \rightarrow \lambda(x)$ for all $x \in V$ (λ_n , as a function of V , converges pointwise to λ)

Proposition 1.4.6: Normed Vector Spaces Weak Topology

Let V be a normed vector-space. Then the weak topology on V and the weak* topology on V^* are both Hausdorff. In particular, we conclude that in the weak and weak* limits, when they exist they are unique

Proof:

Use the Hahn-Banach Theorem

Example 1.4.7: Norm, Weak, Weak* Convergence

Let $V = c_0(\mathbb{N})$, $V^* = \ell^1(\mathbb{N})$, and $V^{**} = \ell^\infty(\mathbb{N})$. Let $e_1, e_2, \dots$ be the standard basis of any of these vector-spaces.

1. The sequence (e_i) converges weakly in V to zero, but not converge in norm in V
2. (easy to think up some examples)

A natural generalization of the weak* topology on X^* can be imposed $B(X, Y)$ for some TVS Y . Just like for X^* , we can impose a weak topology on it, however there is another natural topology we may induce on $B(X, Y)$:

Definition 1.4.8: Strong And Weak Operator Topology

Let $B(X, Y)$ be a all continuous operators between Fréchet spaces. Then

1. the *weak operator topology* is the topology induced by the seminorm $T \mapsto |\lambda(T(x))|$ for all $x \in X$ and $\lambda \in Y^*$
2. If furthermore Y is a Banach space, then the *strong operator topology* (or *operator norm topology*) is the topology induced by the seminorms $T \mapsto \|T(x)\|_Y$ for all $x \in X$

Hence, a sequence $(T_n) \subseteq B(X, Y)$ converges in the strong operator topology to a limit T if and only if $T_n(x) \rightarrow T(x)$ in the norm topology on Y , and T_n converges to T in the weak operator topology if $T_n(x) \rightarrow T(x)$ in the weak topology on x . Hence, we see that the weaker operator topology is weaker than the strong operator topology, which in turn is (somewhat confusingly) weaker than the operator norm topology.

A final result we shall point out is that we can rephrase the uniform boundedness principle for convergence now as follows:

Proposition 1.4.9: Uniform Boundedness Principle Reformulation

Let $(T_N) \subseteq B(X, Y)$ be a sequence of bounded linear operators from a Banach space X to a normed space Y , and let $T \in B(X, Y)$ be another bounded linear operator, and let D be a dense subspace of X . Then the following are equivalent:

1. T_n converges in the strong operator topology of $B(X, Y)$ to T
2. T_n is bounded in the operator norm (i.e. $\|T_n\|_{op}$ is bounded) and the restriction of T_n to D converges in the strong operator topology of $B(D, Y)$ to the restriction of T to D

1.5 Fréchet Spaces

When constructing a space, we often cannot define a space simply through a norm; not all elements may be “finite”. However, we can often construct a space using countably many normed spaces, or usually semi-normed space $\|x\| = 0$ does not imply $x = 0$, for example if f is an integrable function). Due to the countable nature of this construction, most types of convergences are accomplished by a family of semi-norms. For example, convergence on compact sets, pointwise convergence, uniform convergence, and smooth convergence are all convergence in Fréchet spaces. In analysis, there will be many spaces that will Fréchet spaces, notably when you encounter distributions and require to define a topology on the dual space. Hence, a moment to get a better understanding of them is worth it.

Theorem 1.5.1: Semi-Norms Generate TVS

Let $\{p_\alpha\}_{\alpha \in A}$ be a family of norms on a vector space V . If $x \in X$, $\alpha \in A$, and $\epsilon > 0$, let

$$U_{x,\alpha,\epsilon} = \{y \in V \mid p_\alpha(y - x) < \epsilon\}$$

and let $\mathcal{T}$ be the topology generated by the sets $U_{x,\alpha,\epsilon}$. Then:

1. For each $x \in X$ the finite intersections of the sets $U_{x,\alpha,\epsilon}$ ($\alpha \in A, \epsilon > 0$) form a neighborhood base at x
2. if $\langle x_i \rangle_{i \in I}$ is a net in V , then $x_i \rightarrow x$ if and only if $p_\alpha(x_i - x) \rightarrow 0$ for all $\alpha \in A$.
3. $(V, \mathcal{T})$ is a locally convex topological vector-space

Proof:

1. If $x \in \bigcap_1^k U_{x_j, \alpha_j, \epsilon_j}$, let $\delta_j = \epsilon_j - p_{\alpha_j}(x - x_j)$. By the triangle inequality

$$x \in \bigcap_1^k U_{x, \alpha_j, \delta_j} \subseteq \bigcap_1^k U_{x-j, \alpha_j, \epsilon_j}$$

Thus, the assertion now follows from the fact that the topology generated consists of $\emptyset$, X , and all unions of finite intersections.

2. By part (1), it suffices to see that $p_\alpha(x_i - x) \rightarrow 0$ if and only if $\langle x_i \rangle$ is eventually in $U_{x,\alpha,\epsilon}$ for all $\epsilon > 0$.
3. Continuity is easy from part (b), for if $x_i \rightarrow x$ and $y_i \rightarrow y$, then

$$p_\alpha((x_i + y_i) - (x + y)) \leq p_\alpha(x_i - x) + p_\alpha(y_i - y) \rightarrow 0$$

so $x_i + y_i \rightarrow x + y$. If furthermore $\lambda_i \rightarrow \lambda$, then eventually $\lambda_i \leq C + |\lambda| + 1$, so

$$p_\alpha(\lambda_i x_i - \lambda x) \leq p_\alpha(\lambda_i(x_i - x)) + p_\alpha((\lambda_i - \lambda)x) \leq C p_\alpha(x_i - x) + |\lambda_i - \lambda| p_\alpha(x)$$

Hence, $\lambda_i x_i \rightarrow \lambda x$. Furthermore, the sets $U_{x,\alpha,\epsilon}$ are convex, for if $y, z \in U_{x,\alpha,\epsilon}$, then

$$p_\alpha(x - [t + (1-t)z]) \leq p_\alpha(tx - ty) + p_\alpha((1-t)x + (1-t)z) < t\epsilon + (1-t)\epsilon = \epsilon$$

Then local convexity follows from (1)

Definition 1.5.2: Fréchet Space

A complete Hausdorff topological vector-space whose topology is defined by a countable family of seminorms is called a *Fréchet space*.

Proposition 1.5.3: Continuity in Fréchet Space

Let $T : V \rightarrow W$ be a linear map between two vector spaces. Let V be a topology induced by a family of semi-norms $(\| \cdot \|_{V_\alpha})_{\alpha \in A}$ and W the topology induced by a family of semi-norm $(\| \cdot \|_{W_\beta})_{\beta \in B}$. Then T is continuous if and only if for each $\beta \in B$, there exists a finite subsets $A_\beta \subseteq A$ and a constant C_β such that

$$\|T(x)\|_{W_\beta} \leq C_\beta \sum_{\alpha \in A_\beta} \|x\|_{V_\alpha} \quad (1.1)$$

for all $x \in V$.

Proof:

Let equation 1.1 hold and consider a convergent net $\langle x_i \rangle \subseteq X$. Then by theorem 1.5.1 $p_\alpha(x_i - x) \rightarrow 0$ for all α , thus by the inequality $\|T(x_i) - T(x)\|_{W_\beta} \rightarrow 0$ for all β , which implies $T(x_i) \rightarrow T(x)$. Since T preserves convergent nets, T is continuous.

Conversely, if T is continuous for every $\beta \in B$ choose a neighbor U_β of 0 in V such that $\|T(x)\|_{W_\beta} < 1$ for all $x \in U_\beta$. By theorem 1.5.1, we may assume that $U_\beta = \cap_{1 \leq k} U_{x\alpha_{\epsilon_j}}$. Let $\epsilon = \min(\epsilon_1, \epsilon_2, \dots, \epsilon_k)$ so that $\|T(x)\|_{W_\beta} < 1$ whenever $p_{\alpha_j}(x) < \epsilon$ for all j . Now, for $x \in X$, either:

1. $p_{\alpha_j}(x) > 0$ for some j . Then let $y = \epsilon x / (\sum_1^k p_{\alpha_j}(x))$ so tht $p_{\alpha_j}(y) < \epsilon$. Then:

$$\|T(x)\|_{W_\beta} = \sum_1^k \epsilon^{-1} p_{\alpha_j}(x) \|T(y)\|_{W_\beta} \leq \epsilon^{-1} \sum_1^k p_{\alpha_j}(x)$$

2. If $p_{\alpha_j}(x) = 0$ for all j , then $p_{\alpha_j}(rx) = 0$ for all j and all $r > 0$. Thus $r\|T(x)\|_{W_\beta} = \|T(rx)\|_{W_\beta} < 1$ for all $r > 0$, and os $\|T(x)\|_{W_\beta} = 0$. Then certainly

$$\|T(x)\|_{W_\beta} \leq \epsilon^{-1} \sum_1^k p_{\alpha_j}(x)$$

covering both cases, we are done.

The following proposition is an interesting characterization of Fréchet spaces:

Proposition 1.5.4: Hausdorff and invariant Metric

Let X be Fréchet space given by $\{p_\alpha\}_{\alpha \in A}$. Then

1. X is Hausdorff if and only if for each $x \neq 0$, there exists $\alpha \in A$ such that $p_\alpha(x) \neq 0$
2. If X is Hausdorff and A is countable, then X is metrizable with a translation invariant metric (i.e. $\rho(x, y) = \rho(x + z, y + z)$ for all $x, y, z \in X$)

Proof:

exercise 43 Folland chapter 5

Note that topologically, all these spaces are the same (as the bellow theorem will sketch). So what really matters is what extra information these semi-norms (or this invariant metric) provides:

Theorem 1.5.5: Anderson-Kadec Theorem

Every infinite-dimensional separable Fréchet space is homeomorphic to the product space $\mathbb{R}^{\mathbb{N}}$ (the countable product of real lines equipped with the product topology).

Note that L^{∞} is *inseparable* and hence is not included in this theorem.

Proof:

(Sketch). The proof is historically a combination of results by Kadec (1966) and Anderson (1966), establishing a chain of homeomorphisms. We outline the two main steps:

1. The Banach Case: Kadec proved that every infinite-dimensional separable Banach space is homeomorphic to the Hilbert space ℓ_2 . The construction typically involves re-norming the space with a strictly convex equivalent norm (often called a Kadec norm) and defining a homeomorphism via radial projection and coordinate mapping.
2. The Product Case: Anderson proved that ℓ_2 is homeomorphic to $\mathbb{R}^{\mathbb{N}}$. This is often shown using “separable absorption” theorems or the “notback-and-forth” method to manage the coordinate systems.

Since every Fréchet space is a projective limit of Banach spaces (and using the assumption of separability), one can extend the homeomorphism to the general case. Consequently, despite their distinct linear and geometric properties, spaces such as the Schwartz space $\mathcal{S}(\mathbb{R})$, the space of holomorphic functions $H(\mathbb{C})$, and $C^{\infty}[0, 1]$ are all topologically indistinguishable from $\mathbb{R}^{\mathbb{N}}$.

Proposition 1.5.6: dense convergence, then strong convergence

Let $(T_n)_1^{\infty} \subseteq L(V, W)$ where $\sup_n \|T_n\| < \infty$ and $T \in L(V, W)$. If $\|T_n(x) - T(x)\| \rightarrow 0$ for all x in a dense subset D of X , then $T_n \rightarrow T$ strongly.

Proof:

We want to show that $\|T_n(x) - T(x)\| \rightarrow 0$ for all x . Let $C = \sup\{\|T\|, \|T_1\|, \|T_2\|, \dots\}$. Then given $x \in X$ and $\epsilon > 0$, choose $x' \in D$ such that $\|x - x'\| < \epsilon/3C$. Since $T_n(x') \rightarrow T(x')$, choose n large enough so that $\|T_n(x') - T(x')\| < \epsilon/3$. Then

$$\begin{aligned} \|T_n(x) - T(x)\| &\leq \|T_n(x) - T_n(x')\| + \|T_n(x') - T(x')\| + \|T(x') - T(x)\| \\ &\leq 2C\|x - x'\| + \frac{1}{3}\epsilon \\ &< \epsilon \end{aligned}$$

but then $T_n(x) \rightarrow T(x)$ strongly, as we sought to show.

One important result is that in the weak operator topology, unit balls are always compact:

Theorem 1.5.7: Alaoglu's Theorem

Let X be a normed space, and B^* the closed unit ball of X^* , $B^* = \{f \in X^* \mid \|f\| \leq 1\}$. Then B^* is compact in the weak*-topology.

Proof:

We shall show that B^* is a closed subspace of a compact space. For each $x \in X$, define

$$D_x = \{z \in \mathbb{C} \mid \|z\| \leq \|x\|\}$$

and take $D = \prod_{x \in X} D_x$. Then by Tychonoff's Theorem, D is compact. Now, we may exactly identify the elements of D the complex-valued function $\varphi : X \rightarrow \mathbb{C}$ such that $|\varphi(x)| \leq \|x\|$ for all $x \in X$. Then B^* would consist of the elements of D that are linear. Hence, we must just show that B^* is closed. The induced topology B^* from the product topology on D and the weak* topology on X^* are both the topology of pointwise convergence (for D by the product construction), and hence checking for closedness in X is equivalence to checking closedness of D . Take a net $\langle f_\alpha \rangle$ in B^* that converges to $f \in D$ for any $x, y \in X$ and $a, b \in \mathbb{C}$. Then:

$$f(ax + by) = \lim f_\alpha(ax + by) = \lim[af_\alpha(x) + bf_\alpha(y)] = af(x) + bf(y)$$

showing that f is linear, and certainly $\|f\| \leq 1$, hence $f \in B^*$, making it compact and completing the proof.

1.5.8 Note Since X^* is a vector space, one might hope to conclude that it is locally compact (via the “it suffices to show the result at 0” reduction). This is *not* the case, as Exercise 49(b) of Folland's *Real Analysis* shows.

(The following was copied from nLab, and is very informative!)

Every complete locally convex topological vector space T is the cofiltered projective limit of Banach spaces in the category of locally convex spaces. (Note that Fréchet spaces are additionally required to be metrisable, so this is more general.)

(see [this here](#)) Now given that a Fréchet space admits a decreasing sequence of convex balanced and absorbing neighborhoods, it follows immediately that: Every Fréchet space is a sequential projective limit of Banach spaces.

(this is also really cool on homomorphisms of Fréchet space [here](#))

2

L^p space

Look at this blog!! <https://terrytao.wordpress.com/2009/01/09/245b-notes-3-1p-spaces/>
In this chapter, we will be studying Banach spaces which are defined through the integral: L^p spaces. These spaces are a generalization of L^1 spaces, where if $f \in L^p$, then it must decay “faster” at infinity, and f doesn’t blow up too fast at any finite point. In other words, L^p spaces capture a type of “speed of growth”. As we have seen in the previous chapters, limits of integrals are central to many constructions. For some constructions, being in L^1 was enough, but for others (like Fourier series) we will require that function be more than finite in integral (i.e. more than $f \in L^1$). L^p spaces will give a nice set of examples of Banach spaces that are relatively simple to define.

My current intuition is that they are treated as the intermediate to being able to bound a function: it is sometimes nontrivial to immediately say a function is bounded, but we can build-up to it by saying it’s L^3 , which will imply L^5 , which will imply L^7 , and so on allowing us to conclude it’s L^∞ , which means a functions is bounded up to a zero-measure set. This is a common phenomena in analysis where we may not be able to bet a function to fit exactly in one space or another (in our case L^1 or L^∞ which are bounded a.e. functions), but we can put the function in intermediary buckets.

2.1 Basics

So far, we have limited our attention to measurable functions f such that $\int |f| < \infty$ and have labelled them L^1 . Often, what we are looking are function with more precise “convergent properties”, in particular:

$$\int |f|^p < \infty$$

for some $p \geq 1$, including ∞ (which we will define shortly). We start by labelling the collection of all these functions:

Definition 2.1.1: L^p Space

Let f be measurable. for $p \in (1, \infty)^a$, define the p -norm to be:

$$\|f\|_p = \left[\int |f|^p d\mu \right]^{1/p}$$

From this norm, define:

$$L^p(X, \mathcal{M}, \mu) := \left\{ f : X \rightarrow \mathbb{C} \mid f \text{ is measurable and } \|f\|_p < \infty \right\}$$

^awe will define L^∞ later

As with L^1 , we often abbreviate to $L^p(\mu)$, $L^p(X)$, or just L^p . As with L^1 , we'll let L^p be the equivalence class of the set of functions such that $|f|^p$ is integrable where $f \sim g$ if and only if $\int |f - g|^p = 0$. Just like L^1 , two functions will be equal in L^p (i.e. belong in the same equivalence class) if they are equal almost everywhere. Furthermore, we claimed that $\|\cdot\|_p$ is a norm. That $\|f\|_p = 0$ if and only if $f = 0$ a.e. comes almost immediately, since

$$\left(\int |f|^p \right)^{1/p} = 0 \quad \text{iff} \quad \int |f|^p = 0 \quad \text{iff} \quad \int |f| = 0 \quad \text{iff} \quad f = 0 \text{ a.e.}$$

For $\|cf\|_p = |c| \|f\|_p$, we take advantage of our normalisation:

$$\|cf\|_p = \left(\int |cf|^p \right)^{1/p} = \left(\int |c|^p |f|^p \right)^{1/p} = (|c|^p)^{1/p} \left(\int |f|^p \right)^{1/p} = |c| \|f\|_p$$

However, the triangle inequality is not so trivial. We will in fact take good part of this section to show that the triangle inequality is satisfied. For now, we take a moment to explore an example of L^p spaces and L^p functions to gain an intuition:

Example 2.1.2: L^p spaces

1. Let $f : [0, \infty) \rightarrow \mathbb{R}$ $f(x) = \frac{1}{x^{1/p}}$. Then $f \in L^q$ for $q > p$. This comes down to the fact that:

$$x > x^{1/p} \Leftrightarrow \frac{1}{x} < \frac{1}{x^{1/p}}$$

and by the fact that $\int \left| \frac{1}{x^n} \right| < \infty$ if $n > 1$. We have therefore found a class of functions that fits into all possible $p \in [1, \infty)$. In fact, we can find functions that fit into any connected subinterval of $[1, \infty)$! In particular, let $(X, \mathcal{M}, \mu)$ be a measure space and $f : X \rightarrow \mathbb{C}$ a complex measurable function on X . Let

$$E := \left\{ p \in [1, \infty) \mid \int_X |f|^p < \infty \right\}$$

- (a) Then E is connected, that is, if $p, q \in E$ such that $p < r < q$ for some r , then $r \in E$

Proof. Clearly, if $E = \emptyset$ or $|E| = 1$, then E is connected because the only non-trivial clopen set is $\emptyset$ and E , so let $|E| > 1$.

Pick $p, q \in E$ such that $p < q$, and consider a point r such that $p < r < q$. We'll show that $\int_X |f|^r < \infty$ using the following trick:

$$|f| = \chi_{|f| \geq 1} |f| + \chi_{|f| < 1} |f|$$

that is, we're splitting $|f|$ to where f is greater than or equal to 1 and strictly less than 1. The equality holds since $A = \{x \mid |f(x)| \geq 1\}$ and $B = \{x \mid |f(x)| < 1\}$ are disjoint. Importantly, notice that:

$$|f|^p = \chi_{|f| \geq 1} |f|^p + \chi_{|f| < 1} |f|^p \quad |f|^q = \chi_{|f| \geq 1} |f|^q + \chi_{|f| < 1} |f|^q$$

Thus, by the linearity of the integral, we get that:

$$\int |f|^p = \int \chi_{|f| \geq 1} |f|^p + \int \chi_{|f| < 1} |f|^p \quad \int |f|^q = \int \chi_{|f| \geq 1} |f|^q + \int \chi_{|f| < 1} |f|^q$$

It thus suffice show that $\int \chi_{|f| \geq 1} |f|^r < \infty$ and $\int \chi_{|f| < 1} |f|^r < \infty$. This is essentially immediate:

$$\chi_{|f| \geq 1} |f|^r \leq \chi_{|f| \geq 1} |f|^q < \infty \quad \chi_{|f| < 1} |f|^r \leq \chi_{|f| < 1} |f|^p < \infty$$

Therefore

$$\int |f|^r < \infty$$

showing that $r \in E$, as we sought to show. $\square$

- (b) Show that we can pick a measure and an f such that $E = (a, b)$, $[a, b)$, $[a, b]$, or $\{a\}$ for $1 \leq a \leq b \leq \infty$. In other words, all connected subsets of $[1, \infty)$ are possible for E (You may freely use the fact that any Riemann integrable function is also integrable with respect to the Lebesgue measure and its Riemann integral is equal to the Lebesgue integral)

Hint: Think about integrability properties of functions like x^{-a} and $\frac{-1}{\log^2(x)x} = \frac{d}{dx} \frac{1}{\log(x)}$

Proof. Throughout this hole proof we'll be using Riemann integration as a substitute for Lebesgue integration (we showed earlier that this is allowed). We'll be using these function's to construct our intervals:

- i. $f(x) = x^{-1/a}$ on $(1, \infty)$: Notice that

$$\int_1^\infty x^{-1/a} dx = \lim_{x \rightarrow \infty} \frac{x^{-1/a+1}}{-1/a+1} - \frac{1}{-1/a+1}$$

so $\|f\|_p < \infty$ if and only if $p \in (a, \infty]$.

- ii. $f(x) = x^{-1/a}$ on $(0, 1)$: Notice that

$$\int_0^1 x^{-1/a} dx = \frac{1}{-1/a+1} - \lim_{x \rightarrow 0} \frac{x^{-1/a+1}}{-1/a+1}$$

so $\|f\|_p < \infty$ if and only if $p \in [1, a)$.

- iii. $f(x) = \frac{1}{(\ln^2(x)x)^{1/a}}$ on $(1, \infty)$: This integral doesn't have an elementary solution, however we can still ascertain some basic convergence properties. In particular, if $p = a$, then:

$$\int_1^\infty |f(x)|^a dx = \lim_{x \rightarrow \infty} \left[-\frac{1}{\ln(x)} \right] - \lim_{x \rightarrow 1} \left[-\frac{1}{\ln(x)} \right]$$

so $\|f\|_p < \infty$ if and only if $p \in [a, \infty]$

- iv. $f(x) = \frac{1}{(\ln^2(x)x)^{1/a}}$ on $(0, 1)$: This result is similar to the other 3 cases we've proven.

The result for this integral shows that $\|f\|_p < \infty$ if and only if $p \in [1, a]$.

Thus, we have proven all possible “sub-rays” of $[1, \infty]$ can exist. To show all possible sub-intervals (closed, open, or half), we simply add the appropriate function's to produce the result. For example, if we want to have the interval $[a, b)$, simply take:

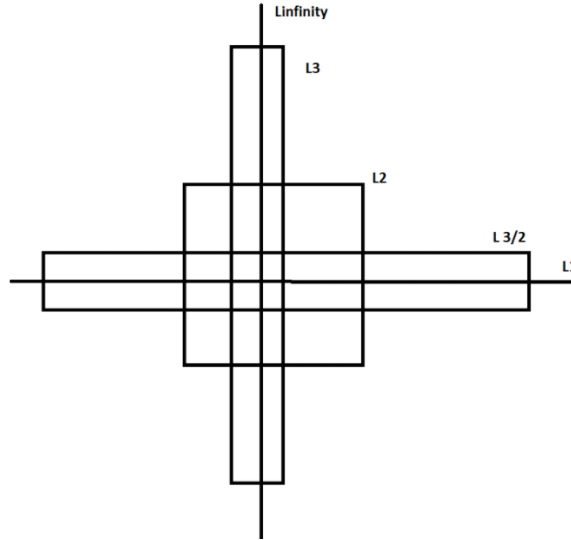
$$f(x) = \frac{1}{(\ln^2(x)x)^{1/a}} \chi_{(1, \infty)} + x^{-1/b} \chi_{(0, 1)}$$

which clearly has interval of convergence of $[a, b)$. Similarly, to get a singleton $\{a\}$, take:

$$f(x) = \frac{1}{(\ln^2(x)x)^{1/a}} \chi_{(1, \infty)} + \frac{1}{(\ln^2(x)x)^{1/a}} \chi_{(0, 1)}$$

Thus, we have all possible connected subintervals of $[1, \infty]$. $\square$

So if $f \in L^p$, it does not imply that $f \in L^q$ for $p > q$ or $p < q$! Later on, we will show some of the possible relations between L^p spaces. A more accurate intuition is the following ^a: If $f \in L^p$ for any p , then f must decay fast enough at infinity, and f must not blow up too fast at any finite point. As p increases, the first requirement gets less strict, and the second gets more strict. If X is bounded, the first requirement is vacuous, and if X is discrete, the second is vacuous. Sometimes this visual is used to represent these relations:



2. A particular case that will be common for analyzing sequences is when μ is the counting measure on A . In that case, we usually denote $L^p(\mu)$ by $\ell^p(A)$. If $A = \mathbb{N}$, we often simply

abbreviate notation to ℓ^p . If we write $f : \mathbb{N} \rightarrow \mathbb{R}$, $f(k) = a_k$, then the sequence f (i.e. $\{a_k\}$) is in ℓ^p if:

$$\sum_{k=1}^{\infty} |a_k|^p < \infty$$

Thus, ℓ^1 is the set of convergence series.

^acredit to this post on stack-exchange: <https://mathoverflow.net/questions/314050/intuition-about-lp-spaces>

Notice that we are “normalizing” each integral by a factor of $1/p$. This normalization is done to make sure that the constant can be brought out of the norm. It might seem tempted to say that:

$$\left(\int |f|^p \right)^{1/p} \text{ “=” } \int |f|$$

However, this is most certainly not true (simply take $f(x) = x$). As mentioned before, this makes it non-trivial to show that $\|\cdot\|_p$ is a norm on L^p . However, it is easy to see that L^p is a vector-space. If $f, g \in L^p$, then:

$$|f + g|^p \leq [2 \max(|f|, |g|)]^p \leq 2^p (|f|^p + |g|^p)$$

which is finite by assumption, and so $f + g \in L^p$. It is also immediate that $cf \in L^p$. Thus, we may freely add and multiply by L^p functions without worrying about leaving L^p . Because of this, it makes it meaningful to ask whether

$$\|f_1 + f_2\|_p \leq \|f_1\|_p + \|f_2\|_p$$

What we will now do is show that $\|\cdot\|_p$ does indeed satisfy the triangle inequality for $p \geq 1$. If $p < 1$, then the triangle inequality fails in general:

Example 2.1.3: $p < 1$ then no Triangle Inequality

Take $a, b > 0$ and $0 < p < 1$. Then for any $t \in \mathbb{R}$, we have that:

$$t^{p-1} > (a+t)^{p-1}$$

Integrating both sides from 0 to b , we get:

$$\begin{aligned} \int_0^b t^{p-1} &> \int_0^b (a+t)^{p-1} \\ \frac{b^p}{p} &> \frac{(a+b)^p}{p} \\ b^p &> (a+b)^p \\ a^p + b^p &> (a+b)^p \\ (a^p + b^p)^{1/p} &> a + b \end{aligned}$$

Now, if E and F are disjoint union with positive finite measures in X such that $\mu(E)^{1/p} = a$ and $\mu(F)^{1/p} = b$, then $\chi_E, \chi_F \in L^p$. Taking advantage of the fact that $\chi_E \chi_F = 0$ (i.e. the zero function) since $E \cap F = \emptyset$, then:

$$\|\chi_E + \chi_F\|_p = (a^p + b^p)^{1/p} > a + b = \|\chi_E\|_p + \|\chi_F\|_p$$

Thus, the $\|\cdot\|_p$ for $0 < p < 1$ cannot be a norm.

Thus, we stick to showing the case where $p \geq 1$. First, we will state a simple inequality fact similar Jensen's inequality. This is in fact the key idea behind the entire proof:

Lemma 2.1.4: Young's Inequality

If $a, b \geq 0$ and $0 < \lambda < 1$, then:

$$a^\lambda b^{1-\lambda} \leq \lambda a + (1-\lambda)b$$

with equality if and only if $a = b$. In particular, notice that if $\frac{1}{p} + \frac{1}{q} = 1$, then if $\lambda = p^{-1}$, then $(1-\lambda) = q^{-1}$ and so:

$$a^{1/p} b^{1/q} \leq \frac{1}{p}a + \frac{1}{q}b$$

Proof:

This proof is trying to capture what this image is doing in formal language:

<https://www.desmos.com/calculator/jdznjzqcju>

If $a = b$, equality clearly holds. If $a = 0$ or $b = 0$, inequality clearly holds. If $a, b \neq 0$, then dividing by b and letting $t = a/b$, we get

$$a^\lambda b^{-\lambda} \leq \lambda(a/b) + (1-\lambda) \Leftrightarrow t^\lambda \leq \lambda t + (1-\lambda) \Leftrightarrow t^\lambda - \lambda t \leq (1-\lambda) \quad (2.1)$$

From here, we use some calculus: we see that $t^\lambda - \lambda t$ is strictly increasing for $0 < t < 1$ (since $a, b \geq 0, t \geq 0$) and strictly decreasing for $t > 1$, so the maximal value occurs at $t = 1$, and in fact we get that the maximal value is $1 - \lambda$:

$$1^\lambda - \lambda 1 = 1 - \lambda \leq 1 - \lambda$$

and so the inequalities in equation (2.1) hold! Equality happens when $t = a/b = 1$ which implies $a = b$, completing the proof.

Usually, Young's inequality is stated as

$$a^\alpha b^\beta \leq \alpha a + \beta b \quad 0 \leq \alpha, \beta \leq 1, \quad \alpha + \beta = 1$$

which is exactly what we have, since if we fix some α , then it must be that $\beta = 1 - \alpha$. Another formulation of Young's inequality is:

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

with $p^{-1} + q^{-1} = 1$. This proof is a "Jensen equality" like proof that more clearly shows the role of concavity and so I'll dedicate a moment to prove this one too:

Proof:

If $a = 0$ or $b = 0$, the result is immediate, so assume $a, b > 0$. Then notice that since log is monotone:

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q} \Leftrightarrow \log(ab) \leq \log\left(\frac{a^p}{p} + \frac{b^q}{q}\right)$$

which is great, since we know $\log$ is concave and so Jensen's inequality applies. Since $p^{-1} + q^{-1} = 1$, we get $t = 1/p$ and $(1 - t) = 1/q$, and so by Jensen's inequality:

$$\log(ta^p + (1 - t)b^q) \geq t \log(a^p) + (1 - t) \log(b^q) = \frac{p}{p} \log(a) + \frac{q}{q} \log(b) = \log(ab)$$

completing the proof.

The second interpretation of $p^{-1} + q^{-1} = 1$ in Young's inequality is a nice way to write the straight line (or sub-straight line) equation without the $1 - \lambda$ term, but just two numbers. It furthermore let's us relate two numbers that can be much larger. For example, if I choose $p = 100$, then it must be that $q = 100/99$ so that

$$p^{-1} + q^{-1} = 1/100 + 99/100 = 1$$

which, if we consider p and q to be the exponents associated to L^p and L^q , we see that this gives us a way to relate different L^p space!

To see this more succinctly, we use this convexity property given by Young's inequality in the “build-up” lemma towards the triangle inequality. This lemma is important later on in functional analysis since it let's us bound the operation of multiplication of tow functions fg , (namely, $\|fg\|_1$ will be less than a value on the right hand side that seperates f and g). For our current purposes, it is important for the triangle inequality:

Theorem 2.1.5: Hölder's Inequality

Let $1 \leq p, q \leq \infty$ satisfy:

$$\frac{1}{p} + \frac{1}{q} = 1$$

or equivalently $q = p/(p - 1)$, where we will interpret $\frac{1}{\infty}$ as 0. Then if f and g are measurable functions on X then:

$$\|fg\|_1 \leq \|f\|_p \|g\|_q$$

In particular, if $f \in L^p$ and $g \in L^q$ then $fg \in L^1$. Equality holds if and only if $\alpha|f|^p = \beta|g|^q$ a.e. for some constants α, β with $\alpha\beta \neq 0$

Notice that when $p = q = 2$ (if $p = 2$, it must be that $q = 2$ to satisfy the equation), then this inequality is called the *Cauchy-Schwartz inequality*¹. Notice too that this is the only time when $p = q$; this will become important when we analyze the $p = 2$ case later.

I also found this intuition on a stack-exchange: the slower f decays, the faster g needs to decay, and the faster f blows up, the more we need g to not blow up. The previous sentence is equally true if we switch f and g , which convinces us that L^p spaces should be reflexive, and by considering $f = g$, we see that the restrictions on f and g should balance at exactly $p = 2$, so L^2 should be self-dual.

Notice too that since we are taking the absolute value of the function on the inside, the integral can be thought of as a sort of “concave function”, where the input would be, say in the special case of $\mathbb{R}$ as the domain, some interval. This makes it easier to see why this inequality holds (notice that convex functions must preserve monotonicity, and so does the integral).

¹Technically we need to define an inner product on L^2 to satisfy this inequality. However, as well see in chapter 6, L^2 is indeed a Hilbert space

Proof:

If $\|f\|_p = 0$ or $\|g\|_p = 0$, then $f = 0$ or $g = 0$ a.e., and so equality holds. Similarly if $\|f\|_p = \infty$ or $\|g\|_p = \infty$, inequality immediately holds. Furthermore, notice that if we find a particular f and g for which the result holds, then the result holds for af and bg since

$$\begin{aligned} \Leftrightarrow \|afbg\|_1 &\leq |ab| \|af\|_p \|bg\|_q \\ \Leftrightarrow |ab| \|fg\|_1 &\leq |ab| \|f\|_p \|g\|_q \\ \Leftrightarrow \|fg\|_1 &\leq \|f\|_p \|g\|_q \end{aligned}$$

So, it suffices to show the result when $\|f\|_p = \|g\|_q = 1$, and equality holds of $|f|^p = |g|^q = 1$. To this end, we use the previous lemma and set $a = |f(x)|^p$ and $b = |g(x)|^q$ and $\lambda = p^{-1}$ (so that $1 - \lambda = 1 - p^{-1} = q^{-1}$). Then by the previous lemma we have:

$$|f(x)|^{p \cdot p^{-1}} |g(x)|^{q \cdot q^{-1}} \leq p^{-1} |f(x)|^p + q^{-1} |g(x)|^q \Leftrightarrow |f(x)g(x)| \leq p^{-1} |f(x)|^p + q^{-1} |g(x)|^q \quad (2.2)$$

Now, integrating both sides, we get:

$$\|fg\|_1 \leq p^{-1} \int |f|^p + q^{-1} \int |g|^q = p^{-1} + q^{-1} = 1 = \|f\|_p \|g\|_q$$

where equality holds when equality holds in equation (2.2) a.e., which holds only when $|f|^p = |g|^q$ a.e., as we sought to show.

For a visual conformation, you can check out:

<https://www.desmos.com/calculator/gwqvfv4tie>

When p and q satisfy the equality in the equation, we call them *conjugate exponents*. As mentioned before, the conjugate exponents is simply another way of writing the equation of a straight line to apply Young's inequality. From this, we can take advantage of this convexity property yet again to prove the triangle inequality of $\|\cdot\|_p$, which has a different name since for the special case of the p -norm due to the fact it is used on other places and needs to be referenced:

Theorem 2.1.6: Minkowski's Inequality

If $1 \leq p < \infty$ and $f, g \in L^p$. Then:

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

Before proceeding with the proof, a quick word on notation to avoid confusion:

$$\|f\|_p^p = \left(\int |f|^p \right)^{p/p} = \int |f|^p$$

We often work with $\|f\|_p^p = \int |f|^p$ to get rid of the $1/p$ exponent to be able to work with the linearity of the integral, and re-introduce it at the end to get the desired result. We did not have to do this for Hölder's inequality since we reduced to a very nice special case.

Proof:

If $p = 1$, then the result immediately follows by the regular triangle inequality and linearity of the integral. If $f + g = 0$ a.e., then the result also follows, so assume that $f + g \neq 0$ a.e., which using the triangle inequality:

$$|f + g|^p \leq (|f| + |g|)|f + g|^{p-1} = |f||f + g|^{p-1} + |g||f + g|^{p-1}$$

So, we will apply Hölder's inequality using the fact that $q = \frac{p}{p-1}$ and integrating:

$$\begin{aligned} \|f + g\|_p^p &= \int |f + g|^p d\mu \\ &= \int |f + g| \cdot |f + g|^{p-1} d\mu \\ &\leq \int (|f| + |g|)|f + g|^{p-1} d\mu \\ &= \int |f||f + g|^{p-1} d\mu + \int |g||f + g|^{p-1} d\mu \\ &= \| |f|(|f + g|^{p-1}) \|_1 + \| |g|(|f + g|^{p-1}) \|_1 \\ &\leq \| |f| \|_p \| |f + g|^{p-1} \|_q + \| |g| \|_p \| |f + g|^{p-1} \|_q \quad \text{Hölder's Inequality} \\ &= \left(\left(\int |f|^p d\mu \right)^{\frac{1}{p}} + \left(\int |g|^p d\mu \right)^{\frac{1}{p}} \right) \left(\int |f + g|^{(p-1)(\frac{p}{p-1})} d\mu \right)^{\frac{p-1}{p}} \quad \text{recall } q = \frac{p}{p-1} \\ &= \left(\left(\int |f|^p d\mu \right)^{\frac{1}{p}} + \left(\int |g|^p d\mu \right)^{\frac{1}{p}} \right) \left(\int |f + g|^p d\mu \right)^{\frac{p-1}{p}} \\ &= (\|f\|_p + \|g\|_p) \frac{\|f + g\|_p^p}{\|f + g\|_p} \end{aligned}$$

Thus, multiplying both sides by $\frac{\|f + g\|_p}{\|f + g\|_p^p}$, we get

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

As we sought to show

Thus, we get that $\|\cdot\|_p$ is indeed a norm! Thus L^p is a normed vector space. In fact, just like L^1 , it is a complete normed vector space, i.e., a Banach space. We first recall a lemma to simplify our task. To understand the lemma, notice that every norm defines a metric:

$$\rho(x, y) = \|x - y\|$$

and so defines a topology, in particular a normal space, that has the notion of sequences well-defined. Next, recall that if $\{x_n\}$ is a sequence in V , it's series is said to converge if there exists an $x \in V$ such that $\sum_{n=1}^{\infty} x_n = x$, and is said to be absolutely convergence if $\sum_{n=1}^{\infty} \|x_n\| < \infty$ (this is lemma 1.1.6). If you did not cover this result yet, it is important to go see it now to understand the following result:

Theorem 2.1.7: L^p is a Banach Space

For $1 \leq p < \infty$, L^p is a complete normed vector-space, i.e., a Banach space. Moreover, if $f_n \rightarrow f$ in L^p , then there exists a subsequence that converges pointwise to f a.e.

Proof:

Let $1 \leq p < \infty$. By the previous lemma, it suffices to show that every absolutely convergent series converges:

$$\sum_{k=1}^{\infty} \|f_k\|_p = B < \infty \quad \Rightarrow \quad \exists f \text{ s.t. } \sum_{k=1}^{\infty} f_k = f \in L^p$$

Let $G_n = \sum_{k=1}^n |f_k|$ be the partial sums of the series and $G = \sum_{k=1}^{\infty} |f_k|$ be the series. Then by Minkowski's inequality, we get:

$$\|G_n\|_p \leq \sum_{k=1}^n \|f_k\|_p \leq B \quad \forall n \in \mathbb{N}$$

and so $\|G\|_p = (\int |G|^p)^{1/p} < \infty$ and so $(\int |G|^p) < \infty$. Since $G_n \leq G_{n+1}$, and $G_n : X \rightarrow [0, \infty)$, by the Monotone Convergence Theorem:

$$\int \lim_{n \rightarrow \infty} G_n^p = \lim_{n \rightarrow \infty} \int G_n^p \leq B^p < \infty$$

and so $G \in L^p$. By proposition ??, $G(x) < \infty$ a.e., so the series $\sum_{k=1}^{\infty} f_k$ converges a.e. If we let $F = \sum_{k=1}^{\infty} f_k$, we have that $|F| \leq G$ a.e., and so $F \in L^p$. From this, we can see that:

$$\left\| F - \sum_{k=1}^{\infty} f_k \right\|_p^p = \int \left| F - \sum_{k=1}^{\infty} f_k \right|^p \rightarrow 0$$

Thus, the series $\sum_{k=1}^{\infty} f_k$ converges (to F) in the p -norm, and so by lemma 1.1.6, L^p is complete, as we sought to show.

This is the proof the prof was doing in class, but I got stuck so found a different proof (the above proof) (Commented it out for now)

This also finally showed that L^1 is a complete space under $\|\cdot\|_1$! Note the importance of taking a subsequence: since $f_n \rightarrow f$ in the p -norm, i.e. the “sum-averaging” of the function converges, then there can be lots of wiggle room for how the functions f_n approach f :

Example 2.1.8: Taking A Subsequence

Construct a sequence of functions such that all the following hold:

1. $f_n : [0, 1] \rightarrow \mathbb{R}$ is continuous
2. $0 \leq f_n(x) \leq 1$ for every x
3. $\lim_{n \rightarrow \infty} \int_0^1 f_n dm = 0$
4. For every $x \in [0, 1]$ we have that $\lim_{n \rightarrow \infty} f_n(x)$ does not exist

(that is, it is definitely important to take a subsequence in order to deduce pointwise convergence from converge in L^1)

Proof. Essentially, we'll be taking advantage that $\sum_n \frac{1}{n}$ does not converge. Let $f_1(x) = 1$, that is, that constant function with constant 1.

For g_2 , we split the interval into 2 equal parts. Define

$$g_2(x) = \begin{cases} 1 & x \in [0, 1/2] \\ 0 & x \in (1/2, 1] \end{cases}$$

This function is not continuous. We will make it continuous by simply climbing to it (and for other values, we'll be descending back down too). At the point $1/2$, slope down at a speed of 2^2 until we reach 0:

$$f_2(x) = \begin{cases} 1 & x \in [0, 1/2] \\ -2^2(x - (1 - \frac{1}{4})) & x \in [1/2, 3/4] \\ 0 & x \in (3/4, 1] \end{cases}$$

From here, we will define our function in a sort of "recursive" sense. Starting from the interval $[0, 1/2]$, we keep the right end point, and add an interval of length $1/3$ to it: $[1/2, 5/6]$. Now, define f_3 to be 1 on $[1/2, 5/6]$, and zero everywhere else. Next, to define f_4 , repeat the process by adding $1/4$ to $5/6$, giving us the interval $[5/6, 13/12]$. Note that $13/12$ goes beyond 1. We thus wrap it around back to 0: $[0, 1/12] \cup [5/6, 1]$. Now, define a function f_3 to be 1 on this set, and zero everywhere else.

In each of these functions, we extend them to be continuous by attaching them a slope which has 2^n as a factor. In this way, the slope shrinks in size quicker than the size of the rectangles, and so we can morally imagine the problem to be one of indicator functions.

We can continue doing this, defining every f_k by appending $1/k$ and wrapping around when we go past 1 and then making the function continuous. Clearly, $\int f_k \rightarrow 0$ as $k \rightarrow \infty$ since each interval will be of total length $1/k + 1/2^k$. However, f_k does not converge anywhere: for any $x \in [0, 1]$, we can choose a subsequence from $f_k(x)$ (say $f_{k_l}(x)$) such that $f_{k_l}(x) = 1$. or $f_{k_l}(x) = 0$

To construct such a sequence, it is easier to imagine our construction without the wrapping around, that is, we keep adding intervals of length $1/k$ as though our co-domain was $[0, \infty)$. Then, our first value of l will be the value for which $x + 1$ is in some interval of length $1/k$ for appropriate k . This interval will exist, since $\sum_k 1/k$ diverges. Similarly, our next value of l for which $x + 2$ will be in some interval of length $1/k$ for appropriate k . We can continue doing this indefinitely, showing that there exists a subsequence of f_k that doesn't converge to 0, meaning f_k cannot converge to 0. On the other hand, it is easy to see that we can choose a subsequence that converges to 0 (since f_k is zero after the trapezoid passes by our point).

Since x was arbitrary, this will work any $x \in [0, 1]$. Thus, f does not converge anywhere, as we sought to show. $\square$

Returning to analyzing L^p spaces, we can now extend theorem ?? to show that simple function are dense in every L^p !

Theorem 2.1.9: Simple Functions Dense in L^p

For $1 \leq p < \infty$, the set of simple functions $f = \sum_{i=1}^n a_i \chi_{E_i}$ where $\mu(E_i) < \infty$ for all i is dense in L^p

Proof:

First of all, the set of simple functions is in L^p for every $1 \leq p < \infty$. Next, choosing any $f \in L^p$, choose some standard representation $\{f_n\}$ where $f_n \rightarrow f$ a.e.^a. Then since $|f_n| \leq |f|$, we get that $|f_n| \in L^p$. Furthermore

$$|f - f_n|^p \leq |2f|^p = 2^p |f|^p \in L^1$$

and so by the dominated convergence theorem, we get:

$$\int |f - f_n|^p \rightarrow 0 \quad \Rightarrow \quad \left(\int |f - f_n|^p \right)^{1/p} = \|f - f_n\|_p \rightarrow 0$$

Moreover, if $f_n = \sum_{k=1}^{\infty} a_k \chi_{E_k}$ was a simple function where all E_i are pairwise disjoint and a_i are nonzero, it must be that $\mu(E_i) < \infty$ since

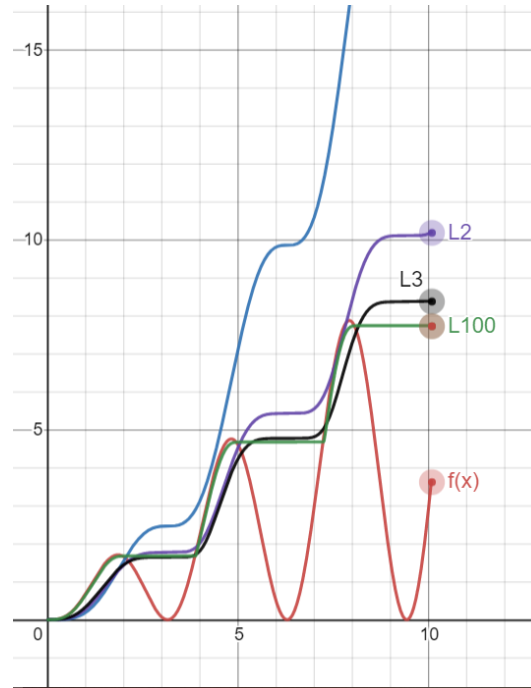
$$\sum_{k=1}^{\infty} |a_k|^p \mu(E_k) = \int |f_n|^p < \infty$$

completing the proof

^aRecall that the any measurable function has a standard representation, not just integrable functions

Some further remarks about which sets are dense can be enlightening. For $1 \leq p, q < \infty$, L^p is dense in L^q , since each simple set of the form we described are subsets of L^p for all $p \geq 1$. If we modify the characteristic function to make them continuous (simply draw a connecting line as we've done in example 2.1.8), then we see that $C_0(\mathbb{R})$ (continuous functions that vanish at infinity) are dense in L^p for every $1 \leq p < \infty$. If we restrict to closed intervals, then $C([a, b])$ is dense in $L^p([a, b])$ since the standard representation becomes uniformly convergent on bounded domains (see theorem ??).

So far, we have defined L^p spaces where $0 < p < \infty$ and showed that for $1 \leq p < \infty$, L^p is a Banach space. Interestingly, if we take a look at a graph of $\|f\|_p$ as $p \rightarrow \infty$, we'll find that there is a striking pattern that appears:



Notice that as $p \rightarrow \infty$, the resulting $\|f\|_p$ value got closer to a max function. In fact, this is almost the case! We might be tempted to then say that $\|f\|_\infty = \sup f$. However, the problem with this is that we might have a measure 0 set that diverges, while the supremum of the rest of the function is attained by a value $k < \infty$. The idea of $\sup f$ on all but a zero measure set will be called the *essential supremum*, and all functions that have a finite essential supremum will be denoted as L^∞ . In this way, L^∞ is simply a natural extension of our definition of L^p :

To see the intuition behind this, let us simply consider the p -norm of $\|\vec{v}\|_p = (|3|^p + |10|^p)^{1/p}$

1. Case $p = 1$: $(3^1 + 10^1)^{1/1} = 13$. (The 3 contributes about 23% to the total)
2. Case $p = 2$ (Euclidean): $\sqrt{3^2 + 10^2} = \sqrt{9 + 100} = \sqrt{109} \approx 10.44$. (The 3 matters less now. The 10 is dominating)
3. Case $p = 10$: $(3^{10} + 10^{10})^{1/10} = (59,049 + 10,000,000,000)^{1/10}$. Notice that 3^{10} is negligible dust compared to 10^{10} . $\approx (10^{10})^{1/10} = 10.00000something$
4. Case $p = 100$: The 3^{100} is mathematically invisible next to 10^{100} . The sum is basically just 10^{100} . When you take the $1/100$ root, you are just returning the base number: 10.

Definition 2.1.10: L^∞ Space

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $f : X \rightarrow [0, \infty]$ be a measurable function. Let

$$S = \{r \in \mathbb{R} \mid \mu(f^{-1}((r, \infty])) = 0\}$$

Then the *essential supremum* of f , denoted $\|f\|_\infty$ or $\text{ess sup } f$ is :

$$\|f\|_\infty := \begin{cases} \infty & \text{if } S = \emptyset \\ \inf S & \text{if } S \neq \emptyset \end{cases}$$

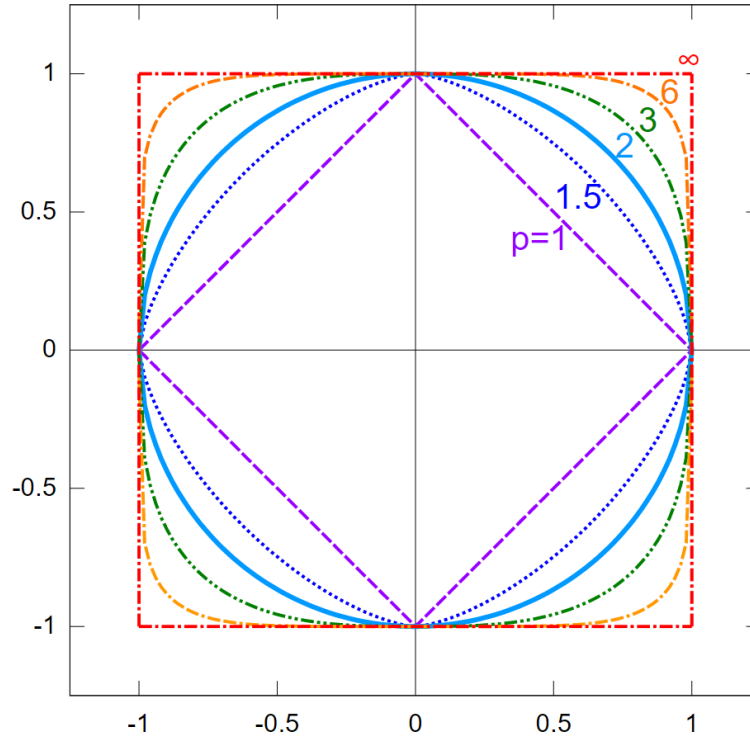
or, equivalently:

$$\|f\|_\infty = \inf_{\substack{E \in \mathcal{M} \\ \mu(X \setminus E) = 0}} \left(\sup_{x \in E} |f(x)| \right)$$

If f is a complex measurable, function, then $\|f\|_\infty = \||f|\|_\infty$

Essentially, $\|f\|_\infty$ is the least upper bound of f , ignoring measure zero sets worth of points. Using this definition, we can define L^∞ to be the equivalence classes of functions f such that $\|f\|_\infty < \infty$.

Another quick way of visualizing the relation between all p -norms, $1 \leq p \leq \infty$ is by taking the ℓ^p norm on $\mathbb{R}^2$, taking the function $x \mapsto \|x\|_p = \sqrt[p]{x^p + y^p}$, and considering when $\|x\| = 1$, or in terms of set notation $S_p = \{x \in \mathbb{R}^2 \mid \|x\|_p = 1\}$. We'd get:



A couple more quick remarks: $L^\infty(\mu)$ depends on μ only insofar as μ determines which sets have measure zero. If μ and ν are mutually absolutely continuous ($\mu \ll \nu$ and $\nu \ll \mu$ ²) then $L^\infty(\mu) = L^\infty(\nu)$. Secondly, if μ is not semi-finite, it is useful to adopt a slightly useful definition of L^∞ (Folland left this as exercise 23-25 in chapter 6).

Most properties of L^p for $1 \leq p < \infty$ generalises naturally to the L^∞ case:

Proposition 2.1.11: Properties of L^∞

Let $(X, \mathcal{M}, \mu)$ be a measure and L^∞ the set of measurable function satisfying $\|\cdot\|_\infty < \infty$. Then:

1. If f, g are measurable functions on X , then

$$\|fg\|_1 \leq \|f\|_1 \|g\|_\infty$$

If $f \in L^1$ and $g \in L^\infty$, then $\|fg\|_1 = \|f\|_1 \|g\|_\infty$ if and only if $|g(x)| = \|g\|_\infty$ a.e. on the set where $f(x) \neq 0$

2. $\|\cdot\|_\infty$ is a norm on L^∞
3. $\|f_n - f\|_\infty \rightarrow 0$ if and only if there exists $E \in \mathcal{M}$ such that $\mu(E^c) = 0$ and $f_n \rightarrow f$ uniformly on E
4. L^∞ is a Banach space
5. The simple functions are dense in L^∞

As a consequence of the well-definedness of L^∞ , $1^{-1} + \infty^{-1} = 1$ is indeed a well-defined notion (so $\infty^{-1} = 1/\infty = 0$). Thus, 1 and ∞ are conjugate exponents!

Proof:

We will prove (1), from which the rest of the results follow.

1. We'll be using the $\|f\|_\infty = \inf_{\substack{E \in \mathcal{M} \\ \mu(X \setminus E) = 0}} (\sup_{x \in E} |f(x)|)$ definition of the essential supremum to take advantage of the supremum within the definition. Our goal is to show that

$$\int |fg| = \|fg\|_1 \leq \|f\|_1 \|g\|_\infty$$

First, we will limit our attention of $\int |fg|$ to some $E \in \mathcal{M}$ such that $\mu(X \setminus E) = \mu(E^c) = 0$ (i.e. ignoring a zero-measure set). Then:

$$\int |fg| = \int_E |fg| + \int_{E^c} |fg| = \int_E |fg|$$

Next, we take advantage the fact that we are working with an infimum to see that:

$$\|f\|_\infty \leq \sup_{x \in A} |f(x)| < \|f\|_\infty + \epsilon$$

²see definition ??

Therefore, we can take:

$$\int_E |fg| \leq \int_E \left| f \cdot \sup_{x \in E} |g(x)| \right| = \int_E |f| \cdot \left| \sup_{x \in E} |g(x)| \right| = \left| \sup_{x \in E} |g(x)| \right| \int_E |f|$$

where the last equality comes from the fact that the supremum is a constant. Finally, by the property of the infimum, we have:

$$\left| \sup_{x \in E} |g(x)| \right| \int_E |f| \leq (\|g\|_\infty + \epsilon) \int_E |f|$$

Notice that this inequality does not depend on ϵ , and so:

$$\|fg\|_1 = \int_E |fg| \leq \|f\|_1 \|g\|_\infty$$

as we sought to show

Notice how similar $\|\cdot\|_\infty$ is to the uniform metric $\|f\|_u = \sup_{x \in X} f(x)$. If we are dealing with Borel measures that assign a positive value to all open sets, then if f is continuous we in fact have $\|f\|_\infty = \|f\|_u$ (since $\{x \mid |f(x)| > a\}$ is open). In this way, we can consider the space of bounded continuous functions as a (closed!) subspace of L^∞ .

We conclude with some final remarks about the density of simple functions in L^∞ . Notice that for L^p where $p < \infty$, simple functions where $\mu(E_i) < \infty$ where dense. This condition is no longer present for density in L^∞ ; we in fact require simple functions where $\mu(E) = \infty$ now since the function $1(x) = 1$ is in $L^\infty(\mathbb{R})$. This also means that $C_0(\mathbb{R})$ cannot be dense in L^∞ . In the case of $L^\infty([a, b])$ $C([a, b])$, not all functions can be converged to uniformly dense: consider $f : [0, 1] \rightarrow \mathbb{R}$, $f(x) = 0$ for $x \in [0, 1/2]$ and $f(x) = 1$ for $x \in [1/2, 1]$. The no continuous function uniformly converges to f in the ∞ -norm.

2.1.1 Relating L^p spaces: Interpolation

In this section, we will show how we can relate L^p spaces with each other. Fundamentally, the way we get any order with L^p spaces is through Hölder's inequality. We take full advantage of it in the following proofs. The end-goal is to show that if $p < q < r$ where we allow $r = \infty$, then:

$$L^p \cap L^r \subseteq L^q \subseteq L^p + L^r$$

This should reflect on the example that we have demonstrated in example 2.1.2. In particular, the larger p is, the less f has decay at infinity, but the more f must not explode at any point. In other words p gets bigger, we can let p decay slower, but it must behave better at points where it can explode. If our space X is bounded, then we don't have to care about decay (this will give us that $L^p \supseteq L^q$ where $p < q$). If X is discrete, we don't have to care about exploding at any point (this will give us that $\ell^p \subseteq \ell^q$ if $p < q$). We thus explore these relation's in this section. These types of relations between space via addition of functions or intersection of spaces is called *interpolation*.

Proposition 2.1.12: L^p subset

Let $0 < p < q < r \leq \infty$. Then

$$L^q \subseteq L^p + L^r$$

In particular, each $f \in L^q$ is the sum of a function in L^p and a function in L^r

Proof:

This proof is really similar to that of example 2.1.2[1]. Let $f \in L^q$. Then consider:

$$|f| = \chi_{|f| \geq 1} |f| + \chi_{|f| < 1} |f|$$

that is, we're splitting $|f|$ to where f is greater than or equal to 1 and strictly less than 1. The equality holds since $A = \{x \mid |f(x)| \geq 1\}$ and $B = \{x \mid |f(x)| < 1\}$ are disjoint. Importantly, notice that:

$$|f|^p = \chi_{|f| \geq 1} |f|^p + \chi_{|f| < 1} |f|^p \quad |f|^r = \chi_{|f| \geq 1} |f|^r + \chi_{|f| < 1} |f|^r$$

Thus, by the linearity of the integral, we get that:

$$\int |f|^p = \int \chi_{|f| \geq 1} |f|^p + \int \chi_{|f| < 1} |f|^p \quad \int |f|^r = \int \chi_{|f| \geq 1} |f|^r + \int \chi_{|f| < 1} |f|^r$$

Then we get:

$$\chi_{|f| \geq 1} |f|^p \leq \chi_{|f| \geq 1} |f|^q < \infty \quad \chi_{|f| < 1} |f|^r \leq \chi_{|f| < 1} |f|^q < \infty$$

Therefore, $\chi_{|f| \geq 1} f \in L^p$ and $\chi_{|f| < 1} f \in L^r$. If $r = \infty$, then clearly $\|\chi_{|f| \leq 1} f\|_\infty \leq 1$, showing the result still holds, as we sought to show.

Proposition 2.1.13: L^p superset

Let $0 < p < q < r \leq \infty$. Then

$$L^p \cap L^r \subseteq L^q$$

In particular, if if we find that f is bounded in growth from above and below by p and r , $\|f\|_p < \infty$ and $\|f\|_r < \infty$, then f is bounded in growth by q .

Proof:

Let $\|f\|_r < \infty$ and $\|f\|_q < \infty$. We need to show that $\|f\|_p < \infty$. Notice that it is equivalent to show that $\|f\|_r^n < \infty$ and $\|f\|_q^m < \infty$ then $\|f\|_p^\ell < \infty$ for $n, m, \ell \in \mathbb{R}_{\geq 0}$ (since if it's finite, then all powers will be finite). We'll prove it in the case where $r = \infty$ and $r < \infty$. In the infinite case, we simply take advantage of bounding properties of the essential supremum, and in the other case we use Hölder's inequality.

Starting with the $r = \infty$ case, notice that

$$|f|^q = |f|^{q-p+p} = |f|^q |f|^{q-p} \leq \|f\|_\infty^{q-p} |f|^p$$

and so, taking the integral and the q th root, we get:

$$\begin{aligned}
\|f\|_q &= \left(\int |f|^q \right)^{1/q} \\
&\leq \left(\int \|f\|_\infty^{q-p} |f|^p \right)^{1/q} \\
&= \|f\|_\infty^{\frac{q-p}{q}} \left(\int |f|^p \right)^{1/q} \\
&= \|f\|_\infty^{\frac{q-p}{q}} \left(\int |f|^p \right)^{p/pq} \\
&= \|f\|_\infty^{1-\frac{p}{q}} \|f\|_p^{p/q} < \infty
\end{aligned}$$

The fact that the last inequality is less than infinity comes from the that $\|f\|_p < \infty$ and $\|f\|_\infty < \infty$, so any power is less than infinity, and so $\|f\|_q < \infty$. Furthermore, notice that the sum of the power's is 1; this fact will come back for another intuition of $\|f\|_\infty$.

For the $r < \infty$ case the key is to find a way to integrate p and r using conjugate exponents. Since

$$p < q < r$$

then

$$\frac{1}{p} > \frac{1}{q} > \frac{1}{r}$$

therefore, define the continuous function $\lambda \mapsto \frac{1-\lambda}{p} + \frac{\lambda}{r}$. Since $(0, 1)$ is open, and $1/p > 1/r$, there exists a λ_0 such that

$$\frac{1-\lambda_0}{p} + \frac{\lambda_0}{r} = \frac{1}{q}$$

manipulating the equation around to get conjugate exponents, notice that:

$$\frac{(1-\lambda_0)q}{p} + \frac{\lambda_0 q}{r} = 1 \quad \Leftrightarrow \quad \frac{1}{\frac{p}{(1-\lambda_0)q}} + \frac{1}{\frac{r}{\lambda_0 q}} = 1$$

Thus, we have that $\frac{p}{(1-\lambda_0)q}$ and $\frac{r}{\lambda_0 q}$. With conjugate exponents established, we proceed with trying to bound $\|f\|_q$. As we commented, it is equivalent to bound $\|f\|_q^m$, so let $m = q$ (this will be a common trick to apply Hölder's inequality). Also, since $1 - \lambda_0 + \lambda_0 = 1$, we have that

$q = \lambda_0 q + (1 - \lambda_0)q$, giving us the ability to split up our integral:

$$\begin{aligned}
 \|f\|_q^q &= \int |f|^q \\
 &= \int |f|^{\lambda_0 q + (1-\lambda_0)q} \\
 &= \int |f|^{\lambda_0 q} |f|^{(1-\lambda_0)q} \\
 &= \left\| |f|^{\lambda_0 q} |f|^{(1-\lambda_0)q} \right\|_1 \\
 &\leq \left\| |f|^{\lambda_0 q} \right\|_{\frac{p}{(1-\lambda_0)q}} \left\| |f|^{(1-\lambda_0)q} \right\|_{\frac{r}{\lambda_0 q}} \quad \text{Hölder's inequality} \\
 &= \left(\int |f|^p \right)^{\frac{(1-\lambda_0)q}{p}} \left(\int |f|^r \right)^{\frac{\lambda_0 q}{r}} \\
 &= \|f\|_p^{(1-\lambda_0)q} \|f\|_r^{\lambda_0 q} < \infty
 \end{aligned}$$

Since the last two terms are finite, any power of them are also finite, and so we have bounded $\|f\|_q^q < \infty$, and so $\|f\|_q < \infty$, as we sought to show.

Proposition 2.1.14: ℓ^p Covariant Inclusion

If $0 < p < q \leq \infty$, then $\ell^p(A) \subseteq \ell^q(A)$

Proof:

They key part is to take advantage of the properties of the counting measure. Since the only set of measure 0 in the counting measure is the empty-set, we have that the essential supremum is actually the supremum!

Let's say $f \in \ell^p$, so that $\sum_{a \in A} |f(a)|^p = \|f\|_p^p < \infty$. We'll start by showing that $\|f\|_\infty < \infty$. Since the $\|f\|_\infty$ is in fact the supremum, we have that:

$$\|f\|_\infty^p = \sup_{a \in A} |f(a)|^p \leq \sum_{a \in A} |f(a)|^p = \|f\|_p^p < \infty$$

Therefore, $\|f\|_\infty < \infty$.

For the $q < \infty$ case, using proposition 2.1.13 with $\lambda = p/q$, we get:

$$\|f\|_q \leq \|f\|_p^\lambda \|f\|_\infty^{1-\lambda} \leq \|f\|_p$$

therefore, $\|f\|_q < \infty$, completing the proof.

Proposition 2.1.15: L^p Contravariant Inclusion in finite measure

Let $(X, \mathcal{M}, \mu)$ be a measure space where $\mu(X) < \infty$. If $0 < p < q \leq \infty$, then $L^p \supseteq L^q$

Proof:

We need to show that if $f \in L^q$ then $f \in L^p$. Since $f \in L^q$, $\|f\|_q < \infty$, so $\|f\|_q^n < \infty$ for all $n \in \mathbb{N}$. If we can show that $\|f\|_p^m \leq \|f\|_q^n$ for some $m, n \in \mathbb{N}$, then the proof is complete.

We'll apply Hölder's inequality in the cases of $q = \infty$ and $q < \infty$. If $q = \infty$, then:

$$\|f\|_p^p = \int |f|^p = \int |f|^p \cdot 1 \stackrel{\text{H.}}{\leq} \|f\|_\infty^p \int 1 = \|f\|_\infty^p \mu(X) < \infty$$

since $\int 1 = \mu(X)$. If $q < \infty$, then we do a similar argument. First, since $p < q$, notice that:

$$\frac{1}{\frac{q}{p}} + \frac{1}{\frac{q}{q-p}} = \frac{p}{q} + \frac{q-p}{q} = \frac{q}{q} = 1$$

and so $\frac{q}{p}$ and $\frac{q}{q-p}$ are conjugate exponents. Therefore, by Hölder's inequality:

$$\begin{aligned} \|f\|_p^p &= \int |f|^p \\ &= \int |f|^p \cdot 1 \\ &= \| |f|^p \cdot 1 \|_1 \\ &\leq \| |f|^p \|_{\frac{q}{p}} \|1\|_{\frac{q}{q-p}} && \text{Hölder's Inequality} \\ &= \left(\int |f|^{p \frac{q}{p}} \right)^{\frac{p}{q}} \left(\int 1 \right)^{\frac{q-p}{q}} \\ &= \|f\|_q^p \mu(X)^{\frac{q-p}{q}} < \infty \end{aligned}$$

therefore, $\|f\|_p^p < \infty$, so $\|f\|_p < \infty$, showing that $f \in L^p$, as we sought to show.

Proposition 2.1.16: Relating L^p and L^∞

Let $p \in [1, \infty)$ and assume $\|f\|_p < \infty$. Show that:

$$\lim_{p \rightarrow \infty} \|f\|_p = \|f\|_\infty$$

Proof:

For ease of typing, we simplify notation: $\|f\|_p := \|f\|_{L^p(\mu)}$

We'll show that $\lim_{p \rightarrow \infty} \|f\|_p \leq \|f\|_\infty$ and $\lim_{p \rightarrow \infty} \|f\|_p \geq \|f\|_\infty$. For the $\leq$ direction, note that $|f| \leq \|f\|_\infty$ a.e., and since x^n is a monotone function on $[0, \infty)$, we have $|f|^n \leq \|f\|_\infty^n$. Then for

some fixed λ_0 :

$$\begin{aligned}\|f\|_p &= \left(\int |f|^p \right)^{1/p} \\ &= \left(\int |f|^{p+\lambda_0-\lambda_0} \right)^{1/p} \\ &= \left(\int |f|^{p-\lambda_0} |f|^{\lambda_0} \right)^{1/p} \\ &\leq \|f\|_\infty^{\frac{p-\lambda_0}{p}} \|f\|_{\lambda_0}^{\frac{\lambda_0}{p}}\end{aligned}$$

where the last inequality is the same trick as part (a) of this question. Since λ_0 is fixed and this inequality holds true for all p , as $p \rightarrow \infty$:

$$\lim_{p \rightarrow \infty} \|f\|_p \leq \lim_{p \rightarrow \infty} \|f\|_\infty^{\frac{p-\lambda_0}{p}} \|f\|_{\lambda_0}^{\frac{\lambda_0}{p}} = \|f\|_\infty$$

For the $\geq$ direction, we have to bound $\|f\|_p$ from below by something that includes $\|f\|_\infty$, but ultimately the inequality will be independent of the choices we make. In particular, let $\epsilon > 0$. Take all points which are epsilon close to the essential supremum (which exists by the infimum property). Let M_ϵ be our set, in particular $M_\epsilon = \{x \mid |f(x)| \geq \|f\|_\infty - \epsilon\}$. Since $\|f\|_\infty = k$, then $\mu(M_\epsilon) > 0$. Thus:

$$\|f\|_p = \left(\int |f|^p \right)^{1/p} \geq \left(\int_{M_\epsilon} (\|f\|_\infty - \epsilon)^p \right)^{1/p} \stackrel{!}{=} (\|f\|_\infty - \epsilon) \mu(M_\epsilon)^{1/p}$$

where $\stackrel{!}{=}$ comes from the fact that we are integrating over a constant region. Since $\mu(M_\epsilon) > 0$, we have that $\mu(M_\epsilon)^{1/p} > 0$ for all p , and $\mu(M_\epsilon)^{1/p} \xrightarrow{p \rightarrow \infty} 1$. Furthermore, notice our inequalities are independent of choice of ϵ , and so since we can let ϵ be as small as we want:

$$\lim_{p \rightarrow \infty} \|f\|_p \geq \|f\|_\infty$$

and since we've shown the $\leq$ direction, we have:

$$\lim_{p \rightarrow \infty} \|f\|_p = \|f\|_\infty$$

as we sought to show

Now that we have examined L^p spaces in more detail, we finish off this section with some comments on the usefulness these spaces:

1. L^1 is natural for integration; we will essentially work with variation's of L^1 when we integrate. From an algebraic perspective, it can be thought of as a generalization of the notion of "counting" all points in a space, and working in spaces where the count is finite.
2. L^∞ is useful as a broadening of the space of uniform-bound functions. It is a generalization of the idea of working with functions that are bounded.

3. L^p interpolates in between L^1 and L^∞ . Many times, L^1 or L^∞ spaces have pathological behaviors (examples of which I yet do not know). L^p spaces behave more nicely, and can sometimes be used to go from L^1 to L^∞ step by step using the inequalities we constructed (ex. we got a function in L^1 , we want to show it's L^∞ , and we use the proceeding inequalities to build our way up)
4. L^2 is particularly nice since it is a Hilbert space, and so many euclidean intuitions work on L^2 and help us construct many nice properties. A lot of Fourier analysis happens on L^2 , though sometimes we do use L^p .

Another way of thinking of L^2 is that it is a generalization of the notion of a space that *can* have an inner product (this will be made rigorous with a particular universal property, see theorem 6.1.24). In ℓ^2 , we would simply get a natural generalization of an inner product on a vector-space, while in L^2 , we get the continuous equivalent.

2.2 The dual of Lebesgue Space

By Hölder's inequality, we have that if $f \in L^p$, then for all $g \in L^q$, $fg \in L^1$. We can thus define a function:

$$\varphi_g : L^p \rightarrow \mathbb{R} \quad \varphi_g(f) = \int fg$$

and the operator norm of φ_g is always at most $\|g\|_q$ ³. With the operator norm, the mapping $g \mapsto \varphi_g$ will almost always be an isometry:

Proposition 2.2.1: Operator Norm And Isometry

Let p and q be conjugate exponents, $1 \leq q < \infty$. If $g \in L^q$, then:

$$\|g\|_q = \|\varphi_g\| = \left\{ \left| \int fg \right| \mid \|f\|_p = 1 \right\}$$

if μ is semifinite, this result also holds for $q = \infty$

Proof:

Folland p. 188

Conversely, if $f \mapsto \int fg$ is a bounded linear functional on L^p , then $g \in L^q$ in almost all cases:

³if $p = 2$, it is usually defined as mapping to $\int f\bar{g}$. This can also be done to $p \neq 2$, there it is just easier to work by conjugating for complex numbers

Theorem 2.2.2: L^p Linear Functional Implies G In L^q

Let p and q be conjugate exponents. Suppose that g is a measurable function on X such that $fg \in L^1$ for all f in the space Σ of simple functions that vanish outside a set of finite measure, and the quantity

$$M_q(g) = \{\|fg\|_1 \mid f \in \Sigma, \|f\|_p = 1\}$$

is finite. Also, suppose either that $S_g = \{x \mid g(x) \neq 0\}$ is σ -finite or that μ is semi-finite. Then $g \in L^q$ and $M_q(g) = \|g\|_q$.

Proof:

Folland p. 189

Finally, perhaps the most striking thing is that the maps $g \mapsto \int fg$ are almost always a surjection of $(L^p)^*$!

Theorem 2.2.3: Description Of Dual Of L^p

Let p and q be conjugate exponents. If $1 < p < \infty$, for each $\varphi \in (L^p)^*$ there exists $g \in L^q$ such that $\varphi(f) = \int fg$ for all $f \in L^p$, and hence L^q is isometrically isomorphic to $(L^p)^*$. The same conclusion holds for $p = 1$ provided μ is σ -finite

Proof:

Folland p. 190

Corollary 2.2.4: L^p Is Reflective

If $1 < p < \infty$, L^p is reflective, that is $(L^p)^{**} \cong L^p$

(Some comment in the case of $p = 1$ and $p = \infty$)

2.3 Some Useful Inequalities

In this section, we explore more inequalities, since they are at the heart of working with L^p spaces. We first present a common inequality that shows up often in statistics:

Theorem 2.3.1: Chebychev's Inequality

Let $f \in L^p$ ($0 < p < \infty$). Then for any $\alpha > 0$

$$\mu(\{x \mid |f(x)| > \alpha\}) \leq \left(\frac{\|f\|_p}{\alpha} \right)^p$$

Proof:

Let $E_\alpha = m(\{x \mid |f(x)| > \alpha\})$. Then:

$$\|f\|_p^p = \int_{\Omega} |f|^p \geq \int_{E_\alpha} |f|^p \geq \int_{E_\alpha} \alpha^p = \alpha^p m(E_\alpha)$$

re-arranging gives us:

$$\mu(\{x \mid |f(x)| > \alpha\}) \leq \left(\frac{\|f\|_p}{\alpha} \right)^p$$

as we sought to show

The next inequality gives us a general statement on being able to get a function from $L^p(\mu)$ “transformed” into a function of $L^p(\nu)$:

Theorem 2.3.2: Transitioning L^p Spaces

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ -finite measures spaces, K be a $M \otimes N$ -measurable function on $X \times Y$, and let's say there exists a $C > 0$ such that

$$\int |K(x, y)| d\mu(x) \leq C \quad \int |K(x, y)| d\nu(y) \leq C$$

for a.e. x and y . Then if $f \in L^p(\nu)$ ($1 \leq p \leq \infty$), the integral:

$$Tf(x) = \int K(x, y) f(y) d\nu(y)$$

converges absolutely for a.e. $x \in X$, $Tf \in L^p(\mu)$, and $\|Tf\|_p \leq C\|f\|_p$

Proof:

Let $p \in [1, \infty]$, and let q be its conjugate exponent. Then:

$$|K(x, y) f(y)| = |K(x, y)| |f(y)| = |K(x, y)|^{\frac{1}{p} + \frac{1}{q}} |f(y)| = (|K(x, y)|^{1/p}) (|K(x, y)|^{1/q} |f(y)|)$$

and applying Hölder to the functions in parenthesis, we get:

$$\begin{aligned} \int |K(x, y) f(y)| d\nu(y) &\leq \left(\int |K(x, y)| d\nu(y) \right)^{1/q} \left(\int |K(x, y)| |f(y)|^p d\nu(y) \right)^{1/p} \\ &\leq C^{1/q} \left(\int |K(x, y)| |f(y)|^p d\nu(y) \right)^{1/p} \end{aligned}$$

for a.e. $x \in X$. Hence, taking the integral over $X \times Y$, using Tonelli's theorem, and substituting

the result we just found, we get:

$$\begin{aligned}
 \int \left[\int |K(x, y) f(y) d\nu(y)|^p d\mu(x) \right] &\leq \int \int C^{p/q} |K(x, y)| |f(y)|^p d\nu(y) d\mu(x) \\
 &= C^{p/q} \int \int |K(x, y)| |f(y)|^p d\mu(x) d\nu(y) \quad \text{Tonelli} \\
 &\leq C^{p/q} \int |f(y)|^p \int K(x, y) d\mu(x) d\nu(y) \\
 &= C^{p/q+1} \int |f(y)|^p d\nu(y)
 \end{aligned}$$

Since $f \in L^p(\nu)$, the last integral is finite. Since this is true for a.e. x , $K(x, \cdot) f \in L^1(\nu)$ for a.e. x . Thus, Tf is well-defined a.e., meaning we can integrate it, and by the inequality we just showed:

$$\int |Tf(x)|^p d\mu(x) = \|Tf\|_p^p \leq C^{p/q+1} \|f\|_p^p$$

taking the p th root of each side get's us the desired inequality

The next result we work towards is a generalization of Minkowski: instead of summing we will be taking integrals. You can also interpret part (a) as you can move the $(|\cdot|^p)^{1/p}$ “down” a layer of integrals, and part (b) as the generalization of monotonicity $(|\int f| \leq \int |f|)$.

Theorem 2.3.3: Minkowski's Inequality For Integrals

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ -finite measure spaces and let f be a [real?] $M \otimes N$ -measurable function on $X \times Y$. Then:

1. if $f \geq 0$ and $1 \leq p \leq \infty$, then:

$$\left[\int \left(\int f(x, y) d\nu(y) \right)^p d\mu(x) \right]^{1/p} \leq \int \left[\int f(x, y)^p d\mu(x) \right]^{1/p} d\nu(y)$$

2. If $1 \leq p \leq \infty$, $f(\cdot, y) \in L^p(\mu)$ for a.e. y and the function $y \mapsto \|f(\cdot, y)\|_p$ is in $L^1(\nu)$, then $f(x, \cdot) \in L^1(\nu)$ for a.e. x , the function $x \mapsto \int f(x, y) d\nu(y)$ is in $L^p(\mu)$ and

$$\left\| \int f(\cdot, y) d\nu(y) \right\|_p \leq \int \|f(\cdot, y)\|_p d\nu(y)$$

Proof:

If $p = 1$, then this becomes Tonelli's Theorem, so consider $1 < p < \infty$ (the case for $p = \infty$ comes later). Let q be it's conjugate exponent so that $1/p + 1/q = 1$. Suppose $\mu, \nu < \infty$ and $\|f\|_\infty < \infty$ (i.e. it's essential supremum is finite). For notational simplicity, define $H(x) = \int_Y f(x, y) d\nu(y)$.

Since the measure is finite and f is essentially bounded, we have that $\|H\|_p < \infty$. Then:

$$\begin{aligned}
 \|H\|_p^p &= \int_X \left(\int_Y f(x, y) d\nu(y) \right)^p d\mu(x) \\
 &= \int_X \left(\int_Y f(x, y) d\nu(y) \right) (H(x))^{p-1} d\mu(x) && x \text{ indep. of } y \\
 &= \int_X \int_Y f(x, y) H(x)^{p-1} d\mu(x) d\nu(y) && \text{Fubini} \\
 &\stackrel{!}{\leq} \int_X \|f(x, \cdot)\|_p \|H\|_p^{p-1} d\mu(x) && \text{Hölder, } q = p/(p-1) \\
 &\leq \|H\|_p^{p-1} \int_X \|f(x, \cdot)\|_p d\mu(x)
 \end{aligned}$$

where $\stackrel{!}{\leq}$ I'm not sure how Hölder's inequality was used. Finally, dividing by $\|H\|_p^{p-1}$ get's us:

$$\|H\|_p \leq \int_X \|f(x, \cdot)\|_p d\mu(x)$$

For the general case, Yakov left it as an exercise to do $X = \bigcup_n X_n$, $Y = \bigcup_n Y_n$, $E_n = \{x \mid f(x) < n\}$, and

$$\tilde{X}_n = X_n \cap E_n \quad \tilde{Y}_n = Y_n \cap E_n$$

and apply the result to each of these, and the monotone convergence theorem at the end (feels similar to how it was done for Fubini Tonelli).

For $p = \infty$, This comes down to monotonicity of integrals:

$$\left\| \int f(x, y) d\nu(y) \right\|_\infty \leq \left| \int f(x, y) d\nu(y) \right| \leq \int |f(x, y)| d\nu(y) = \int \|f(x, \cdot)\|_\infty d\mu(x)$$

where the last inequality comes from the fact that the integral ignores zero-measure sets.

There are a few more results in the book, but Yakov decided to prove the generalized Hölder's inequality instead for now:

Theorem 2.3.4: Generalized Hölder's Inequality

Let $1 \leq p_i \leq \infty$ for $1 \leq i \leq n$ and $\sum_{i=1}^n p_i^{-1} = r^{-1} \leq 1$. If $f_i \in L^{p_i}$, then $\prod_{i=1}^n f_i \in L^r$ and

$$\left\| \prod_{i=1}^n f_i \right\|_r \leq \prod_{i=1}^n \|f_i\|_{p_i}$$

Proof:

If $r = \infty$, then $r^{-1} = 0$, so all the $p_i^{-1} = 0$, so all the $p_i = \infty$, in which case we simply take:

$$\left| \prod_{i=1}^n f_i \right| \leq \prod_{i=1}^n |f_i|$$

For $r < \infty$, just like for Hölder's inequality, it suffice to assume $r = 1$, since we can do:

$$\sum \frac{1}{\frac{p_i}{r}} = 1$$

In that case, notice that:

$$\begin{aligned} \left\| \prod_i^n f_i \right\|_r^r &= \int |f_1 f_2 \cdots f_n|^r \\ &= \int f_1^r f_2^r \cdots f_n^r \\ &\leq \|f_1\|_{p_1}^r \|f_2\|_{p_2}^r \cdots \|f_n\|_{p_n}^r \end{aligned}$$

Notice that each f_i is raised to the power of p_i/r . (TBD)

Now, we do induction on i . Let the induction hypothesis be:

$$\left(\sum_i^{n-1} p_i^{-1} \right) + p_n^{-1} = r^{-1} + p_n^{-1} = 1$$

Then by Hölder's inequality:

$$\int f_1 \cdots f_n \leq \|f_1 \cdots f_{n-1}\|_r \|f\|_{p_n} \stackrel{\text{I.H.}}{\leq} \|f_1\|_{p_1} \cdots \|f_n\|_{p_n}$$

giving our desired result

The final property we'll present a way to represent functions in terms of the integral:

Proposition 2.3.5: Layer Cake Representation

Let $\varphi(t) : [0, \infty] \rightarrow [0, \infty]$ be an increasing function which is C^1 on $(0, \infty)$, satisfies $\varphi(0) = 0$ and satisfies $\lim_{t \rightarrow \infty} \varphi(t) = \varphi(\infty)$. Let (X, M, μ) be a σ -finite measure space and $f : X \rightarrow [0, \infty]$ be a non-negative measurable function. Then:

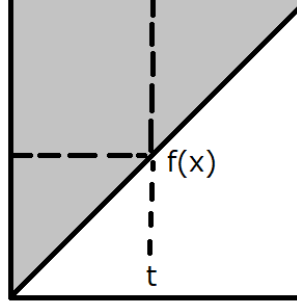
$$\int_X \varphi(f(x)) d\mu(x) = \int_0^\infty \mu(\{x \mid f(x) > t\}) \varphi'(t) dt$$

One consequence of this equation is that with $\varphi(t) = t^p$ for $1 < p < \infty$ is the following formula for the L^p norm of f :

$$\|f\|_p = \left(p \int_0^\infty \mu(\{x \mid f(x) > t\}) t^{p-1} \right)^{1/p} dt$$

Proof:

The proof can be captured pictorially as integrating over the following area:



Let $S \subseteq X \times [0, 1]$, $S = \{(x, t) \mid f(x) \geq t\}$. With this set, define $F : X \times [0, \infty] \rightarrow [0, \infty]$, $F(x, t) = \varphi'(t)\chi_S$. That is, we are going to integrate the “upper triangle”. Since φ is an increasing C^1 function and $\varphi'(t) > 0$, we have $F \in L^+(X, [0, \infty])$. Therefore, we can apply Tonelli’s Theorem and see that it’s equivalent to integrate over the iterated integrals to compute integrals.

To compute it, we need to be clever with the bounds we choose which depend on which iterated integral we choose. The different direction of the iterated integral will ultimately lead to the desired result. If we integrate with respect to $d\mu(x)dt$, then we take slices across t : $(\varphi'(t)\chi_S)_t = \varphi'(t)\chi_{S_t}$, where $(\chi_S)_t = \chi_{S_t}(x, t) = \chi_{S_t}$, with $S_t = \{x \mid f(x) \geq t\}$. And so we get:

$$\int_0^\infty \int_X \chi_{S_t} \varphi'(t) d\mu(x) dt = \int_0^\infty \int_{\{x \mid f(x) \geq t\}} \varphi'(t) d\mu(x) dt = \int_0^\infty \mu(\{x \mid f(x) \geq t\}) \varphi'(t) dt$$

Now, integrating with respect to $td\mu(x)$, we take x slices: $(\varphi'(t)\chi_S)_x = \varphi'(t)\chi_{S_x}$ where $(\chi_S)_x = \chi_{S_x}(x, t) = \chi_{S_x}$, with $S_x = \{t \mid 0 \leq f(x)\}$. And so we get:

$$\int_X \int_0^\infty \chi_{S_x} \varphi'(t) t d\mu(x) = \int_X \int_0^{f(x)} \varphi'(t) t d\mu(x)$$

and since $[0, f(x)]$ is bounded and $\varphi(0) = 0$, the Lebesgue integral matches the Riemann integral, and so we get that:

$$\int_X \varphi(f(x)) d\mu(x)$$

and by Tonelli’s theorem:

$$\int_X \varphi(f(x)) d\mu(x) = \int_0^\infty \mu(\{x \mid f(x) \geq t\}) \varphi'(t) dt$$

As we sought to show. Using this, if we let $\varphi(t) = t^p$, then we see that φ is C^1 on $(0, \infty)$ and $\varphi(0) = 0$ for $1 \leq p < \infty$. Thus, the value of $\|f\|_p$ (where we take f^p instead of $|f|^p$ since $f \geq 0$) is:

$$\begin{aligned} \|f\|_{L^p} &= \left(\int |f|^p \right)^{1/p} = \left(\int f^p \right)^{1/p} = \left(\int_X \varphi(f(x)) d\mu(x) \right)^{1/p} \\ &= \left(p \int_0^\infty \mu(\{x \mid f(x) > t\}) t^{p-1} dt \right)^{1/p} \end{aligned}$$

Another use of this formula is by taking $\varphi(t) = t$. Then we will get that:

$$\int f(x) d\mu(x) = \int_0^\infty \mu(\{x \mid f(x) \geq t\}) dt$$

This can be useful if we have information of what is happening to the growth of f as f gets larger, in particular if we know that somehow, the amount of points such that $\mu(\{x \mid f(x) \geq t\})$ is shrinking rapidly enough say that $\mu(\{x \mid f(x) \geq t\}) \leq \frac{C}{t^r}$ for some constant C .

2.4 Interpolation: The Riesz-Thorin Theorem

As we saw in the previous section, if a function is in L^p and L^r , it is also in L^q for all $p < q < r$. This is a type of “interpolation” of spaces. A natural question arises: if we have a linear operator T that is bounded on L^p and bounded on L^r , is it also bounded on L^q ? The answer is yes, and the Riesz-Thorin theorem provides the precise quantitative relationship between the operator norms.

The proof relies on a powerful complex analysis technique known as the Phragmén-Lindelöf principle, specifically a version called the Three Lines Lemma.

Lemma 2.4.1: Three Lines Lemma

Let φ be a bounded continuous function on the strip $S = \{z \in \mathbb{C} \mid 0 \leq \operatorname{Re}(z) \leq 1\}$ that is holomorphic on the interior of S . If $|\varphi(z)| \leq M_0$ for $\operatorname{Re}(z) = 0$ and $|\varphi(z)| \leq M_1$ for $\operatorname{Re}(z) = 1$, then for all $t \in [0, 1]$ and z with $\operatorname{Re}(z) = t$:

$$|\varphi(z)| \leq M_0^{1-t} M_1^t$$

Proof:

For $\epsilon > 0$, define $\varphi_\epsilon(z) = \varphi(z) M_0^{z-1} M_1^{-z} e^{\epsilon(z^2-1)}$. The term $e^{\epsilon(z^2-1)}$ helps handle the boundedness at infinity within the strip. By the maximum modulus principle applied to rectangles in the strip as the height goes to infinity, one can show $|\varphi_\epsilon(z)| \leq 1$ in the strip. Letting $\epsilon \rightarrow 0$ yields the result.

Theorem 2.4.2: Riesz-Thorin Interpolation Theorem

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be measure spaces. Let $p_0, p_1, q_0, q_1 \in [1, \infty]$. Define p_t and q_t for $t \in (0, 1)$ by:

$$\frac{1}{p_t} = \frac{1-t}{p_0} + \frac{t}{p_1}, \quad \frac{1}{q_t} = \frac{1-t}{q_0} + \frac{t}{q_1}$$

Suppose T is a linear operator defined on the set of simple functions Σ such that for all $f \in \Sigma$:

$$\|Tf\|_{q_0} \leq M_0 \|f\|_{p_0} \quad \text{and} \quad \|Tf\|_{q_1} \leq M_1 \|f\|_{p_1}$$

Then for all $t \in (0, 1)$, T extends to a bounded operator from $L^{p_t}(\mu)$ to $L^{q_t}(\nu)$ satisfying:

$$\|Tf\|_{q_t} \leq M_0^{1-t} M_1^t \|f\|_{p_t}$$

Proof:

Let $t \in (0, 1)$ be fixed. By duality (Proposition 2.2.1), it suffices to estimate $|\int (Tf)g \, d\nu|$ where f and g are simple functions with $\|f\|_{p_t} = 1$ and $\|g\|_{q'_t} = 1$ (where q'_t is the conjugate exponent of q_t).

We construct a family of functions to apply the Three Lines Lemma. Let $f = |f|e^{i\theta}$ and $g = |g|e^{i\psi}$ be the polar decompositions. For any complex number z in the strip $0 \leq \operatorname{Re}(z) \leq 1$, define:

$$\alpha(z) = (1-z)\frac{1}{p_0} + z\frac{1}{p_1}, \quad \beta(z) = (1-z)\frac{1}{q'_0} + z\frac{1}{q'_1}$$

Note that for $z = t$, $\alpha(t) = 1/p_t$ and $\beta(t) = 1/q'_t$. Now define deformations of f and g :

$$f_z = |f|^{p_t \alpha(z)} e^{i\theta}, \quad g_z = |g|^{q'_t \beta(z)} e^{i\psi}$$

(If $f = 0$ or $g = 0$, set the function to 0). Note that $f_t = f$ and $g_t = g$.

Define the function $\Phi(z)$ on the strip $0 \leq \operatorname{Re}(z) \leq 1$:

$$\Phi(z) = \int_Y (Tf_z)g_z \, d\nu$$

Since f and g are simple, f_z is a finite linear combination of terms like $c_k \alpha^z$, making $\Phi(z)$ holomorphic and bounded.

We check the boundary conditions. For $\operatorname{Re}(z) = 0$, $f_z \in L^{p_0}$ and $g_z \in L^{q'_0}$. Specifically:

$$\|f_{iy}\|_{p_0}^{p_0} = \int \|f|^{p_t(\frac{1}{p_0} + i\operatorname{Im}(\alpha))}\|_{p_0}^{p_0} = \int |f|^{p_t} = \|f\|_{p_t}^{p_t} = 1$$

Similarly $\|g_{iy}\|_{q'_0} = 1$. Thus by Hölder's inequality and the hypothesis on T :

$$|\Phi(iy)| = \left| \int T(f_{iy})g_{iy} \right| \leq \|Tf_{iy}\|_{q_0} \|g_{iy}\|_{q'_0} \leq M_0 \|f_{iy}\|_{p_0} \cdot 1 = M_0$$

Similarly, for $\operatorname{Re}(z) = 1$, we find $|\Phi(1 + iy)| \leq M_1$.

By the Three Lines Lemma, $|\Phi(t)| \leq M_0^{1-t} M_1^t$. Since $\Phi(t) = \int (Tf)g$, and this holds for all unit simple functions f, g , we conclude:

$$\|T\|_{L^{p_t} \rightarrow L^{q_t}} \leq M_0^{1-t} M_1^t$$

Counter-Example: The Linear Transport Equation

Consider the simple advection equation in one dimension:

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0$$

where $c \neq 0$ is a real constant velocity.

1. **The Operator:** The solution is a translation of the initial state $f(x)$:

$$u(x, t) = f(x - ct)$$

Therefore, the time-shift operator S_t acts as:

$$(S_t f)(x) = f(x - ct)$$

2. **The Adjoint:** To check if S_t is self-adjoint, we compute its adjoint S_t^* using the standard L^2 inner product $\langle f, g \rangle = \int_{-\infty}^{\infty} f(x) \overline{g(x)} dx$.

$$\langle S_t f, g \rangle = \int_{-\infty}^{\infty} f(x - ct) \overline{g(x)} dx$$

Perform a change of variables $y = x - ct$, so $x = y + ct$ and $dx = dy$:

$$\langle S_t f, g \rangle = \int_{-\infty}^{\infty} f(y) \overline{g(y + ct)} dy$$

$$\langle S_t f, g \rangle = \langle f, S_{-t} g \rangle$$

This shows that the adjoint is the inverse shift:

$$S_t^* = S_{-t}$$

3. **Conclusion:** For S_t to be self-adjoint, we require $S_t = S_t^*$. This would imply $S_t = S_{-t}$, or $f(x - ct) = f(x + ct)$ for all functions f . This is clearly false for $c \neq 0$ and $t \neq 0$. Thus, S_t **is not self-adjoint**.

General Context

- **Unitary vs. Self-Adjoint:** In conservative systems (like Quantum Mechanics or the Wave Equation), the evolution operator S_t is typically **unitary**, meaning $S_t^* = S_t^{-1} = S_{-t}$. A unitary operator is rarely self-adjoint (unless $S_t^2 = I$, which is not true for general time evolution).
- **When is it Self-Adjoint?** S_t *is* self-adjoint in specific dissipative systems, such as the **Heat Equation** ($u_t = \Delta u$). In this case, $S_t = e^{t\Delta}$. Since the Laplacian Δ is a self-adjoint operator (with appropriate boundary conditions), the operator exponential $e^{t\Delta}$ is also self-adjoint for real t .

Zak Katz: I see. Does that mean that S_t is not always “diagonalizable”?

Raycast AI: No, that is a common misconception! “**Not self-adjoint**” **does not mean** “**not diagonalizable**.”

The time-shift operator S_t for the Transport Equation (which I used as a counter-example previously) is diagonalizable, even though it is not self-adjoint.

Here is the breakdown of why this works and when it fails.

1. The Key Concept: Normal Operators

In linear algebra and functional analysis, a deeper property governs diagonalizability: **Normality**.

An operator A is **Normal** if it commutes with its adjoint:

$$AA^* = A^*A$$

The **Spectral Theorem** states that **any Normal operator is diagonalizable** (specifically, unitarily diagonalizable).

- **Self-Adjoint Operators** are Normal ($A = A^*$, so $AA = AA$).
- **Unitary Operators** are Normal ($A^* = A^{-1}$, so $AA^{-1} = A^{-1}A = I$).

2. Revisiting the Transport Equation (The “Yes” Case)

Let’s look at the Transport Equation again ($u_t + u_x = 0$), where S_t shifts the function: $f(x) \rightarrow f(x-t)$.

- We found that $S_t^* = S_{-t}$ (the inverse shift).
- Does it commute?

$$S_t S_t^* = S_t S_{-t} = S_0 = I$$

$$S_t^* S_t = S_{-t} S_t = S_0 = I$$

- Because $S_t S_t^* = S_t^* S_t$, the operator is **Normal**.

How do we “diagonalize” it? We use the **Fourier Transform**. In the frequency domain (Fourier space), the shift operator becomes multiplication:

$$\mathcal{F}(S_t f)(k) = e^{-ikt} \hat{f}(k)$$

Multiplication by a scalar (or function) is the infinite-dimensional equivalent of a diagonal matrix. The Fourier basis functions e^{ikx} are the “eigenvectors” of the time-shift operator.

Conclusion: The Transport operator is not self-adjoint, but it is normal, so it **is** diagonalizable.

3. When is S_t NOT Diagonalizable? (The “No” Case)

For S_t to fail to be diagonalizable (to be “defective”), the system must be **Non-Normal** in a specific way that creates “Jordan Blocks.” This happens when eigenvectors coalesce and you lose a complete basis (this is called an **Exceptional Point** in physics).

Example: A “Defective” System of PDEs Consider a coupled system of two variables $u(x, t)$ and $v(x, t)$:

$$\frac{\partial}{\partial t} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}$$

Here, the change in u depends on v , but v is constant in time.

1. Solve it:

- $v(x, t) = v(x, 0)$ (constant)
- $u_t = v(x, 0) \implies u(x, t) = u(x, 0) + t \cdot v(x, 0)$

2. The Time-Evolution Operator S_t : We can write S_t as a matrix acting on the vector $\begin{pmatrix} f \\ g \end{pmatrix}$:

$$S_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

3. Is it Diagonalizable? Check the matrix $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$.

- The eigenvalues are $\lambda = 1$ (with algebraic multiplicity 2).
- However, there is only **one** independent eigenvector: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.
- Because we are missing a second eigenvector, this operator is **not diagonalizable** for any $t \neq 0$.

Summary Table

Type of PDE	Operator Property	Self-Adjoint?	Diagonalizable?
Heat Eq. ($u_t = \Delta u$)	Self-Adjoint	Yes	Yes (Real eigenvalues)
Transport / Wave	Unitary	No	Yes (Complex eigenvalues)
Defective System	Non-Normal	No	No (Jordan Block)

Bottom Line: Most standard textbook PDEs (Heat, Wave, Schrödinger) have diagonalizable time-evolution operators because they are typically governed by Normal operators. You usually have to construct specific coupled systems or non-Hermitian boundary conditions to break diagonalizability.

Zak Katz: Let me be clear: S_t shifts *in time* not in space. So to be clear: let a PDE $F(x_1, \dots, x_n, t) = 0$ PDE w.r.t time t , if $u(x, t)$ is a solution and $f_t(x) = u(x, t)$ is a “state” at time t , and Let S_t be the “time-shift” operator so that $S_t(f_a) = f_{a+t}$ for some state f_a at time a . (NOT the space shift operator $f(x) \rightarrow f(x+t)$). Is S_t always diagonalizable?

Raycast AI: Thank you for the clarification. I understand the distinction you are making: you are asking about the **time-evolution operator** (often called the Propagator, $U(t)$ or S_t) that maps the initial state $u(\cdot, 0)$ to the state at time t , $u(\cdot, t)$.

The answer remains: **No, S_t is not always diagonalizable.**

While many standard textbook PDEs (Heat, Wave, Schrödinger) have diagonalizable time-evolution operators because their generators are Normal (Self-Adjoint or Skew-Adjoint), it is very easy to construct physical systems where S_t is **defective** (non-diagonalizable).

This happens specifically when the system exhibits **Linear Instability** or **Critical Damping** (often associated with “Exceptional Points” in modern physics).

The Mathematical Reason: Jordan Blocks

If a PDE can be written in the form $\frac{\partial}{\partial t}\psi = L\psi$, the time-evolution operator is formally $S_t = e^{Lt}$.

For S_t to be diagonalizable, the generator L must possess a complete basis of eigenfunctions. If L has a **Jordan Block** (i.e., it has generalized eigenvectors but not enough true eigenvectors), then the exponential $S_t = e^{Lt}$ will also possess a Jordan block structure for any $t \neq 0$, making it non-diagonalizable.

Here are two concrete counter-examples.

Counter-Example 1: The “Resonant” System (Algebraic Growth)

Consider a system of two fields, $u(x, t)$ and $v(x, t)$, on a domain Ω . The PDE is:

$$\frac{\partial}{\partial t} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}$$

where λ is a constant. This is a standard linear PDE system.

1. **Find the Solution:** This is a coupled system.

- $v_t = \lambda v \implies v(x, t) = e^{\lambda t}v(x, 0)$.
- $u_t = \lambda u + v$. Substituting the solution for v : $u_t = \lambda u + e^{\lambda t}v(x, 0)$.
- The solution for u involves a secular term (a linear t):

$$u(x, t) = e^{\lambda t}u(x, 0) + te^{\lambda t}v(x, 0)$$

2. **The Time-Evolution Operator S_t :** Writing this as an operator matrix acting on the initial vector $\psi_0 = \begin{pmatrix} u_0 \\ v_0 \end{pmatrix}$:

$$S_t = e^{\lambda t} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

3. **Why it is NOT Diagonalizable:** Look at the matrix part $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$.

- **Eigenvalues:** It has a single eigenvalue $\mu = 1$ (repeated).
- **Eigenvectors:** It has only **one** linearly independent eigenvector: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.
- Because we need two independent eigenvectors to span the 2D space of fields at every point x , and we only have one, S_t is **not diagonalizable** for any $t \neq 0$.

Physical Interpretation: The term $te^{\lambda t}$ represents “algebraic growth” (or transient growth). Whenever you see a solution of the form $t \times (\text{exponential})$, it is the signature of a non-diagonalizable operator. A diagonalizable operator would only have pure sums of exponentials.

Counter-Example 2: Critical Damping (Classical Mechanics / PDE)

Consider the Damped Harmonic Oscillator equation for a field $u(x, t)$, which is a second-order PDE in time:

$$u_{tt} + 2\gamma u_t + \gamma^2 u = 0$$

(Here $\gamma > 0$ is the damping coefficient).

To analyze S_t , we convert this to a first-order system (state-space form) by defining $v = u_t$:

$$\frac{\partial}{\partial t} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\gamma^2 & -2\gamma \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}$$

The Critical Case: The eigenvalues of the matrix $L = \begin{pmatrix} 0 & 1 \\ -\gamma^2 & -2\gamma \end{pmatrix}$ are found by $\det(L - \omega I) = 0$.

$$\omega^2 + 2\gamma\omega + \gamma^2 = (\omega + \gamma)^2 = 0$$

We have a repeated eigenvalue $\omega = -\gamma$.

Because the matrix L is not the identity matrix but has a repeated eigenvalue, it is a non-trivial Jordan block. The solution is:

$$u(t) = C_1 e^{-\gamma t} + C_2 t e^{-\gamma t}$$

The presence of the $t e^{-\gamma t}$ term confirms that the propagator S_t cannot be diagonalized. If it were diagonalizable, we could write the solution purely as $u(t) = A e^{\omega_1 t} + B e^{\omega_2 t}$, which implies strictly exponential decay without the linear polynomial pre-factor.

Summary

- **Is S_t always diagonalizable? No.**
- **When does it fail?** It fails at **Exceptional Points**, where two or more eigenmodes of the system coalesce (become parallel), making the set of eigenvectors incomplete. This results in solutions that grow or decay with polynomial factors (like t , t^2) multiplying the exponentials.

Zak Katz: Okay. Now, when it *is* diagonalizable, is it *always* the case that we can use Fourier transforms to find an eigenbasis?

Raycast AI: No. Absolutely not.

Just because an operator is diagonalizable does not mean the Fourier Transform provides the basis.

The **Fourier Transform** is a very specific tool that only diagonalizes operators that are **Translation Invariant** (i.e., operators with **Constant Coefficients** defined on the entire space $\mathbb{R}^n$).

If your PDE has **variable coefficients** or is defined on a **bounded domain**, the Fourier basis (e^{ikx}) will generally fail to diagonalize the operator, yet a different eigenbasis might still exist perfectly fine.

Here is the hierarchy of logic:

1. **Diagonalizable:** Means *there exists* a basis $\{\varphi_n(x)\}$ such that the operator acts as multiplication on these functions.
2. **Fourier Diagonalizable:** Means that specific basis is $\{\varphi_k(x) = e^{ikx}\}$.

When does Fourier work?

Fourier analysis works **only** if your operator L commutes with the translation operator T_a (where $T_a f(x) = f(x + a)$).

- **Example:** The Heat Equation $u_t = u_{xx}$.
 - The second derivative behaves the same at $x = 0$ as it does at $x = 100$. Space is homogeneous.
 - Therefore, the eigenfunctions are plane waves e^{ikx} .

When does Fourier fail (but the operator is still diagonalizable)?

Fourier fails whenever **spatial symmetry is broken**. This happens in three common scenarios:

1. **Variable Coefficients (Space is not homogeneous)** Consider the **Quantum Harmonic Oscillator** (one of the most famous diagonalizable systems in physics).

$$i \frac{\partial \psi}{\partial t} = -\frac{\partial^2 \psi}{\partial x^2} + x^2 \psi$$

- **Is it Diagonalizable?** Yes. The Hamiltonian is Self-Adjoint. It has a discrete, perfect set of eigenfunctions.
- **Are they Fourier Modes?** No. The term x^2 breaks translation invariance (physics looks different at $x = 0$ vs $x = 100$).
- **The Actual Basis:** The eigenfunctions are **Hermite Functions** (a Gaussian multiplied by a polynomial):

$$\varphi_n(x) = H_n(x) e^{-x^2/2}$$

If you tried to apply a Fourier Transform to this equation, the x^2 multiplication term turns into a Laplacian in k -space ($-\partial_k^2$), so you effectively just swap the PDE for an identical PDE in frequency space without solving it.

2. **Bounded Domains (The “Boundary Condition” Problem)** Consider the heat equation $u_t = u_{xx}$ on a finite interval $x \in [0, L]$ with Dirichlet boundary conditions $u(0) = u(L) = 0$.

- **Is it Diagonalizable?** Yes.
- **Is it the Fourier Transform?** No. The Continuous Fourier Transform uses e^{ikx} for any real k .
- **The Actual Basis:** The basis is strictly **Sine functions**: $\sin(\frac{n\pi x}{L})$.

- While related to Fourier *Series*, the Continuous Fourier *Transform* basis (e^{ikx}) is invalid here because e^{ikx} does not satisfy the boundary conditions (it doesn't vanish at 0 and L). You are forced onto a discrete subset of the frequency spectrum.

3. Different Geometries If your PDE is on a sphere (like weather patterns on Earth):

$$\frac{\partial u}{\partial t} = \Delta_{\text{sphere}} u$$

- **Is it Diagonalizable?** Yes.
- **The Actual Basis:** The eigenfunctions are **Spherical Harmonics** $Y_l^m(\theta, \varphi)$.
- Standard planar Fourier modes (e^{ikx}) make no sense here.

Summary Table

PDE Feature	Is S_t Diag.?	Fourier?	The Actual Eigenbasis
Constant Coeffs on $\mathbb{R}$ ($u_t = u_{xx}$)	Yes	Yes	Plane Waves (e^{ikx})
Variable Coeffs ($u_t = xu_x$)	Yes	No	Power functions (x^λ)
Potential Well ($u_t = i\Delta u + iV(x)u$)	Yes	No	Hermite / Legendre / Bessel
Bounded Domain ($x \in [0, 1]$)	Yes	No	Fourier <i>Series</i> (Sines/Cosines)

Conclusion: Diagonalizability guarantees that *some* coordinate system exists where the operator is simple. It does not guarantee that the coordinate system is the one given by Fourier.

Zak Katz: Okay. Is it a necessary and sufficient condition that S_t is diagonalizable by the Fourier basis iff the diagonalizable operator is translation invariant? Just to make sure I understand what translation invariant means here, we are talking about *space* translation, and we mean that if S_t is our operator and τ_a is our space-translation, then:

$$S_t(\tau_a(f)) = \tau_a(S_t(f))$$

Raycast AI: Yes, precisely.

For a linear operator S_t acting on functions defined on the entire space $\mathbb{R}^n$ (or a periodic domain), the condition is **necessary and sufficient**:

Theorem: A bounded linear operator S_t is diagonalized by the Fourier Transform if and only if it is **Translation Invariant**.

Your definition of translation invariance is exactly correct:

$$[S_t, \tau_a] = S_t \tau_a - \tau_a S_t = 0 \quad \text{for all } a \in \mathbb{R}^n$$

where τ_a is the spatial shift operator.

Here is the logic for why this is an “if and only if” relationship.

1. Why is it Sufficient? (Translation Invariant $\implies$ Fourier Diagonal)

This relies on a fundamental principle of linear algebra: **Commuting operators share a common eigenbasis.**

1. Consider the family of translation operators $\{\tau_a\}_{a \in \mathbb{R}}$.
2. What are the eigenfunctions of a translation? Let’s verify that the plane waves e^{ikx} are eigenfunctions of τ_a :

$$\tau_a(e^{ikx}) = e^{ik(x+a)} = e^{ika} \cdot e^{ikx}$$

The eigenvalue is the constant number e^{ika} , and the eigenfunction is e^{ikx} .

3. Since your time-evolution operator S_t commutes with **every** translation operator τ_a (for all shifts a), it must share the same set of eigenfunctions.
4. Therefore, the eigenfunctions of S_t must also be e^{ikx} .
5. If the eigenfunctions are e^{ikx} , then by definition, the operator is diagonalized by the Fourier Transform.

2. Why is it Necessary? (Fourier Diagonal $\implies$ Translation Invariant)

If an operator is diagonal in Fourier space, it acts as a simple multiplication. Let’s call the Fourier transform $\mathcal{F}$.

1. Assume S_t is a Fourier Multiplier: $\widehat{S_t u}(k) = m(k)\widehat{u}(k)$ for some multiplier function $m(k)$.
2. The translation operator τ_a in Fourier space is also just a multiplication (by the phase factor e^{ika}).
3. Since regular multiplication of scalars is commutative ($a \cdot b = b \cdot a$), the multiplication by $m(k)$ commutes with the multiplication by e^{ika} .
4. Therefore, S_t and τ_a commute.

Summary

This concept is the bedrock of signal processing and linear systems theory (where these are called **Linear Time-Invariant (LTI)** systems, or in your PDE context, “Linear Space-Invariant” systems).

- **Does S_t commute with space shifts?**

- **Yes:** You can use Fourier Transforms (e.g., Heat Eq, Wave Eq, Schrödinger with constant potential).

- **No:** You cannot use Fourier Transforms to find the eigenbasis (e.g., Schrödinger with Harmonic Oscillator potential, variable coefficient Wave Eq).

Zak Katz: Right. Is there a “theory” for diagonalizable operators S_t ?

Raycast AI: Yes, there is a vast and rich “theory” for this, but it doesn’t have a single name like “The Theory of Diagonalizable Operators.” Instead, it is split into different fields of **Functional Analysis** depending on *why* the operator is diagonalizable.

If you are looking for the generalized version of “Fourier Analysis” for operators that are **not** translation invariant (i.e., not solvable by Fourier Transform), here is the map of the theory.

1. The “Gold Standard”: The Spectral Theorem

This is the master theory. It applies if your operator S_t (or its generator L) is **Normal** ($LL^* = L^*L$).

- **The Theory: Spectral Theory of Normal Operators.**
- **What it says:** Any normal operator on a Hilbert space is unitarily equivalent to a multiplication operator.
- **Physical Meaning:** There *always* exists an orthonormal basis of eigenfunctions (or a “continuous basis” / spectral measure) that diagonalizes the operator.
- **How to find the basis:** This is the hard part. It usually requires solving the **Sturm-Liouville Problem** (for 1D) or using **Scattering Theory** (for unbounded domains).

2. The “Variable Coefficient” Theory: Sturm-Liouville Theory

If your PDE has variable coefficients but is in 1D (or separable), this is the standard replacement for Fourier analysis.

- **The Setup:** Your generator looks like $L = \frac{d}{dx}[p(x)\frac{d}{dx}] + q(x)$.
- **The Result:** Just like Fourier series uses $\sin(nx)$ and $\cos(nx)$, Sturm-Liouville theory guarantees a set of orthogonal eigenfunctions $\varphi_n(x)$ weighted by a density function.
- **Example:** Legendre polynomials, Bessel functions, Hermite polynomials (Quantum Harmonic Oscillator). These are the “custom Fourier bases” for specific geometries or potentials.

3. The “Abstract” Theory: Semigroup Theory (C_0 -Semigroups)

This is the modern mathematical framework for time-evolution operators S_t .

- **The Setup:** You treat S_t as a “one-parameter semigroup” $T(t)$ acting on a Banach space.
- **The Theorem:** The **Hille-Yosida Theorem**.

- **What it does:** It characterizes exactly when a differential operator L generates a stable time-evolution S_t . While Hille-Yosida doesn't guarantee *diagonalizability* (it only guarantees existence and uniqueness of the flow), it is the starting point for analyzing the spectrum of S_t .
- **Spectral Mapping Theorem:** This theorem relates the eigenvalues of the generator L to the eigenvalues of the propagator S_t (basically $\sigma(S_t) = e^{t\sigma(L)}$). If you know the spectrum of L , you know the spectrum of S_t .

4. The “Non-Normal” Theory: Riesz Bases and Spectral Operators

What if S_t is diagonalizable but the eigenvectors are **not orthogonal**? (This happens in non-conservative systems, like convection-diffusion with non-standard boundary conditions).

- **The Theory: Non-Self-Adjoint Spectral Theory** (or the theory of **Riesz Bases**).
- **What it says:** You can still expand solutions in a basis of eigenfunctions, but because the basis is not orthogonal ($\langle \varphi_n, \varphi_m \rangle \neq \delta_{nm}$), you need a “dual basis” $\{\psi_n\}$ to find the coefficients.
 - Expansion: $f = \sum a_n \varphi_n$
 - Coefficients: $a_n = \langle f, \psi_n \rangle$ (instead of $\langle f, \varphi_n \rangle$).
- **Why it matters:** This is currently a very hot topic in fluid dynamics and laser physics, where systems are often “almost” normal but mathematically non-self-adjoint.

Summary: Which “Theory” do you need?

If your operator S_t is...	The Theory is called...	The Basis is...
Translation Invariant	Harmonic Analysis	Fourier Transform
Self-Adjoint / Normal	Spectral Theory	Orthonormal Eigenbasis
1D Variable Coeffs	Sturm-Liouville Theory	Special Functions
Not Normal (but diag.)	Non-Self-Adjoint Theory	Riesz Basis (Biorthogonal)
Not Diagonalizable	Jordan Form	Generalized Eigenvectors

2.5 Convolution and Approximate Identities

In this section, we introduce a new way of multiplying two functions that will be very useful and very natural. The naturality comes from extending the addition of $\mathbb{R}$ to functions on $\mathbb{R}$, for simplicity we stick to real functions however nothing changes if we use complex functions instead. Consider the function $\delta_a : \mathbb{R} \rightarrow \mathbb{R}$

$$\delta_a(x) = \begin{cases} 1 & x = a \\ 0 & \text{otherwise} \end{cases}$$

The naturality comes from the discrete case, and so we will start with it: recall that if we have two polynomials $p(x)$ and $q(x)$ (which can be thought of as formal sums, bringing in the naturality), then

the multiplication formula which groups the terms of the same degree is:

$$\left(\sum_{i=0}^{\deg(p)} p_i x^i \right) \cdot \left(\sum_{j=0}^{\deg(q)} q_j x^j \right) = \sum_{k=0}^{\deg(p)+\deg(q)} \left(\sum_{i+j=k} p_i q_j \right) x^k = \sum_{k=0}^{\deg(p)+\deg(q)} \left(\sum_{l=0}^k p_l q_{k-l} \right) x^k$$

If we set p_k or q_{l-k} to zero if k or $l-k$ is negative, then we can re-write the sum as:

$$\left(\sum_{i=0}^{\deg(p)} p_i x^i \right) \cdot \left(\sum_{j=0}^{\deg(q)} q_j x^j \right) = \sum_{k=0}^{\deg(p)+\deg(q)} \left(\sum_{l=-\infty}^{\infty} p_l q_{k-l} \right) x^k$$

If we let $p[n]$ and $q[n]$ represent the coefficient of the n th degree term, and $h = pq$, then:

$$h[n] = \sum_{k=-\infty}^{\infty} p_k q_{n-k}$$

We can write this as:

$$h[n] = [p * q][n] = \sum_{k=-\infty}^{\infty} p_k q_{n-k}$$

where $p * q$ is a compact notation the right hand side. The summation can be thought of as representing the discrete measure. Generalized to the continuous case, this is called *convolution*:

Definition 2.5.1: Convolution

Let f, g be measurable on $\mathbb{R}^n$. Then the *convolution* of f and g is a function $f * g : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by:

$$f * g(x) := \int f(x-y)g(y)dy$$

for all x such that the integral exists

To see this in action, Let $f = \chi_{[0,1]}$ and $g = \chi_{[0,1]}$, and you'll find that $f * g$ is a triangle. You can imagine sliding the graph of f and finding the area of the intersection of f and g , and plotting it. There are many ways to make sure that $f * g$ is a.e. defined. For example, if f was bounded and compactly supported, g can be locally integrable. Note that in many cases, we will need that $K(x, y) = f(x-y)$ be measurable on $\mathbb{R}^n \times \mathbb{R}^n$. If we let $s(x, y) = x - y$, then since s is continuous, if f is Borel measurable, so is K . (Folland makes a comment here saying we can “assume” that this is the case, which I’m guessing is f being Borel measurable in the definition of convolution, because of his theorem 2.12, and he says the same can happen for Lebesgue measurability). Usually, we will use convolution to somehow combine the property of two functions f and g . We will soon see that convolution often preserves the “nice” properties the “nicer” function, so that we would convolution, say, a smooth function f with an integrable function g to get a smooth function $f * g$.

I would also be remiss if I don't mention that convolution is like multiplying two functions in a way that respects operation the *domain*. If f, g are real functions, then $f \cdot g$ respects the multiplication in the *codomain* as it multiplies the outputs. As $f * g$ is the continuous case of polynomial multiplication which multiplies, it respects the multiplication happening in the *domain*. To motivate why this is worth pointing out, consider a finite group G , $f \in \text{Hom}_{\text{Set}}(G, \mathbb{R})$, and interpret it as an element of $\mathbb{C}[G]$ so that $f = \sum_i a_i g_i$. Then we can always make $r \in \mathbb{R}$ act on f via:

$$r \cdot f(g_i) = r f(g_i)$$

However, we can also make $g \in G$ act on f :

$$g \cdot f(g_i) = f(g^{-1}g_i)$$

where we inverse for associativity:

$$\begin{aligned} (g \cdot (h \cdot f))(x) &= (h \cdot f)(g^{-1}x) \\ &= f(h^{-1}(g^{-1}x)) \\ &= f((h^{-1}g^{-1})x) \\ &= f((gh)^{-1}x) \\ &= ((gh) \cdot f)(x) \end{aligned}$$

This is the same thing as defining $g \cdot f$ given $f \in \mathbb{C}[G]$ since:

$$g \cdot f = g \cdot \sum_{g_i \in G} a_i g_i = \sum_{g_i \in G} a_i (gg_i) = \sum_{g^{-1}g_i \in G} a_i g_i$$

Thus, we may think of the translation operation as an action from the structure of the *domain*, when the domain has a group structure. It shall turn out that the way of characterizing the orthonormal basis for L^2 that gives rise to the Fourier transform is to choose the one that's invariant under τ_y .

Definition 2.5.2: Translation Operator

If $f : \mathbb{R}^n \rightarrow \mathbb{R}$, then

$$y * f(x) = \tau_y f(x) = f(x - y)$$

Notice that:

$$\|\tau_y f\|_p = \|f\|_p \quad \|\tau_y f\|_u = \|f\|_u$$

Using this, we can give another definition of uniformly continuous: f is uniformly continuous if:

$$\|\tau_y f - f\|_u \rightarrow 0$$

Lemma 2.5.3: Compact Support Functions And UC

If $f \in C_c(\mathbb{R}^n)$, then f is uniformly continuous

Proof:

should be easy proof (Folland p.238)

Proposition 2.5.4: Translation Is Continuous In L_p

If $1 \leq p < \infty$, then translation is continuous in the L^p norm, in particular if $f \in L^p$ and $z \in \mathbb{R}^n$, then

$$\lim_{y \rightarrow 0} \|\tau_{y+z} f - \tau_z f\|_p = 0$$

Note this proposition is false when $p = \infty$

Proof:

Folland p. 238

The following results all work over $\mathbb{R}^n$ or $\mathbb{T}^n$ (interpreted as the unit cube)

Proposition 2.5.5: Algebraic Properties Of Convolution

Assume all the following integrals are defined and converge absolutely. Then:

1. $f * g = g * f$
2. $(f * g) * h = f * (g * h)$ (this result in particular requires absolute convergence)
3. $f * (g + h) = f * g + f * h$
4. $a(f * g) = (af) * g$ for all $a \in \mathbb{C}$
5. for all $z \in \mathbb{R}^n$, $\tau_z(f * g) = (\tau_z f) * g = f * (\tau_z g)$
6. If A is the closure of $\{x + y \mid x \in \text{supp}(f), y \in \text{supp}(g)\}$, then $\text{supp}(f * g) \subseteq A$

Proof:

1.

$$\begin{aligned}
 f * g(x) &= \int f(x - y)g(y)dy \\
 &= \int f(z)g(x - z)dz && z = x - y, y = x - z \\
 &= \int g(x - z)f(z)dz \\
 &= g * f(x)
 \end{aligned}$$

2. Follows from (1) from Fubini's theorem
3. Follows from direct computation
4. Follows from direct computation
5. Follows from direct computation and (1)
6. If $x \notin A$, then for any $y \in \text{supp}(g)$, we have $x - y \notin \text{supp}(f)$, thus $f(x - y)g(y) = 0$, i.e. $f * g(x) = 0$

One of the first properties important to establish is the relation between convolution of a function and p -norms:

Theorem 2.5.6: Young's Convolution Inequality

Let $1 \leq p, q, r \leq \infty$ be such that $p^{-1} + q^{-1} = r^{-1} + 1$, and let $f \in L^p$ and $g \in L^q$. Then $f * g(x)$ exists for a.e. x , $f * g \in L^r$ and

$$\|f * g\|_r \leq \|f\|_p \|g\|_q$$

Notice in the statement that we have something a little different than the usual conjugate exponents. To see why, there is another thing worth pointing out, which is that this condition is stable under scaling. In particular, if $f_\lambda(x) = f(\lambda x)$ and $g_\lambda(x) = g(\lambda x)$. Then:

$$\begin{aligned} \|f_\lambda\|_p &= \left(\int_{\mathbb{R}^n} f(\lambda x)^p \right)^{1/p} \\ &= \left(\int_{\mathbb{R}^n} \lambda^{-n} f(y)^p \right)^{1/p} & y = \lambda x, \, dy = \lambda^n dx \\ &= \lambda^{-n/p} \|f\|_p \end{aligned}$$

where λ^n is the determinant of the Jacobian matrix. So:

$$\begin{aligned} (f_\lambda * g_\lambda)(x) &= \int_{\mathbb{R}^n} f(\lambda x - \lambda y) g(\lambda y) dy \\ &= \lambda^{-n} (f * g)(\lambda x) \end{aligned}$$

Thus we have:

$$\begin{aligned} \|f_\lambda * g_\lambda\|_r &= \lambda^{-n} \|(f * g)_\lambda\|_r \\ &= \lambda^{-n/r-n} \|f * g\|_r \\ &\leq \lambda^{-n/p-n/q} \|f\|_p \|g\|_q \end{aligned}$$

Notice that this inequality must hold for *all* $\lambda > 0$. If the exponents of λ on the left and right sides were different, we could force a contradiction by taking $\lambda \rightarrow 0$ or $\lambda \rightarrow \infty$ (one side would vanish while the other explodes). Thus, the exponents must be equal:

$$-n \left(1 + \frac{1}{r} \right) = -n \left(\frac{1}{p} + \frac{1}{q} \right)$$

Dividing out the $-n$, we recover the necessary condition:

$$1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}$$

Intuitively, the “extra” $+1$ comes from the integration dy in the definition of convolution, which adds n dimensions of volume (unlike pointwise multiplication which preserves the exponent balance of Hölder's inequality).

Proof:

First, consider $r = \infty$, meaning $r^{-1} = 0$, $p^{-1} + q^{-1} = 1$. So, we want to show $\|f * g\|_\infty \leq \|f\|_p \|g\|_q$. So:

$$\begin{aligned} \|f * g\|_\infty &= \sup_x \left| \int f(x-y)g(y)dy \right| \\ &\leq \|f(x-\cdot)\|_p \|g\|_q \\ &\stackrel{!}{=} \|f\|_p \|g\|_q \end{aligned}$$

where $\stackrel{!}{=}$ comes from $\int |h(x-y)|dy = \int |h(y)|dy$ where $z = x-y$, $dz = (-1)^n dz$ so the $(-1)^n$ disappears in the absolute value. (Note: I corrected the L^1 typo to L^∞ here to match the case $r = \infty$).

Next, if p or q is 1 (say $p = 1$), then $1 + q^{-1} = r^{-1} + 1$, so $q^{-1} = r^{-1}$. Then we would have:

$$\|f * g\|_q \leq \|f\|_1 \|g\|_q$$

To see that this happens, we use theorem 2.3.2. In particular, let $K(x, y) = f(x-y)$. Then if $\|g\|_q$ exists, $\|K(x, \cdot)\|_1 = \|f\|_1$ and $\|K(\cdot, y)\| = \|f\|_1$, giving us the result.

Now, assume $1 < p < r$, $1 < q < r$. First, we manipulate it to be able to use Hölder's inequality:

$$|f(y)g(x-y)| = (|f(y)|^p |g(x-y)|^q)^{1/r} |f(y)|^{1-p/r} |g(x-y)|^{1-q/r}$$

Next, notice that:

$$\frac{1}{r} + \frac{r-q}{rq} + \frac{r-p}{rp} = \frac{1}{r} \left(1 + \frac{r}{q} - 1 + \frac{r}{p} - 1 \right) = 1$$

So, we can apply general Hölder's inequality since $r-p, r-q \neq 0$ which gives us:

$$\begin{aligned} |(f * g)(x)| &\leq \left(\int_{\mathbb{R}^n} |f(y)|^p |g(x-y)|^q dy \right)^{1/r} \cdot \|f\|_p^{(r-p)/r} \|g\|_q^{(r-q)/r} \\ \Rightarrow |(f * g)(x)|^r &\leq \left(\int_{\mathbb{R}^n} |f(y)|^p |g(x-y)|^q dy \right) \cdot \|f\|_p^{r-p} \|g\|_q^{r-q} \end{aligned}$$

And now, we can integrate the left hand side with respect to x and use Tonelli's theorem:

$$\Rightarrow \|(f * g)\|_r^r \leq \|f\|_p^p \|f\|_p^{r-p} \|g\|_q^q \|g\|_q^{r-q} = \|f\|_p^r \|g\|_q^r$$

completing the proof.

If we focus on the special case of $p = q = r = 1$, then $f * g \in L^1(\mathbb{R}^n)$. Then proposition 2.5.5 and Young's inequality (theorem 2.5.6) turns $L^1(\mathbb{R}^n)$ into a “commutative associative algebra”. Note that this algebra has no multiplicative identity (i.e. We are working over a rng). It does have a notion of an identity in a limit, which we will shortly explore. In fact, if we slightly correctly loosen our notion of function to *distribution*, we shall naturally recover an identity (explored in chapter 4). Note that if $f, g \in L^2(\mathbb{R}^n)$ then $f * g \in L^\infty(\mathbb{R}^n)$, namely convoluting forces the functions the result to be bounded.

Before exploring a notion of identity, it must be shown that convoluting two functions where one

nicer than the other usually keeps the properties of the nicer function:

Lemma 2.5.7: Convolution Preserves C_c^k

Let $f \in L^1$ and $g \in C_c^k$ (more generally, $\partial^\alpha g$ is bounded for $|\alpha| \leq k$). Then

$$f * g \in C^k$$

and

$$\partial^\alpha (f * g) = (f * \partial^\alpha g)$$

Proof:

In Folland's textbook this follows from differentiation under the integral sign.

This is a matter of computation. We'll do the case of $k = 1$, since the rest is a matter of induction. First, write what you know about $f * g$:

$$\begin{aligned} f * g(x) &= \int_{\mathbb{R}^n} f(x-y)g(y)dy \\ &= \int_{\mathbb{R}^n} f(y)g(x-y)dy \end{aligned}$$

So consider

$$\lim_{h \rightarrow 0} \frac{f * g(x+h) - f * g(x)}{h} = \lim_{h \rightarrow 0} \int_{\mathbb{R}^n} f(y) \left(\frac{g(x+h-y) - g(x-y)}{h} \right) dy$$

Notice that:

$$\left| \frac{g(x+h-y) - g(x-y)}{h} \right| \leq \frac{|g'(c)| \cdot h}{h}$$

meaning we have a dominating function, and so we can apply dominated convergence theorem to move the limit in!

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f * g(x+h) - f * g(x)}{h} &= \lim_{h \rightarrow 0} \int_{\mathbb{R}^n} f(y) \left(\frac{g(x+h-y) - g(x-y)}{h} \right) dy \\ &= \int_{\mathbb{R}^n} f(y) \lim_{h \rightarrow 0} \left(\frac{g(x+h-y) - g(x-y)}{h} \right) dy \end{aligned}$$

and since $g \in C_c^1$ (in particular, g is differentiable), this limit exists, and since g has compact support, the integral is still well-defined and is C^1 (see the remarks we made at the beginning).

From here we can apply the induction, and the proof is done.

Proposition 2.5.8: Smooth Function Dense in L^p

Compact smooth functions are dense in L^p .

Proof:

hint: simple functions are dense in L^p . Convolute them with a smooth bump function.

With this information, we are going to define a notion of “multiplicative identity”

Definition 2.5.9: Good Kernels

Let $\{\varphi_t\}_{t>0}$ be a set of functions of the form $\varphi_t : \mathbb{R}^n \rightarrow \mathbb{C}$. Then they are called a *good kernel* if $t \in (0, \epsilon_0)$ and:

1. $\sup_{t>0} \|\varphi_t\|_1 < \infty$
2. $\int_{\mathbb{R}^n} \varphi_t(x) dx = 1$
3. $\lim_{t \rightarrow 0} \int_{|x|>\delta} |\varphi_t(x)| dx = 0$ for all $\delta > 0$

Theorem 2.5.10: Good Kernels Like Mult. Identity

Let $f \in L^p$ for $1 \leq p < \infty$ (note it is $<$, not $\leq$), then:

$$f * \varphi_t \rightarrow f$$

in L^p as $t \rightarrow \infty$. Furthermore, if f is continuous at some x_0 , then:

$$f * \varphi_t(x_0) \rightarrow f(x_0)$$

in L^p

Proof:

It is equivalent to show $f * \varphi_t - f \rightarrow 0$, so for all x :

$$\begin{aligned} f * \varphi_t(x) - f(x) &= \int_{\mathbb{R}^n} f(x-y) \varphi_t(y) dy - f(x) \cdot 1 \\ &= \int_{\mathbb{R}^n} f(x-y) \varphi_t(y) dy - f(x) \cdot \int_{\mathbb{R}^n} \varphi_t(y) dy \\ &= \int_{\mathbb{R}^n} f(x-y) \varphi_t(y) dy - \int_{\mathbb{R}^n} f(x) \varphi_t(y) dy \\ &= \int_{\mathbb{R}^n} [f(x-y) - f(x)] \varphi_t(y) dy \\ &= \int_{|y| \leq \delta} [f(x-y) - f(x)] \varphi_t(y) dy + \int_{|y| > \delta} [f(x-y) - f(x)] \varphi_t(y) dy \end{aligned}$$

and so, taking the p -norm we get:

$$\|f * \varphi_t - f\|_p \leq \int_{|y| \leq \delta} \|f(\cdot - y) - f(\cdot)\|_p |\varphi_t(y)| dy + \int_{|y| > \delta} 2\|f\|_p |\varphi_t(y)| dy$$

at this point, we are almost done. If we let $c = \sup_{t>0} \int_{\mathbb{R}^n} |\varphi_t(y)| dy$, then we get:

$$\limsup_{t \rightarrow 0} \|f * \varphi_t - f\|_p \leq c \limsup_{\delta \rightarrow 0, |y| \leq \delta} \|f(\cdot - y) - f(\cdot)\|_p = 0$$

where the last inequality comes from the shrinking delta.

It was left as an exercise to do the point-convergence in the continuous case.

Note that if f is uniformly continuous and bounded, then $f * \varphi_t \rightarrow f$ uniformly, and if $\int \varphi = a$, then $f * \varphi_t \rightarrow af$.

Example 2.5.11: Good Kernels

Let $X(x) \in C_c^\infty(\mathbb{R}^n)$ where $\int X = 1$. Then:

$$\varphi_t(x) = t^{-n} X(x/t)$$

is a good kernel (ex. let $X(x) = \frac{1}{\sqrt{\pi}} e^{-x^2}$). This is just a matter of computation. Note that $\varphi_t = 0$ if $\frac{x}{t} \geq c_0$ for some c_0 , and so $t \leq \frac{x}{c_0}$. It [somehow] follows that C_c^∞ is *dense* in L^p ($1 \leq p < \infty$) (note the $<$ and not $\leq$)

For another Another example of a family of good kernel, take:

$$K_\delta(x) = \delta^{-1/2} e^{-\frac{\pi x^2}{\delta}}$$

These can be visualized here:

<https://www.desmos.com/calculator/e8xc37r3ck>

Notice that since the function $X \in C_c^\infty(\mathbb{R})$ is smooth on compact support then by lemma 2.5.7, $f * X$ is also smooth, and will remain smooth for every φ_n . Since $f * \varphi_n \rightarrow f$, we see that a sequence of smooth functions converges to an L^p function, meaning smooth functions are *dense* in L^p !

I found this interesting stack-exchange post on the importance of good Kernels:

<https://math.stackexchange.com/questions/3294744/good-kernels-that-exist-in-the-real-world>

Using the fact that convolution preserves the “nicer” structure, we strengthen’s Urysohn’s lemma (or, if you know some differential geometry, generalizes partitions of unity)

Theorem 2.5.12: Smooth Urysohn’s Lemma

Let $K \subseteq \mathbb{R}^n$ be compact and $U \subseteq \mathbb{R}^n$ be open and contain K ($U \supseteq K$). Then there exists a $f \in C_c^\infty$ such that $0 \leq f \leq 1$ and $f \equiv 1$, K and $\text{supp}(K) \subseteq f$.

Proof:

Since K is compact, then $\delta = \rho(K, U^c)$ (the shortest distance from K and U^c) is non-zero. We are going to use a mollification trick: let $V = \{x \mid \rho(x, K) < \frac{\delta}{3}\}$. Choose some $\varphi \in C_c^\infty$ such that $\int \varphi = 1$ and $\varphi(x) = 0$ for $|x| \geq \delta/3$ (as an example, $\{\psi_t\}$ to be a good kernel and take

$$\varphi = \left(\int \psi \right)^{-1} \psi_{\delta/3}.$$

Now, let $f = \chi_V * \varphi$. Then as we've shown before, $f \in C_c^\infty$, and it is easy to see that $0 \leq f \leq 1$, $f \equiv 1$ on K , and $\text{supp}(f) \subseteq \{x \mid \rho(x, K) \leq 2\delta/3\} \subseteq U$

Radon Measures and Dual Theory

(I really like the results of this chapter and am adamant to write it down here, especially the duality results. This chapter is a place-holder for now)

(intuition here once I have my own)

3.0.1 Convention In this chapter X will be a *locally compact Hausdorff space* (LCH space).

3.1 Positive Linear Functionals on Compact Support Continuous Functions

(I think there is a reason we start with $C_c(X)$ which I will eventually write down either here or in the chapter intro)

Definition 3.1.1: Positive Functional

Let $C_c(X)$ denote the real continuous functions with compact support. A linear functional $I : C_c(X) \rightarrow \mathbb{R}$ is called *positive* if $I(f) \geq 0$ whenever $f \geq 0$.

A positive functional was not defined to necessarily be continuous, however the positivity condition gives a sort of continuity that with $C_c(X)$ “having” compact-open topology, however this topology is in fact too coarse since as we’ll see we shall want $\text{supp}(f) \subseteq K$ for compact $K \subseteq X$ which is not necessarily the case for the compact-open topology. Instead, we can form a Banach space $C_K(X)$ with the sup-norm $\|f\|_{u,K} = \sup_{x \in K} |f(x)|$ and then take:

$$C_c(X) = \cup_{K \subseteq X} C_K(X)$$

and put the usual *filtered colimit topology* on it (in classical literature known as inductive limit topology). By the universal property of the topology, a functional is continuous if it is continuous on each $C_K(X)$.

Proposition 3.1.2: Continuity of Positivity

Let I be a positive linear functional on $C_c(X)$. Then for each compact $K \subseteq X$ there exists a C_K such that

$$|I(f)| \leq C_K \|f\|_u \quad \forall f \in C_c(X) \text{ where } \text{supp} f \subseteq K$$

Proof:

Let $f \geq 0$. Given $K \subseteq X$, by Urysohn's lemma choose $\varphi \in C_c(X, [0, 1])$ such that $\varphi = 1$ on K . Then if $\text{supp}(f) \subseteq K$,

$$|f(x)| \leq \|f\|_u \varphi(x) \quad \text{i.e.} \quad |f| \leq \|f\|_u \varphi$$

Hence $\|f\|_u \varphi - f \geq 0$. As f is positive $\|f\|_u \varphi + f \geq 0$. Thus by positivity of I :

$$\|f\|_u I(\varphi) - I(f) \geq 0 \quad \text{and} \quad \|f\|_u I(\varphi) + I(f) \geq 0$$

Hence:

$$\|I(f)\| \leq I(\varphi) \|f\|_u$$

as we sought to show.

Now, given μ is a Bore measure on X such that $\mu(K) < \infty$ for all compact $K \subseteq X$, then clearly $C_c(X) \subseteq L^1(\mu)$. So the map:

$$f \mapsto \int f d\mu$$

is a positive linear functional on $C_c(X)$. What we shall show is that *every* positive linear functional on $C_c(X)$ is of this form, and with a little more regularity μ can be made unique in some way *feels like this should be said earlier*.

Definition 3.1.3: Regular Borel Measure

Let μ be a Borel measure on X and $E \subseteq X$ a Borel subset. Then μ is *outer-regular* on E if:

$$\mu(E) = \inf \{ \mu(U) \mid U \supseteq E, U \text{ open} \}$$

inner regular E if:

$$\mu(E) = \sup \{ \mu(K) \mid K \subseteq E, K \text{ compact} \}$$

and regular on E if E is inner- and outer-regular. The Borel μ is called outer-regular, inner-regular, and regular if all Borel sets are outer-regular, inner-regular, and regular respectively.

In most spaces, a regular measure shall work. However, we shall have some edge-cases of spaces that are not σ -compact on which we cannot define a regular measure (see [ref:HERE](#) for where this comes up)

Example 3.1.4: Too Big For Regular Measure

(you can make one on the LCH space $X = [0, \omega_1)$ which is the collection of all countable ordinals, which is not σ -compact, and take the Borel σ -algebra given the Dieudonné Measure)

Definition 3.1.5: Radon Measure

Let μ be a measure on X . Then if μ is a Borel measure that is:

1. finite on all compact sets
2. outer-regular on all Borel sets
3. inner-regular on all open sets

then μ is called a *Radon Measure*.

Proposition 3.1.8 shall show that all Radon measure are also inner-regular on all σ -finite sets.

Now, for the following theorem define the following notation: if $U \subseteq X$ is open and $f \in C_c(X)$ then:

$$f \prec U \quad \text{iff} \quad 0 \leq f \leq 1 \text{ and } \text{supp}(f) \subseteq U$$

Note this is note quite the same as $0 \leq f \leq \chi_U$ as we this would imply $\text{supp}(f) \subseteq \bar{U}$.

Theorem 3.1.6: Riesz Representation Theorem

If I is a positive linear functional on $C_c(X)$, there is a unique Radon measure μ on X such that $I(f) = \int f d\mu$ for all $f \in C_c(X)$. Moreover, μ satisfies

$$\mu(U) = \sup \{I(f) \mid f \in C_c(X), f \prec U\} \text{ for all open } U \subseteq X \quad (3.1)$$

and

$$\mu(K) = \inf \{I(f) \mid f \in C_c(X), f \geq \chi_K\} \text{ for all compact } K \subseteq X \quad (3.2)$$

Proof:

We will first tackle the uniqueness of such a measure, which will naturally guide us toward the correct construction for existence.

Suppose such a Radon measure μ exists. Then for any open $U \subseteq X$, if $f \prec U$ (meaning $f \in C_c(X)$, $0 \leq f \leq 1$, and $\text{supp}(f) \subseteq U$), we have:

$$I(f) = \int f d\mu \leq \int \chi_U d\mu = \mu(U)$$

Conversely, for any compact $K \subseteq U$, Urysohn's Lemma guarantees a function $f \in C_c(X)$ such that $f \prec U$ and $f = 1$ on K . Thus:

$$\mu(K) = \int \chi_K d\mu \leq \int f d\mu = I(f)$$

Since μ is a Radon measure, it is inner regular on open sets, meaning $\mu(U) = \sup_{K \subseteq U} \mu(K)$. Combining this with the inequalities above shows that $\mu(U)$ must be exactly the supremum of

$I(f)$ for $f \prec U$, satisfying (3.1). Since a Radon measure is determined by its values on open sets (due to outer regularity), μ is unique.

To prove existence, we simply turn this uniqueness property into a definition. For any open U , define:

$$\mu(U) = \sup \{I(f) \mid f \in C_c(X), f \prec U\}$$

and for an arbitrary set $E \subseteq X$, we define the outer measure μ^* standardly as:

$$\mu^*(E) = \inf \{\mu(U) \mid U \supseteq E, U \text{ open}\}$$

Note that for open sets, $\mu^*(U) = \mu(U)$ since $U \subseteq V \implies \mu(U) \leq \mu(V)$. To complete the theorem, we proceed in four steps:

1. Show μ^* is an outer measure.
2. Show every open set is μ^* -measurable.

Once these two are established, Carathéodory's Theorem (??) implies that all Borel sets are measurable and $\mu = \mu^*|_{\mathcal{B}_X}$ is a Borel measure. We then must show:

3. μ satisfies the inner regularity condition (3.2).

4. The functional is recovered by the integral: $I(f) = \int f d\mu$.

1. μ^* is an outer measure: The only non-trivial property to check is countable subadditivity on open sets (since the extension to arbitrary sets follows from ??). Let $U = \bigcup_{j=1}^{\infty} U_j$. We want to show $\mu(U) \leq \sum \mu(U_j)$. Pick any $f \prec U$. Then $K = \text{supp}(f)$ is a compact subset of U , so it is covered by finitely many U_j , say N . Using a **Partition of Unity** [ref:HERE \(Proposition 4.41 in foland\)](#), we can find $g_1, \dots, g_N \in C_c(X)$ such that $g_j \prec U_j$ and $\sum g_j = 1$ on K . Then $f = \sum_{j=1}^N f g_j$, and since each $f g_j \prec U_j$:

$$I(f) = \sum_{j=1}^N I(f g_j) \leq \sum_{j=1}^N \mu(U_j) \leq \sum_{j=1}^{\infty} \mu(U_j)$$

Taking the supremum over all such f , we get $\mu(U) \leq \sum \mu(U_j)$, as desired.

2. *Open sets are measurable*: we need to show that for any open U and any E with finite outer measure, $\mu^*(E) \geq \mu^*(E \cap U) + \mu^*(E \setminus U)$. First, assume E is open. Given $\epsilon > 0$, we can choose an $f \prec E \cap U$ such that $I(f)$ is close to $\mu(E \cap U)$, specifically $I(f) > \mu(E \cap U) - \epsilon$. Similarly, since $E \setminus \text{supp}(f)$ is open, we choose $g \prec E \setminus \text{supp}(f)$ with $I(g) > \mu(E \setminus \text{supp}(f)) - \epsilon$. Since f and g have disjoint supports, $f + g \prec E$. Thus:

$$\begin{aligned} \mu(E) &\geq I(f + g) = I(f) + I(g) \\ &> \mu(E \cap U) + \mu(E \setminus \text{supp}(f)) - 2\epsilon \\ &\geq \mu^*(E \cap U) + \mu^*(E \setminus U) - 2\epsilon \end{aligned}$$

Letting $\epsilon \rightarrow 0$ gives the result for open E . For a general E , we approximate it from the outside by an open $V \supset E$ such that $\mu(V) < \mu^*(E) + \epsilon$ and apply the previous logic to V .

3. *Inner Regularity:* We verified outer regularity by construction; now we check (3.2) for compact K . If $f \in C_c(X)$ and $f \geq \chi_K$, let $U_\epsilon = \{x \mid f(x) > 1 - \epsilon\}$. This is an open set containing K . If we take any $g \prec U_\epsilon$, then $f \geq (1 - \epsilon)g$, so $I(g) \leq (1 - \epsilon)^{-1}I(f)$. Taking the supremum over g , we get $\mu(K) \leq \mu(U_\epsilon) \leq (1 - \epsilon)^{-1}I(f)$. As $\epsilon \rightarrow 0$, $\mu(K) \leq I(f)$. Conversely, for any open $U \supset K$, Urysohn's Lemma gives us an f where $\chi_K \leq f \prec U$, so $I(f) \leq \mu(U)$. Taking the infimum over U confirms the equality.
4. *Recovering the Integral:* it suffices to prove $I(f) = \int f d\mu$ for functions with range $[0, 1]$ (by linearity). The strategy is to slice the range of f into N strips of height $1/N$, similar to constructing a Lebesgue integral. Let $K_j = \{x \mid f(x) \geq j/N\}$ for $j = 0, \dots, N$. We define a “layer” function f_j that lives between K_j and K_{j-1} :

$$f_j = \min \left\{ \max \left\{ f - \frac{j-1}{N}, 0 \right\}, \frac{1}{N} \right\}$$

Notice that $f = \sum_{j=1}^N f_j$. We can bound f_j by characteristic functions: $\frac{1}{N}\chi_{K_j} \leq f_j \leq \frac{1}{N}\chi_{K_{j-1}}$. Integrating this gives:

$$\frac{1}{N}\mu(K_j) \leq \int f_j d\mu \leq \frac{1}{N}\mu(K_{j-1})$$

Applying the functional I (and using the fact that $Nf_j \prec U$ for any open $U \supset K_{j-1}$) yields the same bounds:

$$\frac{1}{N}\mu(K_j) \leq I(f_j) \leq \frac{1}{N}\mu(K_{j-1})$$

Summing these up for all j , both $\int f d\mu$ and $I(f)$ are trapped in the same interval of size $\frac{1}{N}(\mu(K_0) - \mu(K_N))$. Since $\mu(K_0)$ is finite (it's the support of f) and N is arbitrary, the difference must be zero, so $I(f) = \int f d\mu$, completing the proof.

It is worth noting that the construction in this proof actually yields something stronger than just the statement of the theorem. We obtain not just a Borel measure μ , but an extension $\bar{\mu}$ of μ to the σ -algebra of all μ^* -measurable sets. Recall that μ is outer regular. Because of this, for any $E \subseteq X$, the outer measure is effectively approximated by Borel sets:

$$\mu^*(E) = \inf \{ \mu(B) \mid B \in \mathcal{B}_X, B \supseteq E \}$$

This means that μ^* corresponds exactly to the outer measure induced by μ . Consequently, the extension $\bar{\mu}$ relates to the standard completion of a measure: if μ is σ -finite, $\bar{\mu}$ is precisely the completion of μ . In the general case, it is the saturation of the completion.

Finally, while we have focused on $\mathcal{B}_X$, some authors prefer to work with a slightly smaller σ -algebra, specifically the one generated by the continuous functions themselves. For ease of indexing, we box the definition:

Definition 3.1.7: Baire Sets

The σ -algebra generated by $C_c(X)$ (i.e., the smallest σ -algebra with respect to which every $f \in C_c(X)$ is measurable) is denoted by $\mathcal{B}_X^0$. The elements of $\mathcal{B}_X^0$ are called *Baire sets*.

Proposition 3.1.8: Inner Regularity on σ -finite Sets

Every Radon measure is inner regular on all of its σ -finite sets.

Proof:

Let μ be a Radon measure and E be a σ -finite set. We proceed by cases.

1. Assume $\mu(E) < \infty$. The strategy here is to approximate E from the outside with an open set, and then approximate that open set from the inside with a compact set, ensuring we stay "mostly" inside E . For any $\epsilon > 0$, since μ is outer regular on Borel sets, we can choose an open $U \supseteq E$ such that $\mu(U) < \mu(E) + \epsilon$. Since μ is inner regular on open sets, we can find a compact $F \subseteq U$ such that $\mu(F) > \mu(U) - \epsilon$.

However, F is not necessarily a subset of E . The part of F that "spills over" is contained in $U \setminus E$. Notice that:

$$\mu(U \setminus E) = \mu(U) - \mu(E) < \epsilon$$

Since this "error set" $U \setminus E$ is Borel, we can use outer regularity again to cover it with an open set $V \supseteq U \setminus E$ such that $\mu(V) < \epsilon$. Now, define $K = F \setminus V$. Since V is open, K is a closed subset of the compact set F , and thus K is compact. Furthermore, since we removed the part covering $U \setminus E$, we have $K \subseteq E$. Finally, we check the measure:

$$\mu(K) = \mu(F) - \mu(F \cap V) > (\mu(E) - \epsilon) - \mu(V) > \mu(E) - 2\epsilon$$

Since ϵ is arbitrary, μ is inner regular on E .

2. Assume $\mu(E) = \infty$. Since E is σ -finite, it is an increasing union of sets E_j with $\mu(E_j) < \infty$ and $\mu(E_j) \rightarrow \infty$. For any large number N , there exists some j such that $\mu(E_j) > N$. Since E_j has finite measure, by the argument in Case 1, we can find a compact $K \subseteq E_j \subseteq E$ such that $\mu(K)$ is arbitrarily close to $\mu(E_j)$, and specifically $\mu(K) > N$. Thus, the supremum of the measures of compact subsets is infinite, satisfying inner regularity.

Corollary 3.1.9: Regularity of Radon Measures

Every σ -finite Radon measure is regular. Furthermore, if X is σ -compact, every Radon measure on X is regular.

3.2 Dual of $C_0(X)$

In the previous section, we established the correspondence between positive linear functionals and Radon measures. However, $C_c(X)$ is generally not a Banach space under the uniform norm $\|\cdot\|_u$ (unless X is compact). We need a complete space. The completion of $C_c(X)$ under the uniform norm is $C_0(X)$.

Definition 3.2.1: Functions Vanishing at Infinity

A continuous function $f : X \rightarrow \mathbb{R}$ is said to *vanish at infinity* if for every $\epsilon > 0$, the set $\{x \in X \mid |f(x)| \geq \epsilon\}$ is compact.

The space of all continuous functions vanishing at infinity is denoted $C_0(X)$. Equipped with the uniform norm $\|f\|_u = \sup_{x \in X} |f(x)|$, $C_0(X)$ is a Banach space, and $C_c(X)$ is dense in it.

Now, instead of restricting ourselves to *positive* functionals, we want to look at the continuous dual space $C_0(X)^*$, which consists of *all* bounded linear functionals on $C_0(X)$. Since these functionals can be negative, they will naturally correspond to signed measures.

Definition 3.2.2: Space of Finite Radon Measures $M(X)$

Let $M(X)$ denote the space of all finite signed regular Borel measures on X . A signed measure μ is regular if its positive and negative variations (from the Hahn Decomposition), μ^+ and μ^- , are finite Radon measures.

We equip $M(X)$ with the *total variation norm*, defined as $\|\mu\| = |\mu|(X) = \mu^+(X) + \mu^-(X)$.

We are now ready to state the full Riesz Representation Theorem. This is the exact realization of the “dual theory” philosophy. Just as $(L^p)^* \cong L^q$, we now identify the dual of $C_0(X)$.

Theorem 3.2.3: Riesz Representation for $C_0(X)^*$

Let X be an LCH space. For every bounded linear functional $I \in C_0(X)^*$, there exists a unique finite signed regular Borel measure $\mu \in M(X)$ such that

$$I(f) = \int f d\mu \quad \forall f \in C_0(X)$$

Moreover, the map $\mu \mapsto I$ is an isometric isomorphism, meaning $\|I\| = \|\mu\| = |\mu|(X)$.

Proof:

The strategy here is to split the bounded functional I into a difference of two positive functionals, say $I = I^+ - I^-$. Once we do that, we can just apply our previous theorem 3.1.6 to each piece, get two positive Radon measures, and subtract them.

Let's construct I^+ . For any non-negative $f \in C_c(X)$, we define:

$$I^+(f) = \sup \{I(g) \mid g \in C_c(X), 0 \leq g \leq f\}$$

Since I is bounded, $I(g) \leq \|I\| \|g\|_u \leq \|I\| \|f\|_u$, so this supremum is finite. It requires a standard but slightly tedious algebraic verification to show that I^+ is additive and positive on non-negative functions. Once we have I^+ defined for $f \geq 0$, we extend it linearly to all of $C_c(X)$ by writing $f = f^+ - f^-$.

Next, we simply define $I^- = I^+ - I$. Since $I^+(f) \geq I(f)$ for all $f \geq 0$, it immediately follows that I^- is also a positive functional.

By the Riesz Representation Theorem for positive functionals (theorem 3.1.6), there exist unique

positive Radon measures μ^+ and μ^- such that:

$$I^+(f) = \int f d\mu^+ \quad \text{and} \quad I^-(f) = \int f d\mu^-$$

Let $\mu = \mu^+ - \mu^-$. Because I is bounded on $C_c(X)$ with respect to the uniform norm, one can show that μ^+ and μ^- must be finite measures. Thus $\mu \in M(X)$, and $I(f) = \int f d\mu$ for all $f \in C_c(X)$. Because $C_c(X)$ is dense in $C_0(X)$ and I is continuous, this integral representation uniquely extends to all of $C_0(X)$.

Finally, we must show the isometry $\|I\| = \|\mu\|$. First, we show $\|I\| \leq \|\mu\|$. For any $f \in C_0(X)$ with $\|f\|_u \leq 1$:

$$|I(f)| = \left| \int f d\mu \right| \leq \int |f| d|\mu| \leq \int 1 d|\mu| = |\mu|(X) = \|\mu\|$$

Taking the supremum over all such f gives $\|I\| \leq \|\mu\|$.

To get the reverse inequality, we use the Hahn decomposition $X = P \cup N$ where P is the positive set for μ and N is the negative set. If we could just plug the function $f = \chi_P - \chi_N$ into our functional, we would get exactly:

$$\int (\chi_P - \chi_N) d\mu = \mu(P) - \mu(N) = \mu^+(X) + \mu^-(X) = \|\mu\|$$

While $\chi_P - \chi_N$ is sadly not continuous (so it's not in $C_0(X)$), we can use the regularity of μ and Lusin's Theorem to approximate it from within with a sequence of continuous functions $f_n \in C_c(X)$ such that $\|f_n\|_u \leq 1$. Taking the limit as $n \rightarrow \infty$ yields $\|I\| \geq \|\mu\|$, locking in the exact isometry.

This theorem fully characterizes the geometry of $M(X)$: it tells us that regular Borel measures are the natural algebraic dual to continuous functions. In the specific case where X is compact, $C_0(X) = C(X)$, giving us $C(X)^* \cong M(X)$.

3.2.4 Better Than Smooth Functions When covering distributions in chapter 4, we shall see that it is a good to replace evaluation with integrating against a smooth function with compact support. The results of this section show that for distributions represented by continuous functions, it is natural to replace $C_c^\infty(X)$ with $M(X)$, and in fact this would work even for distributions represented by continuous functions as this would recover evaluation; see [ref:HERE](#).

Distribution Theory

Distributions are a generalization of functions that allow us to give more general solutions than classical solutions, similar to how we found a weak solution last week. Locally integrable functions will be distributions (in the appropriate interpretation), but we'll see that there are generalized solutions that don't even fall under the notion of "function".

More broadly, distribution theory is a further exploration of the "dual theory" philosophy that has been a recurring theme in our study of analysis. In functional analysis, we repeatedly found that passing to the dual space reveals structure invisible in the original space: the Riesz representation theorem identified $(L^p)^*$ with L^q , and the Hahn–Banach theorem showed that the dual space is always rich enough to separate points. Distribution theory pushes this idea one step further. Rather than studying functions directly, we study what they do to test functions—that is, we study the linear functionals they induce. By broadening the class of allowable functionals beyond those induced by honest functions, we arrive at a theory in which every distribution is infinitely differentiable, every linear constant-coefficient PDE has a solution, and the Fourier transform becomes an isomorphism. In short, duality buys us completeness of operations that the classical function spaces lack.

Another use we will soon see is the intimate connection between distributions and the Fourier transform. The theory of tempered distributions will allow us to extend the Fourier transform to objects far beyond L^1 or L^2 , giving us a powerful and unified framework for studying partial differential equations, signal processing, and harmonic analysis.

4.1 Intuition

For now, let $f \in L^p$. Then as we know, $(L^p)^* \cong L^q$ is an isometric isomorphism where $1 \leq p < \infty$ and q is the conjugate exponent. In particular, f is mapped to the bounded linear functional

$$g \mapsto \int fg.$$

Using the Lebesgue differentiation theorem, by choosing $\varphi_r = m(B_r)^{-1} \chi_{B_r}$, where B_r is a ball of radius r centered at x , we can recover the value of $f(x)$ for a.e. x , as $\lim_{r \rightarrow 0} \int f \varphi_r$. In this way, we can think of L^p as being embedded into $(L^q)^*$: the function f is completely determined (up to a.e. equivalence) by the values of the functional it defines.

Now, let us modify the above by letting f be locally integrable (in L^1_{loc}), but restricting our test functions to $\varphi \in C_c^\infty$. Then it is clear that $\varphi \mapsto \int f \varphi$ is still a well-defined linear functional on C_c^∞ : the integral converges since φ has compact support and f is integrable on compact sets. Moreover, the pointwise values of f can be recovered a.e. from this functional by the same approximation argument (see Theorem 8.15 in Folland, the Borel Representation Theorem). So locally integrable functions embed faithfully into the space of linear functionals on C_c^∞ .

Here is the key observation: unlike in the case for L^p spaces, there are in fact *many* linear functionals on C_c^∞ that are *not* of the form $\varphi \mapsto \int f \varphi$ for any locally integrable f . The most familiar example is the Dirac delta, $\varphi \mapsto \varphi(0)$, which “evaluates at a point”—something no L^1_{loc} function can do. These functionals, subject to some mild continuity conditions we will soon specify, will be our “generalized functions”, or *distributions*.

4.2 Definitions

We will first rapid-fire some important definitions to be up to speed with terminology. Recall that $C_c^\infty(E)$ where $E \subseteq \mathbb{R}^n$ is the set of all smooth functions whose support is compact and contained in E . If $U \subseteq \mathbb{R}^n$ is open, $C_c^\infty(U)$ is the union of spaces $C_c^\infty(K)$ where K ranges over all compact subsets of U . Each $C_c^\infty(K)$ is a Fréchet space with the topology defined by the family of seminorms

$$\varphi \mapsto \|\partial^\alpha \varphi\|_u \quad (\alpha \in \{0, 1, 2, \dots\}^n).$$

Thus, a sequence $\{\varphi_i\}$ converges in $C_c^\infty(K)$ if and only if $\partial^\alpha \varphi_i \rightarrow \partial^\alpha \varphi$ uniformly for all multi-indices α . This is a strong form of convergence: we demand that not only the functions themselves, but all of their derivatives of every order, converge uniformly. The completeness of $C_c^\infty(K)$ under this topology is established through Exercise 9 in Folland, Section 5.1 (the key idea is that a uniform limit of smooth functions, where all derivatives also converge uniformly, is again smooth). With these in hand, we take the following definitions:

1. A sequence $\{\varphi_i\}$ in $C_c^\infty(U)$ *converges in $C_c^\infty(U)$* to φ if there exists a compact set $K \subseteq U$ such that $\{\varphi_i\} \subseteq C_c^\infty(K)$ and $\varphi_i \rightarrow \varphi$ in the topology of $C_c^\infty(K)$, that is, $\partial^\alpha \varphi_i \rightarrow \partial^\alpha \varphi$ uniformly for all α . Note the requirement that all the φ_i are supported in a *single* compact set K ; the supports are not allowed to “escape to infinity”.
2. If X is a locally convex topological vector space and $T : C_c^\infty(U) \rightarrow X$ is a linear map, T is *continuous* if $T|_{C_c^\infty(K)}$ is continuous for each compact $K \subseteq U$, that is, if $T(\varphi_i) \rightarrow T(\varphi)$

whenever $\varphi_i \rightarrow \varphi$ in $C_c^\infty(K)$ for some compact $K \subseteq U$. In other words, continuity is tested “one compact set at a time”.

3. A linear map $T : C_c^\infty(U) \rightarrow C_c^\infty(U')$ is *continuous* if for each compact $K \subseteq U$, there is a compact $K' \subseteq U'$ such that $T(C_c^\infty(K)) \subseteq C_c^\infty(K')$ and T is continuous from $C_c^\infty(K)$ to $C_c^\infty(K')$. The first condition says that T does not spread out the support in an uncontrolled way; the second says that the map is continuous in the Fréchet topology.

With these, we will define a distribution:

Definition 4.2.1: Distribution

A *distribution* on U is a continuous linear functional on $C_c^\infty(U)$. The space of all distributions on U is denoted by $\mathcal{D}'(U)$, and we set $\mathcal{D}' = \mathcal{D}'(\mathbb{R}^n)$. We impose the weak* topology on $\mathcal{D}'(U)$, that is, the topology of pointwise convergence on $C_c^\infty(U)$.

The *evaluation* of a distribution at a test function φ is often denoted either as:

$$F(\varphi) \quad \text{or} \quad \langle F, \varphi \rangle$$

A couple of remarks are in order. First, the $\mathcal{D}'$ notation comes from the fact that Schwartz used the notation $\mathcal{D}$ for C_c^∞ , and so $\mathcal{D}'$ denotes the continuous dual of C_c^∞ —the space of all continuous linear functionals on it. Next, there is in fact a locally convex topology on $C_c^\infty(U)$ (the so-called *inductive limit topology*) with respect to which sequential convergence is given by the convergence defined above, and the continuity of linear maps $T : C_c^\infty \rightarrow X$ and $T : C_c^\infty \rightarrow C_c^\infty$ agrees with the definitions in the list above. However, defining this topology rigorously requires the machinery of inductive limits of locally convex spaces, and the construction is somewhat involved without contributing much additional insight to the current exposition of distributions, so it will be omitted. The interested reader may consult Rudin’s *Functional Analysis*, Chapter 6, for the full treatment. Furthermore, when we talk about convergence of distributions $F_i \rightarrow F$, we will always mean pointwise convergence: $\langle F_i, \varphi \rangle \rightarrow \langle F, \varphi \rangle$ for every $\varphi \in C_c^\infty(U)$.

Before giving some examples of distributions, we will remind the reader how to check for continuity of a linear functional in the setting of Fréchet spaces:

Proposition 4.2.2: Distribution Continuity Check

Let $F : C_c^\infty(U) \rightarrow \mathbb{R}$ (or $\mathbb{C}$) be linear. Then $u \in \mathcal{D}'(U)$ if and only if for every compact set $K \subseteq U$, there exist constants $C_K > 0$ and $N_K \in \mathbb{N}$ such that

$$|\langle F, \varphi \rangle| \leq C_K \sum_{|\alpha| \leq N_K} \|\partial^\alpha \varphi\|_{L^\infty}$$

for all $\varphi \in C_c^\infty(K)$.

Proof:

Suppose first that the estimate holds. We wish to show that u is continuous on each $C_c^\infty(K)$. Let $\varphi_i \rightarrow \varphi$ in $C_c^\infty(K)$. Then for every multi-index α , $\partial^\alpha \varphi_i \rightarrow \partial^\alpha \varphi$ uniformly, which means

$\|\partial^\alpha(\varphi_i - \varphi)\|_{L^\infty} \rightarrow 0$ for each α . Applying the estimate to $\varphi_i - \varphi$:

$$|\langle u, \varphi_i \rangle - \langle u, \varphi \rangle| = |\langle u, \varphi_i - \varphi \rangle| \leq C_K \sum_{|\alpha| \leq N_K} \|\partial^\alpha(\varphi_i - \varphi)\|_{L^\infty} \rightarrow 0,$$

so $\langle u, \varphi_i \rangle \rightarrow \langle u, \varphi \rangle$, and u is continuous on $C_c^\infty(K)$. Since K was arbitrary, u is a distribution. Conversely, suppose u is continuous on $C_c^\infty(K)$ but no such estimate holds. Then for each $j \in \mathbb{N}$, there exists $\varphi_j \in C_c^\infty(K)$ with

$$|\langle u, \varphi_j \rangle| > j \sum_{|\alpha| \leq j} \|\partial^\alpha \varphi_j\|_{L^\infty}.$$

Set $\psi_j = \varphi_j / |\langle u, \varphi_j \rangle|$. Then $\langle u, \psi_j \rangle = 1$ for all j (taking absolute values and adjusting the sign if needed, we can arrange $|\langle u, \psi_j \rangle| = 1$). But for $|\alpha| \leq j$:

$$\|\partial^\alpha \psi_j\|_{L^\infty} < \frac{1}{j},$$

so $\psi_j \rightarrow 0$ in $C_c^\infty(K)$. By continuity of u , we should have $\langle u, \psi_j \rangle \rightarrow \langle u, 0 \rangle = 0$, contradicting $|\langle u, \psi_j \rangle| = 1$.

Definition 4.2.3: Order of Distribution

Let $F \in \mathcal{D}'(U)$ be a distribution and compact $K \subseteq U$ so that by proposition 4.2.2 there exists a C_K and N_K such that

$$|\langle F, \varphi \rangle| \leq C_K \sum_{|\alpha| \leq N_K} \|\partial^\alpha \varphi\|_{L^\infty}$$

then N_K is called the *order* of F on K . If there exists an $N \in \mathbb{N}$ such that for all compact $K \subseteq U$:

$$|\langle F, \varphi \rangle| \leq C_K \sum_{|\alpha| \leq N} \|\partial^\alpha \varphi\|_{L^\infty}$$

then N is called the *order* of F . If no such N exists, then F is said to have infinite order.

The way I like to think about the continuity check is as follows: we will soon see that there are many distribution spaces, for example the Schwartz space $\mathcal{S}$ will have a parallel $\mathcal{S}'$ (the *tempered distributions*). What the continuity condition tells us is that a distribution respects the convergence structure of its test function space. If $\varphi_j \rightarrow \varphi$ in the test function topology, then $\langle u, \varphi_j \rangle \rightarrow \langle u, \varphi \rangle$. Hence, we can translate convergence information from the test function space to information about the distribution.

Example 4.2.4: Distributions

1. *Locally integrable functions.* For every $f \in L^1_{\text{loc}}(U)$, we can define a distribution on U , namely the functional $\varphi \mapsto \int f\varphi$. This is indeed continuous: if $\varphi \in C_c^\infty(K)$, then

$$\left| \int f\varphi \right| \leq \|f\|_{L^1(K)} \|\varphi\|_{L^\infty},$$

so the estimate in Proposition 4.2.2 holds with $N_K = 0$ and $C_K = \|f\|_{L^1(K)}$. Moreover,

two functions f and g define the same distribution precisely when they are equal a.e. (this follows from the fact that C_c^∞ is dense in L^q for the appropriate q , or more directly from the Lebesgue differentiation theorem). In this way, $L_{\text{loc}}^1(U)$ embeds into $\mathcal{D}'(U)$, and we often identify a locally integrable function with the distribution it defines.

2. *Radon measures.* Every Radon measure μ on U defines a distribution $\varphi \mapsto \int \varphi d\mu$. The continuity check is similar: for $\varphi \in C_c^\infty(K)$, we have $|\int \varphi d\mu| \leq |\mu|(K) \|\varphi\|_{L^\infty}$, where $|\mu|(K)$ is the total variation of μ on K . Again, this satisfies the estimate with $N_K = 0$. Note that this generalizes the previous example, since a locally integrable function f defines a Radon measure $d\mu = f dm$ via the Radon–Nikodym theorem. In fact, by the Radon–Nikodym theorem, it is the singular part of the measure that create distributions that are not representable by functions.
3. *Derivatives of point masses.* If $x_0 \in U$ and α is a multi-index, the map $\varphi \mapsto \partial^\alpha \varphi(x_0)$ is a distribution. The continuity estimate is immediate: $|\partial^\alpha \varphi(x_0)| \leq \|\partial^\alpha \varphi\|_{L^\infty}$. This distribution does not arise from a locally integrable function: no L_{loc}^1 function can “see” the value of a derivative at a single point. It arises from a Radon measure μ precisely when $\alpha = 0$, in which case $\mu = \delta_{x_0}$ is the point mass at x_0 . For $|\alpha| \geq 1$, these distributions are genuinely “beyond” measures—they are the first examples of distributions that cannot be represented by any kind of integration.
4. *measure on cantor set* Another example of a singular distribution that is not the point-mass above is the measure on the cantor set. Once differentiation is defined, we shall see that this distribution is the derivative of the devil’s staircase function (or cantor functions), see [ref:HERE](#).

As mentioned earlier, we often associate a function f with the distribution it defines, and even label the distribution by f . For this reason, it might be confusing to write $f(\varphi)$ to mean the distribution defined by f evaluated on φ , and so for historical reasons we use the bracket notation $\langle f, \varphi \rangle$ for the distributional pairing. This notation is suggestive of an inner product (and indeed, when $f \in L^2$ and $\varphi \in C_c^\infty$, it literally is the L^2 inner product), though the reader should keep in mind that the pairing is between two different spaces. The use of the inner product notation should bring to mind Riesz-Representation theorem (theorem 3.1.6) where the inner product is the “perfect pairing” between a space and its dual.

Let’s next establish some common notation:

1. First, it is often enough that we will reflect a function. Thus, if we put a tilde over a function, we mean:

$$\tilde{\varphi}(x) = \varphi(-x).$$

This notation will appear frequently in the context of convolutions, where the reflection arises naturally from the change of variables $y \mapsto -y$.

2. Second, the point-mass (or Dirac delta) is a very common distribution, which we will denote by δ :

$$\langle \delta, \varphi \rangle = \varphi(0).$$

More generally, δ_{x_0} denotes the point mass at x_0 , defined by $\langle \delta_{x_0}, \varphi \rangle = \varphi(x_0)$. When $x_0 = 0$, we simply write δ .

As an example of the use of the delta function, and to illustrate how convergence of distributions works in practice:

Proposition 4.2.5: Convergence To Delta Function

Let $f \in L^1(\mathbb{R}^n)$ with $\int f = a$, and for $t > 0$, let $f_t(x) = t^{-n}f(t^{-1}x)$. Then $f_t \rightarrow a\delta$ in $\mathcal{D}'$ as $t \rightarrow 0$.

Before the proof, let us unpack what this says. The rescaling $f_t(x) = t^{-n}f(t^{-1}x)$ concentrates the mass of f near the origin as $t \rightarrow 0$: the factor t^{-n} ensures that $\int f_t = \int f = a$ for all t , while the argument $t^{-1}x$ compresses the support. So f_t becomes a taller and narrower spike near 0, with constant total mass a . The proposition says that in the distributional sense, this spike converges to a times the delta function—exactly what our intuition would predict.

Proof:

Let $\varphi \in C_c^\infty$. By Theorem 8.15 (Folland), we have:

$$\langle f_t, \varphi \rangle = \int f_t(x) \varphi(x) dx = \int t^{-n} f(t^{-1}x) \varphi(x) dx.$$

Performing the substitution $y = t^{-1}x$ (so $x = ty$ and $dx = t^n dy$):

$$\int t^{-n} f(t^{-1}x) \varphi(x) dx = \int f(y) \varphi(ty) dy.$$

Now we recognize the right-hand side as $f_t * \tilde{\varphi}(0)$. As $t \rightarrow 0$, $\varphi(ty) \rightarrow \varphi(0)$ pointwise, and since $|\varphi(ty)| \leq \|\varphi\|_{L^\infty}$ and $f \in L^1$, the dominated convergence theorem gives:

$$\langle f_t, \varphi \rangle = \int f(y) \varphi(ty) dy \rightarrow \varphi(0) \int f(y) dy = a\varphi(0) = a\langle \delta, \varphi \rangle,$$

completing the proof.

For equality of distributions, note that it makes sense to say that $F = G$ on an open set $V \subseteq U$ if and only if $\langle F, \varphi \rangle = \langle G, \varphi \rangle$ for all $\varphi \in C_c^\infty(V)$. If F and G are continuous functions, this implies they are pointwise equal on V , and if F and G are only locally integrable, this means they are equal a.e. on V .

Since a function in $C_c^\infty(V_1 \cup V_2)$ need not be supported in either V_1 or V_2 individually, it is not immediately obvious that if $F = G$ on V_1 and on V_2 , then $F = G$ on $V_1 \cup V_2$. However, this is indeed the case, thanks to partitions of unity:

Proposition 4.2.6: Equality On Support

Let $\{V_\alpha\}$ be a collection of open subsets of U and let $V = \bigcup_\alpha V_\alpha$. If $F, G \in \mathcal{D}'(U)$ and $F = G$ on each V_α , then $F = G$ on V .

Proof:

It suffices to show that $F - G = 0$ on V , so we may assume $G = 0$ and show that $F = 0$ on V . Let $\varphi \in C_c^\infty(V)$. Then $\text{supp}(\varphi)$ is a compact subset of $V = \bigcup_\alpha V_\alpha$. By compactness, finitely many of the V_α cover $\text{supp}(\varphi)$, say $V_{\alpha_1}, \dots, V_{\alpha_m}$. Let $\{\chi_1, \dots, \chi_m\}$ be a smooth partition of unity subordinate to this finite cover, so that each $\chi_j \in C_c^\infty(V_{\alpha_j})$, $0 \leq \chi_j \leq 1$, and $\sum_{j=1}^m \chi_j = 1$ on a neighborhood of $\text{supp}(\varphi)$. Then $\varphi = \sum_{j=1}^m \chi_j \varphi$, and each $\chi_j \varphi \in C_c^\infty(V_{\alpha_j})$. Since $F = 0$ on each V_{α_j} , we have $\langle F, \chi_j \varphi \rangle = 0$ for each j , and by linearity:

$$\langle F, \varphi \rangle = \sum_{j=1}^m \langle F, \chi_j \varphi \rangle = 0.$$

Since $\varphi \in C_c^\infty(V)$ was arbitrary, $F = 0$ on V .

As a consequence of Proposition 4.2.6, if $F \in \mathcal{D}'(U)$, there is a largest open set on which $F = 0$, namely the union of all open subsets of U on which $F = 0$. The complement of this set in U is called the *support* of F , denoted $\text{supp}(F)$. When F is a locally integrable function, this agrees with the usual notion of [closed] support (up to a set of measure zero).

Definition 4.2.7: Distribution With Compact Support

The set of all distributions on U with compact support is denoted $\mathcal{E}'(U)$

It turns out (though we will not prove it here) that $\mathcal{E}'(U)$ is precisely the dual of $C^\infty(U)$ —we no longer need to require that the test functions have compact support, because the distribution itself is compactly supported.

4.3 Operations on Distributions

One of the advantages of generalizing functions to distributions is that even though they are not “functions” in the classical sense, we can perform many of the same operations on them: differentiation, multiplication by a smooth function, convolution, and translation! To achieve this, we use a single unifying principle that we now describe.

Suppose U and V are open subsets of $\mathbb{R}^n$ and T is a linear map from some subspace $\mathcal{V}$ of $L_{\text{loc}}^1(U)$ into $L_{\text{loc}}^1(V)$. Suppose that there is another linear map $T' : C_c^\infty(V) \rightarrow C_c^\infty(U)$ (called the *transpose* of T) such that

$$\int (Tf)\varphi = \int f(T'\varphi) \quad (f \in \mathcal{V}, \varphi \in C_c^\infty(V)).$$

Suppose further that T' is continuous in the sense defined earlier (i.e., for each compact $K' \subseteq V$, there exists a compact $K \subseteq U$ with $T'(C_c^\infty(K')) \subseteq C_c^\infty(K)$ and T' continuous from $C_c^\infty(K')$ to $C_c^\infty(K)$). Then T can be *extended* to a map from $\mathcal{D}'(U)$ to $\mathcal{D}'(V)$, still denoted by T , by defining:

$$\langle TF, \varphi \rangle = \langle F, T'\varphi \rangle \quad (F \in \mathcal{D}'(U), \varphi \in C_c^\infty(V)).$$

Why does this work? First, TF is a well-defined linear functional on $C_c^\infty(V)$ since $T'\varphi \in C_c^\infty(U)$ and F is a linear functional on $C_c^\infty(U)$. Second, TF is continuous: if $\varphi_i \rightarrow \varphi$ in $C_c^\infty(V)$, then $T'\varphi_i \rightarrow T'\varphi$ in $C_c^\infty(U)$ by continuity of T' , and so $\langle TF, \varphi_i \rangle = \langle F, T'\varphi_i \rangle \rightarrow \langle F, T'\varphi \rangle = \langle TF, \varphi \rangle$ since

F is a distribution. Third, this definition is *consistent* with the original map on functions: if F is a locally integrable function in $\mathcal{V}$, then $\langle TF, \varphi \rangle = \langle F, T'\varphi \rangle = \int f(T'\varphi) = \int (Tf)\varphi$, which is exactly the distribution defined by the function Tf .

The continuity of T' also guarantees that the extended map $T : \mathcal{D}'(U) \rightarrow \mathcal{D}'(V)$ is continuous with respect to the weak* topology on distributions: if $F_\alpha \rightarrow F$ in $\mathcal{D}'(U)$, then for any $\varphi \in C_c^\infty(V)$, $\langle TF_\alpha, \varphi \rangle = \langle F_\alpha, T'\varphi \rangle \rightarrow \langle F, T'\varphi \rangle = \langle TF, \varphi \rangle$, so $TF_\alpha \rightarrow TF$ in $\mathcal{D}'(V)$.

Using this, we can construct many operations on distributions by transposing the property onto the test functions

1. (*Differentiation*) Let $Tf = \partial^\alpha f$, defined initially on $C^{|\alpha|}(U)$. If $\varphi \in C_c^\infty(U)$, integration by parts gives

$$\int (\partial^\alpha f)\varphi = (-1)^{|\alpha|} \int f(\partial^\alpha \varphi);$$

there are no boundary terms since φ has compact support. Hence $T' = (-1)^{|\alpha|} \partial^\alpha|_{C_c^\infty(U)}$, and we can define the *distributional derivative* $\partial^\alpha F \in \mathcal{D}'(U)$ of any $F \in \mathcal{D}'(U)$ by:

$$\langle \partial^\alpha F, \varphi \rangle := (-1)^{|\alpha|} \langle F, \partial^\alpha \varphi \rangle.$$

By this procedure, every distribution is infinitely differentiable! We can define the derivatives of arbitrary locally integrable functions, even when they are not differentiable in the classical sense. When F happens to be a $C^{|\alpha|}$ function, the distributional derivative agrees with the classical one (by integration by parts), so this truly is a generalization. We will compute the distributional derivatives of some important locally integrable functions shortly.

A particularly important consequence: distributional derivatives commute. That is, $\partial^\alpha \partial^\beta F = \partial^{\alpha+\beta} F = \partial^\beta \partial^\alpha F$ for all distributions F and all multi-indices α, β . This is immediate from the definition and the fact that classical derivatives of smooth functions commute.

2. (*Multiplication by a smooth function*) Let $\psi \in C^\infty(U)$ and define $Tf = \psi f$. Then $T' = T|_{C_c^\infty(U)}$ (that is, $T'\varphi = \psi\varphi$), since $\int (\psi f)\varphi = \int f(\psi\varphi)$. Evidently T' maps $C_c^\infty(U)$ to $C_c^\infty(U)$ and is continuous (since multiplication by a smooth function preserves both support and uniform convergence of derivatives, by the Leibniz rule). Thus we define $\psi F \in \mathcal{D}'(U)$ for any $F \in \mathcal{D}'(U)$ by:

$$\langle \psi F, \varphi \rangle := \langle F, \psi\varphi \rangle.$$

Note that we require ψ to be C^∞ ; multiplying a distribution by a continuous or L^∞ function is in general not well-defined. The Leibniz rule extends to distributions: $\partial^\alpha(\psi F) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} (\partial^{\alpha-\beta} \psi)(\partial^\beta F)$, which is proved by testing against $\varphi \in C_c^\infty$ and using the classical Leibniz rule on the test function side.

3. (*Translation*) Let $y \in \mathbb{R}^n$ and let $V = U + y = \{x + y : x \in U\}$. Take $T = \tau_y$ (recall that $\tau_y f(x) = f(x - y)$). Via the substitution $x \mapsto x + y$, we get

$$\int f(x - y)\varphi(x) dx = \int f(x)\varphi(x + y) dx,$$

and so $T' = \tau_{-y}|_{C_c^\infty(V)}$. This is evidently continuous, and so for a distribution $F \in \mathcal{D}'(U)$, we define:

$$\langle \tau_y F, \varphi \rangle := \langle F, \tau_{-y}\varphi \rangle.$$

As a quick sanity check: $\langle \tau_y \delta, \varphi \rangle = \langle \delta, \tau_{-y}\varphi \rangle = (\tau_{-y}\varphi)(0) = \varphi(y) = \langle \delta_y, \varphi \rangle$. So translating the delta function by y gives the point mass at y , exactly as expected.

4. (*Composition with linear maps*) Let S be an invertible linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^n$. Let $V = S^{-1}(U)$ and $Tf = f \circ S$. By the change of variables formula (see Chapter 1), for $\varphi \in C_c^\infty(V)$:

$$\int (f \circ S)(x) \varphi(x) dx = \int f(y) \varphi(S^{-1}y) |\det S|^{-1} dy.$$

So $T'\varphi = |\det(S)|^{-1} \varphi \circ S^{-1}$. For $F \in \mathcal{D}'(U)$, we define:

$$\langle F \circ S, \varphi \rangle := |\det S|^{-1} \langle F, \varphi \circ S^{-1} \rangle.$$

As a special case, taking $S = -I$ (the map $x \mapsto -x$) gives $\langle \tilde{F}, \varphi \rangle = \langle F, \tilde{\varphi} \rangle$, which defines the *reflection* of a distribution.

Convolutions are interesting and deserve more attention as they smooth distributions and recover functions. Consider first convoluting with a function. Let $\psi \in C_c^\infty$, and let

$$V = \{x : x - y \in U \text{ for all } y \in \text{supp}(\psi)\}$$

Note that V is open (but may be empty). If $f \in L_{\text{loc}}^1(U)$, then the integral

$$f * \psi(x) = \int f(x - y) \psi(y) dy = \int f(y) \psi(x - y) dy = \langle f, \tau_x \tilde{\psi} \rangle$$

is well-defined for all $x \in V$. Notice that we are inputting a *point*. Since the last expression involves only the distributional pairing, we can define for $F \in \mathcal{D}'(U)$:

$$F * \psi(x) = \langle F, \tau_x \tilde{\psi} \rangle$$

This is well-defined because $\tau_x \tilde{\psi} \in C_c^\infty(U)$ for $x \in V$.

Proposition 4.3.1: Convoluting With Function Gives Function

Let F be a distribution and ψ a smooth function with compact support on V . Then $F * \psi$ is a smooth functions

Proof:

Since $\tau_x \tilde{\psi} \rightarrow \tau_{x_0} \tilde{\psi}$ in C_c^∞ as $x \rightarrow x_0$ (and similarly for all derivatives), continuity of F gives $F * \psi(x) \rightarrow F * \psi(x_0)$. More precisely, one can show by differentiating under the pairing that $\partial^\alpha (F * \psi) = F * (\partial^\alpha \psi)$ for all multi-indices α , which establishes that $F * \psi \in C^\infty(V)$.

A pertinent example is the following:

$$\delta * \psi(x) = \langle \delta, \tau_x \tilde{\psi} \rangle = \tau_x \tilde{\psi}(0) = \tilde{\psi}(-x) = \psi(x)$$

so δ is the identity element for convolution! This is one of the most important properties of the delta function and lies at the heart of the theory of Green's functions: if L is a differential operator and E is a distribution satisfying $LE = \delta$ (a *fundamental solution*), then $u = E * f$ formally solves $Lu = f$.

We can also define convolution using the same transpose technique from earlier. Let $\psi, \tilde{\psi}$, and V be as above. If $f \in L_{\text{loc}}^1(U)$ and $\varphi \in C_c^\infty(V)$, then by Fubini's theorem:

$$\begin{aligned} \int (f * \psi)(x) \varphi(x) dx &= \int \int f(y) \psi(x - y) \varphi(x) dy dx = \int f(y) \left(\int \psi(x - y) \varphi(x) dx \right) dy \\ &= \int f(y) (\varphi * \tilde{\psi})(y) dy \end{aligned}$$

Thus $Tf = f * \psi$ maps $L^1_{\text{loc}}(U)$ to $L^1_{\text{loc}}(V)$ with transpose $T'\varphi = \varphi * \tilde{\psi}$. For a distribution $F \in \mathcal{D}'(U)$, we therefore get:

$$\langle F * \psi, \varphi \rangle = \langle F, \varphi * \tilde{\psi} \rangle$$

We should verify that the two definitions of convolution are consistent:

Proposition 4.3.2: Consistency of Convolution

Let $F \in \mathcal{D}'(U)$ and $\psi \in C_c^\infty$, and let V be as above. Then the smooth function $x \mapsto \langle F, \tau_x \tilde{\psi} \rangle$ and the distribution defined by $\langle F, \varphi * \tilde{\psi} \rangle$ agree, in the sense that the 2nd convolution definition is the distribution defined by the locally integrable function from the 1st convolution definition.

Proof:

We need to show that for all $\varphi \in C_c^\infty(V)$:

$$\int \langle F, \tau_x \tilde{\psi} \rangle \varphi(x) dx = \langle F, \varphi * \tilde{\psi} \rangle.$$

The left-hand side can be written as the limit of Riemann sums. More precisely, for φ supported in a compact set $K' \subseteq V$, the function $x \mapsto \tau_x \tilde{\psi}$ is a continuous map from K' into $C_c^\infty(K)$ for some compact $K \subseteq U$. One checks (using the Riemann sum approximation and continuity of F) that the integral of a continuous $C_c^\infty(K)$ -valued function against a scalar function can be “passed inside” the distributional pairing. But the integral $\int \tau_x \tilde{\psi} \varphi(x) dx$, computed pointwise, is exactly the function $y \mapsto \int \psi(x - y) \varphi(x) dx = (\varphi * \tilde{\psi})(y)$. Hence:

$$\int \langle F, \tau_x \tilde{\psi} \rangle \varphi(x) dx = \left\langle F, \int \tau_x \tilde{\psi} \varphi(x) dx \right\rangle = \langle F, \varphi * \tilde{\psi} \rangle.$$

4.4 Density Results

Next, we will show that smooth compactly supported functions $C_c^\infty(U)$ are dense in $\mathcal{D}'(U)$. This is a remarkable fact: it says that every distribution, no matter how singular, can be approximated (in the weak* topology) by smooth functions. We begin with a preparatory lemma.

Lemma 4.4.1: Build-Up For Density

Suppose that $\varphi \in C_c^\infty$, $\psi \in C_c^\infty$ with $\int \psi = 1$, and let $\psi_t(x) = t^{-n} \psi(t^{-1}x)$ for $t > 0$. Then:

1. Given any open neighborhood U of $\text{supp}(\varphi)$, we have $\text{supp}(\varphi * \psi_t) \subseteq U$ for sufficiently small $t > 0$.
2. $\varphi * \psi_t \rightarrow \varphi$ in C_c^∞ as $t \rightarrow 0$.

Proof:

1. Note that $\text{supp}(\varphi * \psi_t) \subseteq \text{supp}(\varphi) + \text{supp}(\psi_t) = \text{supp}(\varphi) + t \cdot \text{supp}(\psi)$. Since $\text{supp}(\varphi)$ is compact, the set $\text{supp}(\varphi) + t \cdot \text{supp}(\psi)$ is the Minkowski sum of $\text{supp}(\varphi)$ with a set of diameter shrinking to 0 as $t \rightarrow 0$. Since U is an open neighborhood of the compact set $\text{supp}(\varphi)$, there exists $\varepsilon > 0$ such that the ε -neighborhood of $\text{supp}(\varphi)$ is contained in U . For t small enough that $t \cdot \text{supp}(\psi)$ lies within a ball of radius ε , we get $\text{supp}(\varphi * \psi_t) \subseteq U$.
2. We need to show that $\partial^\alpha(\varphi * \psi_t) \rightarrow \partial^\alpha \varphi$ uniformly for every multi-index α . Since differentiation commutes with convolution, $\partial^\alpha(\varphi * \psi_t) = (\partial^\alpha \varphi) * \psi_t$. So it suffices to show that $g * \psi_t \rightarrow g$ uniformly whenever $g \in C_c^\infty$. We have:

$$(g * \psi_t)(x) - g(x) = \int [g(x - y) - g(x)] \psi_t(y) dy = \int [g(x - tz) - g(x)] \psi(z) dz,$$

using the substitution $y = tz$. Since g is smooth and compactly supported, it is uniformly continuous, so for any $\varepsilon > 0$ there exists $\delta > 0$ such that $|g(x - tz) - g(x)| < \varepsilon$ whenever $t|z| < \delta$. For $z \in \text{supp}(\psi)$ and t small enough, this is satisfied, giving $|g * \psi_t(x) - g(x)| \leq \varepsilon \int |\psi|$ for all x . By part (a), all the $\varphi * \psi_t$ are eventually supported in a single compact set, so the convergence is in C_c^∞ .

Proposition 4.4.2: Density Of Smooth Compactly Supported Functions In Distributions

Let $U \subseteq \mathbb{R}^n$ be open. Then $C_c^\infty(U)$ is dense in $\mathcal{D}'(U)$ in the weak* topology.

Proof:

Let $F \in \mathcal{D}'(U)$; we must find a net (or sequence) $\{\varphi_i\}$ in $C_c^\infty(U)$ such that $\langle \varphi_i, \eta \rangle \rightarrow \langle F, \eta \rangle$ for all $\eta \in C_c^\infty(U)$, where $\langle \varphi_i, \eta \rangle = \int \varphi_i \eta$.

Fix $\psi \in C_c^\infty(\mathbb{R}^n)$ with $\int \psi = 1$ and let $\psi_t(x) = t^{-n} \psi(t^{-1}x)$. We also need a family of cutoff functions: let $\{\chi_j\}_{j=1}^\infty$ be a sequence in $C_c^\infty(U)$ with $0 \leq \chi_j \leq 1$ and $\chi_j \rightarrow 1$ pointwise on U (e.g., take $\chi_j = 1$ on $\{x \in U : |x| \leq j \text{ and } \text{dist}(x, U^c) \geq 1/j\}$ and smooth it out). Consider the distributions $F_j = \chi_j F$ (defined by $\langle F_j, \eta \rangle = \langle F, \chi_j \eta \rangle$). Each F_j has compact support in U , and $F_j \rightarrow F$ in $\mathcal{D}'(U)$ since for fixed $\eta \in C_c^\infty(U)$, $\chi_j \eta \rightarrow \eta$ in $C_c^\infty(U)$ for large j (eventually $\chi_j = 1$ on $\text{supp}(\eta)$).

Now consider $F_j * \psi_t$. By our earlier discussion, this is a C^∞ function. For t sufficiently small, it has compact support in U (since F_j has compact support in U and the convolution only slightly enlarges the support). By Proposition 4.2.5 and the properties of convolution, $F_j * \psi_t \rightarrow F_j$ in $\mathcal{D}'$ as $t \rightarrow 0$: for any $\eta \in C_c^\infty(U)$,

$$\langle F_j * \psi_t, \eta \rangle = \langle F_j, \eta * \tilde{\psi}_t \rangle \rightarrow \langle F_j, \eta \rangle$$

by lemma 4.4.1, since $\eta * \tilde{\psi}_t \rightarrow \eta$ in C_c^∞ . By a diagonal argument (choosing $t_j \rightarrow 0$ slowly enough), we obtain a sequence $g_j = F_j * \psi_{t_j} \in C_c^\infty(U)$ with $g_j \rightarrow F$ in $\mathcal{D}'(U)$.

4.5 Distributional Derivatives of Classical Functions

We promised earlier that the distributional derivative lets us differentiate functions that are not classically differentiable, and now we deliver on that promise. The computations in this section are among the most important examples in all of distribution theory, and the reader should internalize them thoroughly.

Recall that for $F \in \mathcal{D}'(\mathbb{R}^n)$ and a multi-index α , the distributional derivative is defined by

$$\langle \partial^\alpha F, \varphi \rangle = (-1)^{|\alpha|} \langle F, \partial^\alpha \varphi \rangle.$$

Example 4.5.1: Derivative Of The Heaviside Function

Let $H : \mathbb{R} \rightarrow \mathbb{R}$ be the Heaviside step function, $H(x) = \mathbf{1}_{[0, \infty)}(x)$. Then $H \in L^1_{\text{loc}}(\mathbb{R})$ and hence defines a distribution. Classically, H is differentiable everywhere except at $x = 0$, where it has a jump discontinuity of height 1. Let us compute H' in the distributional sense. For $\varphi \in C_c^\infty(\mathbb{R})$:

$$\langle H', \varphi \rangle = -\langle H, \varphi' \rangle = -\int_0^\infty \varphi'(x) dx = -[\varphi(x)]_{x=0}^{x=\infty} = -\underbrace{(\varphi(\infty) - \varphi(0))}_{=0} = \varphi(0) = \langle \delta, \varphi \rangle.$$

Therefore $H' = \delta$ in $\mathcal{D}'(\mathbb{R})$. This is the prototypical example: the distributional derivative detects the jump discontinuity and places a point mass there. More generally, if f is piecewise C^1 with jump discontinuities of height c_j at points x_j , then $f' = f'_{\text{classical}} + \sum_j c_j \delta_{x_j}$ in $\mathcal{D}'$.

Example 4.5.2: Derivative Of $|x|$ And The Sign Function

Consider $f(x) = |x|$ on $\mathbb{R}$. This function is continuous but not differentiable at $x = 0$. For $\varphi \in C_c^\infty(\mathbb{R})$:

$$\begin{aligned} \langle f', \varphi \rangle &= -\langle f, \varphi' \rangle = -\int_{-\infty}^\infty |x| \varphi'(x) dx \\ &= -\int_{-\infty}^0 (-x) \varphi'(x) dx - \int_0^\infty x \varphi'(x) dx \\ &= \int_{-\infty}^0 x \varphi'(x) dx - \int_0^\infty x \varphi'(x) dx. \end{aligned}$$

Integrating by parts on each piece (the boundary terms at $\pm\infty$ vanish since φ has compact support, and the boundary terms at 0 cancel):

$$= [x\varphi(x)]_{-\infty}^0 - \int_{-\infty}^0 \varphi(x) dx - [x\varphi(x)]_0^\infty + \int_0^\infty \varphi(x) dx = -\left(\int_{-\infty}^0 \varphi\right) + \left(\int_0^\infty \varphi\right) = \int_{-\infty}^\infty \text{sgn}(x) \varphi(x) dx,$$

where $\text{sgn}(x) = \mathbf{1}_{(0, \infty)}(x) - \mathbf{1}_{(-\infty, 0)}(x)$ is the sign function. Hence $|x|' = \text{sgn}(x)$ in $\mathcal{D}'$. Note that $\text{sgn}(x) = 2H(x) - 1$, which is consistent: differentiating again gives $|x|'' = \text{sgn}'(x) = 2H'(x) = 2\delta$.

The next computation requires introducing an important distribution that is not given by a locally integrable function in the standard sense.

Definition 4.5.3: Principal Value Distribution

The *principal value* of $1/x$, denoted $\text{p.v.}(1/x)$, is the distribution on $\mathbb{R}$ defined by

$$\left\langle \text{p.v.} \frac{1}{x}, \varphi \right\rangle = \lim_{\varepsilon \rightarrow 0^+} \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx.$$

We should verify that the limit exists and that this defines a distribution. Since φ is smooth, we can write $\varphi(x) = \varphi(0) + x\psi(x)$ where $\psi(x) = \int_0^1 \varphi'(tx) dt$ is also smooth. Then

$$\int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx = \varphi(0) \underbrace{\int_{|x| > \varepsilon} \frac{dx}{x}}_{= 0 \text{ by symmetry}} + \int_{|x| > \varepsilon} \psi(x) dx \rightarrow \int_{-\infty}^{\infty} \psi(x) dx$$

as $\varepsilon \rightarrow 0$. Since ψ is smooth and φ has compact support, the integral converges. The continuity estimate $|\langle \text{p.v.}(1/x), \varphi \rangle| \leq C(\|\varphi\|_{\infty} + \|\varphi'\|_{\infty})$ on any fixed compact set follows from this decomposition, confirming that $\text{p.v.}(1/x) \in \mathcal{D}'(\mathbb{R})$. Notice that the order of this distribution is 1 (it requires control of one derivative), unlike the distributions arising from locally integrable functions, which have order 0.

Example 4.5.4: Derivative Of $\log |x|$

The function $\log |x|$ is locally integrable on $\mathbb{R}$ (since $\int_{-1}^1 |\log |x|| dx < \infty$) and hence defines a distribution. We claim that its distributional derivative is $\text{p.v.}(1/x)$. For $\varphi \in C_c^{\infty}(\mathbb{R})$:

$$\begin{aligned} \langle (\log |x|)', \varphi \rangle &= -\langle \log |x|, \varphi' \rangle = -\int_{-\infty}^{\infty} \log |x| \varphi'(x) dx \\ &= -\lim_{\varepsilon \rightarrow 0^+} \int_{|x| > \varepsilon} \log |x| \varphi'(x) dx. \end{aligned}$$

Integrating by parts (with the boundary terms at $\pm\infty$ vanishing):

$$\begin{aligned} &= -\lim_{\varepsilon \rightarrow 0^+} \left(\left[\log |x| \varphi(x) \right]_{|x|=\varepsilon}^{|x|=\infty} - \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx \right) \\ &= -\lim_{\varepsilon \rightarrow 0^+} (-\log \varepsilon [\varphi(\varepsilon) + \varphi(-\varepsilon)]) + \lim_{\varepsilon \rightarrow 0^+} \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx. \end{aligned}$$

Since $\varphi(\varepsilon) + \varphi(-\varepsilon) \rightarrow 2\varphi(0)$ and $\varepsilon \log \varepsilon \rightarrow 0$, the first term requires more care. We have $\log \varepsilon [\varphi(\varepsilon) + \varphi(-\varepsilon)] = 2\varphi(0) \log \varepsilon + O(\varepsilon \log \varepsilon)$. But this term appears with a minus sign outside, so:

$$= \lim_{\varepsilon \rightarrow 0^+} \log \varepsilon [\varphi(\varepsilon) + \varphi(-\varepsilon)] + \lim_{\varepsilon \rightarrow 0^+} \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx.$$

For the boundary term, note that $[\log |x| \varphi(x)]_{-\infty}^{-\varepsilon} + [\log |x| \varphi(x)]_{\varepsilon}^{\infty} = -\log \varepsilon \varphi(-\varepsilon) - \log \varepsilon \varphi(\varepsilon)$. Thus:

$$= \lim_{\varepsilon \rightarrow 0^+} \left[\log \varepsilon (\varphi(\varepsilon) + \varphi(-\varepsilon)) + \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx \right].$$

Since $\varphi(\varepsilon) + \varphi(-\varepsilon) = 2\varphi(0) + O(\varepsilon^2)$ (the odd part cancels) and $\int_{\varepsilon}^1 dx/x = -\log \varepsilon$ (with the symmetric contribution from $[-1, -\varepsilon]$ cancelling), the $\log \varepsilon$ terms cancel exactly, and we are left

with

$$\langle (\log |x|)', \varphi \rangle = \lim_{\varepsilon \rightarrow 0^+} \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx = \left\langle \text{p.v.} \frac{1}{x}, \varphi \right\rangle.$$

Therefore $(\log |x|)' = \text{p.v.}(1/x)$ in $\mathcal{D}'(\mathbb{R})$.

It is also instructive to see how multiplication interacts with the principal value:

Proposition 4.5.5: Multiplication By x

As distributions on $\mathbb{R}$, we have $x \cdot \text{p.v.}(1/x) = 1$, that is, the distribution $x \cdot \text{p.v.}(1/x)$ agrees with the constant function 1.

Proof:

For $\varphi \in C_c^\infty(\mathbb{R})$:

$$\langle x \cdot \text{p.v.}(1/x), \varphi \rangle = \left\langle \text{p.v.} \frac{1}{x}, x\varphi \right\rangle = \lim_{\varepsilon \rightarrow 0^+} \int_{|x| > \varepsilon} \frac{x\varphi(x)}{x} dx = \int_{-\infty}^{\infty} \varphi(x) dx = \langle 1, \varphi \rangle,$$

since φ is integrable and the integrand $\varphi(x)$ has no singularity.

This is a distributional identity with no classical analogue: the “function” $1/x$ is not locally integrable, so the product $x \cdot (1/x)$ is not defined pointwise in L_{loc}^1 . But the distributional version makes perfect sense and gives the expected answer. On the other hand, note that $x \cdot \delta = 0$ (since $\langle x\delta, \varphi \rangle = \langle \delta, x\varphi \rangle = 0$), so x is a zero-divisor in $\mathcal{D}'$ —a phenomenon that does not occur for ordinary functions.

Sobolev Spaces

The spaces of the previous chapters measure the size of a function; the Sobolev spaces measure a function *together with its derivatives*, and they are where the differential equations of analysis and geometry live. The construction completes the smooth functions in a norm that controls not only $\|f\|_{L^p}$ but $\|\partial^\alpha f\|_{L^p}$ up to a fixed order, so that a limit of smooth functions keeps a definite derivative “in the L^p sense” even where it is nowhere classically differentiable. Two ingredients already in hand make this work: the Lebesgue spaces of chapter 1, which supply the norm, and the distributional derivative of chapter 4, which supplies a notion of derivative robust enough to survive the completion.

This chapter develops what a reader needs before the elliptic and parabolic equations of the partial-differential-equations volumes: the definition and completeness of $W^{k,p}$; the density of smooth functions, proved with the mollifiers of section 2.5; the embedding theorems that trade Sobolev regularity for classical regularity or higher integrability; the trace that assigns boundary values; and the Rellich-Kondrachov compactness behind existence proofs. Everything here is Fourier-free, resting only on L^p and the distributional derivative; the complementary scale H^s , defined through the Fourier transform, is built in the harmonic-analysis part.

5.1 Weak Derivatives and the Spaces $W^{k,p}$

Recall from chapter 4 that every distribution $F \in \mathcal{D}'(U)$ has distributional derivatives $\partial^\alpha F$, defined by $\langle \partial^\alpha F, \varphi \rangle = (-1)^{|\alpha|} \langle F, \partial^\alpha \varphi \rangle$ for $\varphi \in C_c^\infty(U)$. A locally integrable function is a distribution, so its derivatives of every order exist as distributions; Sobolev theory is organized by one question: for which α is $\partial^\alpha f$ *again a function*, and how integrable is it?

Definition 5.1.1: Weak Derivative

Let $U \subseteq \mathbb{R}^n$ be open, let $f \in L^1_{\text{loc}}(U)$, and let α be a multi-index. A function $g \in L^1_{\text{loc}}(U)$ is the *weak α -derivative* of f if

$$\int_U f \partial^\alpha \varphi \, dx = (-1)^{|\alpha|} \int_U g \varphi \, dx \quad \text{for every } \varphi \in C_c^\infty(U).$$

Equivalently, the distributional derivative $\partial^\alpha f$ is represented by the function g . When such a g exists we write $\partial^\alpha f := g$, and for first-order derivatives $\partial_i f$ and $\nabla f = (\partial_1 f, \dots, \partial_n f)$.

A classical C^k function is weakly differentiable, its weak and classical derivatives agreeing: integration by parts against $\varphi \in C_c^\infty(U)$ produces no boundary terms, which is precisely the identity above. The weak derivative is thus an extension of the classical one, and like the classical derivative it is determined almost everywhere.

Lemma 5.1.2: Uniqueness of the Weak Derivative

If $g_1, g_2 \in L^1_{\text{loc}}(U)$ are both weak α -derivatives of f , then $g_1 = g_2$ almost everywhere.

Proof:

Put $h = g_1 - g_2 \in L^1_{\text{loc}}(U)$. Subtracting the two defining identities gives $\int_U h \varphi \, dx = 0$ for every $\varphi \in C_c^\infty(U)$. Fix a compact $K \subset U$ and a mollifier $\{\psi_\varepsilon\}$, a good kernel (definition 2.5.9) supported in $B(0, \varepsilon)$. For $x \in K$ and $\varepsilon < \text{dist}(K, \partial U)$ the function $y \mapsto \psi_\varepsilon(x - y)$ belongs to $C_c^\infty(U)$, so

$$(h * \psi_\varepsilon)(x) = \int_U h(y) \psi_\varepsilon(x - y) \, dy = 0 \quad (x \in K).$$

As $\varepsilon \rightarrow 0$, $h * \psi_\varepsilon \rightarrow h$ in $L^1(K)$ by theorem 2.5.10, so $h = 0$ almost everywhere on K . Exhausting U by compact sets gives $h = 0$ almost everywhere on U .

Example 5.1.3: Weak Differentiability of a Corner and a Jump

On $U = (-1, 1)$ the corner $f(x) = |x|$ has weak derivative $\text{sgn}(x)$: for $\varphi \in C_c^\infty(-1, 1)$, integrating by parts over $(-1, 0)$ and $(0, 1)$ separately (the boundary terms at ± 1 vanish by compact support, and those at 0 cancel),

$$\int_{-1}^1 |x| \varphi'(x) \, dx = \int_{-1}^0 \varphi \, dx - \int_0^1 \varphi \, dx = - \int_{-1}^1 \text{sgn}(x) \varphi(x) \, dx,$$

so $|x| \in W^{1,p}(-1, 1)$ for every p . The jump sgn , by contrast, has *no* weak derivative: $\int_{-1}^1 \text{sgn}(x) \varphi'(x) \, dx = -2\varphi(0)$, and no $g \in L^1_{\text{loc}}$ satisfies $-\int g \varphi = -2\varphi(0)$ for all φ (the distributional derivative of sgn is the Dirac mass $2\delta_0$, which is not a function). Hence $|x| \in W^{1,p} \setminus W^{2,p}$: weak differentiability is a real but limited amount of regularity, exactly the feature that makes these spaces useful.

With derivatives in hand we form the spaces. Since the weak derivative is linear in f (immediate from definition 5.1.1), what follows is a linear subspace of L^p .

Definition 5.1.4: Sobolev Space $W^{k,p}$

Let $U \subseteq \mathbb{R}^n$ be open, let k be a nonnegative integer, and let $1 \leq p \leq \infty$. The *Sobolev space* is

$$W^{k,p}(U) = \{f \in L^p(U) \mid \partial^\alpha f \in L^p(U) \text{ for all } |\alpha| \leq k\},$$

the derivatives being weak (definition 5.1.1), normed by

$$\|f\|_{W^{k,p}(U)} = \left(\sum_{|\alpha| \leq k} \|\partial^\alpha f\|_{L^p(U)}^p \right)^{1/p} \quad (p < \infty), \quad \|f\|_{W^{k,\infty}(U)} = \max_{|\alpha| \leq k} \|\partial^\alpha f\|_{L^\infty(U)}.$$

When $p = 2$ we write $H^k(U) := W^{k,2}(U)$, an inner-product space under $\langle f, g \rangle_{H^k} = \sum_{|\alpha| \leq k} \int_U \partial^\alpha f \overline{\partial^\alpha g} dx$.

The norm records a function and all of its weak derivatives up to order k as a single L^p quantity; $W^{0,p} = L^p$, and raising k demands more regularity. That the smooth members of $W^{k,p}$ are dense in it, so that $W^{k,p}$ is the completion of smooth functions promised in the introduction, is the subject of the next section; first we record that the space is already complete as defined.

Theorem 5.1.5: $W^{k,p}$ is a Banach Space

For every open $U \subseteq \mathbb{R}^n$, every nonnegative integer k , and every $1 \leq p \leq \infty$, $W^{k,p}(U)$ is a Banach space, and $H^k(U)$ is a Hilbert space.

Proof:

The map $f \mapsto \|f\|_{W^{k,p}}$ is a norm: absolute homogeneity is clear, the triangle inequality follows from that of the L^p norms (theorem 2.1.6) combined with the triangle inequality of the ℓ^p sum, and $\|f\|_{W^{k,p}} = 0$ forces $\|f\|_{L^p} = 0$, hence $f = 0$. Only completeness remains.

Let (f_m) be Cauchy in $W^{k,p}(U)$. For each $|\alpha| \leq k$,

$$\|\partial^\alpha f_m - \partial^\alpha f_l\|_{L^p} \leq \|f_m - f_l\|_{W^{k,p}},$$

so $(\partial^\alpha f_m)$ is Cauchy in $L^p(U)$. As $L^p(U)$ is complete (theorem 2.1.7), there are $g_\alpha \in L^p(U)$ with $\partial^\alpha f_m \rightarrow g_\alpha$ in L^p ; write $f := g_0$.

It remains to identify g_α as $\partial^\alpha f$. Fix $\varphi \in C_c^\infty(U)$. For each m , the weak-derivative identity reads

$$\int_U f_m \partial^\alpha \varphi dx = (-1)^{|\alpha|} \int_U (\partial^\alpha f_m) \varphi dx.$$

Both φ and $\partial^\alpha \varphi$ are bounded with compact support, so they lie in $L^{p'}(U)$; Hölder's inequality (theorem 2.1.5) then passes the limit under each integral, giving $\int_U f_m \partial^\alpha \varphi \rightarrow \int_U f \partial^\alpha \varphi$ and $\int_U (\partial^\alpha f_m) \varphi \rightarrow \int_U g_\alpha \varphi$. Hence

$$\int_U f \partial^\alpha \varphi dx = (-1)^{|\alpha|} \int_U g_\alpha \varphi dx \quad \text{for all } \varphi \in C_c^\infty(U),$$

which says exactly that $\partial^\alpha f = g_\alpha$ weakly. Thus $f \in W^{k,p}(U)$, and

$$\|f_m - f\|_{W^{k,p}}^p = \sum_{|\alpha| \leq k} \|\partial^\alpha f_m - g_\alpha\|_{L^p}^p \rightarrow 0,$$

so $f_m \rightarrow f$ in $W^{k,p}$ and the space is complete. For $p = 2$ the norm is induced by the inner product of definition 5.1.4, so $H^k(U)$ is a Hilbert space.

The completeness proof used nothing but the completeness of L^p and the stability of the weak-derivative identity under L^p limits, the reward for defining derivatives by duality against test functions rather than pointwise. The next section shows that the members of $W^{k,p}$ are, on reasonable domains, limits of smooth functions, so that theorems can be proved by checking them on C^∞ and passing to the limit.

5.2 Approximation by Smooth Functions

This section proves that smooth functions are dense in $W^{k,p}(U)$, so that a theorem about Sobolev functions can be reduced to the smooth case and recovered by passing to the limit. The mechanism throughout is that mollification commutes with the weak derivative.

Theorem 5.2.1: Local Approximation by Mollification

Let $U \subseteq \mathbb{R}^n$ be open, let $f \in W^{k,p}(U)$ with $1 \leq p < \infty$, and fix a mollifier $\{\psi_\varepsilon\}$, a good kernel (definition 2.5.9) with $\psi_\varepsilon \in C_c^\infty(\mathbb{R}^n)$ supported in $B(0, \varepsilon)$. On any open $V \Subset U$, for $\varepsilon < \text{dist}(V, \partial U)$ the mollification $f * \psi_\varepsilon$ is defined and smooth on V , and for every multi-index $|\alpha| \leq k$,

$$\partial^\alpha (f * \psi_\varepsilon) = (\partial^\alpha f) * \psi_\varepsilon \quad \text{on } V$$

Consequently $f * \psi_\varepsilon \rightarrow f$ in $W^{k,p}(V)$ as $\varepsilon \rightarrow 0$.

Proof:

Fix $V \Subset U$ and write $d = \text{dist}(V, \partial U) > 0$; throughout, $\varepsilon < d$. For $x \in V$ the ball $B(x, \varepsilon)$ lies in U , so

$$(f * \psi_\varepsilon)(x) = \int_U f(y) \psi_\varepsilon(x - y) dy = \int_{B(x, \varepsilon)} f(y) \psi_\varepsilon(x - y) dy$$

is a genuine integral against a function supported in U , and it is finite because $f \in L^1_{\text{loc}}(U)$ and ψ_ε is bounded.

Step 1: The commutation identity. Fix $x \in V$ and $|\alpha| \leq k$. Since $y \mapsto \psi_\varepsilon(x - y)$ is smooth and, for x in a small neighbourhood of any point of V , supported in a fixed compact subset of U , differentiation under the integral sign applies to the parameter x . Each x -derivative ∂_{x_i} falls on $\psi_\varepsilon(x - y)$, so

$$\partial_x^\alpha (f * \psi_\varepsilon)(x) = \int_U f(y) \partial_x^\alpha \psi_\varepsilon(x - y) dy$$

Now $\partial_x^\alpha \psi_\varepsilon(x - y) = (-1)^{|\alpha|} \partial_y^\alpha \psi_\varepsilon(x - y)$, since each factor of $x - y$ contributes a sign under the switch from an x -derivative to a y -derivative. For $x \in V$ the function $\varphi(y) := \psi_\varepsilon(x - y)$

belongs to $C_c^\infty(U)$: it is smooth and supported in $B(x, \varepsilon) \subseteq U$. The weak derivative identity of definition 5.1.1 applied to this φ therefore reads

$$\int_U f(y) \partial_y^\alpha \psi_\varepsilon(x - y) dy = (-1)^{|\alpha|} \int_U (\partial^\alpha f)(y) \psi_\varepsilon(x - y) dy$$

Combining the last three displays, the two factors of $(-1)^{|\alpha|}$ cancel and

$$\partial_x^\alpha (f * \psi_\varepsilon)(x) = \int_U (\partial^\alpha f)(y) \psi_\varepsilon(x - y) dy = ((\partial^\alpha f) * \psi_\varepsilon)(x)$$

which is the claimed identity. Smoothness of $f * \psi_\varepsilon$ on V is now immediate: since every x -derivative falls on ψ_ε , the identity $\partial_x^\alpha (f * \psi_\varepsilon) = f * \partial_x^\alpha \psi_\varepsilon$ holds for *every* multi-index α (not only $|\alpha| \leq k$), so all partial derivatives of $f * \psi_\varepsilon$ exist and, being mollifications of the locally integrable f , are continuous.

Step 2: Convergence in $W^{k,p}(V)$. Since $f \in W^{k,p}(U)$, each $\partial^\alpha f$ with $|\alpha| \leq k$ lies in $L^p(U)$, hence in $L^p(W)$ for any W with $V \Subset W \Subset U$; fix such a W . Restricting to $\varepsilon < \text{dist}(V, \partial W)$, the mollification $(\partial^\alpha f) * \psi_\varepsilon$ is computed on V from the values of $\partial^\alpha f$ on W only. By the approximate-identity convergence of theorem 2.5.10, $(\partial^\alpha f) * \psi_\varepsilon \rightarrow \partial^\alpha f$ in $L^p(V)$ as $\varepsilon \rightarrow 0$. Using the Step 1 identity on the left,

$$\|f * \psi_\varepsilon - f\|_{W^{k,p}(V)}^p = \sum_{|\alpha| \leq k} \|\partial^\alpha (f * \psi_\varepsilon) - \partial^\alpha f\|_{L^p(V)}^p = \sum_{|\alpha| \leq k} \|(\partial^\alpha f) * \psi_\varepsilon - \partial^\alpha f\|_{L^p(V)}^p$$

and each of the finitely many summands tends to 0. Hence $f * \psi_\varepsilon \rightarrow f$ in $W^{k,p}(V)$, completing the proof.

The identity $\partial^\alpha (f * \psi_\varepsilon) = (\partial^\alpha f) * \psi_\varepsilon$ is the whole engine of Sobolev approximation. Mollification is a smoothing operation, so its output is a bona fide C^∞ function to which the classical calculus applies; the identity says that this smooth function carries, as its classical derivatives, exactly the mollified weak derivatives of f . Convergence of each weak derivative in L^p then upgrades automatically to convergence in the Sobolev norm, because that norm is nothing but the L^p norms of the derivatives bundled together. The one limitation is visible in the hypothesis $V \Subset U$: the mollification $(f * \psi_\varepsilon)(x)$ reaches out to the ball $B(x, \varepsilon)$, so near ∂U it would sample points outside U where f is not defined. Local approximation is therefore free; the work in the rest of the section is to recover global statements from it. The restriction $p < \infty$ is inherited from theorem 2.5.10, where the translation $y \mapsto f(\cdot - y)$ fails to be L^∞ -continuous.

Example 5.2.2: Mollifying a Corner

Return to $f(x) = |x|$ on $U = (-1, 1)$, which lies in $W^{1,p}(U)$ with weak derivative $\partial f = \text{sgn}(x)$ (example 5.1.3). Fix the standard mollifier $\psi_\varepsilon(x) = \varepsilon^{-1} \psi(x/\varepsilon)$ with $\psi \in C_c^\infty(-1, 1)$ even, $\psi \geq 0$, $\int \psi = 1$. On any $V = (-a, a)$ with $a < 1$ and $\varepsilon < 1 - a$, the mollification is

$$(f * \psi_\varepsilon)(x) = \int_{-\varepsilon}^{\varepsilon} |x - y| \psi_\varepsilon(y) dy$$

a smooth function that agrees with $|x|$ for $|x| > \varepsilon$ (there $|x - y|$ is affine on the support of ψ_ε and the first moment of the even kernel vanishes) and rounds off the corner on $(-\varepsilon, \varepsilon)$. Its derivative is $(f * \psi_\varepsilon)'(x) = (\text{sgn} * \psi_\varepsilon)(x)$, exactly as theorem 5.2.1 predicts: a smooth function increasing from -1 to 1 across the corner, equal to $\text{sgn}(x)$ outside $(-\varepsilon, \varepsilon)$. As $\varepsilon \rightarrow 0$ the smoothed corner and its

derivative converge to $|x|$ and $\text{sgn}(x)$ in $L^p(V)$, hence $f * \psi_\varepsilon \rightarrow f$ in $W^{1,p}(V)$.

Local approximation leaves two gaps. Mollification only produces smooth functions on the interior, so it does not by itself yield a smooth approximant on all of U ; and even where it does, the approximant need not extend smoothly to the boundary. The first gap is closed by patching local mollifications together with a partition of unity, which is the content of the Meyers-Serrin theorem. Historically, two definitions of a Sobolev space competed: the space $W^{k,p}(U)$ of L^p functions with weak derivatives in L^p , and the space $H^{k,p}(U)$ obtained as the abstract completion of $\{f \in C^\infty(U) \mid \|f\|_{W^{k,p}} < \infty\}$ in the Sobolev norm. That these two coincide for every open U , with no regularity assumption on the boundary, is the theorem's content, and it is why one no longer distinguishes “ H ” from “ W ”.

Theorem 5.2.3: Meyers-Serrin ($H = W$)

Let $U \subseteq \mathbb{R}^n$ be open, let $k \geq 0$, and let $1 \leq p < \infty$. Then $C^\infty(U) \cap W^{k,p}(U)$ is dense in $W^{k,p}(U)$: for every $f \in W^{k,p}(U)$ and every $\varepsilon > 0$ there is a $g \in C^\infty(U) \cap W^{k,p}(U)$ with $\|f - g\|_{W^{k,p}(U)} < \varepsilon$.

Proof:

The idea is to mollify f separately on each shell of an exhaustion of U , where the previous theorem applies, and to glue the local smooth pieces with a partition of unity, arranging the individual errors to sum to less than ε .

Step 1: An open exhaustion and a partition of unity. Set, for integers $m \geq 1$,

$$U_m = \{x \in U \mid \text{dist}(x, \partial U) > 1/m \text{ and } |x| < m\}$$

with the conventions $U_0 = U_{-1} = \emptyset$; then $U_m \Subset U_{m+1}$ and $\bigcup_m U_m = U$. The open sets

$$V_m = U_{m+1} \setminus \overline{U_{m-1}} \quad (m \geq 1)$$

form a locally finite open cover of U : each $V_m \Subset U$, and any point of U lies in V_m for at most three consecutive indices. Let $\{\zeta_m\}_{m \geq 1}$ be a smooth partition of unity subordinate to $\{V_m\}$, so $\zeta_m \in C_c^\infty(V_m)$, $0 \leq \zeta_m \leq 1$, and $\sum_m \zeta_m \equiv 1$ on U (the sum being locally finite). Each $\zeta_m f$ then lies in $W^{k,p}(U)$ with support in V_m , because multiplication by the smooth compactly supported ζ_m preserves weak differentiability: for $|\alpha| \leq k$ the Leibniz rule

$$\partial^\alpha(\zeta_m f) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \partial^\beta \zeta_m \partial^{\alpha-\beta} f$$

holds weakly (the distributional Leibniz rule for multiplication by a smooth function, chapter 4), its right-hand side a finite sum of L^p functions.

Step 2: Mollify each piece with a controlled error. Fix $\varepsilon > 0$. The function $\zeta_m f$ is supported in the compact-in- U set $\overline{V_m}$, so theorem 5.2.1 gives $\zeta_m f * \psi_\delta \rightarrow \zeta_m f$ in $W^{k,p}(U)$ as $\delta \rightarrow 0$ (the mollification is computed on a neighbourhood of $\overline{V_m}$ still compact in U). Choose $\delta_m > 0$ small enough that

$$g_m := (\zeta_m f) * \psi_{\delta_m} \in C_c^\infty(U), \quad \|g_m - \zeta_m f\|_{W^{k,p}(U)} < \frac{\varepsilon}{2^m}$$

and, in addition, δ_m small enough that $\text{supp } g_m \subseteq W_m := U_{m+2} \setminus \overline{U_{m-2}}$ (possible because $\text{supp}(\zeta_m f) \subseteq V_m$ sits well inside W_m). The sets W_m are again locally finite: only finitely many W_m meet any given U_l .

Step 3: Sum and estimate. Define

$$g = \sum_{m \geq 1} g_m$$

On any U_l the sum is finite (only the g_m with $W_m \cap U_l \neq \emptyset$ contribute), so g is well-defined and $g \in C^\infty(U)$. Since $\sum_m \zeta_m f = f$ with the same local finiteness, for every $x \in U$

$$g(x) - f(x) = \sum_{m \geq 1} (g_m(x) - (\zeta_m f)(x))$$

is a finite sum. Fix l . On U_l only the finitely many indices m with $W_m \cap U_l \neq \emptyset$ or $V_m \cap U_l \neq \emptyset$ enter, and the triangle inequality in $W^{k,p}(U_l)$ gives

$$\|g - f\|_{W^{k,p}(U_l)} \leq \sum_{m \geq 1} \|g_m - \zeta_m f\|_{W^{k,p}(U)} < \sum_{m \geq 1} \frac{\varepsilon}{2^m} = \varepsilon$$

the bound being uniform in l . Letting $l \rightarrow \infty$ and applying the monotone convergence theorem to $\sum_{|\alpha| \leq k} |\partial^\alpha (g - f)|^p$ over the increasing exhaustion $U_l \uparrow U$,

$$\|g - f\|_{W^{k,p}(U)} = \lim_{l \rightarrow \infty} \|g - f\|_{W^{k,p}(U_l)} \leq \varepsilon$$

In particular $g - f \in W^{k,p}(U)$, so $g \in W^{k,p}(U)$; and $\|g - f\|_{W^{k,p}(U)} \leq \varepsilon$. As $\varepsilon > 0$ was arbitrary, $C^\infty(U) \cap W^{k,p}(U)$ is dense, completing the proof.

The strength of Meyers-Serrin is that it asks nothing of the domain: for any open set whatsoever, however irregular its boundary, the smooth Sobolev functions are dense. This is what dissolves the historical distinction between the completion-of-smooth-functions space $H^{k,p}$ and the weak-derivative space $W^{k,p}$. The price is that the approximating g is smooth only in the *interior* of U : nothing controls its behaviour as x approaches ∂U , and indeed g and its derivatives may blow up there. For many purposes this is enough, since the norm only sees the interior. But to integrate by parts up to the boundary, or to make sense of boundary values, one needs approximants smooth on the closure $\bar{U}$, and that does require the boundary to be regular.

Example 5.2.4: Interior Smoothness Is Not Boundary Smoothness

Take $n = 1$, $U = (0, 1)$, and $f(x) = x^{1/2}$. Then $f \in W^{1,p}(0, 1)$ for $1 \leq p < 2$, since $f' = \frac{1}{2}x^{-1/2} \in L^p(0, 1)$ exactly when $p/2 < 1$. Mollifying f over the exhaustion of $(0, 1)$ as in theorem 5.2.3 produces smooth approximants on the open interval, but they inherit the growth of f' near 0: the mollified derivative $(f' * \psi_\delta)(x)$ is of order $\delta^{-1/2}$ for x within δ of the endpoint, so no interior mollification is bounded up to $x = 0$, and none extends to a function in $C^\infty([0, 1])$. The Meyers-Serrin conclusion is exactly as advertised and no better: density by functions smooth on the *interior*.

The remedy is the inward shift. For small $t > 0$ the translate $f_t(x) = f(x + t) = (x + t)^{1/2}$ is smooth on all of $[0, 1]$, because its only singularity, at $x = -t$, now sits outside the closed interval; and $\|f_t - f\|_{W^{1,p}(0,1)} \rightarrow 0$ as $t \rightarrow 0$ by L^p -continuity of translation. Thus f_t (already smooth here, no mollification needed) is a $C^\infty([0, 1])$ function within any prescribed $W^{1,p}$ tolerance of f , which is precisely what theorem 5.2.5 delivers in general. The shift trades a function singular *on* the boundary for one singular just outside it, at the cost of a small L^p error, converting interior smoothness into smoothness up to $\bar{U}$.

For a bounded domain with regular boundary, one can do better than interior smoothness. The strategy near a boundary point is to flatten the boundary with a C^1 change of coordinates, translate the function a small distance *into* the domain so that its support pulls away from the boundary, and then mollify at a scale finer than the translation. The translated-and-mollified function is smooth on a neighbourhood of the piece of $\bar{U}$ under consideration, and the translation costs little in $W^{k,p}$ because translation is continuous on L^p .

Theorem 5.2.5: Global Approximation up to the Boundary

Let $U \subseteq \mathbb{R}^n$ be bounded with C^1 boundary ∂U , let $k \geq 0$, and let $1 \leq p < \infty$. Then $C^\infty(\bar{U})$ is dense in $W^{k,p}(U)$: for every $f \in W^{k,p}(U)$ and every $\varepsilon > 0$ there is a $g \in C^\infty(\bar{U})$ with $\|f - g\|_{W^{k,p}(U)} < \varepsilon$.

Proof:

Fix $f \in W^{k,p}(U)$ and $\varepsilon > 0$.

Step 1: Localize to a boundary chart and translate inward. Fix a point $x_0 \in \partial U$. Because ∂U is C^1 , after relabelling coordinates there is $r > 0$ and a C^1 function $\gamma: \mathbb{R}^{n-1} \rightarrow \mathbb{R}$ with

$$U \cap B(x_0, r) = \{x \in B(x_0, r) \mid x_n > \gamma(x_1, \dots, x_{n-1})\}$$

so near x_0 the domain lies above the graph of γ . Set $V = U \cap B(x_0, r/2)$. For a point $x \in V$ and $t > 0$, define the inward-shifted point

$$x^t = x + t e_n$$

and the shifted function $f_t(x) = f(x^t)$ on V . Choose r small enough (using continuity of $\nabla \gamma$, i.e. the C^1 property: in coordinates where $\nabla \gamma(x'_0) = 0$ the graph has small slope over a small ball) that there is a cone opening in the $+e_n$ direction: there is $\lambda > 0$ so that for all $x \in V$ and all $0 < t \leq \lambda$, the ball $B(x^t, t/2)$ lies inside $U \cap B(x_0, r)$. Then $f_t \in W^{k,p}(V)$ with $\partial^\alpha f_t = (\partial^\alpha f)_t$ (translation commutes with weak differentiation, by the change of variables $y \mapsto y - t e_n$ in definition 5.1.1), and by the L^p -continuity of translation (a standard consequence of the density of C_c^∞ in L^p , proposition 2.5.8: approximate each $\partial^\alpha f$ by a uniformly continuous compactly supported function, for which the shift converges uniformly),

$$\|f_t - f\|_{W^{k,p}(V)} = \left(\sum_{|\alpha| \leq k} \|(\partial^\alpha f)_t - \partial^\alpha f\|_{L^p(V)}^p \right)^{1/p} \rightarrow 0 \quad (t \rightarrow 0)$$

Fix $t \leq \lambda$ with $\|f_t - f\|_{W^{k,p}(V)} < \varepsilon/2$.

Step 2: Mollify the shifted function. By construction $B(x^t, t/2) \subseteq U$ for every $x \in V$, so for $\delta < t/2$ the mollification $g^\delta := f_t * \psi_\delta$ samples f only at points of U : for $x \in \bar{V}$,

$$g^\delta(x) = \int_{B(0, \delta)} f(x + t e_n - y) \psi_\delta(y) dy$$

is well-defined because $x + t e_n - y \in B(x^t, t/2) \subseteq U$. Thus g^δ is defined and smooth on a neighbourhood of $\bar{V}$ (the mollification of a locally integrable function is smooth, by lemma 2.5.7), including the boundary piece $\partial U \cap B(x_0, r/2)$. By theorem 5.2.1 applied to f_t on V (whose mollification is interior to the slightly larger set where f_t is defined), $g^\delta \rightarrow f_t$ in $W^{k,p}(V)$ as

$\delta \rightarrow 0$; fix $\delta < t/2$ with $\|g^\delta - f_t\|_{W^{k,p}(V)} < \varepsilon/2$. Then $g^\delta \in C^\infty(\bar{V})$ and, by the triangle inequality,

$$\|g^\delta - f\|_{W^{k,p}(V)} \leq \|g^\delta - f_t\|_{W^{k,p}(V)} + \|f_t - f\|_{W^{k,p}(V)} < \varepsilon$$

Step 3: Patch the boundary charts with the interior. The compact boundary ∂U is covered by finitely many such half-ball neighbourhoods $V_1, \dots, V_N$, on each of which Steps 1 and 2 produce a $g_j \in C^\infty(\bar{V}_j)$ with $\|g_j - f\|_{W^{k,p}(U \cap V_j)} < \varepsilon$. Adjoin one more open set $V_0 \Subset U$ with $\bar{U} \subseteq V_0 \cup V_1 \cup \dots \cup V_N$ and, on it, the interior mollification $g_0 \in C^\infty(V_0)$ of theorem 5.2.1 with $\|g_0 - f\|_{W^{k,p}(V_0)} < \varepsilon$. Let $\{\eta_j\}_{j=0}^N$ be a smooth partition of unity subordinate to $\{V_j\}_{j=0}^N$ over a neighbourhood of $\bar{U}$, and set

$$g = \sum_{j=0}^N \eta_j g_j$$

Each $\eta_j g_j$ is smooth up to $\bar{U}$ (either interior, or smooth on $\bar{V}_j$), so $g \in C^\infty(\bar{U})$. Using $\sum_j \eta_j = 1$ and the Leibniz rule as in the proof of theorem 5.2.3, on U

$$\partial^\alpha(g - f) = \sum_{j=0}^N \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \partial^\beta \eta_j \partial^{\alpha-\beta}(g_j - f)$$

Since the derivatives of the η_j are bounded (there are finitely many, each in C_c^∞) by a constant M depending only on k and the cover, and each g_j satisfies $\|g_j - f\|_{W^{k,p}(U \cap V_j)} < \varepsilon$,

$$\|g - f\|_{W^{k,p}(U)} \leq C(k, N, M) \varepsilon$$

for a constant independent of ε . As $\varepsilon > 0$ was arbitrary, $C^\infty(\bar{U})$ is dense in $W^{k,p}(U)$, completing the proof.

The mechanism to remember is the inward shift. Mollification alone fails at the boundary only because it reaches outside U ; translating f a distance t into the domain first pulls the relevant values away from ∂U , buying room to mollify at the finer scale $\delta < t/2$ without ever leaving U . The C^1 hypothesis enters exactly to guarantee a consistent inward direction near each boundary point, so that a single shift $+te_n$ (after flattening) works throughout a boundary chart. Domains with cusps or inward spikes, where no such uniform inward cone exists, can fail this theorem; the C^1 regularity is what rules them out. This global approximation is the workhorse behind the extension and trace theorems of the next section, and behind every argument in this chapter that proves an estimate first for smooth functions and then extends it by density.

The density theorems distinguish two natural classes of Sobolev functions: those approximable by smooth functions with no constraint at the boundary, which is all of $W^{k,p}(U)$, and those approximable by smooth functions that *vanish near the boundary*. The latter class is where boundary conditions live, and it deserves a name.

Definition 5.2.6: The Space $W_0^{k,p}$

Let $U \subseteq \mathbb{R}^n$ be open, let $k \geq 0$, and let $1 \leq p < \infty$. The space $W_0^{k,p}(U)$ is the closure of $C_c^\infty(U)$ in the norm of $W^{k,p}(U)$:

$$W_0^{k,p}(U) := \overline{C_c^\infty(U)}^{\|\cdot\|_{W^{k,p}(U)}}$$

When $p = 2$ one writes $H_0^k(U) := W_0^{k,2}(U)$.

By construction $W_0^{k,p}(U)$ is a closed subspace of the Banach space $W^{k,p}(U)$, hence a Banach space in its own right, and $H_0^k(U)$ a Hilbert space. Its elements are the Sobolev functions that are limits, in the Sobolev norm, of functions compactly supported inside U , and one reads this as the functions “with zero boundary values” together with, for $k \geq 2$, vanishing normal derivatives up to order $k - 1$. The trace theorem of the next section will make this interpretation precise for $k = 1$ by identifying $W_0^{1,p}(U)$ with the kernel of the boundary-value map (theorem 5.3.9); for now it is a definition, and the one case where it collapses onto the full space is worth recording at once.

Proposition 5.2.7: $W_0^{k,p}(\mathbb{R}^n) = W^{k,p}(\mathbb{R}^n)$

For every $k \geq 0$ and $1 \leq p < \infty$, $W_0^{k,p}(\mathbb{R}^n) = W^{k,p}(\mathbb{R}^n)$.

Proof:

The inclusion $W_0^{k,p}(\mathbb{R}^n) \subseteq W^{k,p}(\mathbb{R}^n)$ is immediate, as $W_0^{k,p}$ is a subspace. For the reverse, it suffices to approximate an arbitrary $f \in W^{k,p}(\mathbb{R}^n)$ by elements of $C_c^\infty(\mathbb{R}^n)$ in the $W^{k,p}$ norm; the smooth approximation runs directly on the whole space, and the only extra work is to make the approximants compactly supported by a cutoff.

Fix $\varepsilon > 0$. First mollify. On $\mathbb{R}^n$ there is no boundary to obstruct the convolution, so the commutation identity holds globally: for $U = \mathbb{R}^n$ every ball $B(x, \delta)$ lies in $\mathbb{R}^n$, so the differentiation-under-the-integral and weak-derivative steps of theorem 5.2.1 Step 1 go through for all x , giving $\partial^\alpha(f * \psi_\delta) = (\partial^\alpha f) * \psi_\delta$ on all of $\mathbb{R}^n$ for $|\alpha| \leq k$. Each $\partial^\alpha f \in L^p(\mathbb{R}^n)$, so the approximate-identity convergence of theorem 2.5.10 on $L^p(\mathbb{R}^n)$ gives $(\partial^\alpha f) * \psi_\delta \rightarrow \partial^\alpha f$ in $L^p(\mathbb{R}^n)$ as $\delta \rightarrow 0$; summing over $|\alpha| \leq k$, $f * \psi_\delta \rightarrow f$ in $W^{k,p}(\mathbb{R}^n)$. Choose $h := f * \psi_\delta \in C^\infty(\mathbb{R}^n) \cap W^{k,p}(\mathbb{R}^n)$ with $\|h - f\|_{W^{k,p}(\mathbb{R}^n)} < \varepsilon/2$. Now cut off. Fix a function $\chi \in C_c^\infty(\mathbb{R}^n)$ with $\chi \equiv 1$ on $B(0, 1)$, $0 \leq \chi \leq 1$, supported in $B(0, 2)$, and set $\chi_R(x) = \chi(x/R)$, so $\partial^\alpha \chi_R$ is bounded by $C_\alpha R^{-|\alpha|} \leq C_\alpha$ for $R \geq 1$ and $\chi_R \equiv 1$ on $B(0, R)$. Then $\chi_R h \in C_c^\infty(\mathbb{R}^n)$, and by the Leibniz rule

$$\partial^\alpha(\chi_R h - h) = \partial^\alpha((\chi_R - 1)h) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \partial^\beta(\chi_R - 1) \partial^{\alpha-\beta} h$$

where the $\beta = 0$ term is $(\chi_R - 1) \partial^\alpha h$ and the $\beta \neq 0$ terms carry a factor $\partial^\beta \chi_R$ supported in the annulus $R \leq |x| \leq 2R$. Since $\chi_R - 1$ vanishes on $B(0, R)$ and is bounded by 1, and each $\partial^\alpha h \in L^p(\mathbb{R}^n)$, dominated convergence gives $\|(\chi_R - 1) \partial^\alpha h\|_{L^p(\mathbb{R}^n)} \rightarrow 0$ as $R \rightarrow \infty$; the annular terms are bounded by $C \sum_{|\gamma| < |\alpha|} \|\partial^\gamma h\|_{L^p(R \leq |x| \leq 2R)} \rightarrow 0$ likewise. Summing over $|\alpha| \leq k$, choose R with $\|\chi_R h - h\|_{W^{k,p}(\mathbb{R}^n)} < \varepsilon/2$. Then $\chi_R h \in C_c^\infty(\mathbb{R}^n)$ and

$$\|\chi_R h - f\|_{W^{k,p}(\mathbb{R}^n)} \leq \|\chi_R h - h\|_{W^{k,p}(\mathbb{R}^n)} + \|h - f\|_{W^{k,p}(\mathbb{R}^n)} < \varepsilon$$

| so f lies in the closure of $C_c^\infty(\mathbb{R}^n)$, as we sought to show.

The proposition says that on the whole space there is no boundary to see, so the constraint defining $W_0^{k,p}$ is empty and the two spaces agree. On a bounded domain the situation is the opposite extreme: $W_0^{k,p}(U)$ is a genuinely proper closed subspace of $W^{k,p}(U)$, precisely because a nonzero constant function belongs to $W^{k,p}(U)$ but cannot be approximated by compactly supported functions without paying for the boundary layer where it must drop to zero. The gap between the two spaces is what the trace measures, and the Poincaré inequality of the last section of this chapter will quantify how the vanishing boundary condition of $W_0^{1,p}$ controls a function by its gradient alone.

5.3 Extensions and Traces

A Sobolev function on a domain U is defined only inside U , and two operations that the elliptic theory needs are not visible from the definition alone: extending such a function to all of $\mathbb{R}^n$ without losing a derivative, and restricting it to the boundary ∂U where a function without pointwise values must still be assigned boundary data. This section builds both, resting on the same device, flattening the C^1 boundary in local charts, and on the density of functions smooth up to the boundary proved in the previous section.

Throughout, $U \subseteq \mathbb{R}^n$ is bounded with C^1 boundary ∂U : near each boundary point there is a ball in which, after relabelling coordinates, ∂U is the graph $x_n = \gamma(x')$ of a C^1 function γ of the first $n-1$ coordinates $x' = (x_1, \dots, x_{n-1})$, with U lying on the side $x_n > \gamma(x')$. The boundary carries its surface measure, and ∂U is a compact $(n-1)$ -dimensional C^1 manifold.

The extension is built one boundary chart at a time. In a chart where the boundary has been straightened to a piece of the hyperplane $x_n = 0$, a $W^{1,p}$ function on the upper half is extended to the lower half by reflection. A naive even reflection $f(x', -x_n)$ keeps f continuous across the interface but jumps in $\partial_n f$; the higher-order reflection below instead matches the first weak derivatives across $\{x_n = 0\}$, the condition for the reflected function to lie in $W^{1,p}$ of the full ball. The two lemmas that follow isolate this half-space reflection and the passage across a flattened C^1 graph; the theorem then patches the charts together with a partition of unity.

Write $\mathbb{R}_+^n = \{x \in \mathbb{R}^n \mid x_n > 0\}$ and $\mathbb{R}_-^n = \{x \in \mathbb{R}^n \mid x_n < 0\}$, and for a ball $B = B(0, r)$ put $B_+ = B \cap \mathbb{R}_+^n$ and $B_- = B \cap \mathbb{R}_-^n$.

Lemma 5.3.1: Higher-Order Reflection Across a Hyperplane

Let $B = B(0, r)$ and let $f \in W^{1,p}(B_+)$ with $1 \leq p < \infty$, and suppose f is supported away from the upper cap: $f = 0$ almost everywhere on $B_+ \setminus B(0, \rho)$ for some $\rho < r/2$. Define Ef on B by

$$(Ef)(x) = \begin{cases} f(x', x_n) & x_n > 0 \\ 3f(x', -x_n) - 2f(x', -2x_n) & x_n < 0 \end{cases}$$

where the reflected values are read off as 0 wherever their argument leaves B_+ . Then $Ef \in W^{1,p}(B)$, it agrees with f on B_+ , is supported in $B(0, \rho)$, and

$$\|Ef\|_{W^{1,p}(B)} \leq C\|f\|_{W^{1,p}(B_+)}$$

with C depending only on n .

Proof:

The plan is to prove the estimate and the extension property for $f \in C^1(\overline{B_+})$, where Ef can be differentiated classically, and then remove the smoothness by the density of such functions.

Step 1: The reflection matches value and normal derivative for smooth f . Suppose first $f \in C^1(\overline{B_+})$ vanishing on $B_+ \setminus B(0, \rho)$ with $\rho < r/2$. On B_- set $g(x) = 3f(x', -x_n) - 2f(x', -2x_n)$, so $Ef = f$ on B_+ and $Ef = g$ on B_- . The support hypothesis makes both reflected pieces well-defined on B_- : $f(x', -x_n)$ is nonzero only when $|(x', -x_n)| = |x| < \rho$, and $f(x', -2x_n)$ only when $|(x', -2x_n)|^2 = |x'|^2 + 4x_n^2 < \rho^2$, which forces $|x| < \rho$ as well, so in both cases the argument lies in $B(0, \rho)_+ \subseteq B_+$ (elsewhere the reflected value is 0). In particular g is supported in $B(0, \rho) \cap B_-$, and Ef in $B(0, \rho)$. As $x_n \rightarrow 0^-$ both reflected arguments tend to $(x', 0)$, and

$$\lim_{x_n \rightarrow 0^-} g(x', x_n) = 3f(x', 0) - 2f(x', 0) = f(x', 0)$$

so Ef is continuous across $\{x_n = 0\}$. Differentiating g classically on B_- ,

$$\partial_n g(x) = -3\partial_n f(x', -x_n) + 4\partial_n f(x', -2x_n)$$

whose boundary limit is $(-3 + 4)\partial_n f(x', 0) = \partial_n f(x', 0)$, matching the upper value $\partial_n f(x', 0)$; the tangential derivatives $\partial_i g$ for $i < n$ carry the factors 3 and -2 through the chain rule with total $3 - 2 = 1$, so they too match $\partial_i f(x', 0)$. Thus $Ef \in C^1(\overline{B})$ and its classical gradient is bounded on B by a multiple of $\sup_{B_+} (|f| + |\nabla f|)$.

Step 2: The classical gradient is the weak gradient. Because Ef is continuous across $\{x_n = 0\}$ and C^1 on each of B_+ , B_- separately, its weak gradient on all of B is the function that equals the classical gradient off the interface. To see this, fix $\varphi \in C_c^\infty(B)$ and integrate by parts on B_+ and B_- separately:

$$\int_B (Ef) \partial_i \varphi \, dx = \int_{B_+} f \partial_i \varphi \, dx + \int_{B_-} g \partial_i \varphi \, dx$$

Integration by parts on B_+ produces $-\int_{B_+} (\partial_i f) \varphi$ plus a boundary integral over $\{x_n = 0\}$ with integrand $f(x', 0) \varphi \nu_i$, where ν is the outward normal of B_+ ; the same on B_- produces $-\int_{B_-} (\partial_i g) \varphi$ plus a boundary integral with integrand $g(x', 0) \varphi (-\nu_i)$. The two boundary integrals cancel because $f(x', 0) = g(x', 0)$ by Step 1. Hence $\int_B (Ef) \partial_i \varphi = -\int_B (\partial_i^{\text{cl}} Ef) \varphi$, so the weak derivative $\partial_i(Ef)$ is the classical one, as claimed.

Step 3: The norm estimate for smooth f . With the weak and classical derivatives identified, change variables in the reflected pieces. Under $x_n \mapsto -x_n$ the map $(x', x_n) \mapsto (x', -x_n)$ has unit Jacobian, and under $x_n \mapsto -2x_n$ the map $(x', x_n) \mapsto (x', -2x_n)$ has Jacobian $\frac{1}{2}$ in the x_n variable. By Step 1 both substitutions carry the region where the corresponding reflected value is nonzero into $B(0, \rho)_+ \subseteq B_+$, so for the value term

$$\int_{B_-} |g|^p dx \leq 2^{p-1} \int_{B_-} (3^p |f(x', -x_n)|^p + 2^p |f(x', -2x_n)|^p) dx \leq C \int_{B_+} |f|^p dx$$

using $(a+b)^p \leq 2^{p-1}(a^p + b^p)$ and the substitutions (the second contributing a factor $\frac{1}{2}$ from its Jacobian, and integrating over a region contained in B_+ because f vanishes outside $B(0, \rho)_+$). The identical estimate applied to the derivative formulas of Step 1, whose coefficients are again bounded by constants depending only on n , gives $\int_{B_-} |\nabla g|^p \leq C \int_{B_+} |\nabla f|^p$. Summing, $\|Ef\|_{W^{1,p}(B)}^p \leq C \|f\|_{W^{1,p}(B_+)}^p$ with $C = C(n)$.

Step 4: Removal of smoothness by density. For general $f \in W^{1,p}(B_+)$ vanishing on $B_+ \setminus B(0, \rho)$, the domain B_+ has C^1 (indeed Lipschitz) boundary, so by theorem 5.2.5 there are $\tilde{f}_m \in C^\infty(\bar{B}_+)$ with $\tilde{f}_m \rightarrow f$ in $W^{1,p}(B_+)$. Fix a cutoff $\chi \in C_c^\infty(B(0, \rho'))$ with $\rho < \rho' < r/2$, $0 \leq \chi \leq 1$, and $\chi \equiv 1$ on $\bar{B}(0, \rho)$, and set $f_m := \chi \tilde{f}_m$. Since $f = \chi f$, the product rule gives $f_m \rightarrow f$ in $W^{1,p}(B_+)$, and each $f_m \in C^\infty(\bar{B}_+)$ vanishes on $B_+ \setminus B(0, \rho')$ with $\rho' < r/2$, so the reflection of Steps 1 to 3 applies to each f_m . The map E is linear, and Step 3 gives $\|Ef_m - Ef_l\|_{W^{1,p}(B)} \leq C \|f_m - f_l\|_{W^{1,p}(B_+)}$, so (Ef_m) is Cauchy in $W^{1,p}(B)$; let F be its limit. Restricting to B_+ , where $Ef_m = f_m$, gives $F = f$ on B_+ . The estimate passes to the limit as $\|F\|_{W^{1,p}(B)} \leq C \|f\|_{W^{1,p}(B_+)}$, and the explicit reflection formula for F on B_- follows because $f_m(x', -x_n) \rightarrow f(x', -x_n)$ and $f_m(x', -2x_n) \rightarrow f(x', -2x_n)$ in $L^p(B_-)$ under the same changes of variable. Thus $F = Ef$ as defined, and $Ef \in W^{1,p}(B)$ with the stated bound, completing the proof.

The reflection is engineered so that both the value and the full gradient are continuous across the interface, which Step 2 shows is exactly the condition making the two half-space derivatives assemble into a single weak derivative on the whole ball; the even reflection fails precisely at the normal derivative, and the coefficients 3, -2 are the unique pair (a, b) correcting it while preserving continuity of the value: matching the value forces $a + b = 1$ and matching the normal derivative forces $-a - 2b = 1$ (differentiating the two reflected pieces at $x_n = 0$), two linear equations solved only by $a = 3$, $b = -2$. The support hypothesis, confining the reflected arguments to the half-ball where f lives, costs nothing in the application: the localized pieces that the extension operator reflects are cut off by a partition of unity and already vanish near the top of their chart. This is the entire content of the extension near a straight boundary. The passage to a curved C^1 boundary is a change of variables that straightens it, recorded in the next lemma.

Lemma 5.3.2: Straightening a C^1 Boundary

Let $w \in \partial U$. There is an open ball V about w and a C^1 diffeomorphism $\Phi: V \rightarrow B(0, r)$ with C^1 inverse $\Psi = \Phi^{-1}$, such that $\Phi(V \cap U) = B(0, r)_+$ and $\Phi(V \cap \partial U) = B(0, r) \cap \{x_n = 0\}$. Composition with Φ and Ψ carries $W^{1,p}(V \cap U)$ boundedly to $W^{1,p}(B(0, r)_+)$ and back, with constants depending on Φ .

Proof:

By the C^1 boundary hypothesis, after relabelling coordinates there is a ball V about w and a C^1 function γ of $x' = (x_1, \dots, x_{n-1})$ with $V \cap \partial U = \{x \in V \mid x_n = \gamma(x')\}$ and $V \cap U = \{x \in V \mid x_n > \gamma(x')\}$. Define

$$\Phi(x) = (x', x_n - \gamma(x')), \quad \Psi(y) = (y', y_n + \gamma(y'))$$

Both are C^1 (as γ is C^1) and mutually inverse, with Jacobian determinant identically 1. The shear image $\Phi(V)$ is an open set containing $\Phi(w) = 0$ but need not be a ball, so fix a radius r small enough that $B(0, r) \subseteq \Phi(V)$ and *re-restrict*, replacing V by $\Phi^{-1}(B(0, r))$; on this smaller neighborhood Φ is a C^1 diffeomorphism onto exactly $B(0, r)$, and Φ sends $x_n > \gamma(x')$ to $y_n > 0$ and $x_n = \gamma(x')$ to $y_n = 0$, giving the stated images. For a function u on $V \cap U$, the composition $u \circ \Psi$ is defined on $B(0, r)_+$; by the chain rule for weak derivatives under C^1 changes of variable, $u \in W^{1,p}(V \cap U)$ if and only if $u \circ \Psi \in W^{1,p}(B(0, r)_+)$, with

$$\nabla(u \circ \Psi)(y) = (D\Psi(y))^\top (\nabla u)(\Psi(y))$$

and since $D\Phi, D\Psi$ are continuous hence bounded on the relevant compact sets and the Jacobians are 1, the L^p norms of u and its gradient are comparable to those of $u \circ \Psi$ up to a constant depending on $\sup |D\Phi|, \sup |D\Psi|$. The inverse map $v \mapsto v \circ \Phi$ is bounded likewise, as we sought to show.

With the local reflection and the local straightening in hand, the global extension is assembled by a finite partition of unity subordinate to boundary charts, extending each localized piece by reflection in its straightened chart and extending the interior piece by zero.

Theorem 5.3.3: Extension Operator

Let $U \subseteq \mathbb{R}^n$ be bounded with C^1 boundary and let $1 \leq p < \infty$. Fix a bounded open set W with $U \Subset W$. There is a bounded linear operator

$$E: W^{1,p}(U) \rightarrow W^{1,p}(\mathbb{R}^n)$$

such that for every $f \in W^{1,p}(U)$:

1. $Ef = f$ almost everywhere on U ;
2. Ef is supported in W ;
3. $\|Ef\|_{W^{1,p}(\mathbb{R}^n)} \leq C\|f\|_{W^{1,p}(U)}$, with $C = C(n, p, U, W)$.

Proof:

Step 1: A finite cover by boundary charts. The boundary ∂U is compact, and by lemma 5.3.2 each $w \in \partial U$ has a ball V_w carrying a straightening diffeomorphism $\Phi_w: V_w \rightarrow B(0, r_w)$. Shrinking each V_w so that $V_w \Subset W$, the balls $\{V_w\}_{w \in \partial U}$ cover ∂U ; extract a finite subcover $V_1, \dots, V_N$ with straightenings $\Phi_1, \dots, \Phi_N$. The compact set $K = \overline{U} \setminus \bigcup_{j=1}^N V_j$ lies in the open set U , so choose an open V_0 with $K \Subset V_0 \Subset U$. Then $V_0, V_1, \dots, V_N$ is a finite open cover of $\overline{U}$, all contained in W . Choose a C^∞ partition of unity $\zeta_0, \zeta_1, \dots, \zeta_N$ subordinate to it: each $\zeta_j \in C_c^\infty(V_j)$, $0 \leq \zeta_j \leq 1$,

and $\sum_{j=0}^N \zeta_j = 1$ on a neighborhood of $\bar{U}$. For each boundary chart $j \geq 1$, the support of ζ_j is a compact subset of V_j , so its image $\Phi_j(\text{supp } \zeta_j \cap \bar{U})$ is a compact subset of $B(0, r_j)$, contained in some $\bar{B}(0, s_j)$ with $s_j < r_j$. Post-composing Φ_j with the dilation $x \mapsto (r_j/3s_j)x$ (a straightening onto $B(0, r_j/3)$, then relabelled to radius r_j) moves this image inside $B(0, r_j/3)$, so that any function supported in it vanishes on $B(0, r_j)_+ \setminus B(0, \rho_j)$ for some $\rho_j < r_j/2$.

Step 2: Extend each localized piece. Given $f \in W^{1,p}(U)$, decompose $f = \sum_{j=0}^N \zeta_j f$ on U . The interior piece $\zeta_0 f$ has compact support in $V_0 \Subset U$; extending it by zero outside U gives a function $E_0 f \in W^{1,p}(\mathbb{R}^n)$ with $\|E_0 f\|_{W^{1,p}(\mathbb{R}^n)} \leq C \|\zeta_0 f\|_{W^{1,p}(U)} \leq C \|f\|_{W^{1,p}(U)}$, the product rule $\nabla(\zeta_0 f) = \zeta_0 \nabla f + f \nabla \zeta_0$ bounding the gradient since $\nabla \zeta_0$ is bounded. For a boundary piece $\zeta_j f$ with $j \geq 1$, transport to the straightened chart: $u_j := (\zeta_j f) \circ \Psi_j$ lies in $W^{1,p}(B(0, r_j)_+)$ by lemma 5.3.2 and, by the radius choice of Step 1, is supported in $B(0, \rho_j)$ with $\rho_j < r_j/2$, so it satisfies the support hypothesis of lemma 5.3.1. That lemma gives a reflection $Eu_j \in W^{1,p}(B(0, r_j))$ satisfying $\|Eu_j\|_{W^{1,p}(B(0, r_j))} \leq C \|u_j\|_{W^{1,p}(B(0, r_j)_+)}$. Transport back: $E_j f := (Eu_j) \circ \Phi_j$, extended by zero outside V_j , is a function on $\mathbb{R}^n$ that agrees with $\zeta_j f$ on $V_j \cap U$, is supported in $V_j \Subset W$, and by the two comparisons of lemma 5.3.2 satisfies

$$\|E_j f\|_{W^{1,p}(\mathbb{R}^n)} \leq C \|Eu_j\|_{W^{1,p}(B(0, r_j))} \leq C \|u_j\|_{W^{1,p}(B(0, r_j)_+)} \leq C \|\zeta_j f\|_{W^{1,p}(V_j \cap U)} \leq C \|f\|_{W^{1,p}(U)}$$

Extension by zero across ∂V_j preserves the $W^{1,p}$ class because Eu_j , hence $E_j f$, already vanishes in a neighborhood of ∂V_j .

Step 3: Assemble and estimate. Set $Ef := \sum_{j=0}^N E_j f$. Each E_j is linear in f (reflection, chart transport, and multiplication by ζ_j are all linear), so E is linear. On U every $E_j f$ agrees with $\zeta_j f$, so $Ef = \sum_j \zeta_j f = f$ almost everywhere on U , giving (1). Each $E_j f$ is supported in $V_j \subseteq W$, so Ef is supported in W , giving (2). Finally, by the triangle inequality and the $(N+1)$ bounds of Step 2,

$$\|Ef\|_{W^{1,p}(\mathbb{R}^n)} \leq \sum_{j=0}^N \|E_j f\|_{W^{1,p}(\mathbb{R}^n)} \leq C(N+1) \|f\|_{W^{1,p}(U)}$$

with C absorbing the chart constants, giving (3) with $C = C(n, p, U, W)$. Thus E is a bounded linear extension operator, completing the proof.

The extension operator makes the domain U transparent to any theorem already known on $\mathbb{R}^n$: prove an inequality for $W^{1,p}(\mathbb{R}^n)$, apply it to Ef , and restrict back to U using $Ef = f$ there, at the cost of the operator norm C . This is exactly how the Rellich-Kondrachov compactness of section 5.5 will descend from a statement on all of $\mathbb{R}^n$ to bounded C^1 domains. The dependence of C on U is real and not an artifact: the constant blows up as the boundary degenerates (a domain with an outward cusp has no bounded extension operator), which is why the C^1 hypothesis, controlling $\sup |D\Phi_j|$, cannot be dropped.

Example 5.3.4: Extension of a Radial Bump on the Ball

Let $U = B(0, 1) \subseteq \mathbb{R}^n$ and $f(x) = 1 - |x|$, which lies in $W^{1,p}(U)$ for every p : it is Lipschitz, with $\nabla f(x) = -x/|x|$ of modulus 1 almost everywhere, so $\|f\|_{W^{1,p}(U)}^p = \int_U (1 - |x|)^p + \int_U 1 < \infty$. The construction straightens the sphere ∂U in each boundary chart; because f is already Lipschitz on $\bar{U}$ with $f = 0$ on ∂U , the simplest admissible extension is $Ef(x) = \max(1 - |x|, 0)$, which is Lipschitz on all of $\mathbb{R}^n$, equals f on U , is supported in $\bar{B}(0, 1) \Subset W = B(0, 2)$, and satisfies $\|Ef\|_{W^{1,p}(\mathbb{R}^n)} = \|f\|_{W^{1,p}(U)}$ since Ef adds no mass outside U . The reflection machinery is needed only when f does not vanish on the boundary, where an even continuation would break the normal

derivative; here $f|_{\partial U} = 0$ lets extension by zero do the job, and the general operator agrees with it up to the boundary chart constants.

The trace is the second boundary operation. A function in $W^{1,p}(U)$ is an equivalence class modulo sets of Lebesgue measure zero, and ∂U has n -dimensional Lebesgue measure zero, so “restrict f to ∂U ” has no meaning on the nose. The trace theorem gives it meaning: on functions smooth up to the boundary, restriction is a well-defined map into $L^p(\partial U)$; the estimate below shows this map is bounded for the $W^{1,p}$ norm, so it extends by density to all of $W^{1,p}(U)$. Before the theorem, one preliminary space records where the trace actually lands, which is finer than $L^p(\partial U)$.

The trace of a $W^{1,p}$ function lands in more than $L^p(\partial U)$: it carries a fractional order of differentiability, one derivative shy of the full jump from U (dimension n) to ∂U (dimension $n - 1$) discounted by the integrability exponent. That fractional regularity is measured by a seminorm with no reference to integer derivatives, integrating a difference quotient of order s against the surface measure of ∂U . Here $d = n - 1$ is the dimension of ∂U .

Definition 5.3.5: Fractional Sobolev Space $W^{s,p}(\partial U)$

Let $U \subseteq \mathbb{R}^n$ be bounded with C^1 boundary, write $d = n - 1$ for the dimension of ∂U , and let $0 < s < 1$ and $1 \leq p < \infty$. The *Gagliardo (Slobodeckij) seminorm* of a measurable $f: \partial U \rightarrow \mathbb{R}$ is

$$[f]_{W^{s,p}(\partial U)} = \left(\int_{\partial U} \int_{\partial U} \frac{|f(x) - f(y)|^p}{|x - y|^{d+sp}} dS(x) dS(y) \right)^{1/p}$$

where dS is the surface measure and $|x - y|$ the Euclidean distance in $\mathbb{R}^n$. The *fractional Sobolev space* is

$$W^{s,p}(\partial U) = \{f \in L^p(\partial U) \mid [f]_{W^{s,p}(\partial U)} < \infty\}$$

normed by $\|f\|_{W^{s,p}(\partial U)} = (\|f\|_{L^p(\partial U)}^p + [f]_{W^{s,p}(\partial U)}^p)^{1/p}$.

The exponent $d + sp$ in the denominator is calibrated so that the seminorm sees exactly s derivatives: replacing f by $f(\lambda \cdot)$ scales $[f]_{W^{s,p}}$ by $\lambda^{s-d/p}$ (the change of variables $u = \lambda x$ turns the p -th power of the seminorm into λ^{sp-d} times itself, matching the scaling of $\|\nabla^s f\|_{L^p}$ for the would-be integer analog), which is what makes s the order of differentiability. Finiteness of the double integral demands that $f(x) - f(y)$ decay faster than $|x - y|^{s+d/p}$ near the diagonal on average, a genuine but fractional smoothness: $W^{s,p}$ sits strictly between L^p and the integer space $W^{1,p}$ as s runs through $(0, 1)$.

Example 5.3.6: The Threshold Exponent of a Power Singularity

Take the unit circle $\partial U = \partial B(0, 1) \subseteq \mathbb{R}^2$, so $d = 1$, parametrized by arc length $\theta \in [0, 2\pi)$, and the function with a mild corner at $\theta = 0$, $f(\theta) = |\theta|^a$ near 0 for a fixed $0 < a < 1$ (extended smoothly elsewhere). Near the diagonal the seminorm integrand behaves like $|f(\theta) - f(\eta)|^p / |\theta - \eta|^{1+sp}$, and a direct estimate of $\iint$ over a neighborhood of $\theta = \eta = 0$ shows the double integral converges precisely when $a > s$: the difference $|\theta^a - \eta^a|$ is comparable to $|\theta - \eta|^a$ for a corner of this shape, because $t \mapsto |t|^a$ with $0 < a < 1$ is a -Hölder continuous ($||\theta|^a - |\eta|^a| \leq |\theta - \eta|^a$ by concavity of t^a) and the bound is saturated when one of the two points sits at the corner (e.g. $\eta \approx 0$, where $||\theta|^a - |\eta|^a| = |\theta|^a \sim |\theta - \eta|^a$), so the integrand is comparable to $|\theta - \eta|^{ap-1-sp}$, integrable across the diagonal exactly when $ap - 1 - sp > -1$, i.e. $a > s$. Thus $f \in W^{s,p}(\partial U)$ if and only if $s < a$: the corner of exponent a has fractional smoothness of every order below a and none above, pinning the seminorm down as a sharp gauge of the singularity.

The trace theorem now states what the boundary restriction is and where it lands. Its heart is a single integration by parts on a boundary chart, bounding the L^p norm of the boundary values by the $W^{1,p}$ norm on the interior; density (theorem 5.2.5) then produces the operator on all of $W^{1,p}(U)$, and a refinement of the same estimate reads off the sharper fractional target.

Theorem 5.3.7: Trace Theorem

Let $U \subseteq \mathbb{R}^n$ be bounded with C^1 boundary and let $1 \leq p < \infty$. There is a bounded linear operator

$$T: W^{1,p}(U) \rightarrow L^p(\partial U)$$

the *trace*, such that $Tf = f|_{\partial U}$ for every $f \in C^1(\overline{U}) \cap W^{1,p}(U)$, and

$$\|Tf\|_{L^p(\partial U)} \leq C\|f\|_{W^{1,p}(U)}$$

with $C = C(n, p, U)$. Moreover T maps $W^{1,p}(U)$ into $W^{1-1/p,p}(\partial U)$ for $1 < p < \infty$, boundedly for the fractional norm of definition 5.3.5.

Proof:

Step 1: The boundary estimate on a flat chart. Suppose first $f \in C^1(\overline{U})$, and work in a boundary chart straightened by lemma 5.3.2, so it suffices to bound $\int_{\{x_n=0\}} |f|^p dx'$ over a piece of the flattened boundary in terms of the $W^{1,p}$ norm on the half-ball B_+ above it. Fix a cutoff $\xi \in C_c^\infty(B)$ with $\xi \equiv 1$ on the smaller half-ball where the boundary piece sits and $0 \leq \xi \leq 1$. For x' in that piece, the fundamental theorem of calculus in the x_n -direction gives, since ξf vanishes for large x_n ,

$$|f(x', 0)|^p = - \int_0^\infty \partial_n (\xi(x', x_n) |f(x', x_n)|^p) dx_n$$

Expanding the derivative, $\partial_n (\xi |f|^p) = (\partial_n \xi) |f|^p + \xi p |f|^{p-1} \text{sgn}(f) \partial_n f$, and integrating in x' over the boundary piece,

$$\int |f(x', 0)|^p dx' \leq \int_{B_+} \left(|\partial_n \xi| |f|^p + p \xi |f|^{p-1} |\partial_n f| \right) dx$$

The first term is bounded by $C \int_{B_+} |f|^p$. For the second, when $p > 1$ Young's inequality $ab \leq \frac{a^{p'}}{p'} + \frac{b^p}{p}$ with $a = |f|^{p-1}$, $b = |\partial_n f|$ (so $a^{p'} = |f|^p$) gives $|f|^{p-1} |\partial_n f| \leq \frac{p-1}{p} |f|^p + \frac{1}{p} |\partial_n f|^p$; the degenerate case $p = 1$ is immediate, the integrand $p \xi |f|^{p-1} |\partial_n f|$ reducing to $\xi |\partial_n f| \leq |\partial_n f|$ with no exponent to split. Either way,

$$\int |f(x', 0)|^p dx' \leq C \int_{B_+} (|f|^p + |\nabla f|^p) dx = C \|f\|_{W^{1,p}(B_+)}^p$$

Summing over a finite cover of ∂U by such charts, and using that the surface measure dS is comparable on each chart to dx' up to the bounded Jacobian factor of the graph parametrization (lemma 5.3.2), yields

$$\int_{\partial U} |f|^p dS \leq C \|f\|_{W^{1,p}(U)}^p \quad (f \in C^1(\overline{U}))$$

Step 2: Extension to all of $W^{1,p}(U)$. Define $Tf = f|_{\partial U}$ for $f \in C^1(\bar{U})$, a linear map into $L^p(\partial U)$ bounded by Step 1. By theorem 5.2.5, $C^\infty(\bar{U})$, hence $C^1(\bar{U})$, is dense in $W^{1,p}(U)$: given $f \in W^{1,p}(U)$ take $f_m \in C^\infty(\bar{U})$ with $f_m \rightarrow f$ in $W^{1,p}(U)$. Then (f_m) is Cauchy in $W^{1,p}(U)$, and Step 1 applied to $f_m - f_l$ gives $\|Tf_m - Tf_l\|_{L^p(\partial U)} \leq C\|f_m - f_l\|_{W^{1,p}(U)}$, so (Tf_m) is Cauchy in the complete space $L^p(\partial U)$; define $Tf = \lim_m Tf_m$. The limit is independent of the approximating sequence: if $g_m \rightarrow f$ as well, interleaving the two sequences is again $W^{1,p}$ convergent to f , so the traces converge to a common limit. The bound $\|Tf\|_{L^p(\partial U)} \leq C\|f\|_{W^{1,p}(U)}$ passes to the limit, and T is linear as a limit of linear maps. For $f \in C^1(\bar{U}) \cap W^{1,p}(U)$ the constant sequence $f_m = f$ shows $Tf = f|_{\partial U}$, so T extends pointwise restriction.

Step 3: The sharp fractional target. Fix $1 < p < \infty$; the claim is $[Tf]_{W^{1-1/p,p}(\partial U)} \leq C\|f\|_{W^{1,p}(U)}$, so that $s = 1 - 1/p$ is the order of the trace. By Step 2 and density it suffices to prove this for $f \in C^1(\bar{U})$, the seminorm passing to the limit as in Step 2. Work again in a straightened chart, where the boundary is $\{x_n = 0\}$ and $g := f|_{x_n=0}$; the seminorm is comparable to the flat-chart integral $\iint |g(x') - g(y')|^p / |x' - y'|^{d+sp} dx' dy'$ with $d = n - 1$; the comparability uses that the C^1 graph is a bi-Lipschitz parametrization, so $|x - y|$ on ∂U and $|x' - y'|$ in the chart agree up to a constant (lemma 5.3.2). For $x' \neq y'$ set $h = |x' - y'|$ and pass from $g(x')$ to $g(y')$ through the value of f at height h above each point, splitting the difference into a vertical defect and a horizontal difference that must be handled separately:

$$g(x') - g(y') = (g(x') - f(x', h)) + (f(x', h) - f(y', h)) - (g(y') - f(y', h))$$

Estimate the two types of piece after raising to the p -th power and dividing by $|x' - y'|^{d+sp} = h^{d+p-1}$ (using $sp = p - 1$). *Vertical defect.* Since $g(x') - f(x', h) = -\int_0^h \partial_n f(x', t) dt$, integrating y' over the sphere $|y' - x'| = h$ in polar coordinates $dy' = h^{d-1} dh d\omega$ cancels the h^{d-1} against $h^{-(d+p-1)}$ to leave the weight $h^{-p} dh$, so the vertical contribution at fixed x' is

$$\int \frac{|g(x') - f(x', h)|^p}{h^{d+p-1}} dy' \leq C \int_0^\infty \frac{1}{h^p} \left| \int_0^h \partial_n f(x', t) dt \right|^p dh \leq C \left(\frac{p}{p-1} \right)^p \int_0^\infty |\partial_n f(x', t)|^p dt$$

the last inequality being the one-dimensional Hardy inequality $\int_0^\infty h^{-p} \left(\int_0^h \psi \right)^p dh \leq (p/(p-1))^p \int_0^\infty \psi^p$ applied to $\psi = |\partial_n f(x', \cdot)|$, whose constant is finite precisely because $p > 1$; this is where the logarithm that a direct column bound $\int_0^h |\partial_n f|^p dt$ against dh/h would produce is absorbed. Integrating in x' over the boundary piece bounds the vertical contribution by $C \int_{B_+} |\partial_n f|^p$. *Horizontal difference.* The middle term $f(x', h) - f(y', h) = \int_{[x', y']} \nabla' f(w', h) \cdot dw'$ runs along the length- h tangential segment at fixed height h , so Hölder gives $|f(x', h) - f(y', h)|^p \leq h^{p-1} \int_{[x', y']} |\nabla' f(w', h)|^p dw'$. Dividing by h^{d+p-1} and integrating y' over $|y' - x'| = h$ leaves the weight $h^{-1} dh$ against the segment integral; because the integrand is evaluated at the same height $h = |x' - y'|$ that sets the scale, each height is visited once and the h -integral reconstructs the slab integral of $|\nabla' f|^p$ by Fubini, giving

$$\iint \frac{|f(x', h) - f(y', h)|^p}{h^{d+p-1}} dx' dy' \leq C \int_{B_+} |\nabla' f|^p$$

with no logarithm, since the horizontal integrand carries no whole-column tail. Combining the vertical and horizontal contributions bounds the flat-chart seminorm by $C \int_{B_+} |\nabla f|^p$. Summing the finitely many charts and using the comparability of dS with dx' ,

$$[Tf]_{W^{1-1/p,p}(\partial U)}^p \leq C \int_U |\nabla f|^p dx \leq C \|f\|_{W^{1,p}(U)}^p$$

Combined with the L^p bound of Step 1, $\|Tf\|_{W^{1-1/p,p}(\partial U)} \leq C\|f\|_{W^{1,p}(U)}$, so T maps $W^{1,p}(U)$ boundedly into $W^{1-1/p,p}(\partial U)$, completing the proof.

The trace theorem is what makes boundary value problems well posed for Sobolev functions: prescribing $Tf = g$ on ∂U is a closed linear constraint (the trace is bounded), so the Dirichlet condition “ $f = g$ on ∂U ” has meaning for $f \in W^{1,p}(U)$ even though f has no pointwise values on a measure-zero set. The two targets record two levels of precision: $L^p(\partial U)$ is enough to state boundary conditions and prove the kernel characterization below, while the sharp target $W^{1-1/p,p}(\partial U)$ is the natural home for the boundary data of the inverse problem of extending g inward. Theorem 5.3.7 proves only one half of what makes it sharp, that T maps $W^{1,p}(U)$ boundedly into $W^{1-1/p,p}(\partial U)$; the converse, that T is also onto this space, so that every fractional datum $g \in W^{1-1/p,p}(\partial U)$ is the trace of some $W^{1,p}(U)$ function, is a separate and deeper theorem, established by constructing a bounded right inverse (a trace-extension operator) and not proved here.¹ Granting it, $W^{1-1/p,p}(\partial U)$ is exactly the image of T . The loss of $1/p$ of a derivative is the price of the codimension-one restriction, and it is sharp.

Example 5.3.8: Trace of a Radial Bump on the Ball

Return to $U = B(0, 1) \subseteq \mathbb{R}^n$ and $f(x) = 1 - |x|$ of example 5.3.4, which lies in $W^{1,p}(U)$ for every p and is continuous up to the boundary, so $f \in C(\bar{U}) \cap W^{1,p}(U)$. The trace is then the pointwise restriction: on $\partial U = \{|x| = 1\}$ every point has $|x| = 1$, so

$$Tf(x) = f|_{\partial U}(x) = 1 - |x| = 0 \quad (|x| = 1)$$

Hence $Tf = 0$, and f lies in the kernel of the trace. This is consistent with the extension example 5.3.4 continued f by zero across ∂U without breaking the $W^{1,p}$ class: a boundary value of 0 is exactly what lets extension by zero preserve one derivative. By the characterization below, $f \in W_0^{1,p}(U)$, so f is a $W^{1,p}$ -limit of functions compactly supported inside the open ball, even though f itself does not vanish in the interior. A function whose boundary trace were a nonzero constant c , by contrast, would have $\|Tf\|_{L^p(\partial U)}^p = |c|^p |\partial U| > 0$, placing it outside the kernel.

The kernel of the trace has a clean description: it is exactly the closure of $C_c^\infty(U)$, the space $W_0^{1,p}(U)$ of definition 5.2.6. This says the two competing notions of “vanishing on the boundary” coincide, the analytic one (zero trace) and the approximation-theoretic one (approximable by functions compactly supported inside U).

¹The extension operator of theorem 5.3.3 does not supply this right inverse: it extends a function already given on U , whereas a right inverse must build a $W^{1,p}(U)$ function from boundary data alone. The standard construction lifts g off ∂U by a convolution extension calibrated to the fractional norm, the boundary trace theorem of Gagliardo; a full proof is beyond the scope of this section.

Theorem 5.3.9: Zero-Trace Characterization

Let $U \subseteq \mathbb{R}^n$ be bounded with C^1 boundary and let $1 < p < \infty$. Then

$$W_0^{1,p}(U) = \ker T = \{f \in W^{1,p}(U) \mid Tf = 0\}$$

where T is the trace of theorem 5.3.7 and $W_0^{1,p}(U)$ is the closure of $C_c^\infty(U)$ in $W^{1,p}(U)$ from definition 5.2.6.^a

^aThe restriction to $p > 1$ enters at the boundary Hardy inequality of Step 3, whose constant $(p/(p-1))^p$ blows up as $p \rightarrow 1$; Hardy's inequality genuinely fails at $p = 1$, so the cutoff argument below does not cover that case. The characterization does still hold for $p = 1$, but by a different route (a truncation-and-mollification argument avoiding Hardy), which is not pursued here.

Proof:

Step 1: $W_0^{1,p}(U) \subseteq \ker T$. For $\varphi \in C_c^\infty(U)$ the trace is the pointwise restriction (Step 2 of theorem 5.3.7), and $\varphi \equiv 0$ near ∂U , so $T\varphi = \varphi|_{\partial U} = 0$. Since T is bounded (theorem 5.3.7) and $C_c^\infty(U)$ is dense in $W_0^{1,p}(U)$ by definition (definition 5.2.6), continuity gives $Tf = 0$ for every $f \in W_0^{1,p}(U)$: if $\varphi_m \rightarrow f$ in $W^{1,p}(U)$ with $\varphi_m \in C_c^\infty(U)$, then $Tf = \lim_m T\varphi_m = 0$. Hence $W_0^{1,p}(U) \subseteq \ker T$.

Step 2: $\ker T \subseteq W_0^{1,p}(U)$, *reduction to compactly supported functions*. Let $f \in W^{1,p}(U)$ with $Tf = 0$; the goal is to approximate f in $W^{1,p}(U)$ by functions in $C_c^\infty(U)$. By theorem 5.2.5 choose $f_m \in C^\infty(\bar{U})$ with $f_m \rightarrow f$ in $W^{1,p}(U)$; then $Tf_m = f_m|_{\partial U} \rightarrow Tf = 0$ in $L^p(\partial U)$ by continuity of T . It suffices to produce, for each ε , a function $g \in W^{1,p}(U)$ with compact support in U and $\|f - g\|_{W^{1,p}(U)} < \varepsilon$, since mollifying g (theorem 5.2.1 applied on a neighborhood of supp g) then yields a $C_c^\infty(U)$ approximant, placing f in the closure $W_0^{1,p}(U)$.

Step 3: *Cutting off near the boundary using the vanishing trace*. Work in a boundary chart straightened by lemma 5.3.2, so near a boundary point the domain is the half-ball B_+ , the boundary is $\{x_n = 0\}$, and $f \in W^{1,p}(B_+)$ has trace $g = Tf = 0$ on $\{x_n = 0\}$. The vanishing trace gives the boundary Hardy estimate: since the smooth approximants satisfy $f_m(x', 0) = Tf_m \rightarrow 0$, the fundamental theorem of calculus $f_m(x', t) = f_m(x', 0) + \int_0^t \partial_n f_m(x', \tau) d\tau$ passes to the limit to give, for f with $Tf = 0$,

$$\int_{B_+ \cap \{0 < x_n < h\}} \frac{|f(x)|^p}{x_n^p} dx \leq C \int_{B_+ \cap \{0 < x_n < 2h\}} |\partial_n f|^p dx$$

the one-dimensional Hardy inequality applied in the x_n -variable (its constant $(p/(p-1))^p$ depends only on p and is finite precisely because $p > 1$), using $f(x', 0) = 0$. In particular $|f(x)|/x_n$ is p -integrable near the boundary. Now let $\eta_h(x_n)$ be a smooth cutoff, $\eta_h = 0$ for $x_n \leq h$, $\eta_h = 1$ for $x_n \geq 2h$, with $|\eta_h'| \leq C/h$, and set $g_h = \eta_h f$. Then g_h vanishes in the strip $0 < x_n < h$, so it has positive distance from the flattened boundary, and

$$\|f - g_h\|_{W^{1,p}(B_+)}^p \leq \int_{\{x_n < 2h\}} |f|^p + \int_{\{x_n < 2h\}} |1 - \eta_h|^p |\nabla f|^p + \int_{\{h < x_n < 2h\}} |\eta_h'|^p |f|^p$$

The first two integrals tend to 0 as $h \rightarrow 0$ by dominated convergence (the domains shrink to a null set). The third is controlled by the Hardy estimate: $|\eta_h'|^p \leq C/h^p \leq C/x_n^p$ on $\{h < x_n < 2h\}$, so $\int_{\{h < x_n < 2h\}} |\eta_h'|^p |f|^p \leq C \int_{\{h < x_n < 2h\}} |f|^p / x_n^p \rightarrow 0$ as $h \rightarrow 0$, again because the domain shrinks

to a null set while the integrand is dominated by the integrable $|f|^p/x_n^p$. Hence $g_h \rightarrow f$ in $W^{1,p}(B_+)$, and each g_h vanishes near the flattened boundary.

Step 4: Globalization. Patch the chart cutoffs with the finite boundary partition of unity $\zeta_1, \dots, \zeta_N$ of theorem 5.3.3 together with the interior cutoff $\zeta_0 \in C_c^\infty(U)$: transporting each g_h back through the charts and combining as $\sum_{j \geq 1} \zeta_j g_h^{(j)} + \zeta_0 f$, the resulting function g lies in $W^{1,p}(U)$, has support at positive distance from ∂U hence compact in U , and satisfies $\|f - g\|_{W^{1,p}(U)} < \varepsilon$ for h small, by Step 3 in each boundary chart and the boundedness of the transport maps. This furnishes the compactly supported approximant required in Step 2; mollifying it produces a $C_c^\infty(U)$ function within 2ε of f . Since ε was arbitrary, $f \in C_c^\infty(U) = W_0^{1,p}(U)$, and $\ker T \subseteq W_0^{1,p}(U)$.

The two inclusions give $W_0^{1,p}(U) = \ker T$, completing the proof.

The characterization anchors $W_0^{1,p}(U)$ as the natural space for the Dirichlet problem: a solution with homogeneous boundary data $f|_{\partial U} = 0$ is sought exactly in $W_0^{1,p}(U)$, and the theorem certifies that this closure of $C_c^\infty(U)$ is the same as the analytically defined set of zero-trace functions. It also explains the caveat in definition 5.2.6 that $W_0^{1,p}(\mathbb{R}^n) = W^{1,p}(\mathbb{R}^n)$: on all of $\mathbb{R}^n$ there is no boundary, the trace is the zero map, and its kernel is everything, so every $W^{1,p}(\mathbb{R}^n)$ function is a limit of compactly supported smooth ones. On a bounded domain the boundary is genuine, the trace is nonzero, and $W_0^{1,p}(U)$ is a proper closed subspace on which the Poincaré inequality of section 5.5 will hold.

5.4 Sobolev Embeddings

The completion of section 5.1 records that a Sobolev function has L^p derivatives; it says nothing about the function itself beyond membership in L^p . This section extracts the extra regularity that possession of a weak gradient forces: control of ∇f in L^p buys either higher integrability of f or, past a threshold, outright continuity. Which of the two occurs is decided by comparing the order of differentiation against the dimension n , and the two regimes are governed by the Gagliardo-Nirenberg-Sobolev inequality when the derivative is too weak to reach continuity and by Morrey's inequality when it is strong enough.

The mechanism behind both is a single scaling count. If f is rescaled to $f_\lambda(x) = f(\lambda x)$, then $\|f_\lambda\|_{L^q(\mathbb{R}^n)} = \lambda^{-n/q} \|f\|_{L^q(\mathbb{R}^n)}$ while $\|\nabla f_\lambda\|_{L^p(\mathbb{R}^n)} = \lambda^{1-n/p} \|\nabla f\|_{L^p(\mathbb{R}^n)}$. An inequality of the form $\|f\|_{L^q} \leq C \|\nabla f\|_{L^p}$ can hold for all f and all $\lambda > 0$ only if the two powers of λ agree, forcing $-n/q = 1 - n/p$, that is $q = \frac{np}{n-p}$. This exponent is the only candidate; the content of the theorem is that the inequality holds for it.

Definition 5.4.1: Sobolev Conjugate

Let $1 \leq p < n$. The *Sobolev conjugate* of p is

$$p^* = \frac{np}{n-p}$$

It satisfies $p^* > p$ and $\frac{1}{p^*} = \frac{1}{p} - \frac{1}{n}$.

The relation $\frac{1}{p^*} = \frac{1}{p} - \frac{1}{n}$ is the form to remember: one weak derivative improves the integrability

exponent by the fixed amount $\frac{1}{n}$, independent of p . As $p \rightarrow n^-$ the conjugate $p^* \rightarrow \infty$, the signal that at $p = n$ integrability alone breaks down and a different phenomenon (borderline embedding into every finite exponent, then continuity for $p > n$) takes over.

Example 5.4.2: The Sobolev Conjugate in Low Dimensions

In dimension $n = 3$ the Hilbert exponent $p = 2$ has $p^* = \frac{3 \cdot 2}{3-2} = 6$, so a function on $\mathbb{R}^3$ with gradient in L^2 automatically lies in L^6 . In dimension $n = 2$ the same $p = 2$ gives $n - p = 0$, so p^* is undefined and $p = n$ is the borderline case excluded from definition 5.4.1. For $p = 1$ in dimension n one finds $p^* = \frac{n}{n-1}$, the exponent that will appear at the base of the induction below.

The strategy is to prove the gradient estimate first on smooth compactly supported functions, where the fundamental theorem of calculus applies directly, and then to remove the smoothness by the density theory of section 5.2. The smooth estimate rests on a one-dimensional bound integrated against itself in every coordinate direction, tied together by an iterated Hölder inequality of Loomis-Whitney type.

Theorem 5.4.3: Gagliardo-Nirenberg-Sobolev Inequality

Let $1 \leq p < n$. There is a constant $C = C(n, p)$ such that

$$\|f\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|\nabla f\|_{L^p(\mathbb{R}^n)} \quad \text{for every } f \in W^{1,p}(\mathbb{R}^n)$$

where $p^* = \frac{np}{n-p}$. In particular the inclusion $W^{1,p}(\mathbb{R}^n) \hookrightarrow L^{p^*}(\mathbb{R}^n)$ is a bounded linear map.

Proof:

Step 1: The case $p = 1$ for $f \in C_c^\infty(\mathbb{R}^n)$. Here $p^* = \frac{n}{n-1}$. Fix $f \in C_c^\infty(\mathbb{R}^n)$ and $x = (x_1, \dots, x_n)$. For each index i , integrating the derivative along the i -th coordinate line from $-\infty$ and using compact support,

$$f(x) = \int_{-\infty}^{x_i} \partial_i f(x_1, \dots, x_{i-1}, t, x_{i+1}, \dots, x_n) dt$$

so, bounding the integrand by its absolute value and extending the range to all of $\mathbb{R}$,

$$|f(x)| \leq \int_{-\infty}^{\infty} |\partial_i f(x_1, \dots, t, \dots, x_n)| dt =: g_i(\hat{x}_i)$$

where g_i depends on all coordinates except x_i (denoted $\hat{x}_i$). Taking the product over the n directions and raising to the power $\frac{1}{n-1}$,

$$|f(x)|^{\frac{n}{n-1}} \leq \prod_{i=1}^n g_i(\hat{x}_i)^{\frac{1}{n-1}}$$

The task is to integrate this inequality over $\mathbb{R}^n$. Notice that each factor g_i is independent of the variable x_i , so integrating in x_1 first leaves the factor $g_1^{1/(n-1)}$ constant and applies the generalized Hölder inequality (with $n-1$ exponents all equal to $n-1$) to the remaining $n-1$ factors, each of which depends on x_1 :

$$\int_{\mathbb{R}} |f|^{\frac{n}{n-1}} dx_1 \leq g_1(\hat{x}_1)^{\frac{1}{n-1}} \prod_{i=2}^n \left(\int_{\mathbb{R}} g_i dx_1 \right)^{\frac{1}{n-1}}$$

Integrating successively in $x_2, \dots, x_n$ and applying the same generalized Hölder inequality at each stage, every factor is eventually integrated in the one variable on which it does not already depend, and one arrives at

$$\int_{\mathbb{R}^n} |f|^{\frac{n}{n-1}} dx \leq \prod_{i=1}^n \left(\int_{\mathbb{R}^n} |\partial_i f| dx \right)^{\frac{1}{n-1}}$$

This is the Loomis-Whitney arrangement: an $L^{n/(n-1)}$ bound on f by the geometric mean of the n one-directional gradient integrals. Bounding each $\int_{\mathbb{R}^n} |\partial_i f| \leq \int_{\mathbb{R}^n} |\nabla f|$ and taking the resulting product of n equal factors,

$$\left(\int_{\mathbb{R}^n} |f|^{\frac{n}{n-1}} dx \right)^{\frac{n-1}{n}} \leq \int_{\mathbb{R}^n} |\nabla f| dx$$

that is $\|f\|_{L^{1^*}(\mathbb{R}^n)} \leq \|\nabla f\|_{L^1(\mathbb{R}^n)}$, with constant $C(n, 1) = 1$.

Step 2: General $1 < p < n$ for $f \in C_c^\infty(\mathbb{R}^n)$ by applying Step 1 to a power of $|f|$. Fix $t > 1$ to be chosen and apply the $p = 1$ estimate to the function $|f|^t$, which is C^1 with compact support and gradient $\nabla(|f|^t) = t|f|^{t-1} \operatorname{sgn}(f) \nabla f$. Step 1 gives

$$\left(\int_{\mathbb{R}^n} |f|^{\frac{tn}{n-1}} dx \right)^{\frac{n-1}{n}} \leq t \int_{\mathbb{R}^n} |f|^{t-1} |\nabla f| dx$$

Apply Hölder's inequality (theorem 2.1.5) to the right side with exponents p and $p' = \frac{p}{p-1}$:

$$\int_{\mathbb{R}^n} |f|^{t-1} |\nabla f| dx \leq \left(\int_{\mathbb{R}^n} |f|^{(t-1)p'} dx \right)^{1/p'} \left(\int_{\mathbb{R}^n} |\nabla f|^p dx \right)^{1/p}$$

Choose t so that the two powers of $|f|$ match, namely $\frac{tn}{n-1} = (t-1)p'$. Solving gives $t = \frac{p(n-1)}{n-p} > 1$, and with this choice

$$\frac{tn}{n-1} = (t-1)p' = \frac{np}{n-p} = p^*$$

Writing $I = \left(\int_{\mathbb{R}^n} |f|^{p^*} \right)^{1/p^*} = \|f\|_{L^{p^*}}$, the left side is $I^{p^* \frac{n-1}{n}}$ and the first factor on the right is $I^{p^*/p'}$, so the inequality reads

$$I^{p^* \frac{n-1}{n}} \leq t I^{p^*/p'} \|\nabla f\|_{L^p}$$

A direct computation of exponents gives $p^* \frac{n-1}{n} - \frac{p^*}{p'} = 1$, so dividing by $I^{p^*/p'}$ (which is finite, since $f \in C_c^\infty$ makes I finite) leaves

$$\|f\|_{L^{p^*}(\mathbb{R}^n)} \leq t \|\nabla f\|_{L^p(\mathbb{R}^n)} = \frac{p(n-1)}{n-p} \|\nabla f\|_{L^p(\mathbb{R}^n)}$$

giving the inequality on $C_c^\infty(\mathbb{R}^n)$ with constant $C(n, p) = \frac{p(n-1)}{n-p}$.

Step 3: Extension to $W^{1,p}(\mathbb{R}^n)$ by density. Let $f \in W^{1,p}(\mathbb{R}^n)$. Since $W_0^{1,p}(\mathbb{R}^n) = W^{1,p}(\mathbb{R}^n)$ (definition 5.2.6), there are $f_m \in C_c^\infty(\mathbb{R}^n)$ with $f_m \rightarrow f$ in $W^{1,p}(\mathbb{R}^n)$; concretely one mollifies and truncates via theorem 5.2.1. Applying Step 2 to the difference $f_m - f_l$,

$$\|f_m - f_l\|_{L^{p^*}(\mathbb{R}^n)} \leq C(n, p) \|\nabla f_m - \nabla f_l\|_{L^p(\mathbb{R}^n)} \leq C(n, p) \|f_m - f_l\|_{W^{1,p}(\mathbb{R}^n)}$$

so (f_m) is Cauchy in $L^{p^*}(\mathbb{R}^n)$ and converges there to some h ; passing to a subsequence, $f_m \rightarrow h$ almost everywhere, while $f_m \rightarrow f$ in L^p forces (along a further subsequence) $f_m \rightarrow f$ almost everywhere, so $h = f$ almost everywhere and $f \in L^{p^*}(\mathbb{R}^n)$. Applying Step 2 to f_m and letting $m \rightarrow \infty$, with $\|f_m\|_{L^{p^*}} \rightarrow \|f\|_{L^{p^*}}$ and $\|\nabla f_m\|_{L^p} \rightarrow \|\nabla f\|_{L^p}$,

$$\|f\|_{L^{p^*}(\mathbb{R}^n)} \leq C(n, p) \|\nabla f\|_{L^p(\mathbb{R}^n)}$$

which is the claimed inequality on all of $W^{1,p}(\mathbb{R}^n)$. Boundedness of the inclusion is then $\|f\|_{L^{p^*}} \leq C \|\nabla f\|_{L^p} \leq C \|f\|_{W^{1,p}}$, completing the proof.

The inequality controls the full L^{p^*} norm of f by the L^p norm of its gradient alone, with no term for f itself on the right. This is possible only on $\mathbb{R}^n$, where there is no room for constants: a nonzero constant has vanishing gradient but infinite L^{p^*} norm, so the estimate would fail were $\mathbb{R}^n$ replaced by a set of finite measure without also restricting to functions that decay or vanish at the boundary. On a bounded domain the corresponding statement carries a lower-order term, or restricts to $W_0^{1,p}$; the Poincaré inequality of section 5.5 supplies exactly the missing control. The gain in integrability, from p up to p^* , is the quantitative content: one weak derivative is worth a definite jump in the integrability exponent, and iterating this jump is what produces the full hierarchy at the end of the section.

Example 5.4.4: Sharpness of the Exponent

No exponent larger than p^* can appear in theorem 5.4.3. The scaling argument of the section opening shows why: if $\|f\|_{L^q(\mathbb{R}^n)} \leq C \|\nabla f\|_{L^p(\mathbb{R}^n)}$ held for a fixed C and some $q \neq p^*$, testing on $f_\lambda(x) = f(\lambda x)$ gives $\lambda^{-n/q} \|f\|_{L^q} \leq C \lambda^{1-n/p} \|\nabla f\|_{L^p}$, that is

$$\|f\|_{L^q} \leq C \lambda^{n/q+1-n/p} \|\nabla f\|_{L^p}$$

If $q > p^*$ then $n/q + 1 - n/p < 0$ and letting $\lambda \rightarrow \infty$ forces $\|f\|_{L^q} = 0$; if $q < p^*$ then the exponent is positive and $\lambda \rightarrow 0$ does the same. Only $q = p^*$, where the exponent of λ vanishes, survives.

At $p = n$ the conjugate p^* diverges and the argument above degenerates. Past this threshold, when $p > n$, a weak gradient in L^p is strong enough to force not just integrability but pointwise continuity, indeed Hölder continuity with a definite exponent. The precise target is a Hölder space, which the following definition records.

Definition 5.4.5: Hölder Space $C^{0,\gamma}$

Let $0 < \gamma \leq 1$. A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is *Hölder continuous* with exponent γ if the seminorm

$$[f]_{C^{0,\gamma}(\mathbb{R}^n)} = \sup_{x \neq y} \frac{|f(x) - f(y)|}{|x - y|^\gamma}$$

is finite. The *Hölder space* $C^{0,\gamma}(\mathbb{R}^n)$ is the space of bounded continuous f with $[f]_{C^{0,\gamma}(\mathbb{R}^n)} < \infty$, normed by $\|f\|_{C^{0,\gamma}(\mathbb{R}^n)} = \sup_x |f(x)| + [f]_{C^{0,\gamma}(\mathbb{R}^n)}$. More generally, for a nonnegative integer m , $C^{m,\gamma}(\mathbb{R}^n)$ consists of the $f \in C^m$ whose derivatives up to order m are bounded with $\partial^\alpha f \in C^{0,\gamma}$ for every $|\alpha| = m$, normed by $\|f\|_{C^{m,\gamma}} = \sum_{|\alpha| \leq m} \sup_x |\partial^\alpha f| + \sum_{|\alpha|=m} [\partial^\alpha f]_{C^{0,\gamma}}$.

The exponent γ measures how rapidly f may oscillate: $\gamma = 1$ is Lipschitz continuity, and smaller γ permits sharper local variation. A member of $W^{1,p}(\mathbb{R}^n)$ is only an equivalence class of functions

agreeing almost everywhere, so the assertion that it embeds into $C^{0,\gamma}$ means each class has a (necessarily unique) continuous representative lying in that space, and it is this representative the oscillation estimate controls.

Example 5.4.6: The Model Holder Function

On $\mathbb{R}$ the function $f(x) = |x|^\gamma$ (say cut off smoothly away from the origin) has $[f]_{C^{0,\gamma}} < \infty$ but is not Lipschitz for $\gamma < 1$: near 0 the difference quotient $\frac{||x|^\gamma - 0|}{|x|^1} = |x|^{\gamma-1}$ blows up, whereas $\frac{||x|^\gamma - 0|}{|x|^\gamma} = 1$ stays bounded. This is the extremal behavior Morrey's inequality permits, and the exponent $\gamma = 1 - \frac{n}{p}$ produced below is exactly the regularity of such a cusp at the borderline.

The route to continuity runs through a comparison between the value of f at a point and its average over a small ball. Write

$$f_{B(x,r)} = \frac{1}{|B(x,r)|} \int_{B(x,r)} f \, dy$$

for the average of f over the ball $B(x,r)$ of radius r centered at x , where $|B(x,r)|$ is its Lebesgue measure. The key estimate bounds the deviation of f from this average by an integral of the gradient, a Sobolev analogue of the mean value theorem.

Lemma 5.4.7: Averaged Gradient Estimate

Let $p > n$ and $f \in C^1(\mathbb{R}^n)$. For every $x \in \mathbb{R}^n$ and $r > 0$,

$$\frac{1}{|B(x,r)|} \int_{B(x,r)} |f(y) - f(x)| \, dy \leq C(n) \int_{B(x,r)} \frac{|\nabla f(y)|}{|x - y|^{n-1}} \, dy$$

Proof:

Fix x and a point $y = x + s\omega$ in $B(x,r)$, where $|\omega| = 1$ and $0 < s \leq r$. The fundamental theorem of calculus along the segment from x to y gives

$$f(x + s\omega) - f(x) = \int_0^s \frac{d}{d\tau} f(x + \tau\omega) \, d\tau = \int_0^s \nabla f(x + \tau\omega) \cdot \omega \, d\tau$$

so $|f(x + s\omega) - f(x)| \leq \int_0^s |\nabla f(x + \tau\omega)| \, d\tau$. Integrate this over the unit sphere $\omega \in S^{n-1}$, whose surface measure is written $d\omega$:

$$\int_{S^{n-1}} |f(x + s\omega) - f(x)| \, d\omega \leq \int_{S^{n-1}} \int_0^s |\nabla f(x + \tau\omega)| \, d\tau \, d\omega \leq \int_{S^{n-1}} \int_0^r |\nabla f(x + \tau\omega)| \, d\tau \, d\omega$$

Insert the factor $\tau^{n-1}/\tau^{n-1} = 1$ into the last integrand and pass to the polar-coordinate variable $z = x + \tau\omega$, for which $dz = \tau^{n-1} \, d\tau \, d\omega$ and $|z - x| = \tau$:

$$\int_{S^{n-1}} \int_0^r |\nabla f(x + \tau\omega)| \, d\tau \, d\omega = \int_{S^{n-1}} \int_0^r \frac{|\nabla f(x + \tau\omega)|}{\tau^{n-1}} \tau^{n-1} \, d\tau \, d\omega = \int_{B(x,r)} \frac{|\nabla f(z)|}{|z - x|^{n-1}} \, dz$$

Now multiply the resulting inequality

$$\int_{S^{n-1}} |f(x + s\omega) - f(x)| \, d\omega \leq \int_{B(x,r)} \frac{|\nabla f(z)|}{|z - x|^{n-1}} \, dz$$

by s^{n-1} and integrate in s from 0 to r . The right side is independent of s , so it picks up the factor $\int_0^r s^{n-1} ds = r^n/n$, while the left side becomes $\int_{B(x,r)} |f(y) - f(x)| dy$ in polar coordinates. Thus

$$\int_{B(x,r)} |f(y) - f(x)| dy \leq \frac{r^n}{n} \int_{B(x,r)} \frac{|\nabla f(z)|}{|z - x|^{n-1}} dz$$

Dividing by $|B(x,r)| = \omega_n r^n$, where ω_n is the volume of the unit ball, gives the claim with $C(n) = \frac{1}{n\omega_n}$, as we sought to show.

With the deviation from an average controlled, the oscillation of f between two points follows by placing both points inside a common ball and comparing each to the average over it.

Theorem 5.4.8: Morrey's Inequality

Let $n < p < \infty$ and set $\gamma = 1 - \frac{n}{p} \in (0, 1)$. There is a constant $C = C(n, p)$ such that every $f \in W^{1,p}(\mathbb{R}^n)$ has a representative in $C^{0,\gamma}(\mathbb{R}^n)$, and this representative satisfies

$$|f(x) - f(y)| \leq C |x - y|^\gamma \|\nabla f\|_{L^p(\mathbb{R}^n)} \quad \text{for all } x, y \in \mathbb{R}^n$$

together with $\sup_x |f(x)| \leq C \|f\|_{W^{1,p}(\mathbb{R}^n)}$. Consequently the inclusion $W^{1,p}(\mathbb{R}^n) \hookrightarrow C^{0,\gamma}(\mathbb{R}^n)$ is a bounded linear map.

Proof:

Step 1: The oscillation estimate for $f \in C^1(\mathbb{R}^n)$. Fix $x, y \in \mathbb{R}^n$, set $r = |x - y|$, and let $W = B(x, r) \cap B(y, r)$, a set of measure $|W| \geq c_n r^n$ for a dimensional constant $c_n > 0$ (both balls have radius r and centers at distance r , so their intersection contains a fixed fraction of a ball). The triangle inequality through the average $f_W = \frac{1}{|W|} \int_W f$ gives

$$|f(x) - f(y)| \leq |f(x) - f_W| + |f_W - f(y)| = \frac{1}{|W|} \left| \int_W (f(x) - f(z)) dz \right| + \frac{1}{|W|} \left| \int_W (f(y) - f(z)) dz \right|$$

Estimate the first term. Since $W \subseteq B(x, r)$ and $|W| \geq c_n r^n \geq \frac{c_n}{\omega_n} |B(x, r)|$,

$$\frac{1}{|W|} \int_W |f(x) - f(z)| dz \leq \frac{\omega_n}{c_n} \cdot \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(x) - f(z)| dz \leq C(n) \int_{B(x, r)} \frac{|\nabla f(z)|}{|x - z|^{n-1}} dz$$

the last step by lemma 5.4.7. Apply Hölder's inequality (theorem 2.1.5) with exponents p and p' to this integral over $B(x, r)$:

$$\int_{B(x, r)} \frac{|\nabla f(z)|}{|x - z|^{n-1}} dz \leq \|\nabla f\|_{L^p(\mathbb{R}^n)} \left(\int_{B(x, r)} |x - z|^{-(n-1)p'} dz \right)^{1/p'}$$

The remaining integral is finite precisely because $p > n$: in polar coordinates centered at x ,

$$\int_{B(x, r)} |x - z|^{-(n-1)p'} dz = |S^{n-1}| \int_0^r \tau^{-(n-1)p'} \tau^{n-1} d\tau = |S^{n-1}| \int_0^r \tau^{(n-1)(1-p')} d\tau$$

and the exponent satisfies $(n-1)(1-p') > -1$ exactly when $p > n$, so the integral converges to a constant times $r^{(n-1)(1-p')+1}$. Raising to the power $1/p'$ and simplifying the exponent,

$\frac{(n-1)(1-p') + 1}{p'} = 1 - \frac{n}{p} = \gamma$, so

$$\left(\int_{B(x,r)} |x-z|^{-(n-1)p'} dz \right)^{1/p'} = C(n,p) r^\gamma$$

Combining, the first term is bounded by $C(n,p) r^\gamma \|\nabla f\|_{L^p}$. The second term is identical with y in place of x and $B(y,r)$ in place of $B(x,r)$, giving the same bound. Adding the two and recalling $r = |x-y|$,

$$|f(x) - f(y)| \leq C(n,p) |x-y|^\gamma \|\nabla f\|_{L^p(\mathbb{R}^n)}$$

Step 2: The sup bound for $f \in C^1(\mathbb{R}^n) \cap W^{1,p}$. Fix x and take $r = 1$. Then

$$|f(x)| \leq |f(x) - f_{B(x,1)}| + |f_{B(x,1)}|$$

The first term is bounded via lemma 5.4.7 and the Hölder step of Step 1 (with $r = 1$) by $C(n,p) \|\nabla f\|_{L^p}$; the second is $|\frac{1}{|B(x,1)|} \int_{B(x,1)} f| \leq \frac{1}{|B(x,1)|} \|f\|_{L^1(B(x,1))} \leq C(n) \|f\|_{L^p(\mathbb{R}^n)}$ by Hölder on the unit ball. Taking the supremum over x ,

$$\sup_x |f(x)| \leq C(n,p) (\|\nabla f\|_{L^p} + \|f\|_{L^p}) \leq C(n,p) \|f\|_{W^{1,p}(\mathbb{R}^n)}$$

Step 3: Passage to $W^{1,p}(\mathbb{R}^n)$ by density. Let $f \in W^{1,p}(\mathbb{R}^n)$ and choose $f_m \in C_c^\infty(\mathbb{R}^n)$ with $f_m \rightarrow f$ in $W^{1,p}(\mathbb{R}^n)$, which is possible since $W_0^{1,p}(\mathbb{R}^n) = W^{1,p}(\mathbb{R}^n)$ (definition 5.2.6) and by theorem 5.2.1. Steps 1 and 2 apply to each f_m ; applied to $f_m - f_l$ they give

$$\sup_x |f_m(x) - f_l(x)| \leq C \|f_m - f_l\|_{W^{1,p}}, \quad [f_m - f_l]_{C^{0,\gamma}} \leq C \|\nabla f_m - \nabla f_l\|_{L^p}$$

so (f_m) is Cauchy in the norm of $C^{0,\gamma}(\mathbb{R}^n)$. This space is complete, so $f_m \rightarrow \tilde{f}$ in $C^{0,\gamma}(\mathbb{R}^n)$, hence uniformly, and $\tilde{f}$ is continuous with $[\tilde{f}]_{C^{0,\gamma}} < \infty$. Uniform convergence gives $f_m \rightarrow \tilde{f}$ pointwise, while $f_m \rightarrow f$ in L^p gives $f_m \rightarrow f$ almost everywhere along a subsequence; thus $\tilde{f} = f$ almost everywhere and $\tilde{f}$ is the sought continuous representative. Letting $m \rightarrow \infty$ in the estimates of Steps 1 and 2, now for $\tilde{f}$, yields the oscillation bound and the sup bound, and boundedness of the inclusion follows, completing the proof.

Morrey's inequality is the mechanism by which analysis recovers classical smoothness from weak hypotheses: a function known only to have a p -th power integrable gradient, $p > n$, is after modification on a null set genuinely continuous, with a quantitative modulus of continuity. The exponent $\gamma = 1 - \frac{n}{p}$ is sharp and degenerates to 0 as $p \rightarrow n^+$, matching the divergence of p^* from the other side. The two theorems together partition the range of p at the dimension: below n one gains integrability, above n one gains continuity, and the borderline $p = n$ sits between, embedding into every finite exponent but into no Hölder space (the model cusp $|x|^\gamma$ with $\gamma \rightarrow 0$ shows why continuity fails there, cf. example 5.4.6).

The two first-order theorems combine to describe the entire scale $W^{k,p}$. The organizing quantity is kp , the total order of differentiation weighted against the dimension, and the outcome is decided by comparing kp to n . When $kp < n$ each derivative can be traded, via theorem 5.4.3, for a fixed gain in the integrability exponent, and iterating k times lands in an L^q space with a computable q ; when $kp > n$ enough derivatives can be spent to cross the Morrey threshold and land in a classical Hölder space; and when $kp = n$ the borderline holds, with membership in every finite L^q but not, in general,

continuity.

Theorem 5.4.9: General Sobolev Embedding

Let $k \geq 1$ be an integer and $1 \leq p < \infty$, and let $f \in W^{k,p}(\mathbb{R}^n)$.

1. If $kp < n$, then $f \in L^q(\mathbb{R}^n)$ for $q = \frac{np}{n-kp}$, and there is a constant $C = C(n, p, k)$ with $\|f\|_{L^q(\mathbb{R}^n)} \leq C\|f\|_{W^{k,p}(\mathbb{R}^n)}$. Consequently $W^{k,p}(\mathbb{R}^n) \hookrightarrow L^q(\mathbb{R}^n)$ for every $p \leq q \leq \frac{np}{n-kp}$.
2. If $kp = n$, then $f \in L^q(\mathbb{R}^n)$ for every $p \leq q < \infty$, with $\|f\|_{L^q(\mathbb{R}^n)} \leq C(n, p, k, q)\|f\|_{W^{k,p}(\mathbb{R}^n)}$.
3. If $kp > n$, write $m = k - \lfloor n/p \rfloor - 1$ and let

$$\gamma = \begin{cases} \lfloor n/p \rfloor + 1 - \frac{n}{p} & \text{if } n/p \notin \mathbb{Z} \\ \text{any } \gamma \in (0, 1) & \text{if } n/p \in \mathbb{Z} \end{cases}$$

Then f has a representative in $C^{m,\gamma}(\mathbb{R}^n)$, with $\|f\|_{C^{m,\gamma}(\mathbb{R}^n)} \leq C(n, p, k, \gamma)\|f\|_{W^{k,p}(\mathbb{R}^n)}$. Hence $W^{k,p}(\mathbb{R}^n) \hookrightarrow C^{m,\gamma}(\mathbb{R}^n)$.

Proof:

The argument iterates the first-order theorems, tracking the exponent one derivative at a time. Throughout, if $f \in W^{j,p}$ then each $\partial^\alpha f$ with $|\alpha| \leq j-1$ lies in $W^{1,p}$, so the first-order estimates apply to f and to all its lower-order derivatives at once.

Step 1: The subcritical case $kp < n$. Apply theorem 5.4.3 to f and to each first derivative. Since $f \in W^{k,p}$ has $f, \partial^\alpha f \in W^{1,p}$ for $|\alpha| \leq k-1$, the theorem gives $f \in W^{k-1,p_1}$ with $\frac{1}{p_1} = \frac{1}{p} - \frac{1}{n}$: each derivative of order at most $k-1$ gains one unit of integrability, so $\|f\|_{W^{k-1,p_1}} \leq C\|f\|_{W^{k,p}}$. Repeating, after j steps $f \in W^{k-j,p_j}$ with

$$\frac{1}{p_j} = \frac{1}{p} - \frac{j}{n}$$

valid as long as $p_j < n$ at each stage, that is $\frac{1}{p} - \frac{j}{n} > \frac{1}{n}$, which holds for $j \leq k-1$ because $kp < n$ means $\frac{1}{p} - \frac{k-1}{n} > \frac{1}{n}$. After k steps $f \in W^{0,p_k} = L^{p_k}$ with $\frac{1}{p_k} = \frac{1}{p} - \frac{k}{n} = \frac{n-kp}{np}$, so $p_k = \frac{np}{n-kp} = q$ and $\|f\|_{L^q} \leq C\|f\|_{W^{k,p}}$. The embedding into intermediate L^q for $p \leq q \leq \frac{np}{n-kp}$ follows by interpolation: $f \in L^p \cap L^q$ gives $f \in L^{q'}$ for every q' between with $\|f\|_{L^{q'}} \leq \|f\|_{L^p}^{1-\theta} \|f\|_{L^q}^\theta \leq C\|f\|_{W^{k,p}}$.

Step 2: The critical case $kp = n$. Iterate as in Step 1 while $p_j < n$. The obstruction is the final step: after $k-1$ iterations $f \in W^{1,p_{k-1}}$ with $\frac{1}{p_{k-1}} = \frac{1}{p} - \frac{k-1}{n} = \frac{1}{n}$, so $p_{k-1} = n$ and theorem 5.4.3 no longer applies with a finite conjugate. The following borderline fact finishes the argument: if $g \in W^{1,n}(\mathbb{R}^n)$ then $g \in L^q(\mathbb{R}^n)$ for every $n \leq q < \infty$, with $\|g\|_{L^q} \leq C(n, q)\|g\|_{W^{1,n}}$. To prove it, run the power substitution behind theorem 5.4.3 at $p = n$. For $g \in C_c^\infty(\mathbb{R}^n)$ and $t > 1$, the $p = 1$ Gagliardo estimate applied to $|g|^t$ gives

$$\left(\int_{\mathbb{R}^n} |g|^{\frac{tn}{n-1}} \right)^{\frac{n-1}{n}} \leq t \int_{\mathbb{R}^n} |g|^{t-1} |\nabla g| \leq t \left(\int_{\mathbb{R}^n} |g|^{(t-1)n'} \right)^{1/n'} \|\nabla g\|_{L^n}$$

by Hölder (theorem 2.1.5) with exponents n and $n' = \frac{n}{n-1}$. Since $(t-1)n' = (t-1)\frac{n}{n-1}$ and $\frac{tn}{n-1}$ differ, this bounds $\|g\|_{L^{tn/(n-1)}}$ by a power of $\|g\|_{L^{(t-1)n/(n-1)}}$ times $\|\nabla g\|_{L^n}$. Starting from $t = n$, where the input exponent $(t-1)n/(n-1) = n$ is controlled by $\|g\|_{W^{1,n}}$, the output exponent is $\frac{n^2}{n-1} > n$; feeding the output exponent back in as the new input and raising t at each stage, the output exponent $\frac{tn}{n-1}$ increases without bound, so after finitely many steps it exceeds any prescribed $q < \infty$, and interpolation between that exponent and n (as in Step 1) yields $\|g\|_{L^q} \leq C(n, q)\|g\|_{W^{1,n}}$. Density of C_c^∞ in $W^{1,n}$ (definition 5.2.6, theorem 5.2.1) extends the bound to all $g \in W^{1,n}(\mathbb{R}^n)$, exactly as in Step 3 of theorem 5.4.3. Applying this to $g = \partial^\alpha f$ for $|\alpha| = k-1$, which lie in $W^{1,n}$, gives $\partial^\alpha f \in L^q$ for every finite q and every $|\alpha| \leq k-1$; combined with $f \in L^p$ this gives $f \in L^q$ for every finite q , with the stated bound.

Step 3: The supercritical case $kp > n$, with $n/p \notin \mathbb{Z}$. Let $\ell = \lfloor n/p \rfloor$, so $\ell < n/p < \ell + 1$ and $\ell < k$ (because $kp > n$ gives $k > n/p > \ell$). Iterate theorem 5.4.3 exactly ℓ times: at each stage the current exponent r_j satisfies $\frac{1}{r_j} = \frac{1}{p} - \frac{j}{n}$, which stays positive and below $\frac{1}{n}$ only while $j < n/p$, so all ℓ steps are legitimate and land $f \in W^{k-\ell, r}$ with $\frac{1}{r} = \frac{1}{p} - \frac{\ell}{n}$. Here $r > n$ precisely because $\frac{1}{r} = \frac{1}{p} - \frac{\ell}{n} < \frac{1}{n}$, using $\ell < n/p$. Now $k - \ell = k - \lfloor n/p \rfloor = m + 1$, so $f \in W^{m+1, r}$ with $r > n$. Each $\partial^\alpha f$ with $|\alpha| \leq m$ then lies in $W^{1, r}$, $r > n$, so theorem 5.4.8 gives $\partial^\alpha f \in C^{0, \gamma}(\mathbb{R}^n)$ with $\gamma = 1 - \frac{n}{r} = \ell + 1 - \frac{n}{p}$ and the estimate $[\partial^\alpha f]_{C^{0, \gamma}} \leq C\|\nabla \partial^\alpha f\|_{L^r} \leq C\|f\|_{W^{k, p}}$, together with $\sup |\partial^\alpha f| \leq C\|f\|_{W^{k, p}}$. Assembling over all $|\alpha| \leq m$ gives $f \in C^{m, \gamma}(\mathbb{R}^n)$ with $\|f\|_{C^{m, \gamma}} \leq C\|f\|_{W^{k, p}}$.

Step 4: The supercritical case $kp > n$, with $n/p \in \mathbb{Z}$. Now $\lfloor n/p \rfloor = n/p$ and $m = k - \frac{n}{p} - 1$. Iterate theorem 5.4.3 exactly $\frac{n}{p} - 1$ times, which is legitimate since each intermediate exponent stays below n ; this lands $f \in W^{k-(n/p-1), n}$, that is, $f \in W^{m+2, n}$. Each $\partial^\alpha f$ with $|\alpha| \leq m+1$ then lies in $W^{1, n}$, so by the borderline fact of Step 2 it lies in $L^s(\mathbb{R}^n)$ for every finite s , with $\|\partial^\alpha f\|_{L^s} \leq C(n, s)\|f\|_{W^{k, p}}$. Fix any $s > n$; then each $\partial^\beta f$ with $|\beta| \leq m$ lies in $W^{1, s}$ (its derivative $\partial^\alpha f$, $|\alpha| = |\beta| + 1 \leq m+1$, is in L^s , and $\partial^\beta f$ itself is in L^s), so theorem 5.4.8 gives $\partial^\beta f \in C^{0, \gamma}(\mathbb{R}^n)$ with $\gamma = 1 - \frac{n}{s}$, any value in $(0, 1)$ being attainable by the choice of $s > n$, and $\|\partial^\beta f\|_{C^{0, \gamma}} \leq C\|\partial^\beta f\|_{W^{1, s}} \leq C\|f\|_{W^{k, p}}$. Assembling over $|\beta| \leq m$ gives $f \in C^{m, \gamma}(\mathbb{R}^n)$ for the chosen $\gamma \in (0, 1)$, with the stated estimate, completing the proof.

The theorem converts the abstract membership $f \in W^{k, p}$ into concrete information a differential equation can use, either an integrability exponent or a number of classical derivatives with a Hölder modulus. The count kp against n is the single dial. In the elliptic regularity theory of the partial-differential-equations volumes, a solution is first shown to lie in some $W^{k, p}$ by energy estimates; the embedding then reads off how many classical derivatives it actually has, and the passage from $kp < n$ to $kp > n$ as k increases is precisely the bootstrap that turns a weak solution into a smooth one. The Rellich-Kondrachov theorem of section 5.5 sharpens the subcritical case further, upgrading the bounded inclusion into L^q to a compact one for $q < p^*$, which is what existence proofs by compactness require.

Example 5.4.10: Reading Off Regularity in Dimension Three

Take $n = 3$ and $p = 2$, the setting of most of mathematical physics. For $k = 1$ one has $kp = 2 < 3$, so theorem 5.4.9(1) gives $H^1(\mathbb{R}^3) = W^{1, 2} \hookrightarrow L^q$ for $2 \leq q \leq \frac{3 \cdot 2}{3-2} = 6$, recovering the L^6 bound of example 5.4.2. For $k = 2$ one has $kp = 4 > 3$, so case (3) applies with $\lfloor n/p \rfloor = \lfloor 3/2 \rfloor = 1$, hence $m = 2 - 1 - 1 = 0$ and $\gamma = 1 + 1 - \frac{3}{2} = \frac{1}{2}$; thus $H^2(\mathbb{R}^3) \hookrightarrow C^{0, 1/2}(\mathbb{R}^3)$, and a function with two weak derivatives in L^2 is Hölder- $\frac{1}{2}$ continuous. This is why H^2 is the natural space in which to seek continuous solutions of second-order equations in three dimensions.

5.5 Compactness and the Poincaré Inequality

The embeddings of section 5.4 are bounded but not compact on $\mathbb{R}^n$: a fixed bump translated to infinity keeps its Sobolev norm while wandering off, so no subsequence converges. On a bounded domain there is no room to escape, and the same one-derivative gain that produced the embedding now produces *compactness*. This section proves that on a bounded domain the embedding of $W^{1,p}$ into L^q is compact below the critical exponent, and extracts the two inequalities that turn this compactness into existence theorems: the Poincaré inequality, which controls a function by its gradient once boundary values are killed, and its mean-zero variant, the Poincaré-Wirtinger inequality.

The compactness statement underlies existence proofs in the calculus of variations and the elliptic theory: a bounded sequence of approximate solutions, bounded in $W^{1,p}$ by an energy estimate, has a subsequence converging *strongly* in L^q , and strong convergence is what lets one pass to the limit in a nonlinear term. What makes it work is that a $W^{1,p}$ bound controls not just the size of a function but the size of its translates: a gradient bound forces $\|\tau_h f - f\|_{L^p}$ to be small uniformly, and uniform smallness of translates is exactly the hypothesis of the Fréchet-Kolmogorov compactness criterion in L^p .

Theorem 5.5.1: Rellich-Kondrachov Compactness

Let $U \subseteq \mathbb{R}^n$ be bounded with C^1 boundary, and let $1 \leq p < n$. Then for every q with $1 \leq q < p^*$ the embedding

$$W^{1,p}(U) \hookrightarrow L^q(U)$$

is compact: every bounded sequence in $W^{1,p}(U)$ has a subsequence converging in $L^q(U)$.

Proof:

Let (f_m) be a sequence with $\|f_m\|_{W^{1,p}(U)} \leq M$ for all m ; the goal is a subsequence Cauchy in $L^q(U)$. Throughout, C denotes a constant that may change from line to line but never depends on m .

Step 1: Extend to compactly supported functions on $\mathbb{R}^n$. By the extension theorem (theorem 5.3.3) there is a bounded linear $E: W^{1,p}(U) \rightarrow W^{1,p}(\mathbb{R}^n)$ with $Ef = f$ almost everywhere on U . A cutoff makes E land in functions supported in a fixed bounded open set V with $\bar{U} \subseteq V$: multiply by a fixed cutoff $\zeta \in C_c^\infty(V)$ equal to 1 on a neighborhood of $\bar{U}$, which changes E only by a bounded operator and does not disturb $Ef = f$ on U . Write $g_m := Ef_m$. Then each g_m is supported in V and

$$\|g_m\|_{W^{1,p}(\mathbb{R}^n)} \leq \|E\| \|f_m\|_{W^{1,p}(U)} \leq CM$$

so (g_m) is bounded in $W^{1,p}(\mathbb{R}^n)$, with all g_m supported in the one bounded set V . A subsequence of (f_m) converges in $L^q(U)$ as soon as the corresponding subsequence of (g_m) converges in $L^q(V)$, since $f_m = g_m$ on U ; it therefore suffices to extract a convergent subsequence of (g_m) in $L^q(V)$.

Step 2: A uniform bound at the critical exponent. Each $g_m \in W^{1,p}(\mathbb{R}^n)$, so by the Gagliardo-Nirenberg-Sobolev inequality (theorem 5.4.3),

$$\|g_m\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|\nabla g_m\|_{L^p(\mathbb{R}^n)} \leq C \|g_m\|_{W^{1,p}(\mathbb{R}^n)} \leq CM$$

Thus (g_m) is bounded in $L^{p^*}(\mathbb{R}^n)$, uniformly in m . This is the reservoir of integrability that the interpolation in Step 5 will spend.

Step 3: Control of translates. Fix $h \in \mathbb{R}^n$ and write $\tau_h g(x) = g(x - h)$. For a smooth compactly supported φ the fundamental theorem of calculus along the segment from x to $x - h$ gives

$$\varphi(x) - \varphi(x - h) = \int_0^1 h \cdot \nabla \varphi(x - th) dt$$

so, taking absolute values and applying Minkowski's integral inequality (theorem 2.1.6),

$$\|\tau_h \varphi - \varphi\|_{L^p(\mathbb{R}^n)} \leq |h| \int_0^1 \|\nabla \varphi(\cdot - th)\|_{L^p(\mathbb{R}^n)} dt = |h| \|\nabla \varphi\|_{L^p(\mathbb{R}^n)}$$

the last equality because translation preserves the L^p norm. Since $W_0^{1,p}(\mathbb{R}^n) = W^{1,p}(\mathbb{R}^n)$ (definition 5.2.6), every element of $W^{1,p}(\mathbb{R}^n)$ is a limit in the $W^{1,p}$ norm of such $\varphi \in C_c^\infty(\mathbb{R}^n)$; both sides of the estimate are continuous in that norm, so it passes to the limit and

$$\|\tau_h g_m - g_m\|_{L^p(\mathbb{R}^n)} \leq |h| \|\nabla g_m\|_{L^p(\mathbb{R}^n)} \leq |h| CM$$

holds for every m and every h . The translates of the g_m are thus small uniformly in m : given $\varepsilon > 0$, taking $|h| \leq \varepsilon/(CM)$ makes $\|\tau_h g_m - g_m\|_{L^p(\mathbb{R}^n)} \leq \varepsilon$ for all m at once.

Step 4: Precompactness in $L^1(W_0)$. The estimate of Step 3, together with the common support in V and the uniform L^p bound (which gives a uniform $L^1(V)$ bound since V is bounded), is precisely the hypothesis of the Fréchet-Kolmogorov criterion for precompactness in L^1 . Rather than invoke it, mollify directly. Let W_0 be the closed 1-neighborhood of V , a fixed compact set, and let $\{\psi_\delta\}_{0 < \delta \leq 1}$ be the good kernel (definition 2.5.9) with $\psi_\delta(x) = \delta^{-n} \psi(x/\delta)$ for a fixed $\psi \in C_c^\infty(\mathbb{R}^n)$, $\psi \geq 0$, $\int \psi = 1$, supported in the unit ball; then g_m and $g_m * \psi_\delta$ are all supported in W_0 . For each m ,

$$(g_m * \psi_\delta)(x) - g_m(x) = \int_{\mathbb{R}^n} (g_m(x - y) - g_m(x)) \psi_\delta(y) dy$$

so integrating and using $\int \psi_\delta = 1$, Minkowski's integral inequality (theorem 2.1.6), and Step 3,

$$\begin{aligned} \|g_m * \psi_\delta - g_m\|_{L^1(\mathbb{R}^n)} &\leq \int_{|y| \leq \delta} \|\tau_y g_m - g_m\|_{L^1(\mathbb{R}^n)} \psi_\delta(y) dy \\ &\leq \int_{|y| \leq \delta} |W_0|^{1-1/p} \|\tau_y g_m - g_m\|_{L^p(\mathbb{R}^n)} \psi_\delta(y) dy \leq C \delta \end{aligned}$$

where Hölder on the common bounded support W_0 turned the L^1 norm into an L^p norm at the cost of $|W_0|^{1-1/p}$, and the bound is uniform in m . Fix $\varepsilon > 0$ and choose δ so that $C\delta \leq \varepsilon/3$. For this frozen δ the mollified family $(g_m * \psi_\delta)_m$ is uniformly bounded and equicontinuous on W_0 : uniformly bounded because

$$|(g_m * \psi_\delta)(x)| \leq \|g_m\|_{L^1(\mathbb{R}^n)} \|\psi_\delta\|_{L^\infty(\mathbb{R}^n)} \leq C\delta^{-n}$$

and equicontinuous because

$$|(g_m * \psi_\delta)(x) - (g_m * \psi_\delta)(x')| \leq \|g_m\|_{L^1(\mathbb{R}^n)} \|\tau_{x'-x} \psi_\delta - \psi_\delta\|_{L^\infty(\mathbb{R}^n)}$$

and ψ_δ is uniformly continuous. By the Arzelà-Ascoli theorem a subsequence of $(g_m * \psi_\delta)_m$ converges uniformly on W_0 , hence in $L^1(W_0)$. Combining with the uniform bound $\|g_m * \psi_\delta - g_m\|_{L^1} \leq \varepsilon/3$, along this subsequence

$$\limsup_{k,l} \|g_{m_k} - g_{m_l}\|_{L^1(W_0)} \leq \frac{2\varepsilon}{3} + \limsup_{k,l} \|g_{m_k} * \psi_\delta - g_{m_l} * \psi_\delta\|_{L^1(W_0)} = \frac{2\varepsilon}{3} < \varepsilon$$

A diagonal argument over $\varepsilon = 1, \frac{1}{2}, \frac{1}{3}, \dots$ then produces a single subsequence, still written (g_m) , that is Cauchy in $L^1(W_0)$, hence in $L^1(V)$ since $V \subseteq W_0$.

Step 5: Interpolate up to $L^q(V)$. Fix q with $1 \leq q < p^*$. Interpolation between L^1 and L^{p^*} writes $\frac{1}{q} = \frac{\theta}{1} + \frac{1-\theta}{p^*}$ for a unique $\theta \in (0, 1]$, and Hölder's inequality (theorem 2.1.5) gives, for any g ,

$$\|g\|_{L^q(V)} \leq \|g\|_{L^1(V)}^\theta \|g\|_{L^{p^*}(V)}^{1-\theta}$$

Applying this to differences of the extracted subsequence, and using the uniform L^{p^*} bound of Step 2 (which controls $\|g_{m_k} - g_{m_l}\|_{L^{p^*}(V)} \leq 2CM$),

$$\|g_{m_k} - g_{m_l}\|_{L^q(V)} \leq \|g_{m_k} - g_{m_l}\|_{L^1(V)}^\theta (2CM)^{1-\theta}$$

The right side tends to 0 as $k, l \rightarrow \infty$ because the sequence is Cauchy in $L^1(V)$ and $\theta > 0$. Hence (g_{m_k}) is Cauchy, so convergent, in $L^q(V)$, and its restrictions to U give a subsequence of (f_m) convergent in $L^q(U)$, as we sought to show.

The mechanism is worth isolating. A $W^{1,p}$ bound buys two things: an integrability gain past p^* , and control of translates through the gradient. The first is a fixed reservoir of smallness at high exponents; the second, via mollification, upgrades the reservoir to genuine L^1 compactness on a bounded set. Interpolation then converts L^1 compactness plus an L^{p^*} bound into L^q compactness for every q strictly below p^* . The exclusion of the endpoint $q = p^*$ is not an artifact of the proof: at the critical exponent, concentration can defeat compactness. Fixing a bump $u \in C_c^\infty(\mathbb{R}^n)$ and rescaling by $u_\lambda(x) = \lambda^{(n-p)/p} u(\lambda x)$ keeps both $\|\nabla u_\lambda\|_{L^p}$ and $\|u_\lambda\|_{L^{p^*}}$ constant in λ while concentrating the mass to a point as $\lambda \rightarrow \infty$, so no subsequence converges in L^{p^*} ; the compact embedding is a subcritical phenomenon.

Example 5.5.2: Compactness on the Interval

Take $U = (0, 1) \subset \mathbb{R}$, so $n = 1$. Here the subcritical range $1 \leq p < n$ is empty, but the compactness is still visible directly through Morrey's inequality (theorem 5.4.8), which for $n = 1$ and $p > 1 = n$ turns a $W^{1,p}(0, 1)$ bound into a uniform Hölder bound. Concretely, a sequence (f_m) bounded in $W^{1,2}(0, 1)$ satisfies, for $x < y$,

$$|f_m(y) - f_m(x)| = \left| \int_x^y f'_m(t) dt \right| \leq \|f'_m\|_{L^2(0,1)} |y - x|^{1/2} \leq M |y - x|^{1/2}$$

by Cauchy-Schwarz, so (f_m) is equicontinuous with a uniform Hölder-1/2 modulus; it is uniformly bounded because the Morrey embedding turns the $W^{1,2}(0, 1)$ bound into a uniform sup-bound $\|f_m\|_{L^\infty(0,1)} \leq CM$. Arzelà-Ascoli then extracts a uniformly convergent subsequence, in particular convergent in $L^q(0, 1)$ for every $q < \infty$: in one dimension a gradient bound is a Hölder bound, and Hölder bounds are precompact. The proof of theorem 5.5.1 replaces this pointwise Hölder estimate, unavailable when $p \leq n$, by the L^p control of translates in Step 3.

The first payoff of compactness is a coercivity estimate. For a function that vanishes at the boundary, the gradient alone controls the full L^p norm: there is nothing left to compensate a large function with a small gradient once constants are forbidden. This is the inequality that makes the Dirichlet energy $\int_U |\nabla f|^2$ an equivalent norm on $W_0^{1,2}(U)$, and with it the existence theory for the Dirichlet problem. Its proof needs no compactness at all, only the one-dimensional fundamental theorem of calculus, so it precedes the Wirtinger variant.

Theorem 5.5.3: Poincaré Inequality

Let $U \subseteq \mathbb{R}^n$ be bounded and let $1 \leq p < \infty$. There is a constant $C = C(n, p, U)$ such that

$$\|f\|_{L^p(U)} \leq C \|\nabla f\|_{L^p(U)}$$

for every $f \in W_0^{1,p}(U)$.

Proof:

Since U is bounded, choose $a > 0$ with $U \subseteq \{x \in \mathbb{R}^n \mid |x_1| < a\}$, the slab of width $2a$ in the first coordinate. By the definition of the zero-trace space (definition 5.2.6), $C_c^\infty(U)$ is dense in $W_0^{1,p}(U)$, so it suffices to prove the inequality for $f \in C_c^\infty(U)$ and then pass to the limit; extend such an f by zero to all of $\mathbb{R}^n$, keeping it smooth and compactly supported inside the slab.

Fix $x = (x_1, x') \in U$ with $x' = (x_2, \dots, x_n)$. Because $f(-a, x') = 0$, the fundamental theorem of calculus in the first variable gives

$$f(x_1, x') = \int_{-a}^{x_1} \partial_1 f(t, x') dt$$

Take absolute values and apply Hölder's inequality (theorem 2.1.5) in t over the interval $(-a, a)$, of length $2a$:

$$|f(x_1, x')| \leq \int_{-a}^a |\partial_1 f(t, x')| dt \leq (2a)^{1/p'} \left(\int_{-a}^a |\partial_1 f(t, x')|^p dt \right)^{1/p}$$

Raise to the p th power and integrate in x_1 over $(-a, a)$. The right side does not depend on x_1 , so integrating contributes a factor $2a$:

$$\int_{-a}^a |f(x_1, x')|^p dx_1 \leq (2a)^{p/p'} (2a) \int_{-a}^a |\partial_1 f(t, x')|^p dt = (2a)^p \int_{-a}^a |\partial_1 f(t, x')|^p dt$$

using $p/p' + 1 = p$. Now integrate in the remaining variables x' and recall f vanishes outside the slab:

$$\|f\|_{L^p(U)}^p = \int_{\mathbb{R}^{n-1}} \int_{-a}^a |f(x_1, x')|^p dx_1 dx' \leq (2a)^p \int_{\mathbb{R}^{n-1}} \int_{-a}^a |\partial_1 f|^p dt dx' = (2a)^p \|\partial_1 f\|_{L^p(U)}^p$$

Since $\|\partial_1 f\|_{L^p(U)} \leq \|\nabla f\|_{L^p(U)}$, taking p th roots gives $\|f\|_{L^p(U)} \leq 2a \|\nabla f\|_{L^p(U)}$ with $C = 2a$. For general $f \in W_0^{1,p}(U)$, choose $f_m \in C_c^\infty(U)$ with $f_m \rightarrow f$ in $W^{1,p}(U)$; the inequality holds for each f_m , and both $\|f_m\|_{L^p(U)} \rightarrow \|f\|_{L^p(U)}$ and $\|\nabla f_m\|_{L^p(U)} \rightarrow \|\nabla f\|_{L^p(U)}$, so it passes to the limit, completing the proof.

The constant depends only on the width of the smallest slab containing U , not on its shape: Poincaré's inequality holds on any bounded domain, and its constant degrades only as the domain grows in one direction. On $W_0^{1,p}(U)$ the estimate says $\|\nabla f\|_{L^p(U)}$ is a norm equivalent to the full $\|f\|_{W^{1,p}(U)}$ norm, since $\|f\|_{W^{1,p}}^p = \|f\|_{L^p}^p + \|\nabla f\|_{L^p}^p \leq (1 + C^p) \|\nabla f\|_{L^p}^p$. The gradient alone therefore measures everything, which is why the Dirichlet energy is the natural object on the zero-trace space. The hypothesis that boundary values vanish is essential: on all of $W^{1,p}(U)$ the inequality is false, since a

nonzero constant function has zero gradient but positive L^p norm. The constant functions are the only obstruction, and quotienting them out restores the estimate, which is the content of the next theorem.

Example 5.5.4: The Optimal Constant on an Interval

On $U = (0, L) \subset \mathbb{R}$ with $p = 2$, the Poincaré inequality reads $\|f\|_{L^2(0,L)} \leq C \|f'\|_{L^2(0,L)}$ for $f \in H_0^1(0, L)$. The proof above gives $C = 2a = L$ after centering the interval, but the sharp constant is smaller. Expand $f \in H_0^1(0, L)$ in the sine basis $e_k(x) = \sqrt{2/L} \sin(k\pi x/L)$, which vanishes at both endpoints and diagonalizes $-d^2/dx^2$ with eigenvalues $(k\pi/L)^2$. Writing $f = \sum_k c_k e_k$, Parseval gives $\|f\|_{L^2}^2 = \sum_k c_k^2$ and $\|f'\|_{L^2}^2 = \sum_k (k\pi/L)^2 c_k^2 \geq (\pi/L)^2 \sum_k c_k^2$, so

$$\|f\|_{L^2(0,L)} \leq \frac{L}{\pi} \|f'\|_{L^2(0,L)}$$

with equality for $f = e_1$. The sharp constant L/π is the reciprocal square root of the first Dirichlet eigenvalue $(\pi/L)^2$; the crude slab estimate $C = L$ overshoots it by the factor π . The general principle, that the best Poincaré constant is the reciprocal square root of the first eigenvalue of the Dirichlet Laplacian, connects these inequalities to spectral theory: the compactness of theorem 5.5.1 is exactly what makes $-\Delta$ with zero boundary values have compact resolvent, so its spectrum is a discrete sequence of Poincaré-type constants.

When boundary values are not prescribed the constant functions must be removed by hand, subtracting the average. The resulting inequality controls the deviation of a function from its mean by the gradient, and it holds on any bounded connected domain with reasonable boundary. Unlike the Poincaré inequality, no explicit constant is available from a one-dimensional estimate: the argument is by compactness and contradiction, and it is here that Rellich-Kondrachov is used.

Theorem 5.5.5: Poincaré-Wirtinger Inequality

Let $U \subseteq \mathbb{R}^n$ be bounded, connected, and with C^1 boundary, and let $1 \leq p < \infty$. Write $f_U = \frac{1}{|U|} \int_U f dx$ for the average of f over U . There is a constant $C = C(n, p, U)$ such that

$$\|f - f_U\|_{L^p(U)} \leq C \|\nabla f\|_{L^p(U)}$$

for every $f \in W^{1,p}(U)$.

Proof:

Suppose the inequality fails for every constant. Then for each m there is $f_m \in W^{1,p}(U)$ with

$$\|f_m - (f_m)_U\|_{L^p(U)} > m \|\nabla f_m\|_{L^p(U)}$$

Normalize by setting

$$g_m := \frac{f_m - (f_m)_U}{\|f_m - (f_m)_U\|_{L^p(U)}}$$

which is legitimate because the left side of the failed inequality is positive. Then $g_m \in W^{1,p}(U)$ has three properties: its average $(g_m)_U = 0$ (subtracting a constant sets the mean to zero, and normalizing preserves it), its L^p norm $\|g_m\|_{L^p(U)} = 1$, and dividing the failed inequality by its

own left side,

$$\|\nabla g_m\|_{L^p(U)} = \frac{\|\nabla f_m\|_{L^p(U)}}{\|f_m - (f_m)_U\|_{L^p(U)}} < \frac{1}{m}$$

so $\nabla g_m \rightarrow 0$ in $L^p(U)$. In particular

$$\|g_m\|_{W^{1,p}(U)}^p = \|g_m\|_{L^p(U)}^p + \|\nabla g_m\|_{L^p(U)}^p \leq 1 + \frac{1}{m^p} \leq 2$$

so (g_m) is bounded in $W^{1,p}(U)$.

The compactness of the embedding $W^{1,p}(U) \hookrightarrow L^p(U)$ furnishes a convergent subsequence in every case. If $p < n$, then $q = p < p^*$ lies in the compact range of the Rellich-Kondrachov theorem (theorem 5.5.1), so a subsequence of (g_m) converges in $L^p(U)$. If $p > n$, the $kp > n$ case of theorem 5.4.9 embeds $W^{1,p}(U)$ into a Hölder space, and a bounded sequence there is precompact in $C(\bar{U})$ by Arzelà-Ascoli, hence in $L^p(U)$. The two boundary cases $p = n$ remain. For $p = n \geq 2$, pick $p' < n$ with $(p')^* > p$ (possible since $(p')^* \rightarrow \infty$ as $p' \uparrow n$); then $W^{1,p}(U) \hookrightarrow W^{1,p'}(U)$ on the bounded set U , and $p < (p')^*$ puts $q = p$ in the compact range for exponent p' , so theorem 5.5.1 again applies. The last case is $n = 1$, $p = 1$: here neither route is available, but the Rellich mechanism applies directly. Extend the g_m to a common bounded support in $\mathbb{R}$ as in Step 1 of the proof of theorem 5.5.1; the translate estimate of its Step 3 gives $\|\tau_h g_m - g_m\|_{L^1(\mathbb{R})} \leq |h| \|\nabla g_m\|_{L^1}$, uniformly bounded in m , which together with the common support is the hypothesis of the Fréchet-Kolmogorov criterion for precompactness in L^1 , yielding a subsequence convergent in $L^1(U) = L^p(U)$. In every case, pass to a subsequence, still called (g_m) , with $g_m \rightarrow g$ in $L^p(U)$ for some $g \in L^p(U)$.

The limit g inherits both normalizations by L^p convergence: $\|g\|_{L^p(U)} = \lim \|g_m\|_{L^p(U)} = 1$, and averaging is L^p -continuous on the bounded set U (by Hölder against the constant $1 \in L^{p'}(U)$), so $g_U = \lim (g_m)_U = 0$. It remains to identify the weak gradient of g . For any $\varphi \in C_c^\infty(U)$ and any index i , the weak-derivative identity (definition 5.1.1) for g_m reads $\int_U g_m \partial_i \varphi = - \int_U \partial_i g_m \varphi$. Let $m \rightarrow \infty$: the left side tends to $\int_U g \partial_i \varphi$ since $g_m \rightarrow g$ in L^p and $\partial_i \varphi \in L^{p'}$, while the right side tends to 0 since $\partial_i g_m \rightarrow 0$ in L^p and $\varphi \in L^{p'}$. Hence $\int_U g \partial_i \varphi = 0$ for all φ , which says $\partial_i g = 0$ weakly for every i , so $\nabla g = 0$. Thus $g \in W^{1,p}(U)$ with vanishing weak gradient on the connected set U .

A $W^{1,p}$ function with zero weak gradient on a connected open set is constant almost everywhere. Indeed, mollifying, $g * \psi_\delta$ is smooth with classical gradient $\nabla(g * \psi_\delta) = (\nabla g) * \psi_\delta = 0$ on any $V \Subset U$ (by theorem 5.2.1, which commutes the derivative past the mollifier), hence constant on each component of V ; letting $\delta \rightarrow 0$ and using connectedness of U shows g equals a single constant almost everywhere. But then $g = g_U = 0$ almost everywhere, contradicting $\|g\|_{L^p(U)} = 1$. The assumed failure is therefore impossible, and the inequality holds with some finite constant C , completing the proof.

The proof is nonconstructive: it produces a finite constant but no formula for it, in contrast to the explicit 2a of the Poincaré inequality. This is the price of the compactness argument, and it is characteristic of the method: an inequality is proved by ruling out its failure, and the witness to failure is destroyed by extracting a convergent subsequence whose limit cannot exist. Connectedness is used exactly once, in the final step, to force a gradient-free function to be a single constant rather than a locally constant one; on a disconnected domain the average over the whole set is the wrong reference value, and the inequality fails with a function equal to different constants on different components. The mean f_U is the natural replacement for the boundary value of the Poincaré inequality: both

the zero-trace condition and the zero-mean condition eliminate the constants, which are the sole functions the gradient cannot see, and the estimate holds on the quotient by constants in either case.

Example 5.5.6: Mean-Zero Functions on a Ball

On the unit ball $U = B(0, 1) \subset \mathbb{R}^n$ with $p = 2$, the Poincaré-Wirtinger inequality $\|f - f_U\|_{L^2(U)} \leq C \|\nabla f\|_{L^2(U)}$ again has a spectral reading. Its sharp constant is the reciprocal square root of the first *nonzero* Neumann eigenvalue μ_1 of $-\Delta$ on the ball: the mean-zero constraint $f_U = 0$ is exactly orthogonality to the constant eigenfunction, whose Neumann eigenvalue is $\mu_0 = 0$, so on the orthogonal complement the Rayleigh quotient $\|\nabla f\|_{L^2}^2 / \|f\|_{L^2}^2$ is bounded below by μ_1 . A mean-zero f therefore satisfies $\|f\|_{L^2}^2 \leq \mu_1^{-1} \|\nabla f\|_{L^2}^2$, giving $C = \mu_1^{-1/2}$. The contradiction argument of theorem 5.5.5 recovers the finiteness of this constant without computing μ_1 : it shows only that the infimum of the Rayleigh quotient over mean-zero functions is positive, which is exactly $\mu_1 > 0$, the statement that 0 is a simple Neumann eigenvalue on a connected domain.

6

Hilbert Spaces

There are many reasons why Hilbert spaces are interesting. We shall start with a motivation stemming from looking for the ideal space where the product of two functions is still integrable, and then take a moment to elaborate on some more advanced intuitions of Hilbert spaces for readers with more background.

We are naturally interested in integrating products of functions:

$$\int_X fg$$

The problem we may encounter when working over L^1 is that $\int_X fg$ need not be finite. We thus may say “let us thus work with the functions in which this *is* well-defined”. If $L'(X)$ is the space of all functions measurable functions where for any choice $f, g \in L'(X)$:

$$\int_X f\bar{g} < \infty \quad \text{i.e.} \quad f\bar{g} \in L^1$$

then we have just made a circular-reasoning mistake, for we are quantifying existence of elements in $L'(X)$ using elements of $L'(X)$. Instead, we may ask: given a set S , what is the subset of measurable functions T such that:

$$T = \left\{ f \in \mathcal{M}(X) \mid \int_X f\bar{g} < \infty, \forall g \in S \right\}$$

The two extremes of this are if $S = \{0\}$, then T is all measurable functions, and if S is all measurable functions, then $T = \{0\}$ only contains the zero function. The ideal is that there is a middle ground choice of S where $S = T$ so that we don't need to keep track of the two spaces we are choosing our functions from, namely we would not like to have $(S, T)(X)$. Fortunately, as will be proven in section 2.2, this is *exactly* the space L^2 . Namely, if $S = L^2$, then:

$$T = \left\{ f \in \mathcal{M}(X) \mid \int_X f\bar{g} < \infty \forall g \in L^2 \right\} = L^2$$

Though we can't prove this here, the high level is that if $f, g \in L^2$, then by the Cauchy-Schwartz inequality¹

$$\int_X |fg| \leq \left(\int_X |f|^2 \right)^{1/2} \left(\int_X |g|^2 \right)^{1/2} = \|f\|_2 \|g\|_2$$

showing that $L^2 \subseteq T$. Conversely, by Hölder's inequality, we have that if $f \in L^p$, then for all $g \in L^q$, $fg \in L^1$. If $p = 2$, then $q = 2$, showing that all functionals $(L^2)^*$ are characterized by L^2 functions:

$$\varphi_g : L^p \rightarrow \mathbb{R} \quad \varphi_g(f) = \int f g$$

showing that $T \subseteq L^2$, hence $T = L^2$. From this perspective, L^2 is the ideal space in which to multiply functions and they remain integrable (i.e. $f, g \in L^2$, $fg \in L^1$)! We shall call such multiplication the *inner product*:

$$\langle f, g \rangle = \int_X f \bar{g}$$

The conjugation in the definition is used so that $f\bar{f} = |f|^2$. Note that we are not looking for the product of two functions to remain within L^1 . In section 2.5, we shall see that *convolution* will have the desired property so that $f, g \in L^1(G)$ then $f * g \in L^1(G)$ for G a locally compact group, in particular that shall be a *group algebra*.

6.0.1 post-Hoc Motivation A lot of the following motivation came *a-posteriori*, that is, I felt much more comfortable with Hilbert spaces after learning about L^p spaces, and generalizing some result from Riemannian Geometry to Banach spaces to notice that L^2 is the only space with flat geometry, hence really is the “correct” generalization of euclidean space/geometry in infinite dimensions.

Banach spaces provide a robust framework for measuring the magnitude of elements, allowing us to quantify convergence and bound operations via the triangle and Minkowski inequalities. However, a fundamental deficiency in general Banach spaces is the lack of a linear measurement theory to quantify *similarity*. While we can measure the total size of a sum $\|f + g\|$, we often lack the granular information required to isolate the contribution of f from g . For instance, determining how much a vector projects onto a specific basis element is straightforward in Euclidean space but algebraically obstructed in general norms.

To make this deficiency precise, consider how the inner product arises in Euclidean geometry. Define the *energy* of a vector as $E(x) = \frac{1}{2} \|x\|_2^2$. Then the directional derivative of this energy at x in the direction of a perturbation y is:

$$\left. \frac{d}{dt} E(x + ty) \right|_{t=0} = \left. \frac{d}{dt} \frac{1}{2} \langle x + ty, x + ty \rangle \right|_{t=0} = \langle x, y \rangle$$

Thus, the inner product $\langle x, y \rangle$ is not just an algebraic artifact; it physically represents the *sensitivity* of the energy of x to a perturbation by y . In particular, the energy satisfies the clean decomposition:

$$E(x + y) = E(x) + E(y) + \langle x, y \rangle$$

¹which shall be proven in this section, but the reader most likely has already encountered it and hence it is used here to give intuition

The interaction term $\langle x, y \rangle$ is *bilinear*—it decomposes the interference between two vectors according to the simple rules of superposition. In general Banach spaces, however, such a clean, linear decomposition of the interaction is impossible due to the non-flat geometry of the unit ball.

The natural question, then, is: *what replaces the inner product in L^p ?* The correct generalization is to replace the quadratic energy with the *p-energy*:

$$F(f) = \frac{1}{p} \|f\|_p^p = \frac{1}{p} \int |f|^p$$

and ask: how does $F(f)$ change if we perturb f by a function g ? To answer this, we must compute the gradient ∇F , which requires first understanding the natural duality of L^p spaces.

The Duality Map. What does it mean to go in the “same direction” in a function space? (For simplicity we work throughout with L^p spaces for $1 < p < \infty$.²) Rigorously, we are asking for the functional φ_f of unit norm that *maximizes* the evaluation at f . We can interpret this φ_f as the *optimal filter* or sensor for f —it captures the full “energy” of f without loss ($\varphi_f(f) = \|f\|_p$ and $\|\varphi_f\|_p = 1$).

By theorem 2.2.3 (and the equality condition of Hölder’s inequality), this functional corresponds to $g \in L^q$ such that:

$$g(x) = |f(x)|^{p-1} \text{sgn}(f(x))$$

Note that $g \in L^q$ since:

$$|g|^q = |f|^{q(p-1)} = |f|^{qp-q} \stackrel{!}{=} |f|^p$$

where $\stackrel{!}{=}$ holds since $1/p + 1/q = 1$ iff $p + q = pq$ iff $p = q(p-1)$. Hence as $f \in L^p$:

$$\int |g|^q = \int |f|^p < \infty$$

Thus, we have a map $f \mapsto \varphi_f$. We define the *duality map*:

$$J_p : L^p \rightarrow L^q, \quad f \mapsto \varphi_f$$

As $(L^q)^* \cong L^p$, we can equivalently view this as a map $J_p : L^p \rightarrow (L^p)^*$.

The Central Geometric Fact. The duality map J_p is not an algebraic accident—it is the *geometric normal* to the level sets of the *p-energy*. To see this, consider the function:

$$F(x) = \frac{1}{p} \|x\|_p^p = \frac{1}{p} \sum |x_i|^p$$

(where the ℓ^p norm is used for visualization, but the logic holds for integrals). From multivariable calculus, the unit sphere is the level set $F(x) = \frac{1}{p}$, and the gradient $\nabla F(x)$ is the vector strictly perpendicular (normal) to the level set at point x . Computing the gradient (the Fréchet derivative):

$$\frac{\partial}{\partial x_i} \left(\frac{1}{p} |x_i|^p \right) = |x_i|^{p-1} \text{sgn}(x_i)$$

²The reasoning can extend to nuclear Fréchet spaces, which admit descriptions as projective limits of Hilbert spaces. For general metric spaces, a different approach is needed via CAT(k) geometry; see [5] for details.

So, the gradient vector is:

$$\nabla F(x) = (|x_1|^{p-1}\text{sgn}(x_1), \dots, |x_n|^{p-1}\text{sgn}(x_n))$$

This is exactly our duality map:

$$\boxed{\nabla F(x) = J_p(x)}$$

This identification is the key to the entire theory. It tells us that the optimal functional for f —the element of $(L^p)^*$ that best “measures” f —is precisely the gradient of the energy at f : the direction of steepest increase in p -energy. The duality map is a *geometric* object, not merely an algebraic one.

The p -Pairing and Its Interpretation. With the gradient in hand, we can now answer our original question: how does the p -energy $F(f)$ change under a perturbation by g ? The first-order variation is³:

$$dF(f; g) = \langle \nabla F(f), g \rangle_{\text{dual}} = \int \nabla F(f) \cdot g$$

Substituting $\nabla F = J_p(f)$, this defines a natural pairing:

$$\langle f, g \rangle_p := \int J_p(f) \cdot g$$

Note that it is important to pass through $J_p(f)$ since $\int fg$ need not be finite in general. However, by Hölder’s inequality:

$$\int |J_p(f) \cdot g| < \infty$$

This pairing has a precise geometric interpretation. The gradient $J_p(f)$ is a cotangent vector—a 1-form $dF|_f \in T_f^*L^p$ —and the perturbation g is a tangent vector $g \in T_fL^p$. Thus the pairing is the evaluation of a 1-form on a tangent vector:

$$\langle f, g \rangle_p = dF|_f(g) \in T_f^*L^p \times T_fL^p \rightarrow \mathbb{R}$$

This canonical evaluation pairing between the tangent and cotangent spaces exists in *every* geometric setting without any choice of metric. What J_p adds is the ability to *produce* a specific cotangent vector from a tangent vector, turning the pairing from a bilinear form on $T_fL^p \times T_f^*L^p$ into a (generally nonlinear) form on $T_fL^p \times T_fL^p$ —that is, internalizing the measurement within the space itself.

The Infinitesimal Meaning of V^* . We are now in a position to make a precise statement about the geometric content of the dual space. In a bare vector space without additional structure, the dual V^* is merely an algebraic object—a collection of linear maps with no geometric significance beyond probing linearly. However, in L^p equipped with its norm, the identity $J_p = \nabla F$ reveals that *every element of $(L^p)^*$ arises as the gradient of the energy functional*—a differential $dF|_f$. This endows $(L^p)^*$ with genuine infinitesimal meaning: its elements are not just linear maps, but *cotangent vectors* that encode the first-order sensitivity of the geometry.

Crucially, cotangent vectors are a different species of infinitesimal object than tangent vectors. A tangent vector $g \in T_fL^p \cong L^p$ represents an *infinitesimal direction of motion*—it answers “which way?”. A cotangent vector $\varphi \in T_f^*L^p \cong L^q$ represents an *infinitesimal measurement*—it answers “how fast is this quantity changing?”. These are related by the evaluation pairing $\varphi(g) = dF|_f(g)$, but they

³The dF here can be thought of as the Gâteaux derivative, commonly defined for functions between locally convex topological vector spaces.

transform in opposite ways: tangent vectors push forward under maps, while cotangent vectors pull back. The duality map $J_p : T_f L^p \rightarrow T_f^* L^p$ provides a (nonlinear) identification between these two types of infinitesimal data, allowing us to transfer the directional interpretation onto the dual space. However, this transfer *depends essentially on the norm geometry*—it is canonical only for $p = 2$.

Geometrically, if we view the identity map $f \mapsto f$ as the radial vector field (a *section* of the tangent bundle TL^p), then the duality map J_p transforms this into a covector field (a *section* of the cotangent bundle T^*L^p). In the language of Riemannian geometry, this transformation is the *musical isomorphism* $\flat$ (flat), which maps a vector field X to a 1-form $X^\flat$ by lowering indices via the metric tensor. The four corollaries that follow are all consequences of this single geometric fact.

Corollary 1: Nonlinearity of the Musical Isomorphism. The duality map is *not* linear:

$$J_p(2f) = |2f|^{p-1} \text{sgn}(2f) = 2^{p-1} |f|^{p-1} \text{sgn}(f) = 2^{p-1} J_p(f)$$

This is only linear in the special case of $p = 2$, since:

$$J_2(nf) = |nf|^{2-1} \text{sgn}(nf) = n J_2(f)$$

This distinction is profound. In a general Banach space ($p \neq 2$), the musical isomorphism is *nonlinear*: the operation of lowering an index depends on the magnitude of the vector itself, meaning the “metric” changes as we move through the space. However, when $p = 2$, this map is linear. This implies a linear bundle isomorphism $TL^2 \cong T^*L^2$, effectively trivializing the distinction between vectors and covectors. Only in L^2 is the musical isomorphism a linear operator, meaning the “force” required to correct an error is linearly proportional to the error itself.

Corollary 2: Non-Symmetry of the Pairing. The pairing $\langle f, g \rangle_p$ is *not* symmetric for $p \neq 2$. This is an immediate consequence of the nonlinearity of J_p : the 1-form $dF|_f$ evaluated on g has no reason to equal the 1-form $dF|_g$ evaluated on f , since the gradient depends nonlinearly on its base point. If however $p = 2$, then $J_p(x)$ maps identically to itself and we recover the symmetric form $\int fg$. This asymmetry is the first signal that the geometry of L^p is *not flat* for $p \neq 2$.

Corollary 3: Failure of Rotational Invariance. This lack of symmetry in the pairing $\langle f, g \rangle_p$ is directly linked to the fact that L^2 is the unique L^p space that is *rotationally invariant*. Geometrically, rotational invariance means that the properties of a vector depend only on its length, not on its alignment with the coordinate axes. Mathematically, this requires that the duality map commutes with rotations. If R is a rotation operator (an isometry), we expect the “normal vector” of the rotated point to be the rotated “normal vector” of the original point:

$$J_p(Rf) \stackrel{?}{=} R(J_p(f))$$

In L^2 , since J_2 is linear (essentially the identity), this holds trivially: $J_2(Rf) = Rf = R(J_2(f))$. However, for $p \neq 2$, the map $f \mapsto |f|^{p-1} \text{sgn}(f)$ applies a non-linear distortion component-wise. Rotating the vector mixes the components, and applying the power law *after* mixing is not the same as applying it *before*.

To see this concretely, consider ℓ^3 ($p = 3$) in $\mathbb{R}^2$ and let $f = (1, 0)$. If we first compute the functional and then rotate by $\pi/4$ (45°), we get:

$$R(J_3(f)) = R(1^2, 0^2) = R(1, 0) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$$

However, if we rotate the vector first, the components become $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$. Applying the duality map J_3 (squaring components) yields:

$$J_3(Rf) = J_3\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \left(\frac{1}{2}, \frac{1}{2}\right)$$

Since $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}) \neq (\frac{1}{2}, \frac{1}{2})$, we have $J_p(Rf) \neq R(J_p(f))$. This confirms that for $p \neq 2$, the geometry is *anisotropic*: the shape of the unit ball looks different depending on the angle from which you view it.

Corollary 4: Birkhoff-James Orthogonality. With this understanding of the position-dependent metric, we can clarify the notion of orthogonality. In Euclidean geometry, orthogonality is symmetric ($x \perp y \iff y \perp x$) and additive. In general Banach spaces, without a global bilinear form, we must rely on *Birkhoff-James orthogonality*: we say $x \perp y$ if moving in the direction of y does not decrease the length of x ($\|x + \lambda y\| \geq \|x\|$). Geometrically, this means the line passing through x in the direction of y is tangent to the unit sphere at x .

Analytically, utilizing the duality map J_p , this condition is equivalent to requiring that the tangent vector y be annihilated by the cotangent vector generated by x :

$$\langle J_p(x), y \rangle = 0$$

This is precisely the statement that y lies in the kernel of the 1-form $dF|_x$ —a natural geometric condition. However, it reveals the fundamental asymmetry in non-Hilbert geometry. In L^2 , because J_2 is linear, the condition is symmetric. However, in L^p ($p \neq 2$), the non-linearity of J_p breaks reciprocity. The set of vectors orthogonal to x (kernel of $J_p(x)$) does not align symmetrically with the set of vectors orthogonal to y . Consequently, it is common to have $x \perp y$ but $y \not\perp x$. This lack of symmetry illustrates why Hilbert spaces are the unique setting where Euclidean geometric intuition remains valid.

The Riemannian Perspective. The corollaries above all concern the first derivative of the energy (the gradient J_p). The full geometry of the space, however, is determined by how this gradient *changes*—the second derivative, which acts as the metric tensor. In this framework, the metric tensor $g_{ij}(x)$ at a point x dictates how to measure the length of tangent vectors. In our context, this metric is generated by the Hessian of the energy function $F(x) = \frac{1}{p}\|x\|_p^p$. For L^2 , this Hessian is proportional to the identity operator everywhere; the metric is *constant* (flat). However, for L^p ($p \neq 2$), the second derivative depends on position x , specifically scaling as $|x|^{p-2}$. This implies that the “metric” changes as we move around the space.

It is worth noting why the theory breaks down at the boundary cases. For L^1 , the unit ball is a diamond with corners; at these corners, the gradient is not unique (it is a set-valued subgradient). For L^∞ , the unit ball has flat faces; on these faces, the gradient is non-unique or vanishes in certain directions. This lack of smoothness (L^1) or rotundity (L^∞) prevents a well-defined duality map, leaving L^p ($1 < p < \infty$) as the sweet spot for smooth geometry.

Topological vs. Geometric Structure. To visualize the practical consequences of this non-flat geometry, it is helpful to consider the “shape” of the unit ball in infinite dimensions. In finite dimensions, all norms are equivalent, meaning the unit ball of one norm can always be fit inside a scaled unit ball of another. In infinite dimensions, this equivalence breaks. The curvature of the norm dictates how “energy” is allowed to concentrate without bound. The L^∞ unit ball acts like a strict box, while the L^1 unit ball allows for infinitely long thin spikes.

Ideally, we would like to know if these distortions are topological or “merely” geometric. This is answered by the *Mazur Map* $M_{p,q} : L^p(B) \rightarrow L^q(B)$ where B is the unit ball. The map:

$$M_{p,q}(f) = \operatorname{sgn}(f) \cdot |f|^{p/q}$$

shows that all $L^p(B)$ spaces are *uniformly homeomorphic* to each other (not just homeomorphic as theorem 1.5.5 shows). This means that we can smoothly deform the “spikes” of L^1 into the “sphere” of L^2 . However, while the Mazur map preserves the topology, it destroys the linear structure (additivity). The Hilbert space L^2 is the unique fixed point where this topological structure aligns perfectly with the linear algebraic structure (where the Mazur map would be the identity).

This geometric disparity explains why convergence depends on the norm. Convergence $f_n \rightarrow 0$ is not about movement, but about entering arbitrarily small ϵ -balls. Consider the sequence $f_n(x) = x^n$ on $[0, 1]$. In the L^1 geometry, this sequence travels down one of the “spikes” of the unit ball: the functions get thinner and thinner, their area shrinking to zero despite their height remaining constant. In the L^∞ geometry, however, these functions are trapped on the surface of the unit sphere; they cannot enter the ϵ -ball because the L^∞ “box” does not extend down the “spikes” of concentration.

Consequences for Hilbert Spaces. The unique geometry of Hilbert spaces—specifically self-duality, isotropy, and the inner product—unlocks powerful results unavailable in general Banach spaces. The preceding analysis explains *why*: in L^2 , the duality map J_2 is the identity, making the musical isomorphism linear, the pairing symmetric, and the geometry flat. The following are key consequences we will explore:

1. *The Projection Theorem (Best Approximation)*: In general Banach spaces, finding the element in a subspace M closest to a point x is difficult (the closest point might not be unique, e.g., the flat edge of L^1). In Hilbert spaces, the strict convexity and “roundness” of the unit ball guarantee that for any closed subspace M , there exists a *unique* element $P_M x$ minimizing the distance $\|x - m\|$. Furthermore, the error vector $x - P_M x$ is orthogonal to the subspace.
2. *The Riesz Representation Theorem (Self-Duality)*: The duality map $J : H \rightarrow H^*$ is linear and bijective. This means we can completely identify linear functionals with vectors. Every continuous linear functional φ acts exactly like an inner product: $\varphi(x) = \langle x, y \rangle$ for a unique y . This collapses the study of H^* to the study of H .
3. *Orthonormal Bases and Fourier Series*: In L^p , we may have a Schauder basis⁴, but determining the coefficients is non-trivial. In Hilbert spaces, orthogonality allows us to decouple coefficients. This is key for the Fourier transform, where any element is reconstructed by projection: $x = \sum \langle x, e_n \rangle e_n$. Additionally, the *Parseval Identity* allows us to measure size by summing squared coefficients: $\|x\|^2 = \sum |\langle x, e_n \rangle|^2$.
4. *Existence of Adjoints*: In Banach spaces, the adjoint of $T : X \rightarrow Y$ maps $Y^* \rightarrow X^*$. Because Hilbert spaces are self-dual, we can pull this operator back to the original space. Thus, for every bounded operator T on a Hilbert space, there is a unique adjoint T^* on the *same space* satisfying $\langle Tx, y \rangle = \langle x, T^*y \rangle$. This is foundational for spectral theory.

In summary, while Banach spaces provide the necessary topological framework (completeness), Hilbert spaces add the rigid geometric structure required to generalize linear algebra to infinite dimensions.

⁴A sequence $\{b_n\}$ where every $v \in V$ has a unique expansion $v = \sum \alpha_n b_n$.

The duality map J_p , understood as the gradient of the p -energy, reveals the precise mechanism: it is the *cotangent structure* of the space—the infinitesimal measurement theory—that determines the geometry. The canonical evaluation pairing between tangent and cotangent vectors is the universal source of “derivative-like” phenomena; J_p is the geometric data that internalizes this pairing within the space itself. In L^2 alone, this internalization is linear, making vectors and covectors—directions and measurements—two aspects of a single object.

6.1 Basic definition and properties

Definition 6.1.1: Inner Product Space

Let H be a complex vector space with an inner product $\langle \cdot, \cdot \rangle : H \times H \rightarrow \mathbb{C}$ where

1. $\langle x, y \rangle = \overline{\langle y, x \rangle}$
2. $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$
3. $\langle ax, y \rangle = a\langle x, y \rangle$, $a \in \mathbb{C}$
4. $\langle x, x \rangle \in (0, \infty)$ for all nonzero $x \in H$
5. $\langle x, x \rangle = 0$ if and only if $x = 0$

We'll define $\|x\|_2 = \sqrt{\langle x, x \rangle}$ where the 2 subscript hints that this will become the L^2 -norm and is added to not hold ambiguity with the L_1 norm. This is naturally a norm since:

1. $\|x\|_2 = \sqrt{\langle x, x \rangle} = 0$ if and only if $x = 0$
2. $\|ax\|_2 = \sqrt{\langle ax, ax \rangle} = \sqrt{a\bar{a}\langle x, x \rangle} = \sqrt{a^2\langle x, x \rangle} = |a|\sqrt{\langle x, x \rangle} = |a|\|x\|_2$

Proving the triangle inequality is a bit more difficult: IF we square both sides, we it is equivalent to show:

$$\|x + y\|_2^2 \leq \|x\|_2^2 + 2\|x\|_2\|y\|_2 + \|y\|_2^2$$

expanding out the left hand side, we get:

$$\|x + y\|_2^2 = \|x\|_2^2 + 2\operatorname{Re}\langle x, y \rangle + \|y\|_2^2$$

so we have to show that $\operatorname{Re}\langle x, y \rangle \leq \|x\|_2\|y\|_2$. In fact, a bit more is true: $|\langle x, y \rangle| \leq \|x\|_2\|y\|_2$. Though we are using this inequality to prove the triangle inequality, notice that this in fact gives us a general bound on the value of the inner product! This inequality is one of the central tools used in the study of Hilbert spaces:

Theorem 6.1.2: Cauchy-Schwartz Inequality

$$|\langle x, y \rangle| \leq \|x\|_2\|y\|_2$$

Proof:

If $\|y\|_2 = 0$ then $y = 0$, and then $|\langle x, 0 \rangle| = 0$, so assume $\|y\|_2 \neq 0$. Then for all $t \in \mathbb{R}$

$$\begin{aligned}
 0 &\leq \langle x - ty, x - ty \rangle \\
 &= \|x\|_2^2 + 2t\operatorname{Re}\langle x, y \rangle + t^2\|y\|_2^2 \\
 &= \|x\|_2^2 + \frac{2\operatorname{Re}\langle x, y \rangle^2}{\|y\|_2^2} + \frac{\operatorname{Re}\langle x, y \rangle^2}{\|y\|_2^2} \quad t = \frac{\operatorname{Re}\langle x, y \rangle}{\|y\|_2^2} \\
 &\Rightarrow |\operatorname{Re}\langle x, y \rangle| \leq \|x\|_2\|y\|_2
 \end{aligned}$$

To finish the proof, pick $\alpha \in \mathbb{C}$ with $|\alpha| = 1$ such that

$$|\operatorname{Re}(\alpha\langle x, y \rangle)| = |\langle x, y \rangle|$$

Notice this does not break the inequality, and so we can repeat the entire process but with α taken into account with t , and so:

$$|\langle x, y \rangle| \leq \|x\|_2\|y\|_2$$

as we sought to show

With this, the rest of the proof that $\|\cdot\|_2$ induced by $\langle \cdot, \cdot \rangle$ is a norm follows

Theorem 6.1.3: Triangle Inequality For Induced Norm

$$\|x + y\|_2 \leq \|x\|_2 + \|y\|_2$$

Proof:

$$\begin{aligned}
 \|x + y\|_2^2 &= \langle x + y, x + y \rangle \\
 &\leq \|x\|_2^2 + 2\operatorname{Re}\langle x, y \rangle + \|y\|_2^2 \\
 &\leq \|x\|_2^2 + 2\|x\|_2\|y\|_2 + \|y\|_2^2 \\
 &= (\|x\|_2 + \|y\|_2)^2
 \end{aligned}$$

The Cauchy-Schwartz inequality is the tool we use to bring in geometry as we know it into Hilbert space. For example, in real inner product spaces, we will define the angle between two vectors to be θ where θ satisfies:

$$\langle u, v \rangle = \cos(\theta)\|u\|_2\|v\|_2$$

Many other cool properties of the Cauchy-Schwartz inequality can be found here: (commented)

Now, notice we can define a metric by doing $\rho(x, y) = \|x - y\|_2$:

1. $\rho(x, x) = \|x - x\|_2 = 0$ if and only if $x - x = 0$ so $x = x$
2. $\rho(x, y) = \|x - y\|_2 \geq 0$
3. $\rho(x, y) = \|x - y\|_2 = \|x - y + y - z\|_2 \leq \|x - y\|_2 + \|y - z\|_2 = \rho(x, y) = \rho(y, z)$

Thus, we can define a natural topology on any pre-Hilbert space. If that topology is complete, we give it a name:

Definition 6.1.4: Hilbert Space

Let H be an inner product space. Then H is a Hilbert space if it is *complete* with respect to the induced metric.

Example 6.1.5: Hilbert Spaces

1. $\mathbb{R}^n$ is a (real) Hilbert space with the inner product being:

$$\langle x, y \rangle = \sum_i x_i y_i$$

verifying this is a Hilbert space is left as an exercise.

2. $L^2_\mu(X)$ for a complex measure μ is a Hilbert space with the inner product being:

$$\langle f, g \rangle_{L^2} = \int_X f(x) \overline{g(x)} d\mu(x)$$

To show it's well-defined, recall that if $a, b \geq 0$:

$$2|ab| = 2|a||b| \leq |a|^2 + |b|^2$$

Thus:

$$|f(x) \overline{g(x)}| = |f(x)||g(x)| \leq |f(x)|^2 + |g(x)|^2$$

As integrating keeps inequality when integrating positive functions:

$$\|f\overline{g}\|_1 \leq \int_X |f(x)|^2 + \int_X |g(x)|^2 < \infty$$

showing it is well-defined. Alternatively, using Cauchy-Schwartz:

$$\|f\overline{g}\|_1 \leq \|f\|_2 \|g\|_2 < \infty$$

3. As an exercise, show that $\ell^2(\mathbb{Z})$ is a Hilbert Space.

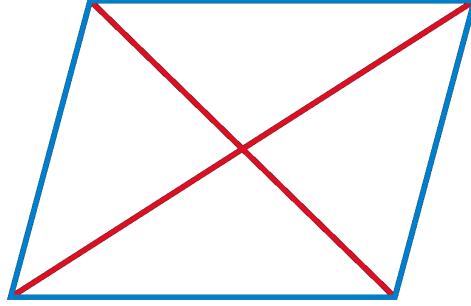
Hilbert spaces are very euclidean in nature, for example:

Theorem 6.1.6: Parallelogram Law

Let $\| - \|$ be a norm. Then $\| - \|$ is induced by an inner product if and only if:

$$\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$$

Visually, this means that the “area” of a parallelogram is can be measured in two ways:



Proof:

$$\begin{aligned}
 \|x + y\|^2 + \|x - y\|^2 &= \langle x + y, x + y \rangle + \langle x - y, x - y \rangle \\
 &= 2\langle x, x \rangle + \langle x, y \rangle - \langle x, y \rangle + \langle y, x \rangle - \langle y, x \rangle + 2\langle y, y \rangle \\
 &= 2\langle x, x \rangle + 2\langle y, y \rangle \\
 &= 2\|x\|^2 + 2\|y\|^2
 \end{aligned}$$

It turns out that a norm satisfying the parallelogram law will in fact define an inner product. This shows how norms define geometries that are more general than euclidean:

Example 6.1.7: Exploring Parallelogram Law

1. (The polarization Identity) $\mathcal{H}$ be a Hilbert space. For any $x, y \in \mathcal{H}$,

$$\langle x, y \rangle = \frac{1}{4}(\|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2)$$

Proof. This is simply a matter of computation:

$$\begin{aligned}
 &\frac{1}{4}(\|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2) \\
 &= \frac{1}{4}(\langle x + y, x + y \rangle - \langle x - y, x - y \rangle + i\langle x + iy, x + iy \rangle - i\langle x - iy, x - iy \rangle) \\
 &= \frac{1}{4}(\langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle - (\langle x, x \rangle - \langle x, y \rangle - \langle y, x \rangle + \langle y, y \rangle) \\
 &\quad + i(\langle x, x \rangle + \langle x, iy \rangle + \langle iy, x \rangle + \langle iy, iy \rangle - (\langle x, x \rangle - \langle x, iy \rangle - \langle iy, x \rangle + \langle iy, iy \rangle)) \\
 &= \frac{1}{4}(2\langle x, y \rangle + 2\langle y, x \rangle + i(2\langle x, iy \rangle + 2\langle iy, x \rangle)) \\
 &= \frac{1}{4}(2\langle x, y \rangle + 2\langle y, x \rangle + i(-i)2\langle x, y \rangle + 2ii\langle y, x \rangle) \\
 &= \frac{1}{4}(2\langle x, y \rangle + 2\langle y, x \rangle + 2\langle x, y \rangle - 2\langle y, x \rangle) \\
 &= \frac{1}{4}4\langle x, y \rangle \\
 &= \langle x, y \rangle
 \end{aligned}$$

□

2. If $\mathcal{H}'$ is another Hilbert space, a linear map from $\mathcal{H}$ to $\mathcal{H}'$ is unitary if and only if it is isometric and surjective

Proof. Let U be unitary so that $\langle Ux, Uy \rangle_2 = \langle x, y \rangle_1$ for any $x, y \in \mathcal{H}$. Letting $x = y$ gives us $\|Ux\|_2 = \|x\|_1$, giving us that it's an isometry. Since U is invertible, it is surjective, and so U is a surjective isometry.

Conversely, let U be a surjective isometry. We need to show its invertible (it suffices to show it's injective), an isometry, and that U and U^{-1} are bounded. Clearly, U is bounded since $\|U(x)\| \leq 1\|x\|$. For injectivity, let $Ux = Uy$, so $Ux - Uy = 0$. Taking the norm on both sides, we get:

$$0 = \|0\|_2 = \|Ux - Uy\|_2 = \|U(x - y)\|_2 = \|x - y\|_1$$

where the last equality is due to U being an isometry. Since $\|z\| = 0$ if and only if $z = 0$, we have $x - y = 0$, i.e. $x = y$, proving that U is injective. Along with surjectivity, that makes U invertible. With this, we can also show that U^{-1} is bounded: for any $y \in \mathcal{H}'$, there exists an $x \in \mathcal{H}$ such that $U(x) = y$. Thus:

$$\|U^{-1}(y)\| = \|x\| \stackrel{!}{=} \|U(x)\| = \|y\|$$

where $\stackrel{!}{=}$ comes from U being an isometry, showing that $\|U^{-1}(y)\| \leq 1 \cdot \|y\|$, making it also bounded.

Finally, to show that $\langle Ux, Uy \rangle_2 = \langle x, y \rangle_1$, we will use part (a):

$$\begin{aligned} \langle x, y \rangle &= \frac{1}{4}(\|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2) \\ &\stackrel{!}{=} \frac{1}{4}(\|U(x + y)\|^2 - \|U(x - y)\|^2 + i\|U(x + iy)\|^2 - i\|U(x - iy)\|^2) \\ &= \frac{1}{4}(\|U(x) + U(y)\|^2 - \|U(x) - U(y)\|^2 + i\|U(x) + iU(y)\|^2 - i\|U(x) - iU(y)\|^2) \\ &= \langle U(x), U(y) \rangle \end{aligned}$$

where $\stackrel{!}{=}$ comes from U being an isometry. Thus:

$$\langle Ux, Uy \rangle = \langle x, y \rangle$$

As we sought to show. □

Proposition 6.1.8: Inner Product is Continuous

Let V be a Hilbert space. Then $\langle -, - \rangle : V \times V \rightarrow \mathbb{R}$ (or $\mathbb{C}$) is continuous

Proof:

As $V \times V$ has the product topology, it is enough to show that if $x_n \rightarrow x$ and $y_n \rightarrow y$ in V then

$$\langle x_n, y_n \rangle \rightarrow \langle x, y \rangle$$

Take:

$$\begin{aligned}
 |\langle x, y \rangle - \langle x_n, y_n \rangle| &= |\langle x, y \rangle - \langle x_n, y \rangle + \langle x_n, y \rangle - \langle x_n, y_n \rangle| \\
 &= |\langle x - x_n, y \rangle + \langle x_n, y - y_n \rangle| \\
 &\leq |\langle x - x_n, y \rangle| + |\langle x_n, y - y_n \rangle| \leq \|x - x_n\| \|y\| + \|x_n\| \|y - y_n\| \\
 &\rightarrow 0
 \end{aligned}$$

where the last line comes since $x_n \rightarrow x$ and $y_n \rightarrow y$, completing the proof.

The next concept we consider is a way of defining two vectors being somehow “unrelated” or going in some notion of “direction” that doesn’t give information for the other vector:

Definition 6.1.9: Orthogonal

We say that x is *orthogonal* to y if $\langle x, y \rangle = 0$. We will write $x \perp y$

If two vectors are orthogonal, the triangle inequality becomes instant:

Theorem 6.1.10: Pythagoras

$$x \perp y \Rightarrow \|x + y\|^2 = \|x\|^2 + \|y\|^2$$

Proof:

$$\|x + y\|^2 = \langle x + y, x + y \rangle = \|x\|^2 + \|y\|^2$$

We might wonder if we take $E \subseteq H$, can we find all vectors in H that are somehow “dual” to E ? Define it’s *orthogonal compliment* to be:

$$E^\perp = \{x \in H \mid \langle x, y \rangle = 0 \ \forall y \in E\}$$

Notice that E is *always* a closed subspace:

Proof. The fact it’s a subspace follows from linearity of the second component.

for closed, suppose there is a sequence $\{x_n\} \subseteq E^\perp$ where $x_n \rightarrow x \in H$. Then:

$$|\langle x, y \rangle| \leq |\langle x - x_n, y \rangle| \leq \|x - x_n\| \|y\| \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

□

We can also find closed (and not closed) subspace of hilbert spaces:

Example 6.1.11: Subspaces Examples

1. If $\mathcal{H}$ is a finite-dimensional Hilbert space, then all subspace are closed
2. If $\mathcal{H}$ is infinite-dimensional, consider $\mathcal{H} = L^1([0, 1])$ with the L^2 inner product. and take $P([0, 1]) \subseteq L^1([0, 1])$ to be the subspace of all polynomial functions. This is certainly not equal to $L^1([0, 1])$, however polynomial functions are dense in $L^1([0, 1])$ (they can arbitrarily

approximate continuous bump characteristic functions, and hence simple functions). However, their closure is $L^1([0, 1])$, showing that $P([0, 1])$ cannot be closed.

3. Assume $E \subseteq \mathcal{H}$ is an open subspace. Then since $0 \in E$, there exists a $r > 0$ such that

$$B_r(0) \subseteq E$$

However, for any nonzero element $x \in \mathcal{H}$,

$$\frac{r}{2\|x\|}x \in B_r(0) \subseteq E$$

Since E is closed under scalar multiplication, this shows that $x \in E$, so $\mathcal{H} \subseteq E$, so $\mathcal{H} = E$, showing that the only open subspace of $\mathcal{H}$ is $\mathcal{H}$ itself.

Thus, we are interested in closed subspace since are closed under limits, meaning we can use our tools from analysis to get the following decomposition result:

Theorem 6.1.12: Hilbert Space Decomposition

Let $E \subseteq \mathcal{H}$ be a closed subspace. Then:

$$\mathcal{H} = E \oplus E^\perp$$

In particular, for any $x \in H$, there is a unique $y \in E$ and $z \in E^\perp$ such that $x = y + z$.

Proof:

Let $x \in \mathcal{H}$, and let $\delta = \inf \{\|x - y\| = \rho(x, y) \mid y \in E\}$ (which exists by completeness of $\mathbb{R}^n$ and $\mathcal{H}$). let $\{y_n\} \subseteq E$ be a sequence such that $\|x - y_n\| \rightarrow \delta$. We will show that $\{y_n\}$ is a cauchy sequence, and then find a limit point for it. By the parallelogram law:

$$2(\|x - y_n\| + \|x - y_m\|) = \|y_n - y_m\|^2 + \|(y_n + y_m) - 2x\|^2$$

Since $y_n + y_m \in E$, for sufficiently large N so that $n, m > N$:

$$\begin{aligned} \|y_n - y_m\|^2 &= 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - 4\left\|\frac{1}{2}(y_n + y_m) - x\right\|^2 \\ &\leq 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - 4\delta^2 \end{aligned}$$

and as $n, m \rightarrow \infty$, this sequence goes to $4\delta^2 - 4\delta^2 = 0$, and so $\|y_n - y_m\| \rightarrow 0$, meaning $\{y_i\}_i$ is a cauchy sequence. Take $y = \lim y_n$. Then since E is closed, $y \in E$. By construction $\|x - y\| = \delta$. Now let $z = x - y$.

I claim that $z \in E^\perp$, that is $\langle z, u \rangle = 0$ for all $u \in E$. So, consider $\langle u, z \rangle$ for any $u \in E$. If $\langle u, z \rangle$ is complex, multiply it by a [complex] unit-length scalar to make it real ($\langle u, z \rangle = 0$ if and only if $a\langle u, z \rangle = 0$). With the fact that it's real, we will do a sneaky trick where we take advantage of optimization of differentiable functions. In particular, let

$$f : \mathbb{R} \rightarrow \mathbb{R} \quad f(t) = \|z + tu\|^2 = \|z\|^2 + 2t\langle z, u \rangle + \|u\|^2 t^2$$

f is certainly a differentiable function, being a polynomial with a square-root on $\mathbb{R}$. Furthermore, $z = x - y$ is the minimal value of $\|x - y\|$, and so the minimal value of $\|x - y\|^2$. It follows

that $z + ut$ is not minimal, and so this equation has a minimum when $t = 0$. Thus, we get $0 = f'(0) = 2\langle u, z \rangle$, implying that $\langle u, z \rangle = 0$. Since u was arbitrary, we get that $z \in E^\perp$.

To show that z is the minimal distance from x , let's say there was another $z' \in E^\perp$ that satisfied the equation. Then by the Pythagorean theorem:

$$\begin{aligned} \|x - z'\| &= \|(x - z) + (z - z')\|^2 \\ &= \|x - z\|^2 + \|z - z'\|^2 & x - y \in E, z - z' \in E^\perp \\ &\geq \|x - z\|^2 \end{aligned}$$

So equality holds if and only if $z = z'$, meaning z is of minimal distance to x . A similar reasoning shows the same for y .

To show uniqueness, let's say $y + z = x = y' + z'$. Then $y - y' = z - z' \in E \cap E^\perp = \{0\}$, so $y - y' = z - z' = 0$, which can be used to show that $y = y'$ and $z = z'$, completing the proof.

We said that orthogonal vector are somehow dual to each other. However, there is the more classical notion of “dual” vector spaces. Let $\mathcal{H}^*$ be set of all linear functions from $\mathcal{H}$ to the underlying field (usually $\mathbb{C}$, but sometimes $\mathbb{R}$); such linear functions are called functionals. It turns out that all linear functionals can be written in the form $f_y(x) = \langle x, y \rangle$, and we can define a norm on $\mathcal{H}^*$ such that $\|f_y\| = \|y\|$. Thus, $y \mapsto f_y$ will define a conjugate-linear isometry from $\mathcal{H}$ to $\mathcal{H}^*$. The following theorem shows that this map is surjective:

Theorem 6.1.13: Natural Isomorphism To $\mathcal{H}^*$ (Riesz Representation)

Let $f \in \mathcal{H}^*$. Then there exists a $y \in \mathcal{H}$ such that $f(x) = \langle x, y \rangle$ for all $x \in X$.

Proof:

It is easy to check uniqueness: if $\langle x, y \rangle = \langle x, y' \rangle$ for all x , then let $x = y - y'$. Manipulating the equation, we get that $\|y - y'\| = 0$, so $y = y'$.

If f is the zero functional, then take $y = 0$. If otherwise, then let $\mathcal{M} = \{x \mid f(x) = 0\}$ be the kernel of f . It is clearly a subspace, and any sequence will also necessarily converge within $\mathcal{M}$, and so it is also closed. Furthermore, $\mathcal{M}$ is a proper subspace, since $f \neq 0$. Thus, $\mathcal{M}^\perp \neq \{0\}$ by the previous proof.

Now, pick any $z \in \mathcal{M}^\perp$ with $\|z\| = 1$. Then if we set $u = f(x)z - f(z)x$, we have that $u \in \mathcal{M}$. Hence:

$$0 = \langle u, z \rangle = f(x)\|z\|^2 - f(z)\langle x, z \rangle = f(x) - \langle x, \overline{f(z)}z \rangle$$

re-arrange and setting $y = \overline{f(z)}z$, we get

$$f(x) = \langle y, z \rangle$$

as we sought to show.

The uniqueness makes this map to be injective, and by linearity of f and the first component of the inner product makes it an isomorphism. Furthermore, since $\|f_y\| = \|y\|$, limits are also preserved, and so this in fact defines an isomorphism between $\mathcal{H} \cong \mathcal{H}^*$. Notice how this differs from vector-spaces where there is no natural isomorphism between V and V^* (there is one between V and V^{**}).

Since we are working over a vector-space, we should be able to define a basis. Since we are studying infinite dimensional vector-spaces, we often care more about “countable linear combination” rather than “finite linear combination” as we have for vector-spaces. To distinguish the two, we will call *Hamel basis* a set of linearly independent elements for which all *finite linear combinations* can represent every element of the vector space. We will call a *Schauder basis* a set of linearly independent elements for which all *countable linear combinations* can represent every element of the vector space. If every element in a Schauder basis has norm 1 and is orthogonal, we call it an *orthonormal basis*. Since we have an inner-product that is complete, it would be nice if this vector-space has an orthonormal basis. We strive to this end in the following build-up:

Theorem 6.1.14: Bessel’s Inequality

If $\{u_\alpha\}_{\alpha \in A}$ is an orthonormal set in $\mathcal{H}$, then for any $x \in \mathcal{H}$:

$$\sum_{\alpha \in A} |\langle x, u_\alpha \rangle|^2 \leq \|x\|^2$$

As a consequence, $\{\alpha \mid \langle x, u_\alpha \rangle \neq 0\}$ is countable.

Proof:

It suffices to show that $\sum_{\alpha \in F} |\langle x, u_\alpha \rangle|^2 \leq \|x\|^2$ for any finite subset $F \subseteq A$, since if it fails for some countable amount, it will fail for some finite amount. Then we see that:

$$\begin{aligned} 0 &\leq \left\| x - \sum_{\alpha \in F} \langle x, u_\alpha \rangle u_\alpha \right\|^2 \\ &= \|x\|^2 - 2\operatorname{Re} \left\langle x, \sum_{\alpha \in F} \langle x, u_\alpha \rangle u_\alpha \right\rangle + \left\| \sum_{\alpha \in F} \langle x, u_\alpha \rangle u_\alpha \right\|^2 \\ &= \|x\|^2 - 2 \sum_{\alpha \in F} |\langle x, u_\alpha \rangle|^2 + \sum_{\alpha \in F} |\langle x, u_\alpha \rangle|^2 && \text{Pythagorean Theorem} \\ &= \|x\|^2 - \sum_{\alpha \in F} |\langle x, u_\alpha \rangle|^2 \end{aligned}$$

and manipulating the final equation we get:

$$\sum_{\alpha \in F} |\langle x, u_\alpha \rangle|^2 \leq \|x\|^2$$

as we sought to show

This in particular is important for uncountable sets. In the following we establish when equality holds, and thus when an orthonormal set is an orthonormal basis

Theorem 6.1.15: Orthonormal Basis Condition

Let $\{u_\alpha\}_{\alpha \in A}$ be an orthonormal set in $\mathcal{H}$. Then the following are equivalent:

1. Finite linear combinations of $\{u_\alpha\}$ are dense in $\mathcal{H}$.
2. (Completeness) If $\langle x, u_\alpha \rangle = 0$ for all α , then $x = 0$
3. For each $x \in \mathcal{H}$, $x = \sum_{\alpha \in A} \langle x, u_\alpha \rangle u_\alpha$ where the right hand side sum has only countably many nonzero terms and converges in the norm topology no matter the order of the terms
4. (Parseval's Identity) $\|x\|^2 = \sum_{\alpha \in A} |\langle x, u_\alpha \rangle|^2$ for all $x \in \mathcal{H}$

Proof:

- (1) $\Rightarrow$ (2) Assume $\langle f, u_\alpha \rangle = 0$. As finite linear combinations of $\{u_\alpha\}$ are dense in $\mathcal{H}$, for any $f \in \mathcal{H}$ there is a sequence g_n such that $\|f - g_n\| \xrightarrow{n \rightarrow \infty} 0$. As $\langle f, u_\alpha \rangle = 0$, then $\langle f, -g_n \rangle = 0$. By bi-linearity and Cauchy-Schwartz:

$$\|f\|^2 = \langle f, f \rangle = \langle f, f - g_n \rangle \leq \|f\| \|f - g_n\| \xrightarrow{x \rightarrow \infty} 0 \quad (6.1)$$

hence $f = 0$

- (2) $\Rightarrow$ (3) As the nonzero portion of the sum is countable, we can reduce to the countable case. For any $f \in \mathcal{H}$, let:

$$S_n(f) = \sum_{k=1}^n a_k u_k \quad a_k = \langle f, u_k \rangle$$

Let's first show $S_n(f) \rightarrow g$ for some $g \in \mathcal{H}$, then show $g = f$. First, we show an alternative proof of Bessel's identity with a satisfying trick. Notice that certainly $S_n(f) \perp (f - S_n(f))$. Thus:

$$\|f\|^2 = \|S_n(f)\|^2 + \|f - S_n(f)\|^2 = \sum_{k=1}^n a_k^2 + \|f\|^2$$

Taking $n \rightarrow \infty$ we get:

$$\|f\|^2 \geq \sum_{k=1}^{\infty} a_k^2$$

Importantly, $\sum_{k=1}^{\infty} a_k^2$ converges, and hence $S_n(f)$ forms a Cauchy sequence in $\mathcal{H}$, so by completeness $S_n(f) \rightarrow g$. To show $f = g$, notice that for any k , for sufficiently large n :

$$\langle f - S_n(f), u_k \rangle = a_k - a_k = 0$$

Thus:

$$\langle f - g, u_k \rangle = 0 \quad \forall k$$

so by (2), $f - g = 0$, so $f = g$.

(3) $\Rightarrow$ (4) Assuming (3) so that $\|f - S_n(f)\| \rightarrow 0$ so plugging this into equation (6.1):

$$\|f\|^2 = \sum_k |a_k|^2$$

(4) $\Rightarrow$ (1) Notice that $S_n(f)$ is a finite linear combinations elements u_k , hence using equation (6.1) again we get the desired implication.

Any orthonormal set satisfying these conditions is given a name:

Definition 6.1.16: Orthonormal Basis

Let $\{u_\alpha\}_{\alpha \in A}$ be an orthonormal set. Then if the set satisfies any of the 3 conditions in theorem 6.1.15, it is called an *orthonormal basis*.

An argument similar to the one for vector-spaces gives that every Hilbert spaces has an orthonormal basis, essentially an orthonormal basis is an orthonormal set that is maximal:

Theorem 6.1.17: Hilbert Space Basis

Every Hilbert space has an orthonormal basis.

Proof:

Application of Zorn's lemma: The collection of ordered orthonormal sets ordered by inclusion has a maximal element, and maximality is equivalent to the first property of theorem 6.1.15.

Example 6.1.18: Orthonormal Basis

Given a Hilbert space $L^2(G)$ where $G = \mathbb{R}^n$ or $\mathbb{T}^n$, we would want to find an orthonormal basis to get decomposition of functions. **go over all of these!**

1. Legendre basis (can be thought of as “global” Taylor approximation) (must add integrate image. Also a good example of orthonormal basis that is not the Fourier one)

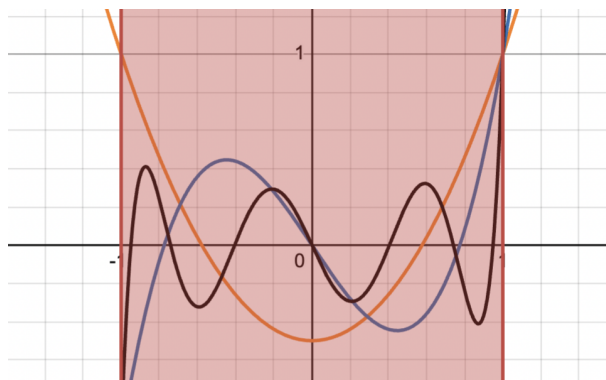


Figure 6.1: Legendre polynomials integrate to 0

2. Hermite basis
3. Jacobi Basis
4. Chebyshev

6.1.19 Basis On Banach Space Without an inner product, we cannot define an orthonormal basis on L^p . We can ask if we can at least define a *Schauder basis*, i.e any $f \in L^p$ is a (potentially) linear of a *single* sequence $\{b_n\}$ with different weights:

$$f = \sum a_n b_n$$

This was actually a very difficult problem and is related to the following question: is every compact operator on a Banach space the limit of finite operators. It isn't too hard to check this is the case in Hilbert spaces (exercise [ref:HERE](#)). However, this was proven in the negative by Per Enflo [2] (for a refinement of the proof see [4]). A consequence of this is that not every Banach space has a Schauder basis [1]. Another consequence of this paper for Banach spaces is that not all Banach space embed into $\ell^p(\mathbb{Z})$, $L^p(\mathbb{R}^d)$, or $L^p(K)$ for a compact set K ; the family of Banach spaces is in fact larger (ex. $C(K)$ Is a Banach space not belonging to any of these spaces) .

Now that we have established a basis for Hilbert spaces, we will separate them into 3 categories: Those with finite, countable, and uncountable basis. In most cases, we will be working with countable basis due to the following property:

Proposition 6.1.20: Separable Hilbert Spaces

A Hilbert space $\mathcal{H}$ is separable if and only if it has a countable orthonormal basis, in which case every orthonormal basis for $\mathcal{H}$ is countable.

Proof:

p. 176

Though all separable Hilbert spaces are “equivalent” (isometric), it is often the specific manifestation of the Hilbert space that matters as it frames how operators on the space transform functions (ex. the choice of basis, the choice of inner product).

Theorem 6.1.21: Enflo’s Rigidity Theorem

A Banach space is uniformly homeomorphic to a Hilbert space if and only if it is linearly isomorphic to a Hilbert space.

Proof:

This was a seminal result by Per Enflo (1969) [3]. It settled the question of whether the “uniform” structure of a space determines its linear structure in the context of Hilbert spaces. Since L^p (for $p \neq 2$) is not isomorphic to L^2 , this theorem implies they are not uniformly homeomorphic either.

6.1.22 Hierarchy of Sameness It is useful to summarize how “same” two infinite-dimensional separable spaces can be. Let X and Y be infinite-dimensional separable Banach spaces (e.g., L^p and L^2).

1. Topological Homeomorphism : They are homeomorphic by the *Anderson-Kadec Theorem*, i.e. all such spaces are homeomorphic to $\mathbb{R}^{\mathbb{N}}$. They are topologically indistinguishable.
2. Uniform Homeomorphism: Generally no. By *Enflo’s Rigidity Theorem*, if X is a Hilbert space and Y is not linearly isomorphic to it (like $L^p, p \neq 2$), they are *not* uniformly homeomorphic. *Note:* Their unit balls *are* uniformly homeomorphic via the **Mazur Map**, but this does not extend to the whole space.
3. Linear Isomorphism (The “Algebraic” level): Is there a linear bijection? Depends. L^p is not isomorphic to L^q for $p \neq q$.
4. Isometric Isomorphism (The “Geometry” level): They don’t they preserve distance. The geometry of the unit ball (round vs. diamond vs. box) is strictly preserved here.

We next talk about the isomorphisms in the category of Hilbert spaces:

Definition 6.1.23: Unitary Map

Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be two Hilbert spaces with inner products $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$, and let $U : \mathcal{H}_1 \rightarrow \mathcal{H}_2$. Then U is called an *isometry* if it is a linear map that preserves inner products:

$$\langle Ux, Uy \rangle_2 = \langle x, y \rangle_1$$

If it is also bijective, it is a *unitary map*.

Note that by taking $x = y$, we get that $\|x\|_1 = \|Ux\|_2$, i.e. it is an isometry. In fact, the converse is true too: every surjective isometry is unitary. With the definition of an isomorphism, we can see that all Hilbert spaces are isomorphic to $\ell^2(A)$ given basis A :

Theorem 6.1.24: Classifying Hilbert Spaces

Let $\{e_\alpha\}_{\alpha \in A}$ be an orthonormal basis for $\mathcal{H}$. Then the correspondence

$$x \mapsto \hat{x} \quad \hat{x}(\alpha) = \langle x, u_\alpha \rangle$$

is a unitary map from $\mathcal{H}$ to $\ell^2(A)$.

Proof:

We’ll show it’s a surjective isometry, which is equivalent to it being unitary.

First off, the map $x \rightarrow \hat{x}$ is linear since the inner product is linear in each component, and it is an isometry by Parseval identity: $\|x\|^2 = \sum |\hat{x}(x)|^2$.

For surjectivity, let $f \in \ell^2(A)$, so $\sum_{\alpha \in A} |f(\alpha)|^2 < \infty$, so the Pythagorean shows that all partial sums of the series $\sum f(\alpha)u_\alpha$ are Cauchy^a, so $x = \sum_{\alpha} f(\alpha)u_\alpha$ exists in $\mathcal{H}$ and $\hat{x} = f$.

^aNote that only countably many terms are nonzero

As an application that ties together several threads from previous chapters, we close off with the proof of a theorem of Grothendieck showing that the Hilbert space structure of L^2 imposes rigid constraints on subspaces of L^p . The result says, roughly, that a closed subspace of L^p cannot consist entirely of bounded functions unless it is finite-dimensional. The proof combines the closed graph theorem from the Baire category section (theorem 1.2.10), the L^p inclusion theory (proposition 2.1.13), and the Bessel inequality developed above.

Theorem 6.1.25: Grothendieck's Theorem on Closed Subspaces of L^p

Let $(X, \mathcal{F}, \mu)$ be a finite measure space, i.e. $\mu(X) < \infty$. Suppose that E is a closed subspace of $L^p(X, \mu)$ for some $1 \leq p < \infty$, and that $E \subseteq L^\infty(X, \mu)$. Then E is finite-dimensional.

Proof:

Step 1: The L^∞ norm is controlled by the L^p norm on E . Consider the inclusion $\iota : (E, \|\cdot\|_p) \rightarrow L^\infty(X, \mu)$ defined by $\iota(f) = f$. We verify that its graph is closed. Suppose $f_n \rightarrow f$ in L^p with $f_n \in E$, and simultaneously $f_n \rightarrow g$ in L^∞ . Since E is closed in L^p , $f \in E$. Convergence in L^p gives $f_{n_k} \rightarrow f$ a.e. along a subsequence, while convergence in L^∞ gives $f_n \rightarrow g$ a.e. Hence $f = g$ a.e., showing the graph is closed. By the closed graph theorem (theorem 1.2.10), ι is bounded: there exists $C > 0$ such that:

$$\|f\|_\infty \leq C\|f\|_p \quad \text{for all } f \in E$$

Step 2: E carries a Hilbert space structure. Since $\mu(X) < \infty$ and $E \subseteq L^\infty$, the L^2 norm is finite on E . We show the L^p and L^2 norms are equivalent on E .

For one direction: $\|f\|_2^2 = \int |f|^2 = \|f\|_\infty^2 \mu(X) \leq C^2 \|f\|_p^2 \mu(X)$, giving $\|f\|_2 \leq C\mu(X)^{1/2} \|f\|_p$.

For the reverse: if $p \leq 2$, then Hölder directly gives

$$\|f\|_p \leq \mu(X)^{1/p-1/2} \|f\|_2$$

If $p > 2$, then interpolate:

$$\|f\|_p^p \leq \|f\|_\infty^{p-2} \|f\|_2^2 \leq (C\|f\|_p)^{p-2} \|f\|_2^2$$

so that $\|f\|_p^2 \leq C^{p-2} \|f\|_2^2$, i.e. $\|f\|_p \leq C^{(p-2)/2} \|f\|_2$.

In either case, the L^p and L^2 norms are equivalent on E . Since E is complete in $\|\cdot\|_p$, it is complete in $\|\cdot\|_2$, so $(E, \langle \cdot, \cdot \rangle_{L^2})$ is a Hilbert space. Write C' for a constant satisfying $\|f\|_\infty \leq C'\|f\|_2$ for all $f \in E$ (obtained by combining Step 1 with the norm equivalence).

Step 3: Bounding the dimension via Bessel's inequality. Let $\{e_1, \dots, e_N\}$ be any orthonormal set in $(E, \|\cdot\|_2)$. Note that N is independent of the choice of C' ; this will be the key fact to bound $\dim(E)$. For almost every $x \in X$, the evaluation functional $\varphi_x : E \rightarrow \mathbb{C}$ defined by $\varphi_x(f) = f(x)$ satisfies:

$$|\varphi_x(f)| \leq \|f\|_\infty \leq C'\|f\|_2$$

so φ_x is a bounded linear functional on the Hilbert space E with $\|\varphi_x\| \leq C'$. By the Riesz representation theorem (theorem 6.1.13), there exists $K_x \in E$ with $\varphi_x(f) = \langle f, K_x \rangle$ and

$\|K_x\|_2 = \|\varphi_x\| \leq C'$. In particular,

$$\varphi_x(e_n) = e_n(x) = \langle e_n, K_x \rangle$$

for each n . Applying Bessel's inequality (theorem 6.1.14) to K_x :

$$\sum_{n=1}^N |e_n(x)|^2 = \sum_{n=1}^N |\langle K_x, e_n \rangle|^2 \leq \|K_x\|_2^2 \leq C'^2$$

Integrating over X and using $\|e_n\|_2 = 1$:

$$N = \sum_{n=1}^N \|e_n\|_2^2 = \int_X \sum_{n=1}^N |e_n(x)|^2 d\mu \leq C'^2 \mu(X)$$

Since N was the size of an arbitrary orthonormal set, $\dim(E) \leq C'^2 \mu(X) < \infty$.

The element K_x appearing in the proof is a *reproducing kernel*: it allows one to recover pointwise values of functions in E via the inner product. The argument shows that the existence of such a kernel, together with the finiteness of the measure, forces the space to be finite-dimensional. The bound $\dim(E) \leq C'^2 \mu(X)$ is quantitative: the larger the ratio $\|f\|_\infty / \|f\|_2$ can be on E , the fewer orthonormal directions the space can accommodate.

It is worth pausing to appreciate how three seemingly unrelated tools conspire here. The closed graph theorem (theorem 1.2.10) upgrades the set-theoretic inclusion $E \subseteq L^\infty$ into a norm inequality. The L^p interpolation inequalities (proposition 2.1.13) transfer this to equivalence of all L^q norms on E , revealing a hidden Hilbert space. And Bessel's inequality (theorem 6.1.14) converts the uniform pointwise bound into a dimension bound via integration.

6.2 Operators on Hilbert Spaces

In this section, we try to generalize many notion's we've seen before, including:

1. adjoints, self-adjoint, and normal operators
2. sign and absolute value and when can we decompose a function into this form
3. diagonalizable
4. polar decomposition (think of how for every $z \in \mathbb{C}$, $z = \arg(z)|z|$)

Adjoints

Hilbert spaces have very simple dual structure, since $\mathbb{H} \cong \mathbb{H}^*$ in a very natural way with the inner product of $\mathbb{H}$. Thus, the problem of finding the structure of dual spaces is fully finished for hilbert spaces. Due to this very simple dual structure, we will be able to properly define an “opposite” operator given any operator $T : \mathbb{H}_1 \rightarrow \mathbb{H}_2$. In essence, there is an interesting symmetry in the structure of any map: Given a map $T : \mathbb{H}_1 \rightarrow \mathbb{H}_2$ between two hilbert spaces, there is a natural dual

map $T^* : \mathbb{H}_2^* \rightarrow \mathbb{H}_1^*$. Hence, we have

$$(-)^* : \text{Hom}(\mathbb{H}_1, \mathbb{H}_2) \rightarrow \text{Hom}(\mathbb{H}_2^*, \mathbb{H}_1^*)$$

and since in Hilbert spaces $\mathbb{H}_i \cong \mathbb{H}_i^*$, we may envision this map as “flipping” the direction of arrows in a natural way:

$$(-)^* : \text{Hom}(\mathbb{H}_1, \mathbb{H}_2) \rightarrow \text{Hom}(\mathbb{H}_2, \mathbb{H}_1)$$

Not only does this map exist, but it is an isometric isomorphism (in **Ban**)! We will start with the special case of $\text{Hom}(\mathbb{H}, k) = \mathbb{H}^*$. We will find that $(-)^*$ is a map

$$(-)^* : \text{Hom}(\mathbb{H}, \mathbb{H}) \rightarrow \text{Hom}(\mathbb{H}, \mathbb{H})$$

The following shows how we can define such a dual and some consequences:

Lemma 6.2.1: Endomorphism and And Inner Products

Let $\mathbb{H}$ be a complex hilbert space. Then there is a bijective isometric correspondence between $\text{End}_k(\mathbb{H})$ and bounded sesquilinear forms on $\mathbb{H}$ given by the map $T \rightarrow B_T$ where

$$B_T(x, y) = \langle x, T(y) \rangle$$

Proof:

Let $T \in \text{End}_k(\mathbb{H})$. Then certainly B_T is a sesquilinear form on $\mathbb{H}$ bounded by $\|T\|$ since

$$\begin{aligned} \|B_T\| &= \sup \{ |B_T(x, y)| \mid \|x\| \leq 1, \|y\| \leq 1 \} \\ &= \sup \{ |\langle x, T(y) \rangle| \mid \|x\| \leq 1, \|y\| \leq 1 \} \\ &\leq \|T\| \end{aligned}$$

On the other hand

$$\begin{aligned} \|T(x)\|^2 &= \langle T(x), T(x) \rangle \\ &= B_T(T(x), x) \\ &\leq \|B_T\| \|T(x)\| \|x\| \\ &\leq \|B_T\| \|T\| \|x\|^2 \end{aligned}$$

showing that $\|T\| \leq \|B_T\|$, and so $\|T\| = \|B_T\|$.

Theorem 6.2.2: Existence Of Adjoint

Let $T \in \text{End}_k(\mathbb{H})$. Then there exists a unique T^* in $\text{End}_k(\mathbb{H})$ such that

$$\langle T(x), y \rangle = \langle x, T^*(y) \rangle$$

for all $x, y \in \mathbb{H}$. The map $T \mapsto T^*$ is conjugate linear, contravariant with respect to multiplication, an isometry of $\text{End}_k(\mathbb{H})$ onto itself, an involution, and satisfies

$$\|T^*T\| = \|T\|^2$$

Proof:

Fix $T \in \text{End}_k(\mathbb{H})$, and let $B_T(x, y)$ be a continuous sesquilinear form defined as $(x, y) \mapsto \langle T(x), y \rangle$. Then by lemma 6.2.1, there exists a $T^* \in \text{End}_k(\mathbb{H})$ such that $T^* \rightarrow B_T$. In particular

$$B_T(x, y) = \langle x, T^*(y) \rangle$$

Thus,

$$\langle T(x), y \rangle = \langle x, T^*(y) \rangle$$

giving us existence of such an operator. We now establish all the properties named in the theorem. By conjugate symmetry

$$\begin{aligned} \langle x, T^{**}(y) \rangle &= \langle T^*(x), y \rangle \\ &= \overline{\langle y, T^*(x) \rangle} \\ &= \overline{\langle T(y), x \rangle} \\ &= \langle x, T(y) \rangle \end{aligned}$$

Thus, T^{**} and T correspond to the same sesquilinear form, thus $T^{**} = T$. This also shows that $T \mapsto T^*$ is an involution. Furthermore:

$$\langle x(ST)^*y \rangle = \langle S(T(x)), y \rangle = \langle T(x), S^*(y) \rangle = \langle x, T^*(S^*(y)) \rangle$$

thus $(S \circ T)^* = T^* \circ S^*$ showing $(-)^*$ is a contravariant endofunctor. Since the operator norm is submultiplicative, we have $\|T^*T\| \leq \|T\|\|T^*\|$. On the other hand, applying $\langle T(x), y \rangle = \langle x, T^*(y) \rangle$ to T^* we get

$$\|T(x)\|^2 = \langle T(x), T(x) \rangle = \langle T^*(T(x)), x \rangle = \|T^*T\|\|x\|^2$$

showing that $\|T\|^2 \leq \|T^*T\|$

. Combining these we get $\|T\| \leq \|T^*\|$, hence $\|T\| = \|T^*\|$ (since $T^{**} = T$). This finishes the proof.

Note that if $\mathbb{H}$ is a finite dimensional complex Hilbert space, then each operator T has a matrix representation, say (α_{ij}) . Then $T^* = (\overline{\alpha_{ji}})$, i.e. $T^* = \overline{T^t}$ where T^t is the transpose given the matrix representation.

Definition 6.2.3: Adjoint And Self-Adjoint

Let $T \in B(\mathbb{H})$ be a continuous linear operator between Hilbert spaces. Then T^* is called the *adjoint* of T . If $T = T^*$, then T is said to be *self adjoint* (or *Hermitian*).

Proposition 6.2.4: Adjoint Properties I

Let $T \in \text{End}_k(\mathbb{H})$. Then

$$\ker(T^*) = (T(\mathbb{H}))^\perp$$

Proof:

Since $\langle T(x), y \rangle = \langle x, T^*(y) \rangle$ we have $y \in \ker(T^*)$ when $y \in (T(\mathbb{H}))^\perp$. Conversely, if $y \in (T(\mathbb{H}))^\perp$, then $T^*(y) \in \mathbb{H}^\perp = \{0\}$.

For ease of reference, we shall list some of the results we have proven in the theorem:

Proposition 6.2.5: Adjoint Properties II

Let $T, S \in B(\mathbb{H})$. Then:

1. $(T + S)^* = T^* + S^*$
2. $(cT)^* = \bar{c}T^*$
3. $(TS)^* = S^*T^*$
4. $(T^*)^* = T$
5. $I^* = I$

Proof:

If you're not convinced of any of these, double check the theorem or try proving it again.

Proposition 6.2.6: Adjoint Properties III

Let $T \in \text{End}_k(\mathbb{H})$. Then the following are equivalent:

1. T is invertible (so $T^{-1} \in \text{End}_k(\mathbb{H})$)
2. T^* is invertible
3. T and T^* are bounded away from zero.
4. T and T^* are injective, and $T(\mathbb{H})$ is closed
5. T is injective and $T(\mathbb{H}) = \mathbb{H}$

Proof:

Since T is invertible, $T^{-1}T = TT^{-1} = I$, and so $(T^{-1})^*$ is the inverse of T^* , giving $(1) \Leftrightarrow (2)$.

For $(1) \Rightarrow (3)$, we say that an operator F is bounded away from zero if

$$\|x\| \leq C\|F(x)\|$$

for some constant C . In our case:

$$\|x\| = \|T^{-1} \circ T(x)\| \leq \|T^{-1}\| \|T(x)\|$$

Since $(1) \Leftrightarrow (2)$, the same holds for T^* .

For $(3) \Rightarrow (4)$, We have $\|T(x)\| \geq \epsilon\|x\|$ and $\|T^*(x)\| \geq \epsilon\|x\|$ for some $\epsilon > 0$ and all $x \in \mathbb{H}$. Thus certainly both T and T^* are injective: if $T(x) = T(y)$ then

$$0 = \|T(x) - T(y)\| = \|T(x - y)\| \geq \epsilon\|x - y\|$$

hence $\|x - y\| = 0$, so $x = y$. Furthermore, $\|T(x) - T(y)\| \geq \|x - y\|$ implies that $T(\mathbb{H})$ is complete, hence a closed subspace of $\mathbb{H}$.

For (4) $\Rightarrow$ (5), by corollary ?? ($\overline{Y} = (Y^\perp)^\perp$) and proposition 6.2.4 we get

$$\overline{T(\mathbb{H})} = (T(\mathbb{H})^\perp)^\perp = (\ker(T^*))^\perp = \{0\}^\perp = \mathbb{H}$$

where the last equality come from T^* bein injective.

Finally, for (5) $\rightarrow$ (1), we use the open mapping theorem.

This allows us to generialize this result to the followin

Proposition 6.2.7: Inversing Maps

Let $\mathbb{H}_1$ and $\mathbb{H}_2$ be hilbert spaces and consider an operator $T \in \text{Hom}_k(\mathbb{H}_1, \mathbb{H}_2)$. Then there exists a unique operator $T^* \in B(\mathbb{H}_2, \mathbb{H}_1)$ such that

$$\langle T(x), y \rangle = \langle x, T^*(y) \rangle$$

show that the map $T \rightarrow T^*$ is conjugate linear isometry of $\text{Hom}_k(\mathbb{H}_1, \mathbb{H}_2)$ to $\text{Hom}_k(\mathbb{H}_2, \mathbb{H}_1)$ and satisfies

$$\|T^*T\| = \|T^2\| = \|TT^*\|$$

Proof:

Let

$$\hat{T} = \begin{pmatrix} 0 & 0 \\ T & 0 \end{pmatrix} \in \text{End}_k(\mathbb{H}_1 \oplus \mathbb{H}_2)$$

By theorem 6.2.2, there is a unique function $\hat{T}^*$ such that

$$\langle \hat{T}(x_1, x_2), (y_1, y_2) \rangle = \langle (x_1, x_2), \hat{T}^*(y_1, y_2) \rangle$$

By definition of $\hat{T}$, we see

$$\hat{T}^* = \begin{pmatrix} 0 & T^* \\ 0 & 0 \end{pmatrix}$$

Hence we have that T^* is a function with domain and codomain $T^* : \mathbb{H}_2 \rightarrow \mathbb{H}_1$. It follows that

$$\langle T(x), y \rangle = \langle x, T^*(y) \rangle$$

and by theorem 6.2.2, T and T^* respect all the desired properties, completing the proof.

For the rest of this seciton, we will focus on $T \in B(\mathbb{H})$ inseat of $T \in B(\mathbb{H}_1, \mathbb{H}_2)$. The reason for this is to find suitable decompositino theorems of T generializing the notion of eigenvalues.

Recall that over finite dimensional vector spaces, $TT^* = T^*T$ implied T was diagonalizable, with the usual special case of $T = T^*$ (self-adjoint) being looked into. We would like to know if this notion of diagonalizability for normal operators remains true for infinite dimensional Hilbert spaces. We shall see that this is not the case, some more conditions is required (for ex. a special case of digaonalizable operators will be comapct normal operators). Before we get into the technicalities, I want to put a word on how I think about $TT^* = T^*T$. In my mind, when two functions commute, it means that

one function is somehow “invariant” with respect to the other. Think of PDE’s with equations that commute with the differential operator that makes it invariant to the Lorentzian metric. In this case, I like to think of T being somehow “symmetric”. Since $TT^*(x) = T^*T(x)$ for all $x \in \mathbb{H}$, and T^* is in some ways the dual (think of [ref:HERE](#), the generalization), we see that T is “invariant” to the effect of the dual T^* . We shall see in a moment that the invariant to the dual is translated to T and T^* being metrically identical. In the simplest case of $T = T^*$, it is evident why this is the case. If $T = T^*$, we can deepen our intuition with complex numbers: if we consider $T : \mathbb{C} \rightarrow \mathbb{C}$ where $T(z) = z$, then $T^*(z) = \bar{z}$. Hence, $T = T^*$ if and only if $T(\mathbb{C}) \subseteq \mathbb{R}$, since only the real numbers are invariant under conjugation.

Definition 6.2.8: Normal Operator

Let $T \in \text{End}_k(\mathbb{H})$. Then T is said to be *normal* if it commutes with its adjoint, i.e. if $T^*T = TT^*$.

As an immediate consequence, we have

$$\|T(x)\| = \langle T^* \circ T(x), x \rangle^{1/2} = \langle T \circ T^*(x), x \rangle^{1/2} = \|T^*x\|$$

showing that T and T^* are metrically identical. In fact, by the polarization law (also known as polarization identity) if $T \in \text{End}_k(\mathbb{H})$ such that T and T^* are metrically identical, we can show $\langle T^* \circ T(x), y \rangle = \langle T \circ T^*(x), y \rangle$ for all x, y and so T is normal by [lemma 6.2.1](#). By [proposition 6.2.6](#), an operator is normal if and only if it is bounded away from zero.

Positive Operators

Definition 6.2.9: Positive Operator

Let $T \in \text{End}_k(\mathbb{H})$. Then T is said to be *positive* if $T = T^*$ and $\langle T(x), x \rangle \geq 0$ for all $x \in \mathbb{H}$. If $k = \mathbb{C}$, the positive condition implies self-adjointness by the polarization identity.

Certainly, the sum of two positive operators is positive (geometrically, positive operators form a cone in $\text{End}_k(\mathbb{H})$). However the product is not positive, and need not be self-adjoint.

Example 6.2.10: Positive And Non-Positive Operators

here

However, if the operators commute (if the product is self-adjoint), we see that there is a product that is positive since $(ST = S^{1/2}TS^{1/2} \geq 0$ by what we’ll show in the next proposition)

Let $B(\mathbb{H})_{sa} \subseteq \text{End}_{\mathbb{R}}(\mathbb{H})$ be the collection of self-adjoint operators. Then this is a closed subspace, this subspace intersected with the cone of positive operators defines an order $S \leq T$ if $T - S \geq 0$. By taking the case of 2 dimensional real subspaces and $\text{End}_{\mathbb{R}}(\mathbb{R}^2)$, this order is not total, not even a lattice order.

We require the following lemma to gain some important results (which will be an immediate consequence of the spectral theorem in [ref:HERE](#))

Lemma 6.2.11: Square Root Lemma

If $S \leq T$ in $B(\mathbb{H})_{sa}$, then $A^*SA \leq A^*TA$ for every $A \in \text{End}_k(\mathbb{H})$. If moreover $S \leq T$, then $\|S\| \leq \|T\|$

Proof:

Since $S \leq T$, we know that $\langle (T - S)(y), y \rangle \geq 0$ for every $y \in \mathbb{H}$. Replacing y with $A(x)$ for appropriate $x \in \mathbb{H}$, it follows that $A^*(T - S)A \geq 0$, so $A^*SA \leq A^*TA$.

If $0 \leq S \leq T$, by the Cauchy-Schwarz inequality on the positive sesquilinear form $\langle S(\cdot), \cdot \rangle$, we get that for every pair of unit vectors $x, y \in \mathbb{H}$

$$|\langle S(x), y \rangle|^2 \leq \langle S(x), x \rangle \langle S(y), y \rangle \leq \langle T(x), x \rangle \langle T(y), y \rangle \leq \|T\|^2$$

Hence, $\|S\|^2 \leq \|T\|^2$ by lemma 6.2.1.

Lemma 6.2.12: Approximation Lemma

There is a sequence $(p_n) \subseteq \mathbb{R}[x]$ of polynomials with positive coefficients such that $\sum p_n$ converges uniformly on $[0, 1]$ to

$$t \mapsto 1 - (1 - t)^{1/2}$$

Proof:

This is immediate from real analysis, but we can also construct it explicitly. If $q_0 = 0$ and $q_n(t) = \frac{1}{2}(t + q_{n-1}(t)^2)$, then we can see that each coefficient of q_n is positive and that $q_n(t) \leq 1$ for every $t \in [0, 1]$. Moreover, since

$$2(q_{n+1} - q_n) = q_n^2 - q_{n-1}^2 = (q_n - q_{n-1})(q_n + q_{n-1})$$

it follows by induction that each polynomial $p_n = q_n - q_{n-1}$ has positive coefficients. Taking the q_n as elements of $C([0, 1])$, we see they converge pointwise to $[0, 1]$ to a function q such that $2q(t) = t + q(t)^2$. Thus, $q(t) = 1 - (1 - t)^{1/2}$. Since $q \in C([0, 1])$ and $[0, 1]$ is compact, by the monotone convergence $q_n \nearrow q$ is in fact uniform by Dini's lemma, hence

$$\sum p_n = \lim q_n = q$$

completing the proof.

Proposition 6.2.13: Square-Rooting Positive Operator

Let $T \in \text{End}_k(\mathbb{H})$ be a positive operator. Then there is a unique positive operator $T^{1/2}$ such that $(T^{1/2})^2 = T$. Moreover, $T^{1/2}$ commutes with every operator commuting with T .

Proof:

Analysis Now p.92

Proposition 6.2.14: Invertible Positive Operator

A positive operator T is invertible in $\text{End}_k(\mathbb{H})$ if and only if $T \geq \epsilon I$ for some $\epsilon > 0$. In that case, $T^{-1} \geq$ and $T^{1/2}$ is invertible and $(T^{-1})^{1/2} = (T^{1/2})^{-1}$. If $T \leq S$, then $S^{-1} \leq T^{-1}$.

Proof:

Analaysis now p.94

Projection

Let $W \subseteq \mathbb{H}$ be a closed subspace of a hilbert space H . Then we know that $\mathbb{H} = W + W^\perp$. In particular, for eah $y \in \mathbb{H}$, we have $y = w + w^\perp$, $w \in W$ and $w^\perp \in W^\perp$. Since

$$\alpha y_1 + \beta y_2 = (\alpha x_1 + \beta x_2) + (\alpha x_1^\perp + \beta x_2^\perp)$$

is the decomposition of a linear combination, it follows that $P : \mathbb{H} \rightarrow \mathbb{H}$ given by $P(y) = x$ is an operator in $\text{End}_k(\mathbb{H})$ with $\|P\| \leq 1$. Note P is idempotent ($P^2 = P$), self-adjoint, and positive since

$$\langle P(y_1), y_2 \rangle = \langle x_1, x_2 + x_2^\perp \rangle = \langle x_1, x_2 \rangle = \langle x_1 + x_1^\perp, x_2 \rangle = \langle y_1, P(y_2) \rangle$$

and

$$\langle P(y), y \rangle = \langle x, x + x^\perp \rangle = \|x\|^2 \geq 0$$

Definition 6.2.15: (Orthogonal) Projection

The map P in the above is called the *(orthogonal) projection*.

If P is a self-adjoint and idempotent in $\text{End}_k(\mathbb{H})$, then

$$W = \{x \in \mathbb{H} \mid P(x) = x\} = P(\mathbb{H})$$

is a closed subsapce of $\mathbb{H}$. If $w^\perp \in W^\perp$, we have (since $P = P^*$) that

$$\|P(w^\perp)\|^2 = \langle w^\perp, P^2(w^\perp) \rangle = 0$$

Thus, P is an orthogonal projection of $\mathbb{H}$ onto W . Note that $I - P$ is the projection on $W^\perp$.

Projections are some of the east operators we may construct, and can be thought as the “characteristic function” from real analysis. The analogue of simple function is then obtained by decomposing $\mathbb{H}$ into a finite orthogonal sum

$$\mathbb{H} = W_1 \oplus W_2 \oplus \cdots \oplus W_n$$

corresponding to the projections $P_1, \dots, P_n$ which are pairwise orthogonal ($P_i P_j = 0$ for $i \neq j$ satisfying $\sum P_i = I$). We may then consider

$$T = \sum \lambda_n P_n$$

for scalars $\lambda_1, \lambda_2, \dots, \lambda_n \in k$. This may look familiar to the notion of *diagonalization*. Indeed, we may generalize the concept to infintie dimensions like so:

Definition 6.2.16: Diagonalizable

Let $T \in \text{End}_k(\mathbb{H})$ be an operator. Then T is said to be *diagonalizable* if there is an orthonormal basis $\{e_i \mid i \in I\}$ for $\mathbb{H}$ and a bounded set $\{\lambda_i \mid i \in I\} \subseteq k$ such that

$$T(x) = \sum \lambda_i \langle x, e_i \rangle e_i$$

The number $\langle x, e_i \rangle$ are the coordinates for x in the basis $\{e_i\}$ and each λ_i is an *eigenvalue* for T corresponding to the eigenvector e_i . Denote P_i to be the projection of $\mathbb{H}$ onto ke_i , we have $P_i(x) = \langle x, e_i \rangle e_i$, so we may write the operator as:

$$T = \sum \lambda_i P_i$$

Note that the above sum is not necessarily convergent in the norm topology on $\text{End}_k(\mathbb{H})$, but only pointwise convergent (i.e. convergent in the strong operator topology). If T is diagonalizable, then T^* is diagonalizable with

$$T^*(x) = \sum \overline{\lambda_i} \langle x, e_i \rangle e_i$$

with eigenvalues $\{\overline{\lambda_i}\}$. Furthermore, if T is diagonalizable, $T^*T = TT^*$ (with eigenvalues $|\lambda_i|^2$ with basis $\{e_i\}$), and hence T is normal. T is self-adjoint if and only if all λ_i are real and T is positive if and only if $\lambda_i \geq 0$ for all i . If $\mathbb{H}$ is finite-dimensional, every normal operator is diagonalizable. This is not generally true for infinite dimensions. In general, most “interesting” (normal) operators on infinite-dimensional Hilbert spaces are nondiagonalizable, and must be handled with a continuous analogue of the concept of a basis (this will be the spectrum, see section 6.3 and chapter 7).

Unitary Operators

Recall $T : \mathbb{H} \rightarrow \mathbb{H}$ is unitary if T is an isometric isomorphism from $\mathbb{H}$ onto itself (if $k = \mathbb{R}$ we say it is an *orthogonal operator*). By the polarization identity, we see that $U^*U = UU^* = I$, so that U is normal and invertible with $U^{-1} = U^*$. Conversely, if $U \in \text{End}_k(\mathbb{H})$ such that $U^{-1} = U^*$ then U is unitary since

$$\|U(x)\|^2 = \langle U^*U(x), x \rangle = \|x\|^2$$

So U is an isometry and is surjective since U is invertible. If $\{e_i \mid i \in I\}$ is an orthonormal basis for $\mathbb{H}$ and U is unitary, then $\{Ue_i \mid i \in I\}$ is a basis for $\mathbb{H}$. Conversely, we saw that every transition between orthonormal bases is given by a unitary operator. For future reference denote the group of unitary operators $U(\mathbb{H}) \subseteq \text{End}_k(\mathbb{H})$ (it should be evident it forms a group).

Definition 6.2.17: Unitary Equivalent

Let $S, T \in \text{End}_k(\mathbb{H})$. Then we say that S and T are *unitarily equivalent* if $S = UTU^*$ for some unitary operator U .

It should be clear that unitary equivalent preserves norm, self-adjointness, normality, diagonalizability, and unitarity.

Definition 6.2.18: Partial Isometry

Let $U \in \text{End}_k(\mathbb{H})$. Then U is said to be a *partial isometry* if there is a closed subspace $X \subseteq \mathbb{H}$ such that $U|_X$ is isometric and $U|_{X^\perp} = 0$

This immediately implies that $X^\perp = \ker(U)$, hence $X = \overline{U^*(\mathbb{H})}$. Setting $P = U^*U$, then

$$\langle P(x), x \rangle = \|U(x)\|^2 = \|x\|^2$$

for every $x \in X$, hence $P(x) = x$ by the Cauchy-Schwarz inequality. Furthermore, $P(x^\perp) = 0$ for every $x^\perp \in X^\perp$, so P is a projection of $\mathbb{H}$ onto X . Conversely, if $U \in \text{End}_k(\mathbb{H})$ such that $U^*U = P$ for some projection P , taking $X = P(\mathbb{H})$ and see from the equation $\|U(x)\|^2 = \langle P(x), x \rangle$ that U is isometric on X and 0 on $X^\perp$, and so U is a partial isometry. Using this, we can also show that U^* is a partial isometry. Since $U(I - P) = 0$,

$$(UU^*)^2 = UU^*UU^* = UPU^* = UU^*$$

so UU^* is self-adjoint idempotent, i.e. a projection. Note that U^* is the projection of $U(\mathbb{H})$ back onto $U^*(\mathbb{H})$ so that U^* is a *partial inverse* of U .

Be weary that there exist isometries that are not unitary (since they need not be surjective). For example, if $\mathbb{H}$ is a separable and has countable basis $\{e_n \mid n \in \mathbb{N}\}$, we have

$$S\left(\sum \alpha_n e_n\right) = \sum \alpha_n e_{n+1}$$

Then S (the unilateral shift operator) is an isometry of $\mathbb{H}$ onto the subspace $\{e_1\}^\perp$ and consequently S^* is a partial isometry of $\{e_1\}^\perp$ onto $\mathbb{H}$. In particular, $S^*S = I$, whereas SS^* is the projection onto $\{e_1\}^\perp$.

Theorem 6.2.19: Polar Decomposition of T

Let $T \in \text{End}_k(\mathbb{H})$. Then there is a unique positive operator $|T| \in \text{End}_k(\mathbb{H})$ such that

$$\|T(x)\| = \||T|(x)\|$$

and $|T| = (T^*T)^{1/2}$. Moreover, there is a unique partial isometry U with kernel $\ker(U) = \ker(T)$ and $U \circ |T| = T$. In particular

$$U^*U|T| = |T| \quad U^*T = |T| \quad UU^*T = T$$

Proof:

Analysis Now p.96

We may think of U as the generalized “sign” of T and $|T|$ as the generalized “absolute value” of T . By the formula

$$T^* = |T|U^* = U^*(U|T|U^*)$$

and the uniqueness of polar decomposition, we see that U^* is the sign of T^* , whereas $U|T|U^*$ is the absolute value of T^* . In general, the absolute value of a sum or product of noncommuting operators

doesn't have much relation to the sum or product of the absolute values. Similarly, the sign (in the finite dimensional case), can't always be chosen to be unitary (the unilateral shift is an immediate counter example). For T to have the form $U|T|$ for some unitary operator U , it is necessary and sufficient that the closed subspaces $\ker(T)$ and $\ker(T^*)$ have the same dimension (finite or infinite). Easy cases of this general theorem are established in the next result

Proposition 6.2.20: Polar Decomposition For Isomorphisms

Let $T \in \text{End}_k(\mathbb{H})$ be invertible. Then the partial isometry in its polar decomposition is unitary

Proof:

Analysis Now p.97

Proposition 6.2.21: Polar Decomposition For Normal Operators

Let $T \in \text{End}_k(\mathbb{H})$ be normal. Then there is a unitary operator W commuting with T, T^* , and $|T|$ such that $T = W|T|$

Proof:

Analysis Now p.97

I want to add one more thing, I just don't know where: for any $T \in B(\mathbb{H})$, $T = T_1 + iT_2$ for self-adjoint operators T_1, T_2 . This lets us generalize the spectral theorem for normal operators, since we will know it for self-adjoint operator.

Lemma 6.2.22: lemma I

If $T = T^*$ and $\|T\| \leq 1$, the operator $U = T + i(I - T^2)^{1/2}$ is unitary and $T = \frac{1}{2}(U + U^*)$

Proof:

Analysis Now p.97

Lemma 6.2.23: Lemma II

Let $S \in \text{End}_k(\mathbb{H})$ with $\|S\| < 1$. Then for every unitary U , there are unitaries U_1, V_1 such that

$$S + U = U_1 + V_1$$

Proof:

Analysis Now p.98

Theorem 6.2.24: Russo-Dye-Gardner Theorem

Let $T \in \text{End}_k(\mathbb{H})$ with

$$\|T\| < 1 - \frac{2}{n} \quad n > 2$$

Then there are unitary operators $U_1, U_2, \dots, U_n$ such that

$$T = \frac{1}{n}(U_1 + U_2 + \dots + U_n)$$

Proof:

Analysis Now p.98

Since the self-adjoint elements span $\text{End}_k(\mathbb{H})$, we see from lemma 6.2.22 that every $T \in \text{End}_k(\mathbb{H})$ is the linear combination of (at least 4) unitary operators. The estimate in the Russo-Dye-Gardner Theorem is much more precise with respect to the norm, and shows that every element in the open unit ball of $\text{End}_k(\mathbb{H})$ is a convex combination (indeed, a mean) of unitary operators. On the surface of the ball this need not be the case, in fact the unilateral shift cannot be expressed as a convex combination of unitaries (exercise 3.2.8 in Gert K. Pederson Analysis Now).

Proposition 6.2.25: Numerical Radius Of Operator

Let $\mathbb{H}$ be a complex hilbert space and $T \in \text{End}_{\mathbb{C}}(\mathbb{H})$. Then we deifn eth *numerical radius* of T as

$$|||T||| = \sup \{ \langle T(x), x \rangle \mid x \in \mathbb{H}, \|x\| \leq 1 \}$$

Then $\|\cdot\|$ is a norm on $\text{End}_{\mathbb{C}}(\mathbb{H})$ satisfying $\frac{1}{2}\|\cdot\| \leq |||\cdot||| \leq \|\cdot\|$. More over, $|||T^2||| \leq |||T|||^2$ for every T and $|||T||| = \|T\|$ if T is normal

Proof:

Analysis Now p.99

6.3 Compact Operators

Throughout this section, let $\mathbb{H}$ be a hilbert space over $k \in \{\mathbb{R}, \mathbb{C}\}$ unless otherwise specified.

Definition 6.3.1: Finite Rank Operator

Let $T \in \text{End}_k(\mathbb{H})$. Then T is said to have *finite rank* if $T(\mathbb{H})$ is finite-dimensional subspace of $\mathbb{H}$. Denote by $B_f(\mathbb{H}) \subseteq \text{End}_k(\mathbb{H})$ the set of all finite rank opeartors.

Clearly, $B_f(\mathbb{H})$ is a subspace, even a subalgebra, even an ideal of $\text{End}_k(\mathbb{H})$. If $T \in B_f(\mathbb{H})$, by proposition 6.2.4, we get an orthogonal decomposition $\mathbb{H} = T(\mathbb{H}) \oplus \ker(T^*)$, which shows that $T^*(\mathbb{H}) = T^*T(\mathbb{H})$, so T^* has finite rank. Thus, $B_f(\mathbb{H})$ is *self-adjoint* ideal in $\text{End}_k(\mathbb{H})$ ($(B_f(\mathbb{H}))^* = B_f(\mathbb{H})$ as a set). The subset $B_f(\mathbb{H}) \subseteq \text{End}_k(\mathbb{H})$ is similar in relation to $C_c(X) \subseteq C_b(X)$ when X is

a locally compact Hausdorff space. In particular, these classes describe the local phenomena on $\mathbb{H}$ and on X . Passing to a limit in norm may destroy the “locality”, but enough structure is preserved to make these “quasilocal” operators and functions very useful and interesting. We shall study the closure of $B_f(\mathbb{H})$ in this section as a noncommutative analogue of $C_0(X)$ in function theory.

Lemma 6.3.2: Net And Compact Operators

There is a net $(P_\lambda)_{\lambda \in \Lambda}$ of projection in $B_f(\mathbb{H})$ such that $\|P_\lambda(x) - x\| \rightarrow 0$ for each $x \in \mathbb{H}$.

that is, the projection are “dense” in $B_f(\mathbb{H})$.

Proof:

We want to find a net P_n such that $\|P_n(x) - x\| \rightarrow 0$. Let $\{e_i \mid i \in I\}$ be an orthonormal basis for $\mathbb{H}$ and let N be a net of finite subsets of I ordered by inclusion. For each $n \in N$, let P_n denote the projection of $\mathbb{H}$ onto $\text{span}\{e_i \mid i \in n\}$ so that $(P_n)_{n \in N}$ is a net in $B_f(\mathbb{H})$.

Now, if $x \in \mathbb{H}$, then $x = \sum a_i e_i$, thus

$$\|P_n(x) - x\|^2 = \sum_{i \notin n} |a_i|^2$$

By Parseval’s identity, this tends to zero, completing the proof.

Theorem 6.3.3: Closed Unit Balls In Hilbert Spaces

Let $B \subseteq \mathbb{H}$ be a closed unit ball in a hilbert space $\mathbb{H}$. Then the following conditions on an operator $T \in \text{End}_k(\mathbb{H})$ are equivalent

1. $T \in \overline{B_f(\mathbb{H})}$
2. $T|_B$ is a weak-norm continuous function from B into $\mathbb{H}$
3. $T(B)$ is compact in $\mathbb{H}$
4. $\overline{T(B)}$ is compact in $\mathbb{H}$
5. each net in B has a subnet whose image under T converges in $\mathbb{H}$

Proof:

Analysis Now p.105

Definition 6.3.4: Compact Operators

Let $T \in B(\mathbb{H})$. Then T is compact if $T(B)$ is compact. The set of compact operators is denoted by $B_0(\mathbb{H})$ to show that these operators “vanish at infinity”.

Lemma 6.3.5: Diagonalizable Operator Is Compact

Let $T \in \text{End}_k(\mathbb{H})$. Then T is compact if and only if its eigenvalues $\{\lambda_i \mid i \in I\}$ belongs to $c_0(I)$

Proof:

Let $T(x) = \sum \lambda_i \langle x, e_i \rangle e_i$. If $T \in B_0(\mathbb{H})$, then for any $\epsilon > 0$, define

$$I_\epsilon = \{i \in I \mid |\lambda_i| \geq \epsilon\}$$

If I_ϵ is infinite, the net $(e_i)_{i \in I_\epsilon}$ converges weakly to zero for any well-ordering of I_ϵ since $\langle e_i, x \rangle \rightarrow 0$ by Parseval's identity. Since $\|T(e_i)\| = |\lambda_i| \geq \epsilon$ for $i \in I_\epsilon$ this contradicts theorem 6.3.3(3). Thus, I_ϵ must be finite for each $\epsilon > 0$, hence λ_i must vanish at infinity.

Conversely, if I_ϵ is finite for all $\epsilon > 0$, let

$$T_\epsilon = \sum_{i \in I_\epsilon} \lambda_i \langle \cdot, e_i \rangle e_i$$

Then T_ϵ has finite rank, and

$$\begin{aligned} \|(T - T_\epsilon)(x)\|^2 &= \left\| \sum_{i \notin I_\epsilon} \lambda_i \langle x, e_i \rangle e_i \right\|^2 \\ &= \sum_{i \notin I_\epsilon} |\lambda_i|^2 |\langle x, e_i \rangle|^2 \\ &\leq \epsilon^2 \|x\|^2 \end{aligned}$$

hence, $\|T - T_\epsilon\| \leq \epsilon$, thus $T \in B_0(\mathbb{H})$ by theorem 6.3.3(1), completing the proof.

Lemma 6.3.6: Eigen Vectors Between Adjoint

Let x be an eigenvector for a normal operator $T \in \text{End}_k(\mathbb{H})$ corresponding to an eigenvalue λ . Then x is an eigenvector for T^* corresponding to the eigenvalue $\bar{\lambda}$. Eigenvectors for T corresponding to different eigenvalues are orthogonal.

Proof:

Note that the operator $T - \lambda I$ is normal and its adjoint is $T^* - \bar{\lambda} I$. Conversely,

$$\|(T^* - \bar{\lambda} I)(x)\| = \|(T - \lambda I)(x)\| = 0$$

If $T(y) = \gamma y$ where $\gamma \neq \lambda$, then we may certainly assume $\lambda \neq 0$ so that

$$\begin{aligned} \langle x, y \rangle &= \lambda^{-1} \langle T(x), y \rangle \\ &= \lambda^{-1} \langle x, T^*(y) \rangle \\ &= \lambda^{-1} \langle x, \bar{\gamma} y \rangle \\ &= \lambda^{-1} \gamma \langle x, y \rangle \end{aligned}$$

Hence, $\langle x, y \rangle = 0$, showing they are different eigenvectors, completing the proof.

Lemma 6.3.7: Normal Compact Operator, then Eivenvalue Exists

Every normal, compact operator T on a complex Hilbert space $\mathbb{H}$ has an eigenvalue λ with $|\lambda| = \|T\|$

Proof:

Analysis now p. 107

Theorem 6.3.8: Normal Compact, Then Diagonalizable

Every normal compact operator $T \in \text{End}_k(\mathbb{H})$ is diagonalizable and its eigenvalues (counted-with multiplicity) vanish at infinite. Conversely, each such operator is normal and compact.

Proof:

Analysis now p.108

It will be convenient for future sections to introduce a new notation: Let $x \odot y$ represent the rank one operator in $\text{End}_k(\mathbb{H})$ determined by the vectors $x, y \in \mathbb{H}$ and the formula

$$(x \odot y)(z) = \langle z, y \rangle x$$

The map $(x, y) \mapsto x \odot y$ is sesquilinear map of $\mathbb{H} \times \mathbb{H}$ into $B_f(\mathbb{H})$. If $\|e\| = 1$ then $e \odot e$ is the one-dimensional projection of $\mathbb{H}$ on $\mathbb{C}e$. Every normal compact operator $\mathbb{H}$ can now be written by the above theorem in the form

$$T = \sum \lambda_i e_i \odot e_j$$

For suitable orthonormal basis $\{e_i \mid i \in I\}$. The sum will converge in norm, since either the set $I_0 = \{i \in I \mid \lambda_i \neq 0\}$ is finite (so that $T \in B_f(\mathbb{H})$) or else countably infinite in which case the sequence $\{\lambda_i \mid i \in I_0\}$ converges to zero.

Definition 6.3.9: Spectrum

We say that the compact set

$$\sigma(T) = \{\lambda_i \mid i \in I_0\} \cup \{0\}$$

is the *spectrum* of T .

For every continuous function f on $\sigma(T)$, define

$$f(T) = \sum f(\lambda_i) e_i \odot e_i$$

Then $f(T)$ is compact if and only if $f(0) = 0$ and the map $f \mapsto f(T)$ is an isometric $*$ -preserving homomorphism of $C(\sigma(T))$ into $\text{End}_k(\mathbb{H})$. If $f(z) = \sum \alpha_{nm} z^n \bar{z}^m$ is a polynomial of two commuting variables z and $\bar{z}$, then $f(T) = \sum \alpha_{nm} T^n (T^*)^m$. This result is the *spectral (mapping) theorem* for normal compact operators. In the next chapter, we shall generalize this to any normal operator.

Calkin Algebra and Index

Since $B_0(\mathbb{H}) \subseteq B(\mathbb{H})$ is a closed ideal, we may define $B(\mathbb{H})/B_0(\mathbb{H})$ with the quotient norm⁵. This quotient is given a name

Definition 6.3.10: Calkin Algebra

Let $B_0(\mathbb{H}) \subseteq B(\mathbb{H})$. Then the algebra

$$B(\mathbb{H})/B_0(\mathbb{H})$$

is called the *Calkin Algebra* of $\mathbb{H}$

We shall see that many properties of $B(\mathbb{H})$ are conveniently expressed in terms of the Calkin algebra.

Definition 6.3.11: Compact Perturbation

If S and T are elements in $B(\mathbb{H})$ and $S - T \in B_0(\mathbb{H})$, we say that S is a *compact perturbation* of T .

This is another way of saying that S and T have the same image in the Calkin algebra. Such “local” perturbations happen frequently in application and properties of an operator that are invariant under compact perturbations are highly used. The best known of these we shall study now is known as the *index*

Theorem 6.3.12: Atkinson’s Theorem

Let $T \in B(\mathbb{H})$. Then the following are equivalent:

1. There is a unique operator $S \in B(\mathbb{H})$ such that ST and TS are the projections on $(\ker(T))^\perp$ and $\ker(T^*)^\perp$, respectively, and both projections have finite co-rank
2. For some operator $S \in B(\mathbb{H})$ both operators $ST - I$ and $TS - I$ are compact
3. the image of T is invertible in the Calkin algebra $B(\mathbb{H})/B_0(\mathbb{H})$
4. Both $\ker(T)$ and $\ker(T^*)$ are finite-dimensional subspaces and $T(\mathbb{H})$ is closed

Proof:

Analysis now p.109

In essence, Fredholm operators are operators that are almost invertible up to some finite-dimension being ignored, in particular they are invertible modulo compact operators.

Definition 6.3.13: Fredholm Operator

The operators satisfying the conditions in Atkinson’s theorem are called *Fredholm operators*, and the class of them is denoted $F(\mathbb{H})$.

⁵This is even a C^* -algebra

Definition 6.3.14: Index

For each $T \in F(\mathbb{H})$, define the *index* of T to be

$$\text{index}(T) = \dim(\ker(T)) - \dim(\ker(T^*))$$

If we choose S for T as in (1) of Atkinson's theorem and write $suchthat = I - P$ and $TS = I - Q$ with P and Q projection of rank, then another way of writing this would be

$$\text{index}(T) = \text{rank}(P) - \text{rank}(Q)$$

By (3) of Atkinson's Theorem, the product of Fredholm operators is again a Fredholm operator. In particular, every product $RT \in F(\mathbb{H})$ if $T \in F(\mathbb{H})$ and R is invertible in $B(\mathbb{H})$, and in this case

$$\text{index}(RT) = \text{index}(TR) = \text{index}(T)$$

since R is bijective. It should also be clear that $T^* \in F(\mathbb{H})$ if $T \in F(\mathbb{H})$ with $\text{index}(T^*) = -\text{index}(T)$. Finally, the partial inverse S to T in (1) of Atkinson's theorem is a Fredholm operator with $\text{index}(S) = -\text{index}(T)$. F

Definition 6.3.15: Indexed Fredholm Operators

For each $n \in \mathbb{Z}$ define

$$F_n(\mathbb{H}) = \{T \in F(\mathbb{H}) \mid \text{index}(T) = n\}$$

All of these subsets are nonempty, since if S is the unilateral shift, then $S^n \in F_{-n}(\mathbb{H})$ and $(S^*)^n \in F_n(\mathbb{H})$ for every $n > 0$.

Lemma 6.3.16: Finite Rank And 0 Index

If $A \in B_f(\mathbb{H})$, then $I + A \in F_0(\mathbb{H})$

Proof:

Analysis now p.110

Corollary 6.3.17: Fredholm Alternative

If $A \in B_0(\mathbb{H})$ and $\lambda \in \mathbb{C} \setminus \{0\}$, then either $\lambda I - A$ is invertible in $B(\mathbb{H})$ or λ is an eigenvalue of A with finite multiplicity, in which case $\bar{\lambda}$ is an eigenvalue for A^* with the same multiplicity

Proof:

Let $T = I - \lambda^{-1}A$. Since $T \in F_0(\mathbb{H})$, by lemma 6.3.16 we know that $T(\mathbb{H})$ is closed and that

$$\dim(\ker(T)) = \dim(\ker(T^*)) < \infty$$

If these dimension are 0, then $T(\mathbb{H}) = (\ker(T^*))^\perp = \mathbb{H}$ and $\ker(T) = \{0\}$, hence T is invertible by proposition 6.2.6, completing the proof.

This lemma implies that the spectrum of a compact operator (i.e. those λ such that $\lambda I - T$ is not invertible, as we'll see in the next chapter) consists of $\{0\}$ and the eigenvalues for T . IN particular, it is a countable subset of $\mathbb{C}$ with 0 as the only possible accumulatpion point.

Theorem 6.3.18: Index Invariant Under Compact Operator

Let $T \in F(\mathbb{H})$ and A be a compact operator. Then

$$T + A) = T)$$

Proof:

Analysis now p.111

Proposition 6.3.19: Fredholm Class Is Open

Each Fredholm class $F_n(\mathbb{H})$ for each $n \in \mathbb{H}$ is open in $B(\mathbb{H})$

Proof:

analysis now p.112

Proposition 6.3.20: Grading Structure Of Freidholm Operators

If $T_1 \in F_n(\mathbb{H})$ and $T_2 \in F_m(\mathbb{H})$ then $T_1 T_2 \in F_{n+m}(\mathbb{H})$

Proof:

Analysis now p. 112

We close with an interesting topological fact that brings some light to the index .Let G denote the group of invertible elements of the Calkin algebra $B(\mathbb{H})/B_0(\mathbb{H})$. and G_0 is the connected component of the identity (which is the group generated by the elements of the form $\exp(A)$ for some $A \in B(\mathbb{H})/B_0(\mathbb{H})$). Then G_0 is an open and closed subgroup of G and G/G_0 is a discrete group that labels the connected components in G . In this case, G/G_0 is isomorphic to $\mathbb{Z}$, and the index of a Fredholm operator T is the image of T under the compsoed quotinet maps from $F(\mathbb{H})$ to G and from G to G/G_0 ($= \mathbb{Z}$)

6.4 Trace

6.5 *Intuition to add

This is a conversation I had with Gemini that I really want to incorporate somewhere as it gave some good insights. I will save it here for now:

Zak Katz: Let $F(x_1, \dots, x_n, t)$ represent a PDE w.r.t time. Let's say $u(x, t)$ represent a solution

and $f_t(x) = u(x, t)$ is a “state” at time 0. Let S_t be the “time-shift” operator so that $S_t(f_a) = f_{a+t}$ for some state f_a at time a .

My question is: Is S_t always self-adjoint? If yes, can you prove it, and if not, can you give a counter-example?

Raycast AI:

Tool: `web_search`

No, the time-shift operator S_t is not always self-adjoint.

In fact, for many common PDEs (like the wave equation, transport equation, or Schrödinger equation), S_t is almost never self-adjoint. It is usually unitary (energy-conserving) or a contraction semi-group (dissipative).

Spectral Theory

In short, Spectral theory generalizes the notion of eigen vectors and values to infinite dimensional operators. In the previous chapter, we saw how there is a notion of diagonalizability for compact normal operators. In this chapter, we extend this result to normal operators. We also along the way learn some important dualities such as the Gelfand duality which has some useful applications, and is an interesting motivator for classical algebraic geometry.

7.1 Banach Algebras

Recall the definition of a banach algebra

Definition 7.1.1: Banach Algebra Revisited

A *Banach Algebra* is an algebra A with a submultiplicative norm (so that it is continuous) st the unerlying vector space is a banach space.

All the same applies for Banach Algebras as we expect. Note that for A/I to be a Banach algebra for an ideal I , I has to be closed, just like for subspaces. If A is unital and $A \neq \{0\}$, then it is certainly unique and $\|I\| \geq 1$ (why not =?? Analysis now p.128) If A has no units, we may try to isometrically embed it into a larger Banach Algebra $\hat{A}$ such that $A \subseteq \hat{A}$ becomes an ideal such that every nonzero ideal of $\hat{A}$ has a nonzero intersection with A (such an ideal is called an *essential ideal*). This can be thought as the algebraic counterpart of the notion of compactification of a topological space. For classical Banach algebra, there is usually a natural way of *adjoining a unit*, but there is always a general method of doing so that is the counter-part of one-point compactification. Take the direct sum $A \oplus k = \hat{A}$ along with the product

$$(x, \alpha)(y, \beta) = (xy + \alpha y + \beta x, \alpha\beta)$$

and the submultiplicative 1-norm. Then $A \oplus k$ is a unital Banach algebra with $I(0, 1)$ and contains A as an ideal of co-dimension 1.

Definition 7.1.2: Regular Representation

Let A be a Banach Algebra. Then the (left) *regular representation* $\rho : A \rightarrow B(A)$ by $\rho(x)(y) = xy$ for $x, y \in A$.

It should be verified that ρ is a norm decreasing algebra homomorphism, and if A is unital, the inequalities

$$\|A\| \leq \|\rho(A)\| \|I\| \leq \|A\| \|I\|$$

show that ρ is a homeomorphism. Renorming A by using the equivalent norm from $B(A)$, we may assume that $\|I\| = 1$ (if $A \neq \{0\}$), which we do.

Definition 7.1.3: Approximate Unit

A net $(E_\lambda)_{\lambda \in \Lambda}$ in the unit ball of a Banach algebra A is an *approximate unit* if

$$\lim E_\lambda x = \lim x E_\lambda = x$$

for all $x \in A$.

Example 7.1.4: Approximate Units

Can you think of approximate units for the non-unital Banach algebra $C_0(X)$, $B_0(\mathbb{H})$ and $L^1(\mathbb{R}^n)$? Show that the existence of an approximate unit guarantees that the regular representation ρ is an isometry

We shall denote $A^\times = \text{GL}(A)$ the set of all invertible elements of A (in algebra, we'd usually call these elements unital, but the term unit in functional analysis is more commonly reserved for the identity)

Lemma 7.1.5: Almost Invertible Elements

Let $x \in A$ where A is a unital Banach algebra with $\|x\| < 1$. Then $I - x \in \text{GL}(A)$ and

$$(I - x)^{-1} = \sum_{n=0}^{\infty} x^n$$

Proof:

Since the norm is submultiplicative $\|x^n\| \leq \|x\|^n$, thus the series $\sum x^n$ converges in x to an element y . Since $xy = yx = y - I$ by manipulating the above power series, it follows that $y = (I - x)^{-1}$, completing the proof.

Proposition 7.1.6: Properties Of $GL(A)$

Let A be a unital algebra and $GL(A)$ be it's multiplicative group. Then $GL(A)$ is an open subset of A , an the map $x \mapsto x^{-1}$ is a homeomorphism of $GL(A)$

Proof:

Let $x \in GL(A)$ and $y \in A$, Then by lemma 7.1.5

$$y = x - (x - y) = x(I - x^{-1}(x - y)) \in GL(A)$$

provided that $\|A^{-1}(A - B)\| < 1$. In particular, the ball $B_\epsilon(x)$ is contained in $GL(A)$ for $\epsilon < \|x^{-1}\|^{-1}$, which proves that $GL(A)$ is open. If $y \rightarrow a$, we see from the above equation and lemma 7.1.5 that $y^{-1} \mapsto a^{-1}$, thus inverting is a continuous process and therefore a homeomorphism since $(A^{-1})^{-1} = A$, completing the proof.

From ehre on out, all Banach algebra will be over $\mathbb{C}$, i.e. all Banach algebras will be *complex*. See exercise ref:HERE on the complexification of general real Banach algebras (or E.2.1.13 in Analysis Now).

Definition 7.1.7: Spectrum

Let A be a complex unital Banach algebra. Then for every $x \in A$, define the *spectrum* of x to be the set

$$\sigma(x) = \{\lambda \in \mathbb{C} \mid \lambda I - x \notin GL(A)\}$$

The complement of $\sigma(x)$ is the *resolvent set*.

On the resolvent set, we can define the resolvent function

$$R(x, \lambda) = (\lambda I - x)^{-1}$$

Definition 7.1.8: Spectral Radius

Let A be a complex unital Banach algebra, $x \in A$, and $\sigma(x)$ be its spectrum. Then the smallest $r \geq 0$ such that $\sigma(x) \subseteq B_r(0)$ is called the *spectral radius* of x , and is denoted $r(x)$. Thus

$$r(x) = \sup \left\{ |\lambda| \mid \lambda \in \sigma(x) \right\}$$

Lemma 7.1.9: Build-Up Lemma I

Let $x \in A$ and $f(z) = \sum \alpha_n z^n$ be a holomorphic function in a region contained in the closed disk $B_r(0)$ with $\|x\| \leq r$. then we can define $f(x) = \sum \alpha_n x^n$ in A . Moreover, if $\lambda \in \sigma(x)$ and $|\lambda| \leq r$, then $f(\lambda) \in \sigma(f(x))$

Proof:

The series $f(x)$ is absolutely convergence ($\sum |\alpha_n| \|x\|^n < \infty$), so certainly $f(x) \in A$. Moreover

$$\begin{aligned} f(\lambda)I - f(x) &= \sum \alpha_n (\lambda^n I - x^n) \\ &= (\lambda I - x) \sum \alpha_n P_{n-1}(\lambda, x) \\ &= (\lambda I - x)y \end{aligned}$$

where $P_{n-1}(\lambda, x) = \sum_{k=0}^{n-1} \lambda^k A^{n-k-1}$ so that $\|P_{n-1}(\lambda, x)\| \leq nr^{n-1}$, which implies the series of polynomials converges in A to an element y commuting with x . We see from this computation that if $f(\lambda)I - f(x) \in \text{GL}(A)$ with inverse v , then yv will be the inverse of $\lambda I - x$, so that $\lambda I - x \in \text{GL}(A)$

Lemma 7.1.10: Build-Up Lemma II

For every x in A , we have

$$r(x) \leq \inf \|A^n\|^{1/n}$$

Proof:

If $|\lambda| > \|A\|$, then by lemma 7.1.5

$$(\lambda I - x)^{-1} = \lambda^{-1}(I - \lambda^{-1}x)^{-1} = \sum \lambda^{-n-1}x^n$$

which shows that $r(x) \leq \|x\|$. Now if $\lambda \in \sigma(x)$, then $\lambda^n \in \sigma(x^n)$ by lemma 7.1.9. Thus $|\lambda^n| \leq \|x^n\|$ from what we've just shown. It follows that $r(x) \leq \|x^n\|^{1/n}$ for every n , completing the proof.

Theorem 7.1.11: Spectrum Is Compact

Let $x \in A$ be an element of a complex unital Banach algebra A . Then the spectrum of x , $\sigma(x)$ is a compact nonempty subset of $\mathbb{C}$, and the spectral radius of x is the limit of the convergence sequence $\|A^n\|^{1/n}$

Proof:

Analysis now p.131, really cool proof that I should take the time to write down!

A cool consequence of this is that we get a classical result of Representation theory known as Artin-Wedderburn for algebraically closed algebras:

Corollary 7.1.12: Algebraically Closed Division Algebra $\mathbb{C}$

Let A be a unital Banach algebra over $\mathbb{C}$. Then if A is a division ring, $A = \mathbb{C}$

Proof:

For each $x \in A$, there is some $\lambda \in sp(x)$ by the above theorem. Thus, $\lambda I - x \notin GL(A)$, hence $x = \lambda I$.

7.2 Continuous Functional Calculus

Let $T \in B(\mathbb{H})$ and consider $\sigma(T)$. Consider $B(\sigma(T))$, which is a $*$ -algebra where $*$ is complex conjugation. We may consider this algebra with the supremum norm. In this section, we'll be looking into various sub $*$ -algebra of $B(\sigma(T))$, for example the polynomials functions on $\sigma(T)$ or the continuous functions on $\sigma(T)$

Definition 7.2.1: Functional Calculus

Let $T \in B(\mathbb{H})$. Then a *functional calculus* of T is a continuous $*$ -homomorphism from $*$ -algebra of functions $(\sigma(T) \rightarrow \mathbb{C})$ to $B(\mathbb{H})$ which sends the constant function 1 to the identity operator, and sends the identity function $x \mapsto x$ to T

Since it's a $*$ -homomorphism, we have

$$\varphi(\bar{f}) = \varphi(f)^*$$

since we have the supremum norm and the $*$ -homomorphism is continuous, $f_n \rightarrow f$ in the supremum norm then $f_n(T) \rightarrow f(T)$ in the norm topology of $B(\mathbb{H})$.

(more here, this is really cool!)